



**GREEN
CLIMATE
FUND**

Meeting of the Board
28 – 31 March 2022
Songdo, Incheon, Republic of Korea
Provisional agenda item 13

GCF/B.31/02/Add.02

11 March 2022

Consideration of funding proposals - Addendum II

Funding proposal package for FP183

Summary

This addendum contains the following seven parts:

- a) A funding proposal titled "Inclusive Green Financing Initiative (IGREENFIN I): Greening Agricultural Banks & the Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Great Green Wall (GGW) Countries – Phase I";
- b) No-objection letter issued by the national designated authority(ies) or focal point(s);
- c) Environmental and social report(s) disclosure;
- d) Secretariat's assessment;
- e) Independent Technical Advisory Panel's assessment;
- f) Response from the accredited entity to the independent Technical Advisory Panel's assessment; and
- g) Gender documentation.

Table of Contents

Funding proposal submitted by the accredited entity	3
No-objection letter issued by the national designated authority(ies) or focal point(s)	121
Environmental and social report(s) disclosure	146
Secretariat's assessment	149
Independent Technical Advisory Panel's assessment	170
Response from the accredited entity to the independent Technical Advisory Panel's assessment	181
Gender documentation	183

Funding Proposal

Project/Programme title:	Inclusive Green Financing Initiative (IGREENFIN I): Greening Agricultural Banks & the Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Great Green Wall (GGW) countries - Phase I
Country(ies):	IGREENFIN I [Components 1 and 2]: Burkina Faso, Côte d'Ivoire, Ghana, Mali and Senegal GCF GGW Regional Support Programme [Component 3]: Burkina Faso, Chad, Djibouti, Eritrea, Ethiopia, Mali, Mauritania, Niger, Nigeria, Senegal and Sudan
Accredited Entity:	International Fund for Agricultural Development (IFAD).
Date of first submission:	<u>[2021/04/19]</u>
Date of current submission	<u>[2021/11/17]</u>
Version number	<u>[V.004]</u>



Contents

Section A	PROJECT / PROGRAMME SUMMARY
Section B	PROJECT / PROGRAMME INFORMATION
Section C	FINANCING INFORMATION
Section D	EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA
Section E	LOGICAL FRAMEWORK
Section F	RISK ASSESSMENT AND MANAGEMENT
Section G	GCF POLICIES AND STANDARDS
Section H	ANNEXES

Note to Accredited Entities on the use of the funding proposal template

- Accredited Entities should provide summary information in the proposal with cross-reference to annexes such as feasibility studies, gender action plan, term sheet, etc.
- Accredited Entities should ensure that annexes provided are consistent with the details provided in the funding proposal. Updates to the funding proposal and/or annexes must be reflected in all relevant documents.
- The total number of pages for the funding proposal (excluding annexes) **should not exceed 60**. Proposals exceeding the prescribed length will not be assessed within the usual service standard time.
- The recommended font is Arial, size 11.
- Under the [GCF Information Disclosure Policy](#), project and programme funding proposals will be disclosed on the GCF website, simultaneous with the submission to the Board, subject to the redaction of any information that may not be disclosed pursuant to the IDP. Accredited Entities are asked to fill out information on disclosure in section G.4.

Please submit the completed proposal to:

fundingproposal@gcfund.org

Please use the following name convention for the file name:

FP-IFAD-Multi-country-2021/03/29

A. PROJECT/PROGRAMME SUMMARY																							
A.1. Project or programme	Programme	A.2. Public or private sector	Public																				
A.3. Request for Proposals (RFP)	If the funding proposal is being submitted in response to a specific GCF <u>Request for Proposals</u> , indicate which RFP it is targeted for. Please note that there is a separate template for the Simplified Approval Process and REDD+. Not applicable																						
A.4. Result area(s)	Check the applicable <u>GCF result area(s)</u> that the <u>overall</u> proposed project/programme targets. For each checked result area(s), indicate the estimated percentage of <u>GCF budget</u> devoted to it. The total of the percentages when summed should be 100%. <table border="1"> <thead> <tr> <th>Mitigation: Reduced emissions from:</th> <th>GCF contribution:</th> </tr> </thead> <tbody> <tr> <td>☒ Energy access and power generation:</td> <td><u>20%</u></td> </tr> <tr> <td>☐ Low-emission transport:</td> <td><u>15%</u></td> </tr> <tr> <td>☐ Buildings, cities, industries, and appliances:</td> <td></td> </tr> <tr> <td>☒ Forestry and land use:</td> <td></td> </tr> <tr> <td colspan="2">Adaptation: Increased resilience of:</td> </tr> <tr> <td>☒ Most vulnerable people, communities, and regions:</td> <td><u>25%</u> <u>25%</u></td> </tr> <tr> <td>☒ Health and well-being, and food and water security:</td> <td><u>15%</u></td> </tr> <tr> <td>☐ Infrastructure and built environment:</td> <td></td> </tr> <tr> <td>☒ Ecosystem and ecosystem services:</td> <td></td> </tr> </tbody> </table>			Mitigation: Reduced emissions from:	GCF contribution:	☒ Energy access and power generation:	<u>20%</u>	☐ Low-emission transport:	<u>15%</u>	☐ Buildings, cities, industries, and appliances:		☒ Forestry and land use:		Adaptation: Increased resilience of:		☒ Most vulnerable people, communities, and regions:	<u>25%</u> <u>25%</u>	☒ Health and well-being, and food and water security:	<u>15%</u>	☐ Infrastructure and built environment:		☒ Ecosystem and ecosystem services:	
Mitigation: Reduced emissions from:	GCF contribution:																						
☒ Energy access and power generation:	<u>20%</u>																						
☐ Low-emission transport:	<u>15%</u>																						
☐ Buildings, cities, industries, and appliances:																							
☒ Forestry and land use:																							
Adaptation: Increased resilience of:																							
☒ Most vulnerable people, communities, and regions:	<u>25%</u> <u>25%</u>																						
☒ Health and well-being, and food and water security:	<u>15%</u>																						
☐ Infrastructure and built environment:																							
☒ Ecosystem and ecosystem services:																							
A.5. Expected mitigation impact	5,637,184 tCo2eq over project life span	A.6. Expected adaptation impact	Direct beneficiaries: 378,600 (~50 % female) Indirect beneficiaries: 2,494,000 (~50 % female) % of population (Adaptation only) Direct beneficiaries: 0.23% Indirect beneficiaries: 1.75%																				
A.7. Total financing (GCF + co-finance)	177,481,678 Euros	A.9. Project size	Medium (Upto USD 250 million)																				
A.8. Total GCF funding requested	<u>104,415,189</u> Euros																						
A.10. Financial instrument(s) requested for the GCF funding	<i>Mark all that apply and provide total amounts. The sum of all total amounts should be consistent with A.8.</i> ☒ Grant 30,444,189 EUR ☐ Equity <u>Enter number</u> ☒ Loan 73,971,000 EUR ☐ Results-based payment <u>Enter number</u> ☐ Guarantee <u>Enter number</u>																						
A.11. Implementation period	6 years	A.12. Total lifespan	<u>25 years</u>																				
A.13. Expected date of AE internal approval	10/3/2022	A.14. ESS category :	B or I-2																				
A.15. Has this FP been submitted as a CN before?	Yes ☒ No ☐	A.16. Has Readiness or PPF support been used to prepare this FP?	Yes ☒ No ☐																				

A.17. Is this FP included in the entity work programme?	Yes ☒ No ☐	A.18. Is this FP included in the country programme?¹	Yes ☒ No ☐
A.19. Complementarity and coherence	Does the project/programme complement other climate finance funding (e.g., GEF, AF, CIF, etc.)? If yes, please elaborate in section B.1. Yes ☒ No ☐		
A.20. Executing Entity information	1. In this funding proposal, “Executing Entity” (EE) refers to any entity that will sign a Subsidiary Agreement with IFAD to execute, carry-out or implement an activity funded by this programme. IFAD has assessed the capacity of each EE in accordance with IFAD policies and procedures (see Section B.4 for information on such policies and procedures and the track record of selected EEs). 2. IFAD (AE) acts as EE and will implement solely the activities related to the programme coordination, supervision and reporting to the GCF for IGREENFIN I through a Regional Coordination Unit (RCU). IFAD will recruit the RCU staff (IFAD staff) in charge of this programme's coordination. IFAD will also carry out all activities related to the GCF GW Regional Support Programme, which are detailed in Component 3 under Output 3.1. Knowledge Management & Communications; Output 3.2 Innovation and digital transformation technologies mapped and ecosystem built and the Support Unit & Governance 3. National Executing Entities (NEEs): The Republic of Burkina Faso, the Republic of Côte d'Ivoire, the Republic of Ghana, the Republic of Mali and the Republic of Senegal, represented by their respective ministries of economy and finance, are the NEEs. The relevant on lending agreements shall incorporate and impose the relevant obligations and requirements established in the Accreditation Master Agreement (AMA) and Funded Activity Agreement (FAA) to the NEEs. The NEEs will be recipients of the GCF financing and guarantee the repayment of the GCF loan portion. Each NEE will sign an on lending agreement with the Local National Agricultural Banks (LNABs) to pass on the GCF loan proceeds. The NEEs will sign the relevant legal agreements with each of the five LNABs (the Agricultural Bank of Burkina Faso, the National Investment Bank of Côte d'Ivoire, ARB Apex Bank Limited of Ghana, the Agricultural Bank of Mali and the Agricultural Bank of Senegal) on the implementation of the GCF loan proceeds. In this programme, the agricultural banks are referred to as Local National Agricultural Banks (LNABs). The NEEs will be responsible for the implementation of activities 1.1.1, 1.1.2 and 1.1.3 under component 1 on behalf on the NEEs. 4. For the implementation of the GCF grant proceeds (Component 2), the NEEs will make arrangements with their respective Ministry of Agriculture who will act through IFAD baseline investments programme management units (PMUs). These ministries are: the Ministry of Agriculture, Hydro-Agricultural Development and Mechanization of Burkina Faso; the Ministry of Agriculture and Rural Development of Côte d'Ivoire; the Ministry of Agriculture of Mali; the Ministry of Agriculture of Ghana, and the Ministry of Agriculture and Rural Equipment of Senegal. They will be responsible for the implementation of the activities of Output 2.1 and 2.2. under Component 2. 5. The NEEs will make the decisions on the use of the loan and grant proceeds with no-objection from the AE.		
A.21. Executive summary (max. 750 words, approximately 1.5 pages)			

¹ All countries have a country work programme.

6. The eleven countries² of the Great Green Wall (GGW)³, Côte d'Ivoire and Ghana⁴ are among the world's poorest and most vulnerable countries to climate change⁵. They ranked at the bottom of the Human Development Index in 2020⁶ and a large majority of their population lacks access to employment, basic health care and education and natural resources, which contributes to food insecurity, youth migration and conflict⁷. Agriculture, livestock and forestry activities are the foundations of their economies and more than 70 per cent of rural communities depend directly on rainfed agriculture (crops, livestock, fishery and forestry)⁸. The region's agriculture, forestry and other land use (AFOLU) and energy sectors are responsible for just under a quarter of anthropogenic greenhouse gas (GHG) emissions generated mainly by deforestation (wood and clearing land for agriculture), livestock raising, soil and nutrient management⁹ and unsustainable rice production¹⁰. The agricultural sector is extremely vulnerable to climate change and climate variability due to these countries' geographical location and their socioeconomic and technological characteristics¹¹. Over recent decades, climate change has increased the frequency, intensity and duration of droughts and floods, temperatures, desertification, water stress and soil erosion, all of which reduce agricultural productivity and food security. Shocks from climate change and adverse weather conditions – resulting in the loss of assets, crops and livestock, disruptions in value chains and soaring food prices – and COVID-19 are pushing millions of smallholder farmers further into poverty¹².
7. Climate projections indicate that precipitation levels will continue to decrease, while temperatures are expected to increase between 1 and 1.72°C for the 2031-2050 period compared to the 1986-2005 reference period¹³. According to these projections, agricultural production is expected to drop by at least 20 per cent in the region, which will reduce food availability and farmers' incomes and fuel competition over diminishing natural capital, greater instability and migration, especially of rural youth¹⁴. The climate models produced using the IFAD Climate Adaptation in Rural Development Assessment Tool¹⁵ indicated that the production of the main crops in the targeted countries will be severely affected by future climate change: average millet production is predicted to decrease by 10 per cent, groundnut by 11 per cent, and rice by 8 per cent over the next 20 years. This will have negative impacts on the 50 million¹⁶ people in the region, including smallholders who already experience high levels of food insecurity and poverty¹⁷. It will also augment these countries' dependence on grain imports, which has been growing over the past three decades, placing a heavy burden on government resources. According to the Ecological Threat Report 2021, climate change also has far-reaching implications for national and regional economic, political and social stability and security in the GGW area, which will increasingly transcend the capacity of each country to manage these issues alone¹⁸. Hence, the urgent need for a regional approach that integrates and coordinates the efforts of multiple stakeholders, including governments, private investors, public and private banks, insurance companies, project developers, cooperatives, producers' organizations, technical institutions and international and regional organizations.
8. Political instability, together with other non-climate stressors such as environmental and land degradation caused by unsustainable land use, overgrazing, pressures placed on communities by climate change, urbanization and pollution, conflict over land tenure, the destruction of biodiversity and the spread of climate-induced pests are all processes that will become more pronounced as climate change progresses¹⁹. Combined, these negative impacts of climate change are exacerbated when smallholders have low adaptive capacity and limited financial resources to withstand disasters and losses²⁰.
9. Climate change also destabilizes local markets, curbs economic growth and investment, and as a result, job creation, particularly for youth and women²¹. It will thus make it more difficult to overcome one of the primary barriers limiting the growth of rural smallholder farmers' organizations, cooperatives and micro, small and medium-sized enterprises (MSMEs) in the GGW: the lack of access to credit²² and technical assistance. Rural financing is key to improving on-farm productivity by providing working capital, allowing for investment in post-harvest value aggregation of products and processing systems and the acquisition of new equipment²³ and, when combined with technical assistance, strengthening capacity and awareness on climate risks and their impacts on the production, trade and marketing of agricultural commodities.
10. To address these increasingly complex and interrelated challenges and support the transition towards climate resilient, low emission agriculture, the Great Green Wall Initiative (GGWI) was launched in 2007 by the African Union. The GGWI's initial objectives were to address land degradation, climate change adaptation and mitigation, and protect biodiversity and forests. Under the GGWI, environmental aspects and natural capital have been integrated into the development agenda and a multi-stakeholder dialogue has been established to ensure country ownership. It has also created opportunities for the deployment of scaled-up investments based on successful experiences on the ground. There is a need to accelerate investment in the region to address these issues, especially since the onset of the Covid-19 pandemic, which has had severe impacts on the region's economic development. In a post-Covid-19 context, the ambition of the GGWI is to pursue a green growth recovery by restoring 100 million ha of currently degraded land, sequestering 250 million tons of carbon and creating 10 million green jobs by 2030. It will support communities along the GGW to: i) increase the fertility of the land, one of humanity's most precious natural assets; ii) create economic opportunities for the world's youngest population; iii) build food security for the millions that go hungry every day; iv) build climate resilience in a region where

temperatures are rising faster than anywhere else on Earth, and v) grow a new world wonder spanning close to 8,000 km across Africa. As Sahelian countries are struggling with limited budgets and funding in a COVID context, UNCCD set up the GGW Accelerator to help meet financial requirements and turbo charge the achievement of its goals with an estimated budget target of US\$33 billion from private, national and international sources. The GGW Accelerator has five pillars:

- Pillar 1: Investment in small and medium-sized farms and strengthening of value chains, local markets, organization of exports
- Pillar 2: Land restoration and sustainable management of ecosystems
- Pillar 3: Climate resilient infrastructures and access to renewable energy
- Pillar 4: Favourable economic and institutional framework for effective governance, sustainability, stability and security
- Pillar 5: Capacity-building.

11. Prior to this, in 2018, the Government of Niger requested IFAD's support to develop the GCF "Inclusive Green Financing for Climate Resilient and Low Emission Smallholder Agriculture" (IGREENFIN) project. In November 2019, the Simplified Approval Process IFAD-Niger IGREENFIN project (SAP012) was approved at the 24th meeting of the GCF Board. **The main objective of IGREENFIN I** is to build and scale up the resilience and adaptive capacity of farmers' organizations (FOs), cooperatives and micro, small and medium-sized enterprises (MSMEs) in Niger by removing key barriers to farmers' access to financial and non-financial services that support the adoption of best climate change adaptation and mitigation practices and solutions. It will contribute to reducing GHG emissions through the promotion of sustainable forest, land, water and energy management and use in selected agricultural value chains, as well as renewable energy technologies (RETs) to power energy efficient processing, storage and packaging equipment and irrigation systems.
12. In early 2020, five West African countries (Burkina Faso, Côte d'Ivoire, Ghana, Mali and Senegal) with similar development challenges requested IFAD's support to replicate the IGREENFIN Niger project in their countries. In response, IFAD initiated the first phase of an IFAD-funded regional approach to draft a preliminary regional concept note (CN) and annexes (Feasibility Studies, Gender Assessment, Preliminary ESMF), which were submitted in July 2020 to the GCF. After the first review of the CN and the virtual Ministerial Conference on the Great Green Wall held on 7 September 2020, it was suggested that the project's geographical scope be expanded to all countries of the GGW, plus Côte d'Ivoire and Ghana. The latter two have been engaged in the talks to replicate the IGREENFIN I approach since the beginning of 2020. In coordination with UNCCD and the GCF, IFAD reached out to all the GGW countries to assess their interest in participating in the IGREENFIN I programme and supporting the elaboration of the GCF GGW Regional Support Programme.

² Burkina Faso, Chad, Djibouti, Eritrea, Ethiopia, Mali, Mauritania, Niger, Nigeria, Senegal and Sudan.

³ The Great Green Wall (GGW) is an African-led movement with an epic ambition to grow an 8,000 km natural wonder of the world across the entire width of Africa. The goal is to restore 100 million hectares of degraded land, sequester 250 million tonnes of carbon and create 10 million green jobs in rural areas across the Sahel by 2030. For more information, see: <https://www.greatgreenwall.org/about-great-green-wall>

⁴ African Forestry and Wildlife Commission (2020), Twenty-second Session: Forests and Wildlife: Africa's diversity for shared prosperity and security, FAO, <http://www.fao.org/3/ca8011en/ca8011en.pdf>

⁵ IPCC (2021), AR6 Climate Change 2021: The Physical Science Basis, Sixth Assessment Report, <https://www.ipcc.ch/report/ar6/wg1/>

⁶ UNDP (2020), Human Development Report, <http://hdr.undp.org/en/content/latest-human-development-index-ranking>

⁷ IEP (2020), Global Peace Index Report 2020, Measuring Peace in a Complex World, <https://imctc.org/en/eLibrary/INTReports/Pages/Reports15012021.aspx>

⁸ OECD/FAO (2016), "Agriculture in Sub-Saharan Africa: Prospects and challenges for the next decade", in *OECD-FAO Agricultural Outlook 2016-2025*, OECD Publishing, Paris, <http://www.fao.org/3/BO092E/BO092E.pdf>

⁹ IPCC (2019), "Food Security" in *Climate Change and Land: an IPCC special report on climate change, desertification, land degradation, sustainable land management, food security, and greenhouse gas fluxes in terrestrial ecosystem*, <https://www.ipcc.ch/srccl/chapter/chapter-5/>

¹⁰ FAO, 207, Rice and climate change, Factsheets

¹¹ OECD (2018), Climate Change Profile: West African Sahel, <https://www.government.nl/binaries/government/documents/publications/2019/02/05/climate-change-profiles/West+African+Sahel.pdf>

¹² IPCC (2019), "Food Security".

¹³ Climate Analytics, 2019.

¹⁴ Cordaid (2020), Saving lives and reducing conflict in the Sahel region, Relief Web, <https://reliefweb.int/report/burkina-faso/saving-lives-and-reducing-conflict-sahel-region>

¹⁵ Information on IFAD's Climate Adaptation in Rural Development (CARD) Assessment Tool is available from <https://www.ifad.org/en/web/knowledge/-/publication/climate-adaptation-in-rural-development-card-assessment-tool>

¹⁶ Marcus Arcanjo (2019), Risk and Resilience: Climate change and Instability in the Sahel, Climate Institute, 24 October, <https://climate.org/risk-and-resilience-climate-change-and-instability-in-the-sahel/>

¹⁷ FAO (2016), Strengthening resilience to food and nutrition insecurity in the Sahel and Western Africa: Good Practices Booklet, <https://www.fao.org/3/i6263en/i6263EN.pdf>

¹⁸ Institute for Economics & Peace (2021). Ecological Threat Report 2021: Understanding Ecological Threats, Resilience and Peace, Sydney, October, <http://visionofhumanity.org/resources>

¹⁹ FAO (2017), The future of food and agriculture – Trends and challenges. Rome, <http://www.fao.org/3/i6583e/i6583e.pdf>

²⁰ FAO (2015), Climate change and food security: risks and responses, <http://www.fao.org/3/i5188e/i5188e.pdf>

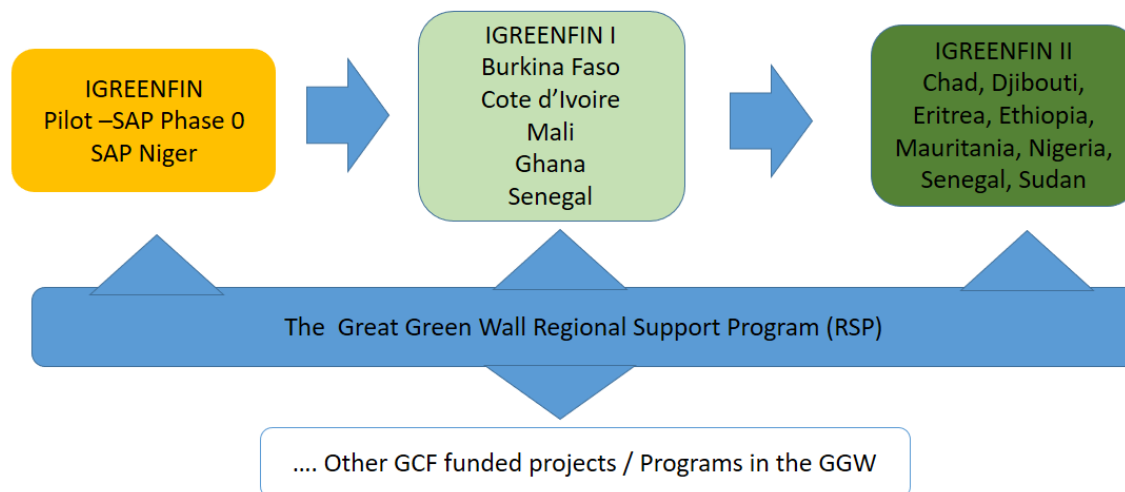
²¹ IMF (2019), Long-Term Macroeconomic Effects of Climate Change: A Cross-Country Analysis.

²² IFAD (2021), IFAD Inclusive Financial Services: Portfolio Stocktaking, Rome, https://www.ifad.org/documents/38714170/43254162/IFS_Web.pdf/150d86c4-a444-26a3-2412-a386bbca3e3e?t=1623839327713

²³ Douglas Mataji and Oniankitan Grégoire Agai (2019), Access to finance for inclusive agri-business development, Paper 11, 2Scale, https://www.2scale.org/upload/650938_2SCALE_paper11.pdf.

13. To roll out this regional approach, IGREENFIN I and the Regional Support Programme will be implemented using a phased approach:
- (i) **Phase 0:** The GCF Board approved the SAP012 (IGREENFIN in Niger) in November 2019. This pilot project was instrumental in providing the design template for the investment model for IGREENFIN and for utilizing this to expand to Phase 1 and Phase 2 countries.
 - (ii) **Phase 1** will be implemented in the first set of three GGW countries (Burkina Faso, Mali and Senegal) plus Côte d'Ivoire and Ghana.
 - (iii) **Phase 2** will be implemented in the second set of seven GGW countries (Chad, Djibouti, Eritrea, Ethiopia, Mauritania, Nigeria and Sudan).
 - (iv) **The Regional Support Programme (RSP)** will provide support (on enhancing knowledge management and exchanges, reporting and monitoring; piloting digital transformation technologies and promoting private sector engagement) only for GCF-funded projects in the GGW submitted in the future by all Accredited Entities, including projects developed under IGREENFIN phases 0, I and II.

Figure 1: Diagram illustrating the phased approach for IGREENFIN and the GGW Regional Support Program



14. IFAD's long-term vision is to implement and expand IGREENFIN (IFAD's climate finance brand) and the Regional Support Programme in African countries outside the GGW area in order to tackle climate change and support smallholder farmers' access to finance and technical assistance. This Funding Proposal is comprised only of IGREENFIN I and the Regional Support Programme. For IGREENFIN II, a separate funding proposal will be prepared; the process of designing the IGREENFIN II funding proposal has been launched.
15. The proposed IGREENFIN I programme has three components (described in detail in Section B.3.) and is structured as follows:
16. **Component 1: Green Business Financing Facility** – will be established to provide concessional loans to foster best adaptation and mitigation practices for green businesses within the selected agricultural value chains in each target country (refer to Section B.1, Table 8). Operated by local national agricultural banks (LNABs) – the Agricultural Bank of Burkina Faso, the National Investment Bank of Côte d'Ivoire, the ARB Apex Bank Limited of Ghana, the Agricultural Bank of Mali and the Agricultural Bank of Senegal – the facility will offer special lines of financing for green business projects prepared by farmer's organizations (FOs), women and youth organizations, cooperatives and MSMEs (including agribusiness dealers and solar operators). The proposed investment criteria and list of specific eligible green business projects are presented in Annex 21 -Operations Manual. This component will contribute simultaneously to pillar 1 (Investment in small and medium-sized farms and strengthening of value chains, local markets and organization of exports), pillar 2 (Sustainable management of ecosystems and land restoration), pillar 3 (Climate resilient infrastructures and access to renewable energy) and pillar 4 (Favourable economic and institutional framework for effective governance, sustainability, stability and security) of the Great Green Wall Accelerator. In this funding proposal, the term "**green business**" refers to investments in the selected agricultural value chains that promote climate resilient, low carbon practices in line with the GGWI objectives and the target countries' climate change adaptation and mitigation agenda.

17. **Component 2: Technical Assistance Facility (TAF)** – will be created to address capacity, knowledge and policy gaps hindering the uptake of green business projects in the selected countries. It will target LNABs, central banks and clients (FOs, MSMEs and cooperatives), with a special emphasis on women and youth. The TAF will provide support for specific technical assistance to the following Direct Access Entities: i) the Agricultural Bank of Senegal, which is to serve as a GCF accredited pilot NEE for the other targeted agricultural banks; ii) Attijariwafa Bank (an accredited commercial bank), whose role will be to crowd in additional financing as a strategic partner of the selected LNABs, and iii) the Centre de Suivi Ecologique (CSE), a strategic IGREENFIN I partner that is to perform activities 2.1.1 and 2.1.8 with all selected countries and activity 3.2 with IFAD. The TAF will provide necessary technical assistance and policy and regulatory support to the Central Bank of the West Africa Monetary Union (WAMU) for Burkina Faso, Côte d'Ivoire, Mali and Senegal, and the Central Bank of Ghana on the creation of an enabling environment for financing the uptake of green business projects (refer to section B.2 for the theory of change) and CSE. This component will contribute to pillar 2 (Sustainable management of ecosystems and land restoration), pillar 3 (Climate resilient infrastructures and access to renewable energy) and pillar 5 (Capacity-building) of the Great Green Wall Accelerator.
18. **Component 3: GGW Regional Support Programme** - This component will increase the collective impacts of the individual GCF projects and programmes (including IGREENFIN I and II) through two outputs: i) **Output 3.1.** Enhanced knowledge management and exchanges to accelerate the uptake of good practices, increase learning and inform policy and investments across GCF and other projects, and ii) **Output 3.2.** Innovation and digital transformation technologies are mapped and a digital and innovation ecosystem built.
19. IGREENFIN I seeks EUR 73, 971,000 million in loans and EUR 30,444,189 million in technical assistance grants from the GCF. Additional co-financing is being mobilized from IFAD (EUR 37, 724, 790 million), the Islamic Development Bank (EUR12,328,500 million) and the African Development Bank (EUR 14,794,500 million) and Local National Agricultural Banks (EUR 8,219,000 million). The GCF resources will be innovatively structured to stimulate and create a market for green business and will provide additional climate finance to complement IFAD baseline investments²⁴. The programme will support: i) the acceleration of the transfer of climate resilient, low carbon technologies; ii) the development of innovative green business models for agriculture value chain projects, pipeline planning and deployment, and iii) the initial alignment of the targeted LNABs' portfolios and pipelines with each country's National Determined Contributions (NDCs) through capacity-building, policy dialogue and knowledge-building activities. The Funding Proposal is only for IGREENFIN Phase I and the Regional Support Programme.
20. IGREENFIN I intends to directly build the resilience and adaptive capacity of 378,600 smallholder farmers organized around 1,500 MSMEs and 2,500 FOs or cooperatives, and approximately 2,494,000 indirect beneficiaries in the five selected countries (Burkina Faso, Côte d'Ivoire, Ghana, Mali and Senegal). The programme is expected to reduce 225,487 tCO_{2e} annually (**5,637,184** million tCO_{2e} over a 25year lifespan of investments) from AFOLU, energy use and power generation.

²⁴ IFAD baseline investments refers to IFAD projects that have or are currently being designed to mobilize additional climate finance for IGREENFIN I to complement IFAD work. The list is presented in the Table 3 on IFAD baseline investments.

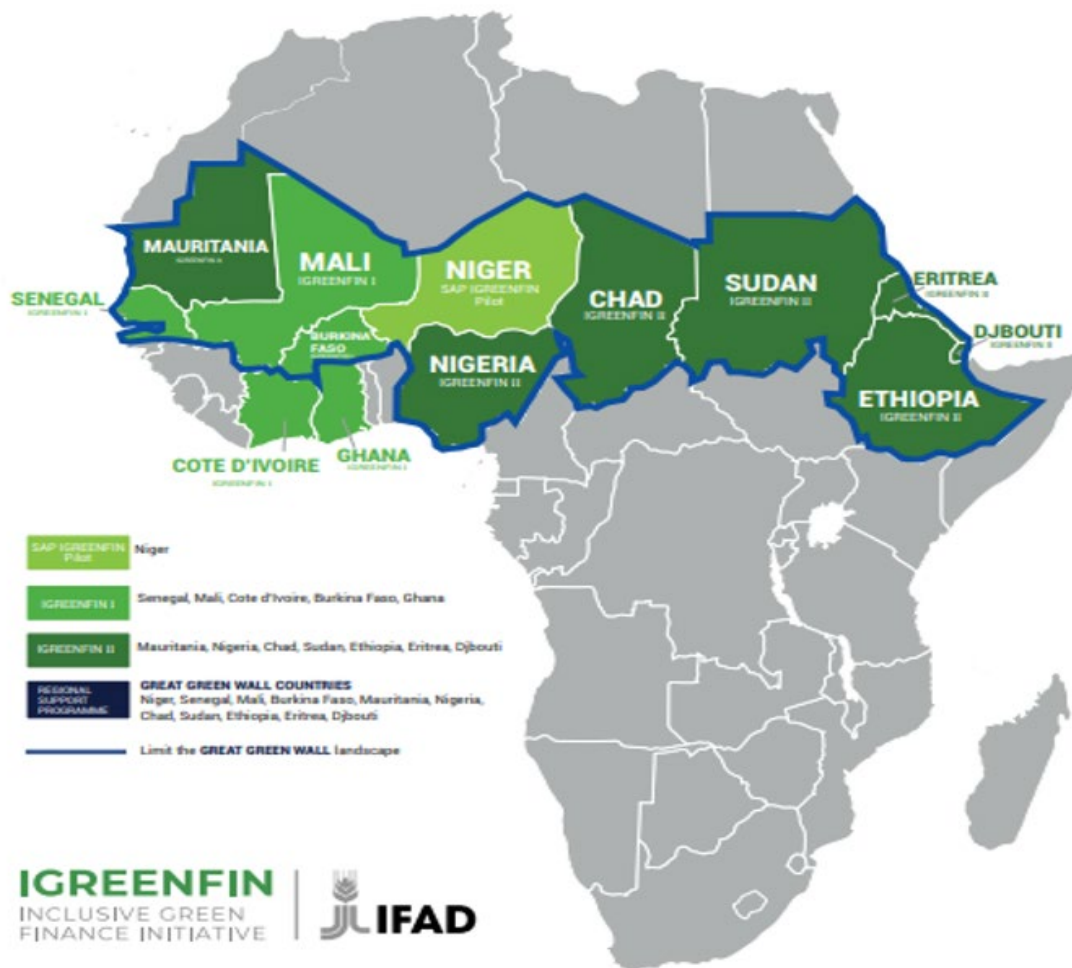
B. PROJECT/PROGRAMME INFORMATION

B.1. Climate context (max. 1000 words, approximately 2 pages)

Programme region: The Sahel, the Great Green Wall and the targeted countries

21. This section begins with a geographic representation of the 13 targeted countries of the interventions of IGREENFIN I and the Regional Support Programme (Figure 2), followed by the climate analysis for the entire region and a detailed analysis of the IGREENFIN I targeted countries.

Figure 2: The 13 targeted countries for IGREENFIN I and the Regional Support Programme



Source: Elaborated by IFAD (2021).

22. The Sahel stretches from the Atlantic Ocean to the Horn of Africa across all eleven countries of the Great Green Wall and the northern part of coastal countries such as Benin, Côte d'Ivoire, Ghana, Guinea and Togo. One of the largest semi-arid and arid areas in the world, it is wedged between the Sudanian Savannah to the south and the Sahara Desert to the north, which is expanding southward at a rate of 1-10 km per year²⁵, depending on the country. It is considered one of the poorest and most environmentally degraded regions in the world and therefore, one of the most vulnerable to climate change, as the increase in temperatures, rainfall variability and the frequency of extreme weather events will only worsen the already harsh living and growing conditions²⁶. According to the IPCC 6th Assessment Report²⁷, monsoon precipitation is projected to increase over the central Sahel and decrease over the far western Sahel, and the monsoon season is expected to have a

²⁵ USAID (2017), Climate Change Risk in West Africa Sahel: Regional Fact Sheet, 4.

²⁶ Kirsty Lewis and Carlo Buontempo (2016), "Climate Impacts in the Sahel and West Africa: The Role of Climate Science in Policy Making", *West African Papers*, No. 2, OECD Publishing, Paris, <https://doi.org/10.1787/5j1smktwjd0-en>.

²⁷ IPCC (2021), Regional Fact Sheet – Africa, Sixth Assessment Report, https://www.ipcc.ch/report/ar6/wg1/downloads/factsheets/IPCC_AR6_WGI_Regional_Fact_Sheet_Africa.pdf

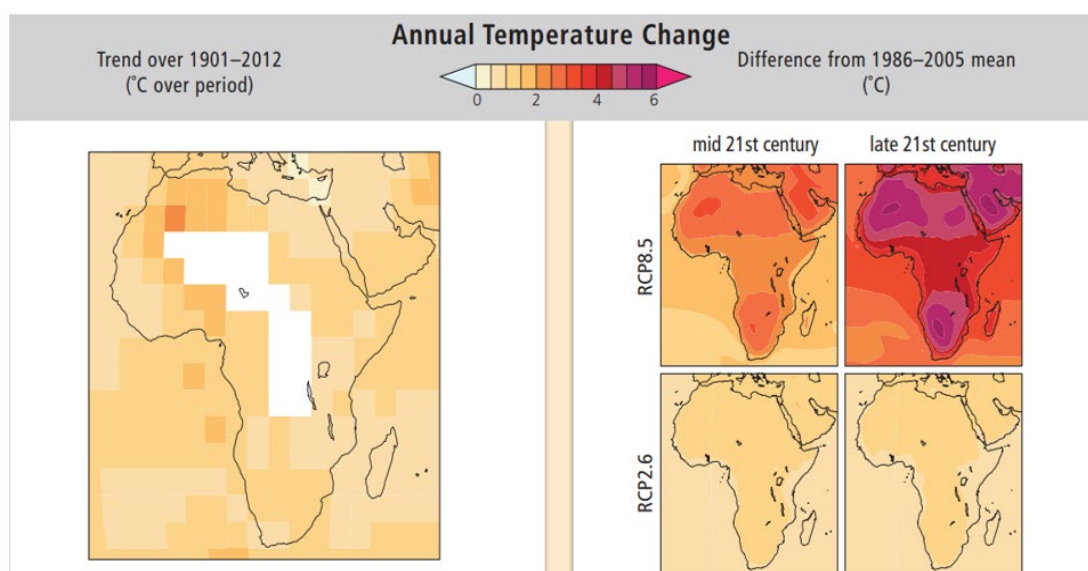
delayed onset and retreat. The IPCC also reported an observed increase in river flooding, drying and agricultural and ecological droughts, and projected increases in meteorological droughts at GWL 4° (mostly in seasonal timescales), as well as mean wind speed, heavy precipitation and pluvial flooding.

23. The Sahel is home to over 500 million people and its population is expected to double by 2040²⁸. The region has a range of ecosystems and agricultural zones, such as the savannah, semi-arid regions, sub humid Guinea and extensive coastal areas. Farmers in the Sahel produce various tradable commodities such as maize, soybean, dairy and livestock across the Guinea Savannah; rice and cassava in humid and sub-humid zones; and tree crops (cocoa, coffee, cashew, mango and oil palm), horticulture and fish in other regions. In the Sahel, 70 per cent of people rely on agriculture as a major source of food and livelihood²⁹. The region is endowed with great potential for renewable energy sources, primarily solar, which can be used to power the agricultural sector and accelerate its industrialization³⁰. It has one of the most diverse cultural bases in the world and a large, vibrant and creative youth population. The countries of the GGW are committed to achieving the goals of the Paris Climate Agreement, as expressed in their INDCs (conditional and unconditional), by strengthening the mitigation and adaptation capacity of the agricultural sector.

Climate Change in the Sahel: Observed trends and projected impacts

24. **Rising temperatures:** As GHG emissions warm the Earth's surface around the world, the impacts of this human-influenced driver of climate change are already visible in the Sahel. Temperatures there have increased nearly 1°C since 1970, at a rate twice the global average³¹. In West Africa and the Sahel, 2019 and 2020 were the seventh warmest year on record³².

Figure 3: Observed and projected changes in annual average temperature and precipitation from 1901–2012, derived from a linear trend [WGI AR5 Figures SPM.1 and 2.21]



Source: I. Niang et al. (2014), Africa, in *Climate Change 2014: Impacts, Adaptation, and Vulnerability. Part B: Regional Aspects. Contribution of Working Group II to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change*, https://www.ipcc.ch/site/assets/uploads/2018/02/WGIIAR5-Chap22_FINAL.pdf.

25. According to the IPCC's 6th Report, near surface temperatures have increased over the last 50 years for the 11 countries of the GGW. Collins (2011) shows statistically significant warming of between 0.5°C and 0.8°C between 1970 and 2010 over the Sahel region using remotely sensed data, with a greater magnitude of change in the last 20 years of the said period compared to the first two decades.

²⁸ United Nations, Department of Economic and Social Affairs, Population Division (2019), World Population Prospects 2019: Data Booklet, https://population.un.org/wpp/Publications/Files/WPP2019_DataBooklet.pdf

²⁹ GEF (2003), Sahel Integrated Lowland Ecosystem Management (SILEM), Phase I.

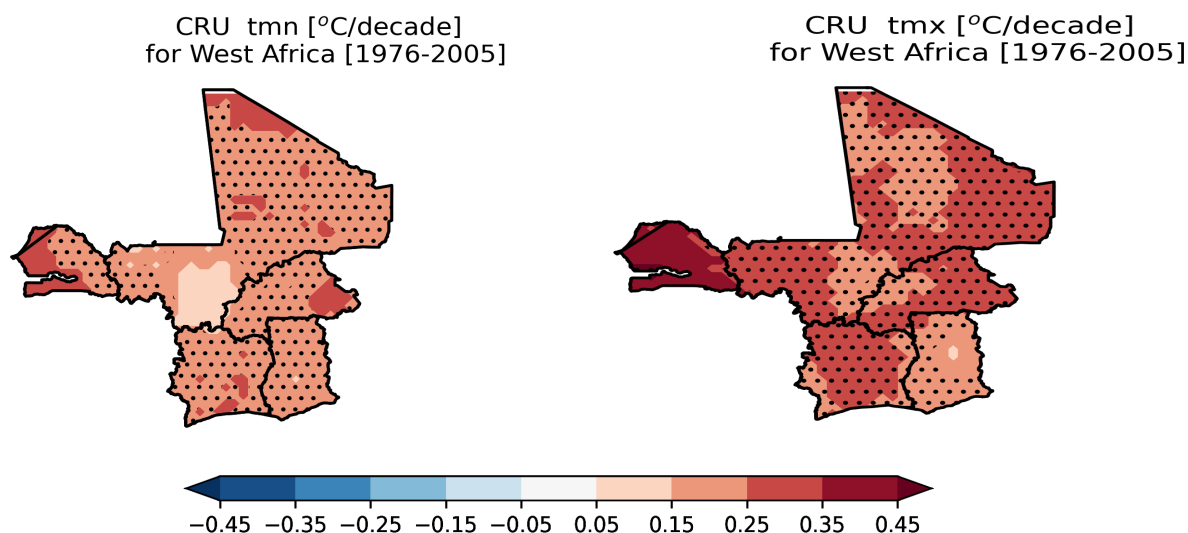
³⁰ IRENA (2018), Planning and prospects for renewable power: West Africa, https://www.irena.org/-/media/Files/IRENA/Agency/Publication/2018/Nov/IRENA_Planning_West_Africa_2018.pdf

³¹ Alec Crawford (2015), Climate Change and State Fragility in the Sahel, IISD, <https://www.iisd.org/publications/climate-change-and-state-fragility-sahel>

³² Sixth Assessment Report, IPCC, 2021- Climate Change 2021 The physical Science Basis

26. For IGREENFIN I countries, data from the last 50 years show drier conditions and hotter temperatures in Burkina Faso, Mali and Senegal and the northern part of Côte d'Ivoire and Ghana. The average temperature increase ranges between 0.6-0.8°C, which is slightly higher than the global average³³. The rate of warming is higher during the day than at night, as shown in Figure 4 below.

Figure 4: Daily trends of minimum and maximum temperatures in the region



Source: Authors, 2021

Note: Minimum temperatures are shown on the left and maximum on the right. The black dots represent regions with statistically significant trends (at a 0.05 significance level).

27. According to climate projections, average temperature increases in the Sahel region are projected to be high, with an extremely marked north-south gradient, directly affecting soils and ecosystems due to a higher level of evapotranspiration. By mid-century (2031-2050), temperatures are expected to rise between 1 and 1.72°C above those of the reference period 1986-2005³⁴. Projections for the Sahel at the end of the 21st century from both the CMIP3 GCMs (SRES A2 and A1B emission scenarios) and CMIP5 GCMs (RCP4.5 scenarios) show temperatures that are 3°C above the late 20th century baseline³⁵. It is estimated that temperature increases of more than 2°C will cause millet and sorghum yields to decrease 15-25 per cent by 2080 under RCP8.5, with substantially higher impacts on sorghum yields³⁶. The highest increase in the daily maximum temperature is expected over southern Mali and western Burkina Faso, where it is projected to reach >5°C above current temperatures by the end of the century under the RCP8.5 scenario³⁷.

28. **Shift in precipitation patterns:** In the 1970s and 1980s, the region experienced one of the most severe multi-year droughts in the last hundred years, resulting in a 30 per cent decrease in rainfall³⁸. Since the 1980s, rainfall has not returned to pre-1960s levels and droughts have become recurrent. The lengthening of the dry season and more frequent dry spells combined with less frequent and more intense rainfall over shorter wet seasons have affected the balance of the water cycle, resulting in a greater frequency of extreme rainfall events and severe floods³⁹.

³³ USAID (2017), Climate Change Risk in West Africa Sahel.

³⁴ Climate Analytics, (2021); State of Climate Action 2021

³⁵ Joseph Yaro and Jan Hesselberg (eds.), *Adaptation to Climate Change and Variability in Rural West Africa*, Springer International Publishing, Switzerland.

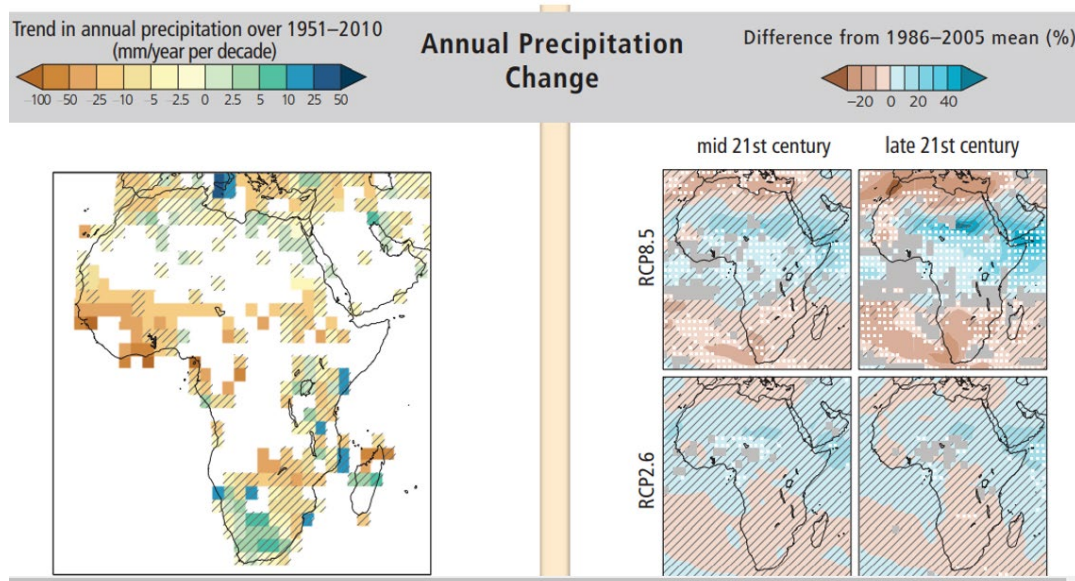
³⁶ Ibid.

³⁷ Ibid.

³⁸ IPCC (2018), Summary for Policymakers: Global Warming of 1.5°C. An IPCC Special Report on the impacts of global warming of 1.5°C above pre-industrial levels and related global greenhouse gas emission pathways, in the context of strengthening the global response to the threat of climate change, sustainable development, and efforts to eradicate poverty [Masson-Delmotte et al. (eds.)]. World Meteorological Organization, Geneva, Switzerland, 32 pp.

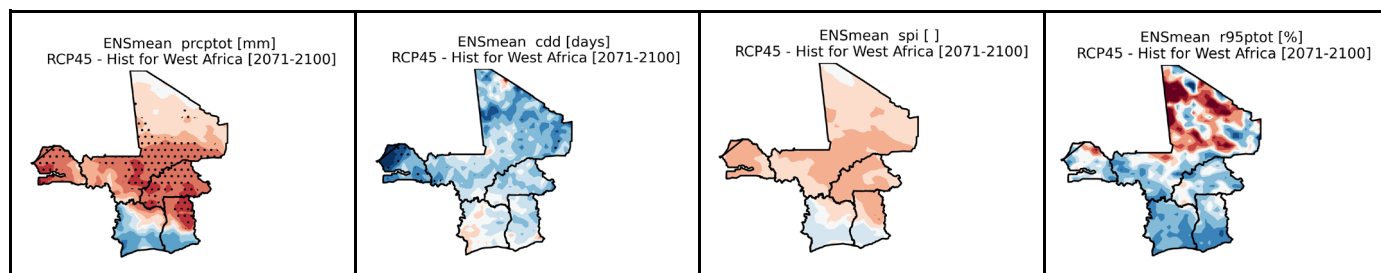
³⁹ IPCC (2021), Sixth Assessment Report.

Figure 5: Trends in annual precipitation levels from 1951-2010



Source: I. Niang et al. (2014), Africa.

29. While projections indicate drying and increased GHG emissions across all five target countries, the countries that will be affected the most are Senegal, southern Mali and western Burkina Faso⁴⁰. Over Senegal and parts of Mali, precipitation is projected to decline by as much as 30 per cent by the end of the century under the RCP8.5 scenario. While model uncertainty exists on the intensity and size of change over the southern part of the target region, the decline of precipitation over Senegal and southern Mali is consistent with other studies who used a different set of models⁴¹. In general, however, impact models predict **a general reduction of crop yields by 10 to 20 per cent by 2050 due to drier and warmer climate conditions** and much more severe yield losses in certain places⁴². Studies by UNEP suggested that as a result of **changing rainfall patterns and land degradation**, Chad and Niger could potentially lose **its entire rainfed agriculture sector** by 2100, while in Mali, cereal harvests might decline by as much as 30 per cent⁴³.
30. Projections of **extreme rainfall** related to floods indicate a north-south spatial gradient (figure 6) with **increased heavy precipitation (amount and occurrence) in the south and a decline in the north**. This suggests that there will be a southward shift in the frequency and intensity of heavy precipitation events that produce floods. For the northern part of the target region (part of Burkina Faso, Mali and Senegal), the decline in the intensity and frequency of heavy precipitation events will be accompanied by an increased in dry spell length (consecutive dry days or CDD, column b below) and droughts (standardized precipitation index, SPI, column c below), implying that the region will be more vulnerable to strain on water resources which will affect rainfed agricultural production. CORDEX RCM projections of a regional decrease in the length of the rainy season, an increase in the length of dry spells and in heatwave frequency, intensity and duration highlight the importance of determining the right type and timing for planting and harvesting key crops, especially those prone to drought⁴⁴.



⁴⁰ Ibid.

⁴¹ See, for example: Mouhamadou Sylla et al. (2016), "Climate Change over West Africa: Recent Trends and Future Projections" in *Adaptation to Climate Change and Variability in Rural West Africa*, Joseph A. Yaro and Jan Hesselberg (eds.), Springer International Publishing; or I. Diallo et al. (2016), "Projected changes of summer monsoon extremes and hydroclimatic regimes over West Africa for the twenty-first century", *Climate Dynamics*, 47.

⁴² Peter Jones and Philip Thornton (2003), "The potential impacts of climate change on maize production in Africa and Latin America in 2055", *Global Environmental Change*, 13 (1), April.

⁴³ USAID (2017), *Climate Change Risk in West Africa Sahel*.

⁴⁴ Ibid.

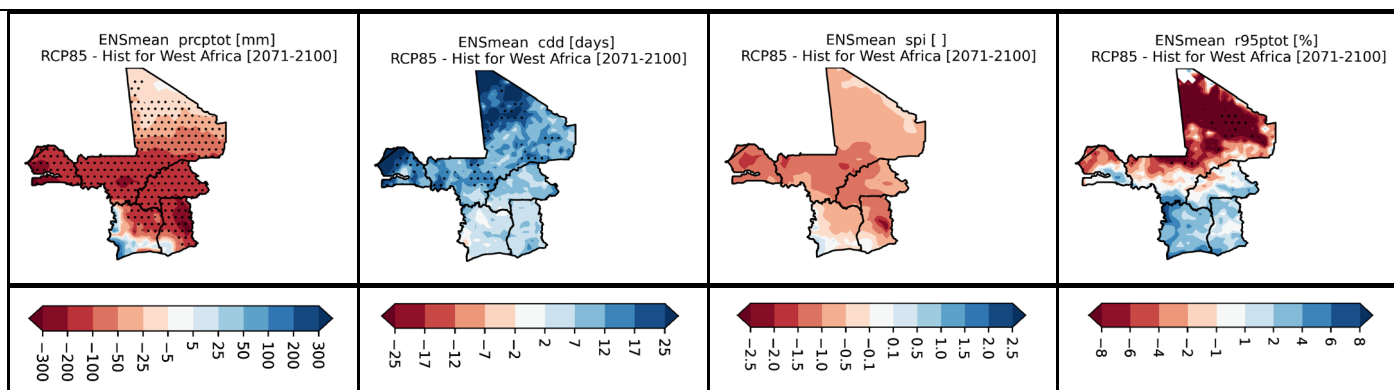
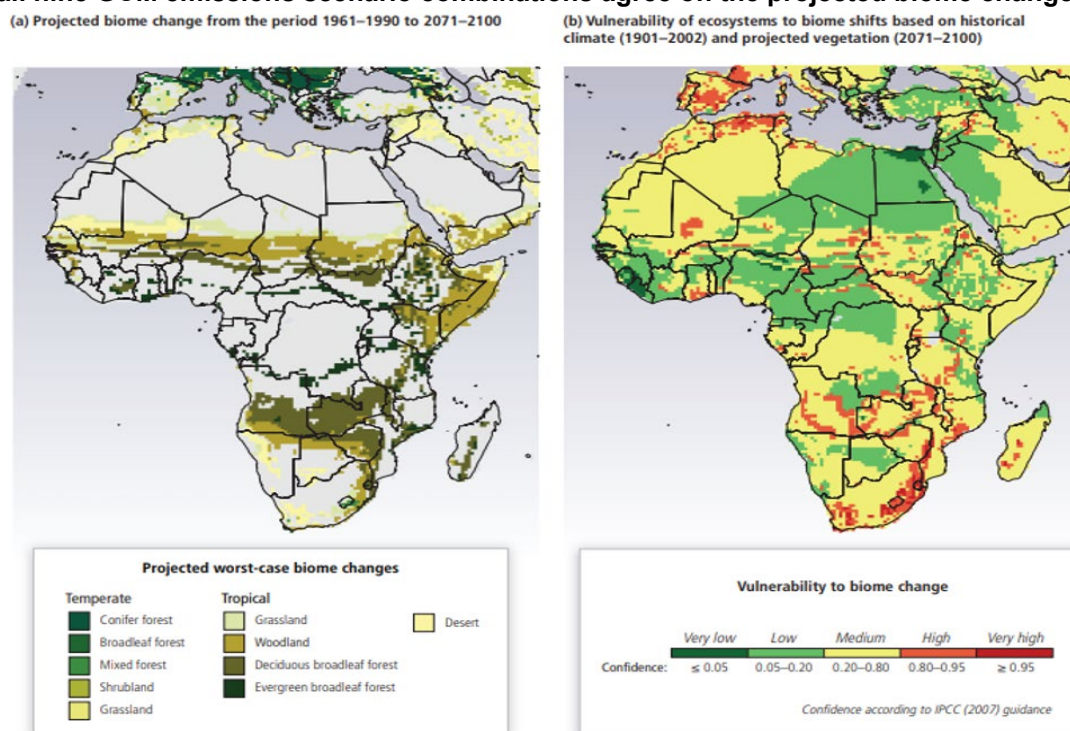


Figure 6: Projected change in (column a) annual total wet-day rainfall [prcptot], (column b) consecutive dry days [cdd], (column c) standardized precipitation index [spi] and (column d) contribution from very wet days [r95ptot] for the end of 21st century (i.e., 2071-2100) under RCP4.5 (low-medium, first row) and RCP8.5 (high, second row) emission scenario. The black dots represent regions with statistically significant change at a 0.05 significance level.
Source: Authors, 2021

31. **Shift of agro-climatic zones due to climate change:** The combined effect of changes in precipitation and temperature will strongly affect the state of agro-climatic zones. Prevailing warmer and drier conditions in the projected future climate will lead to a shift in the agro-climatic zone (Figure 7) from dry to semi-arid zones in parts of the target region, which will have major implications for the native vegetation and farming. A recent study (EXFAM, 2017) found that one in six trees in the region has died since the 1950s and a fifth of all species has disappeared locally because of rising temperatures and lower rainfall linked to climate change. It also found that in some sites in the Sahel, average temperatures rose by 0.8°C and rainfall decreased by 48 per cent and, as a result, trees have shifted southward towards wetter areas, which could affect rural communities engaged in the extraction of NTFPs⁴⁵.

Figure 7: (a) Projected biome change from the periods 1961–1990 to 2071–2100 using the MC1 Dynamic Vegetation Model. (b) Vulnerability of ecosystems to biome shifts based on historical climate (1901–2002) and projected vegetation (2071–2100), where all nine GCM emissions scenario combinations agree on the projected biome change



Source: I. Niang et al. (2014), Africa.

⁴⁵ USAID (2017), Climate Change Risk in West Africa Sahel.

Notes: Change is indicated if any of nine combinations of three General Circulation Models (GCMs: Commonwealth Scientific and Industrial Research Organisation, CSIRO Mk3), Met Office Hadley Centre climate prediction model 3 (HadCM3) and Model for Interdisciplinary Research on Climate 3.2 (medres) and three emissions scenarios (B1, A1B, A2) project change and is thus a worst-case scenario. Colours represent the future biome predicted.

32. **Desertification** is a process by which productive land becomes progressively drier and its soil nutrients, increasingly depleted, until vegetation is no longer able to grow. Both human hands and natural forces can cause and worsen desertification. Some scientists argue that by increasing global temperatures and altering rainfall patterns, climate change is lengthening the naturally occurring droughts in the Sahel. Extended droughts can turn naturally arid, low-yield land into areas completely unsuited for natural or human-sown vegetation. While climate change may be partly to blame for this, human activity – primarily deforestation (firewood and clearing land for farming) – has also contributed significantly. Without trees and other vegetation to protect the soil (including soil life) from the sun's direct heat and rainfall, it dries and eventually turns to dust, which is blown away by the wind. In recent years, **vast areas of the Sahel have succumbed to desertification**. As a result, water is becoming increasingly scarce for people, animals and plants in the region. Without water and productive land, farmers and herders are unable to sustain their traditional livelihoods. As many lack training for other types of work, they face poverty and have little choice except to migrate elsewhere to survive.
33. **Extreme weather events:** The three selected GGW countries under IGREENFIN I (Burkina Faso, Mali and Senegal) plus Côte d'Ivoire and Ghana face a range of natural hazards and risks, particularly droughts, flooding, wildfires, water scarcity, sandstorms and extreme heat. The main climate-related natural hazards observed in the targeted countries between 1970 and 2015 were drought (> 50 per cent)⁴⁶. As shown in Table 1 below, all countries have at least one administrative area in the basin where the risk of flooding is high, and nearly all countries are considered to be at high or medium risk for wildfires.

Table 1: Risks for climate hazards for IGREENFIN I

Country	River flood	Landslide	Extreme heat	Wildfire	Water scarcity
Burkina Faso	High	Very Low	High	High	High
Mali	High	Very Low	High	High	High
Senegal	High	Very Low	High	High	High
Côte d'Ivoire	High	Low	High	High	High
Ghana	High	Medium	High	High	High

Source: GFDRR, n/d, <https://thinkhazard.org/>

34. In the GGW countries, Côte d'Ivoire and Ghana, the consequences of warming temperatures are already manifesting themselves in the form of recurrent extreme weather events such as floods, droughts, heatwaves, sandstorms, wildfires and desertification⁴⁷. These, together with soil erosion and increases in water stress, diseases and locust outbreaks, have a heavy toll on rural communities' livelihoods and production systems⁴⁸. Over the past 30 years, over 75 per cent of the West African population has been affected at least once every two years by this type of hazard. Four major drought-related emergencies have been reported in less than 10 years in Burkina Faso and Mali, causing decreases in yields of 25 per cent on average and increases in staple crop prices of up to 50 per cent. Furthermore, the cost of responding to the severe drought events experienced in Burkina Faso, Côte d'Ivoire, Ghana, Mali and Senegal has been estimated at over US\$100 million, as indicated in Table 2 below. Prolonged droughts have forced many pastoral communities to abandon their traditional way of life to become semi-agricultural⁴⁹. The intensity of extreme weather phenomena is projected to increase, causing negative impacts on agricultural production.

⁴⁶ Africa Risk Capacity (2020), Independent Evaluation Report.

⁴⁷ For more, see the Great Green Wall website: www.greatgreenwall.org

⁴⁸ IPCC 5th Report (2018), Africa.

⁴⁹ Kanta Kumari Rigaud et al. (2021), Groundswell Africa: Internal Climate Migration in West African Countries.

Table 2: Drought frequency and response by country for IGREENFIN I

Country	Period	Number of severe droughts	Frequency of droughts (1 in x years)	Cost of response (average) US\$ million
Burkina Faso	2000-2017	4.0	4.0	20.0
Côte d'Ivoire	1972-2016	4.0*	3.0	10.0**
Ghana	1983-2015***	3.0***	3.0***	10.0****
Mali	1983-2017	14.0	4.0	20.0
Senegal	1983-2017	12.0	5.0	56.0

* Based on the statement that the effects of drought (especially those of 1983, 1998 and 2010) have weakened the Ivorian economy. Source: ARC, National Contingency Plan in Côte d'Ivoire (2018), https://www.africanriskcapacity.org/wp-content/uploads/2020/01/C%C3%B4te-d'Ivoire_-_Operations-Plan_201901_EN_modifs.pdf

** Conducting systematic evaluations of the impact of catastrophic events is not yet a regular practice in Côte d'Ivoire. However, in the 2018 contingency plan for drought, it was estimated that a payment of US\$10 million would provide food assistance to 90,000 people and cover the costs of distributing rice seed and agricultural inputs to 20,000 vulnerable smallholder farmers in the regions of Poro, Kabadougou, Tokpi, Bounkani, Worodougou, Gontougo and Bafing. The total population of these regions is 6.5 million people. Source: SODEXAM. Drought conditions and strategies for their management: Case of Côte d'Ivoire.

**** Source: Development of Drought Early Warning System implemented by the Climate Technology Centre and Network (CTCN) with support from the Green Climate Fund (GCF). Available from https://knowledge.unccd.int/sites/default/files/country_profile_documents/1%2520FINAL_NDP_Ghana.pdf

Programme location for IGREENFIN I

35. **Selected countries:** The IGREENFIN I (phase 1) programme focuses on the five selected IFAD developing member states (Burkina Faso, Côte d'Ivoire, Mali, Ghana, and Senegal). The criteria are:

- Be an IFAD member state with resource allocations under IFAD 11 (2019-2021)
- High level of poverty, food insecurity and malnutrition
- Rural gender disparities and youth unemployment
- Vulnerability of the agricultural sector (including livestock and forestry) and key agricultural value chains to climate change and climate variability
- Country's needs and request for IFAD support to mobilize GCF resources through the National Designated Authorities
- Opportunities for green business development in the country as defined in the IFAD Country Opportunity Strategic Papers (COSOP and country national development plans and adaptation plans).

Table 3: Selected countries, IFAD baseline investments and additionally of IGREENFIN I

	Project name	Interventions areas for both IFAD baseline investments and IGREENFIN I	Beneficiaries	
			Direct	Indirect
IFAD baseline investments (Country)	Agricultural Value Chains Support Project (PAFA) - Burkina Faso	Boucle du Mouhoun, Haut Bassin, Cascades	80,000	330,000
	Agricultural Emergency Support Project (AESP) – Côte d'Ivoire	Bagoue, Poro, Tchologo, Hambol and Gbeke Tonpki, Kabadougou, Folon, Bafing, du Worodougou and Bere	53,600	294,800
	Affordable Agricultural Financing for Resilient Rural Development Project (AFFORD) – Ghana	Northern, Savannah and North-east Regions, Bono, Bono East and Regions	155,000	930,000
	Inclusive Finance in Agricultural Value Chain Project (INCLUSIF) - Mali	Koulikoro, Sikasso Kayes, Segou,	40,000	440,000
	Rural Youth Agripreneur Support Project (AGRIJEUNES)- Senegal	Louga, Thiès, Diourbel, Fatick, Kaolack, Kaffrine, Sédhiou and Ziguinchor	50,000	500,000

36. IGREENFIN I introduces innovative green business models to improve the risk-return profile of baseline projects, makes much needed concessional capital available and crowds in private sector capital, while providing technical assistance for subproject preparation, as well as

incentives to beneficiaries to adopt best adaptation and mitigation practices and technologies, capacity-building and reporting & monitoring of GHG emissions.

36. **Selection of country sites (IGREENFIN I):** A full description of each region per country is provided in Annex 2 Feasibility Study.

Figure 8: Map indicating the regions of targeted interventions in each country

Burkina Faso



Côte d'Ivoire



Ghana



Mali



Senegal



37. **Regional selection criteria:** The target regions in each country have been selected for their vulnerability to climate change, the historical and projected impacts of climate change (temperature and precipitation), and climatic events (droughts, floods, heatwaves and sandstorms). The methodology used for site selection consisted of the following steps: (1) define selection criteria based on ones previously established in the IFAD country strategy for each country; (2) develop a list of potential sites and obtain supporting information from the countries' national development plans, adaptation plans and agricultural

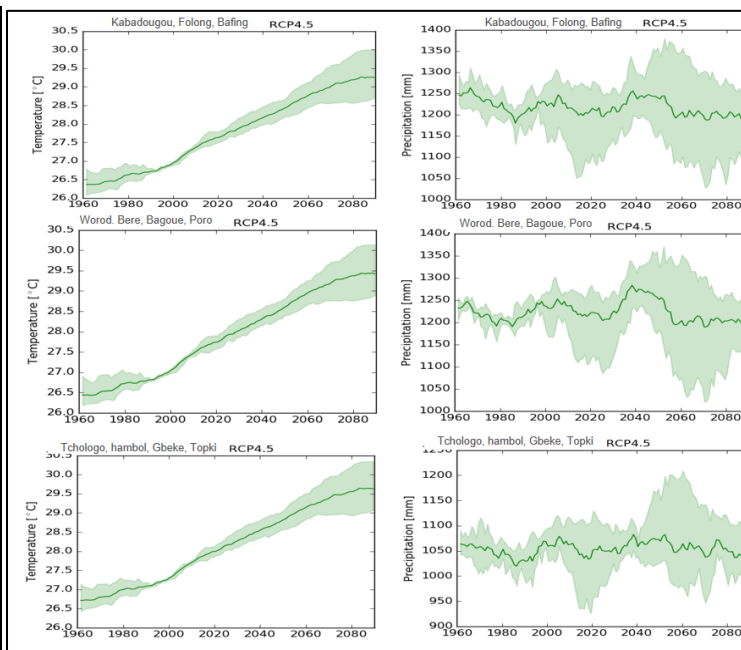
sector plan; (3) apply the site selection filter using social and economic criteria (poverty level, remoteness, food insecurity and nutrition, rural gender disparities, youth unemployment, absence or lack of rural infrastructure including energy, opportunities for job creation both for youth and women within key prioritized climate resilient agricultural value chains) and climate criteria (climate vulnerability and natural hazards risks) to identify potential target sites; (4) conduct site visits during IFAD design baseline and IGREENFIN I missions, and (5) select target site(s) through team discussions and in consultation with the government of each country. The maps below present the historical and projected impacts of climate change (temperature and precipitation) for each site in the five IGREENFIN I target countries.

Figure 9: Historical and projected impact of climate (temperature and precipitation) 1960-2080 in the selected sites per country

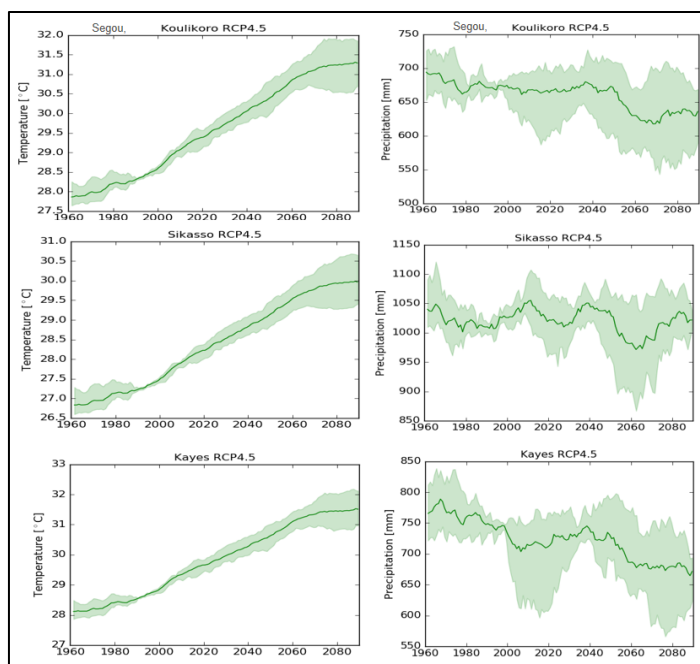
Burkina Faso



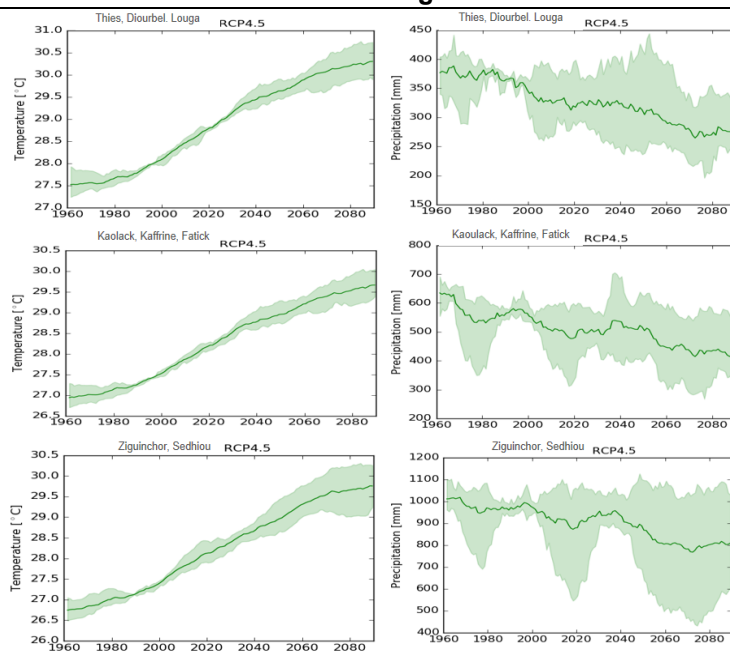
Cote d'Ivoire

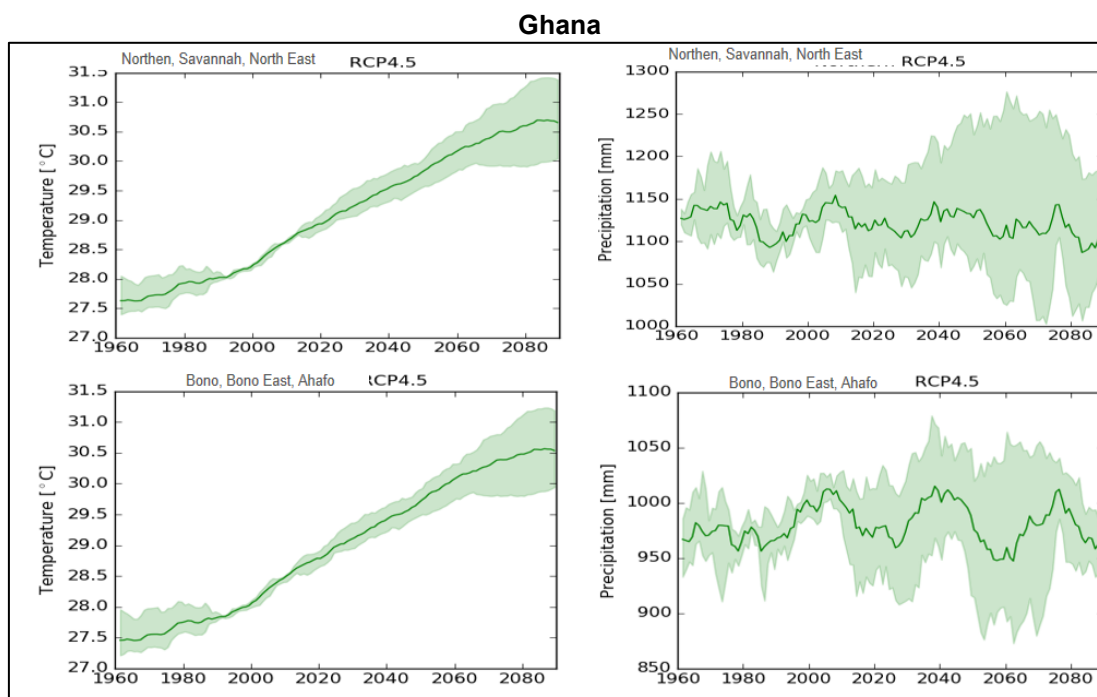


Mali



Senegal





Source: Climate analytics, 2021.

Climate change vulnerability and impacts

38. **Vulnerability to climate change:** The countries included in IGREENFIN I (Burkina Faso, Côte d'Ivoire, Ghana, Mali and Senegal) are among the world's least resilient countries to climate change and the least ready to mobilize investments and convert them into adaptation actions, as evidenced by their low ranking on the ND-GAIN index (Table 4 below). Rural populations in these countries are among the poorest and most vulnerable to climatic variability and land degradation⁵⁰, and depend heavily on healthy ecosystems capable of sustaining their rainfed agriculture, fisheries and livestock activities. These sectors are the primary source of employment in the region and generate at least 40 per cent of GDP in most of the countries⁵¹. Furthermore, these ecosystems provide products that are essential for smallholders' livelihoods, such as food, fuelwood, bushmeat, non-timber forest products (NTFPs) and fresh water, and perform essential services such as waste treatment, soil erosion prevention and recreation, among others. The rapidly growing population in the GGW area and in Côte d'Ivoire and Ghana puts even greater pressure on natural resources, as rural communities struggle to obtain food, fodder and fuelwood in an already vulnerable environment. These activities accelerate the deterioration of natural resources, notably vegetation cover. Climate variability along with frequent droughts, floods, heatwaves, wildfires and poorly managed land and water resources (surface and underground) have caused rivers and lakes to dry up and contribute to increased soil erosion⁵², crop losses, disruption of production, higher food prices and rising food insecurity⁵³.

Table 4. Adaptive capacity, vulnerability and readiness scoring for the five countries

Countries	Adaptive capacity scoring	Vulnerability ranking	ND-Gain Index 11 (out of 181 countries)	
			Ranking	Vulnerability and readiness
Burkina Faso	0.687	0.572	161	19 th most vulnerable, 34 th least ready
Côte d'Ivoire	0.667	0.514	145	47 th most vulnerable, 32 nd least ready
Ghana	0.532	0.468	74	8 th most vulnerable and 81 st least ready
Mali	0.731	0.609	166	9 th most vulnerable and 38 th least ready
Senegal	0.595	0.535	131	42 nd most vulnerable and 65 th least ready

⁵⁰ Africa Forestry and Wildlife Commission (2020), Twenty-Second Session.

⁵¹ Ibid.

⁵² Ibid.

⁵³ Philipp Heinrigs (2011), Security Implications of Climate Change in the Sahel Region: Policy considerations, <https://www.oecd.org/swac/publications/47234320.pdf>

Source: ND-GAIN Country Index rankings, <https://gain.nd.edu/our-work/country-index/>

Climate change impact on key sectors

39. **Agriculture and food security:** Agricultural production and productivity levels in the region are already low by global standards and they are expected to decline even further due to the impacts of climate change and COVID-19 because of not only biophysical changes, but also these countries limited economic, financial and institutional capacity to cope with key risks (climate, biological, price, labour/health, policy and political)⁵⁴. Agriculture in the West Sahel is almost entirely rainfed and reliant on three to four months of variable summer rainfall (June-September; annual precipitation between 200 mm to 1200 mm), making it highly vulnerable to increasing climate variability and putting the large percentage of the region's people who depend on agriculture at risk⁵⁵. FAO estimates that 20-80 per cent of the inter-annual variability of crop yields is associated with weather phenomena and 5-10 per cent of national agricultural production losses is associated with climate variability⁵⁶. In addition, agriculture suffers 26 per cent of damages and losses caused by climate-related disasters. In the Sahel, dry spells and droughts lead to increased evaporation, which can reduce water resources and diminish soil moisture and fertility, with negative implications for agricultural yields. In tropical and coastal zones, where famine is already widespread, heavy rainfall events, flooding and reduced overall annual rainfall are among the main climate threats to agricultural production⁵⁷. The generally nutrient-limited soils in the region, which are being degraded by overgrazing, continuous cropping and deforestation, will be further threatened by desertification and sand intrusion brought about by reduced precipitation and higher temperatures. This will lead to increased food prices and food insecurity. Within the agriculture and food security sectors, there are also several transboundary issues that may be intensified by climate change such as the spread of animal diseases, land degradation and pollution, food contamination and pressure on natural resources⁵⁸.
40. The main subsistence crops grown in the region are maize, pearl millet and sorghum while important cash crops include rice, groundnuts, cashew and mango⁵⁹. Smallholders also exploit a range of NTFPs such as Arabic gum, shea butter, tamarind, baobab, etc. for their direct subsistence value and as an important source of cash income⁶⁰. NTFPs play a very important role in the local economies in Burkina Faso, Côte d'Ivoire, Ghana, Mali and Senegal⁶¹. Climate change is expected to lead to significant declines in yields of the main crops in the area by 2100. The climate models generated using the IFAD Climate Adaptation in Rural Development Assessment Tool⁶² (CARD) indicate that in the targeted countries, the production of millet is predicted to decrease on average by 10 per cent; groundnut, 11 per cent, and rice, 7.82 per cent over the next 20 years in Burkina Faso, Mali and Senegal and the northern dry part of Côte d'Ivoire and Ghana. This will have negative impacts on livelihoods and production systems, which will push millions more people into poverty and generate acute levels of food insecurity. Climate change is also a threat to livestock production because of the impact on quality of feed crop and forage, water availability, animal and milk production, livestock diseases, animal reproduction and biodiversity⁶³. Multi-hazard analyses (heat waves, extreme rain events and dry spells) conducted on each of the five countries at project sites for the RCP4.5 and RCP8.5 scenarios produce heterogeneous results for the different locations, but all point to impacts that affect crop suitability (see Annex 2C Climate Hazards Targeted Areas Analysis).
41. The Covid-19 pandemic has had severe impacts on the already vulnerable economies in the region and undermined efforts to strengthen the resilience of smallholder farmers to climate change⁶⁴. For instance, the prices of Senegal and Côte d'Ivoire's cashews dropped a record 47 per cent due to the decline in demand in domestic markets, while demand for cocoa beans from Côte d'Ivoire and Ghana (the two largest cocoa producers in the world) fell across Europe⁶⁵ in 2020-2021. Unemployment increased while household monetary incomes declined due to the closure of several MSMEs in key agricultural value chains⁶⁶. A drop in household income affects savings, as low-income households are forced to use their savings or resort to other negative coping strategies to buy food and meet other basic needs. This, in turn, hinders their

⁵⁴ FAO (2021), The impact of disasters and crises on agriculture and food security: 2021, Rome. <https://doi.org/10.4060/cb3673en>

⁵⁵ A. Elbehri et al. (2013), "West Africa food systems: An overview of trends and indicators of demand, supply, and competitiveness of staple food value chains", in *Rebuilding West Africa's Food Potential*, A. Elbehri (ed.), FAO/IFAD.

⁵⁶ FAO (2020), Impact of COVID-19 on agriculture, food systems and rural livelihoods in Eastern Africa: policy and programmatic options, Accra, <http://www.fao.org/3/cb0552en/CB0552EN.pdf>

⁵⁷ USAID (2018). Climate Change Risk in West Africa Sahel.

⁵⁸ FAO (2020), Impact of COVID-19.

⁵⁹ J. Dixon, A. Gulliver and D. Gibbon (2001), *Farming Systems and Poverty: Improving Farmers' Livelihoods in a Changing World*, FAO & World Bank, Rome.

⁶⁰ Terry Sunderland and Ousseyun Ndoye (eds.) (2004), *Forest Products, Livelihoods and Conservation: Case Studies of Non-timber Forest Products. Volume 2 – Africa*, CIFOR, <https://www.cifor.org/publications/ntfbsite/pdf/NTFP-Africa-R.PDF>

⁶¹ FAOSTAT, NTFP in West Africa, 2015

⁶² For information on CARD, see <https://www.ifad.org/en/web/knowledge/-/publication/climate-adaptation-in-rural-development-card-assessment-tool>

⁶³ M. Melissa Rojas-Downing et al. (2017), "Climate change and livestock: Impacts, adaptation, and mitigation", *Climate Risk Management* (16), 145-163.

⁶⁴ CIRAD (2021), COVID and Food Security: How African Agriculture could hold its own in the face of the crisis, <https://www.cirad.fr/en/cirad-news/news/2020/science/covid-food-security-african-agriculture-resilient-in-face-of-crisis>

⁶⁵ World Bank (2020), Côte d'Ivoire 10th Economic Update: Taking Stock and Looking Ahead: Cote d'Ivoire and the COVID-19 Pandemic, <https://openknowledge.worldbank.org/bitstream/handle/10986/34559/Taking-Stock-and-Looking-Ahead-Cote-d-Ivoire-and-the-COVID-19-Pandemic.pdf?sequence=10&isAllowed=y>

⁶⁶ Inter-Agency Standing Committee Reference Group on Risk, Early Warning and Preparedness and the European Commission (2020), INFORM COVID Risk Index, April.

already limited chances of gaining access to much-needed credit. The unanticipated shock of COVID-19 underscores the need for a shift from “business as usual” practices to a more forward-looking approach that invests in the productivity, sustainability and resilience of food systems⁶⁷.

IGREENFIN I will provide technical skills in climate resilient, low emission agriculture (crop and livestock production).

42. **Forestry:** These immense challenges and associated land use pressures in the Sahel have led to major changes to the Sahelian environment. Since 1975, farmland has doubled in West Africa (to 100 million ha), while savannah and grazing areas have been halved, trees have been reduced by 40 per cent and wildlife has largely disappeared in most areas⁶⁸. Forests and woodlands play a key role in sustaining livelihoods by contributing to the conservation of biological diversity and water sources; the prevention, control and reversal of land degradation; the combat against desertification and the sustainable production of wood and non-wood forest products and services⁶⁹. Sustainable forest and woodland management has the potential to address the objectives of protecting biodiversity and tackling climate change and land degradation using a programmatic approach.
43. In West Africa, the **land use change and forestry sector** was the main source of GHG emissions, responsible for 31.5 per cent of total emissions, followed by the energy, agriculture, waste and the industrial processing sectors, which contributed 27.1 per cent, 22.7 per cent, 10.9 percent and 7.9 percent, respectively. This trend is expected to continue due to the growing population and demand for energy in the coming decades⁷⁰. The main cause of deforestation is clearing for agriculture but uncontrolled logging and the use of wood for energy and cooking are also taking their toll. Deforestation exposes soil to high temperatures, which breaks down organic matter, increases evaporation and makes soils vulnerable to erosion and further degradation⁷¹. Overgrazing and land cover changes often lead to the deterioration of natural grassland ecosystems and the degradation of catchments in arid and semi-arid regions⁷². Burkina Faso, Côte d'Ivoire, Ghana, Mali and Senegal are each responsible for less than 0.1 per cent of total global emissions⁷³.
44. In the **energy sector**, Nigeria is the highest emitter, as it is responsible for more than half of West Africa's regional energy emissions (60 per cent). Nigeria is followed by Equatorial Guinea, Ghana, Côte d'Ivoire, Cameroon and Senegal, whose combined energy emissions account for more than 85 per cent of the region's GHG emissions from energy activities. Energy is the leading source of emissions in five countries: Equatorial Guinea, Ghana, Côte d'Ivoire, Cape Verde, and Sao Tome & Principe⁷⁴.

Figure 10: Historical Emissions from AFOLU and Energy

⁶⁷ FAO (2021), The impact of disasters and crises on agriculture and food security.

⁶⁸ S.E. Cotillon and G. G. Tappa (2016), Landscapes of West Africa: A Window on a Changing World, US Geological Survey, <https://pubs.er.usgs.gov/publication/70176549>

⁶⁹ IUCN (2020), Identifying Ecosystems Services in Protected Areas.

⁷⁰ Climate links (2019), Green House Gas Emissions in the West Africa Region Fact Sheet, USAID, https://www.climatelinks.org/sites/default/files/asset/document/2019_USAID_West%20Africa%20Regional%20GHG%20Emissions%20Factsheet.pdf

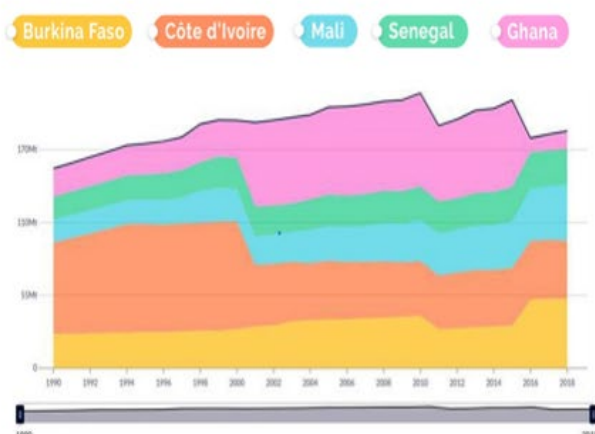
⁷¹ FAO (2020), Main causes of deforestation: causes and scenarios.

⁷² J. J. Kashaigili and A. M. Majaliwa (2013). Implications of land use and land cover changes on hydrological regimes of the Malagarasi River, Tanzania. *Journal of Agric. Sci. Appl.*, 2, 45–50.

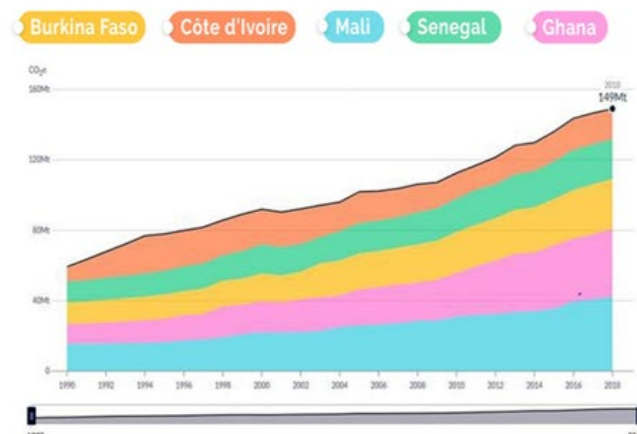
⁷³ Based on data from the World Resources Institute Climate Analysis Indicators Tool (WRI CAIT) 2.0, 2017.

⁷⁴ Climatelinks (2020), Green House Gas Emissions in the West Africa Region.

Historical Emissions for Agriculture and Energy



Historical Emissions for Agriculture, Energy and LUCF



Source: Prepared by the IFAD Project Design Team based on World Resources Institute Climate Analysis Indicators Tool (WRI CAIT 2.0, 2017).

45. The following table gives an overview of the selected countries' share of global GHG emissions.

Table 5: Contribution to global GHG emissions – target countries plus France and the UK for comparison

Country	Total GHG emissions (MtCO ₂ e)	Percent of global emissions	Population	tCO ₂ e per capita	GDP (billion US\$)	tCO ₂ e/ million US\$ GDP	Change in GHG emissions (1990–2014) (MtCO ₂ e)
Burkina Faso	32.6	0.07%	17,585,977	1.85	10.908	2989	10.13 (+45%)
Cote d'Ivoire	39.21	0.08%	22,531,350	1.74	31.204	1257	21.53 (+122%)
Ghana	38.57	0.08%	26,962,563	1.43	44.752	862	5.45 (+16%)
Mali	38.7	0.08%	16,962,846	2.28	11.972	3233	15.36 (+66%)
Senegal	30.45	0.06%	14,546,111	2.09	14.838	2052	9.20 (+43%)
France	361.37	0.97%	67,413,000	5.27	2,700.0	n/a	9.64 billion (35.76%)
UK	441.13	1%	68,237,070	5.30	2,800.0	n/a	13.18 billion (21.01%)

Note: Emissions are in million metric tons of carbon dioxide equivalent (MtCO₂e). WRI CAIT uses global warming potentials (GWP) from the Intergovernmental Panel on Climate Change (IPCC) Second Assessment Report (SAR).

Source: World Resources Institute Climate Analysis Indicators Tool 2.0, 2017.

IGREENFIN I will foster sustainable use of NTFP and restore land through the promotion of agroforestry and sustainable natural resource management methodologies to arrest land degradation and desertification.

46. **Water resources:** Climate change will significantly affect water resources and therefore, it is crucial to ensure that agricultural production and irrigation systems are better adapted to shorter rainy seasons. Climate models also project large declines in the flows of certain transboundary rivers in West African countries – a major issue, as 40 per cent of these countries' surface water resources originates outside their borders⁷⁵. Studies suggest that the five countries have significant groundwater reserves in dry areas, but these are far from urban centres⁷⁶. Declines in rainfall, rising temperatures and more

⁷⁵ IPCC (2014), Africa.

⁷⁶ Isabelle Ramdoo (2019), L'Afrique des ressources naturelles, IISD, novembre, <https://www.iisd.org/system/files/publications/afrique-ressources-naturelles-vangaurdia-fr.pdf>

frequent droughts contribute to a decrease in surface and groundwater availability and accessibility. However, total renewable water resources per capita range from 745,600 m³/year in Burkina Faso to 6,818,000 m³/year in Mali⁷⁷. An extensive analysis of groundwater availability and recharge rates (see Annex 2 Feasibility Study) shows that the risk of maladaptation will not occur within the context of the proposed intervention, as some parts of the study area can even sustain commercial irrigation schemes utilizing appropriate management regimes, water infrastructures and technologies to expand production beyond the rainy season. Nonetheless, it is projected that by 2025, even areas such as Burkina Faso that have low, but sufficient water resources per capita are expected to experience physical water scarcity (defined as when the water supply falls below 1,000 m³ per person per year)⁷⁸. Furthermore, increasing demand from a growing population and planned irrigation schemes along the Niger and Senegal Rivers have led to 25-60 per cent reductions in flows over the last 30 years, causing increasingly severe low water levels, with frequent pauses in water flows, depleted reservoirs and reduced water supplies for cities⁷⁹. Caution is required to ensure that planned irrigation systems use water sustainably and efficiently. As rains are often seasonal, groundwater is a primary source of water for many people in the region⁸⁰. As a result, disputes over access to water, fish catches and ownership of land exposed by receding waters have increased dramatically. Climate variability and change, together with non-climate stressors such as population and economic growth, inefficient water management and infrastructure, failure to jointly manage basin resources, decreases in groundwater levels and land use/land cover changes, are expected to further reduce river basin water supplies in the future⁸¹. All the above indicate that the implementation of sustainable practices such as low emission, climate-resilient agriculture that utilizes water in a sustainable manner is urgently required to mitigate the effects of climate change.

47. The development of irrigation in the Sahel, including the northern part of Côte d'Ivoire and Ghana, is a key intervention for improving the adaptation and climate resilience of agriculture and an instrument for fostering food security⁸². Access to irrigation can expand farmers' production opportunities in the off-season and even when used in low quantities, it mitigates production risks and saves crops. By reducing risk, it encourages farmers to make more intensive use of inputs and land. However, the type of irrigation systems needs to be carefully assessed to reduce the risk of depleting water sources and soil salinization. Past – and not always successful – efforts were based on large public irrigation schemes (i.e., river dams and related canals) and in a growing number of contexts, investments in small and micro-irrigation systems are more desirable and sustainable than conventional large-scale schemes⁸³. Renewable energy technologies (RETs) now offer effective, climate resilient solutions for farmers in remote areas with little access to clean energy. Existing small-scale irrigation systems in the region are being developed, although limited systematized evidence exists to date. The realization of this potential is extremely sensitive to the costs of irrigation, among the highest in the world, with some technologies more sensitive than others (i.e., small river diversions)⁸⁴. The hydrological assessment in Annex 2 Feasibility Study presents the impact of climate on surface and groundwater resources (namely a consistent rise in the demand for water and irrigation requirements, but a decline in water availability associated with increased GHGs). Global warming and the over-abstraction of groundwater could also pose a threat to groundwater resources, as they contribute to decreases in recharge rates and increases in salinity and the depth of the water level. This means that future irrigation development plans should consider the threats from climate change. Therefore, such plans would require local irrigation water requirements analysis to be undertaken for specific project conditions. The analysis in Annex 2- Feasibility also identifies opportunities for irrigation in the region. Risk management and mitigation must be fully integrated into all support for the development of irrigated agriculture.

IGREENFIN I aims to promote climate resilient, low emission agriculture practices and the sustainable use of scarce water resources.

48. **Financial system:** Access to credit is crucial to building resilience and improving productivity, but it remains a key constraint for smallholder farmers, FOs, cooperatives and MSMEs in the targeted countries, particularly in COVID-19 recovery efforts and for rural women and youth who rarely benefit from financial services⁸⁵. Despite its crucial role, financial markets (both insurance and credit markets) in the Sahel region are often underdeveloped, with over 80 per cent of the population being financially excluded⁸⁶. The inherent risks in smallholder farming discourage the private sector, especially the banking sector, from investing in agriculture. Most banks, including LNABs in the target area, still face numerous internal obstacles including: i) the perception of small-scale agriculture as too risky, fostering reluctance to lend farmers money and the practice of

⁷⁷ USAID (2017), Climate Risks in Food for Peace Geographies - Burkina Faso, https://www.climate-links.org/sites/default/files/asset/document/20170807_USAID%20ATLAS_FFP_BurkinaFaso.pdf

⁷⁸ Climat Analytics (2019), Etats des lieux des connaissances scientifiques sur les ressources en eau et l'impact des changements climatiques sur ces ressources.

⁷⁹ USAID (2017), Climate Change Risk in West Africa Sahel.

⁸⁰ K Lewis and C. Buontempo (2016), "Climate Impacts in the Sahel and West Africa: The Role of Climate Science in Policy Making", West African Papers, No. 02, OECD Publishing, Paris. <http://dx.doi.org/10.1787/5jsmktwjd0-en>

⁸¹ Ibid.

⁸² Abdulai Jalloh et al. (2013), West African Agriculture and Climate Change: A comprehensive analysis, IFPRI, <http://www.fao.org/family-farming/detail/en/c/413652/>

⁸³ Ibid.

⁸⁴ Ibid.

⁸⁵ IFAD (2021), IFAD Inclusive Financial Services Portfolio Stocktaking.

⁸⁶ Terfa Abraham (2018), Estimating the effects of financial access on poor farmers in rural northern Nigeria", Financial Innovation 4 (25), <https://fin-swufe.springeropen.com/articles/10.1186/s40854-018-0112-2>

charging them higher interest rates; ii) a lack of awareness on the potential benefits of investing in climate adaptation and mitigation; iii) an overestimation of risks due to a lack of technical knowledge; iv) a lack of access to appropriate capital for on-lending⁸⁷; v) the absence of financial equity and limited financial literacy among smallholders, especially women, and vi) land tenure systems, practices and culturally and legally recognized rights which limit women and youth's access to land. Secure tenure rights provide the foundation for investment and economic growth, enhanced food security and nutrition, reduced conflict and improved natural resource management⁸⁸. Safeguarding tenure rights thus improves the overall investment environment, thereby stimulating investment.

49. Other limiting factors are: i) high transaction costs when dealing with frequent small loans, and unclear and unfavourable land tenure and property rights; ii) the broader business and legal environment that may affect women and men differently; (iii) personal profile of the entrepreneurs (such as education and skills); (iv) characteristics of the firm (size, area of specialization, location, formal/informal sector, as the majority of smallholders operate informally), and (v) constraints within financial institutions, primarily the lack of familiarity with rural entrepreneurs, especially women and youth, and the agricultural sector in general. These constraints limit the ability of the selected countries to leverage more private resources for the implementation of the objectives of the GGW. In a post COVID-19 context, contributing to the elimination of these technical and financial barriers would allow these institutions to finance transformational change towards climate compatible investments at a time when local financial institutions, FOs, cooperatives, MSMEs and governments are recovering from the impacts of the pandemic.
50. Annex 2 Feasibility Study and Annex 8 Gender Assessment both demonstrate that access to finance is a bigger constraint for youth and women farmers. In West Africa, men have better access to finance than women. Only 20 per cent of women have a bank account compared to 27 percent for men⁸⁹. Moreover, due to cultural beliefs and practices, only a small minority of women own land or have adequate and stable access to land and agricultural inputs. The lack of assets in their names is an important impediment to developing viable business plans and obtaining loans, as they have nothing to offer as collateral. The negative impacts of climate change will worsen this situation, as they affect farmers' yields and income and thus, their ability to repay financial obligations, leading to a growing number of loan defaults⁹⁰. Special attention must thus be paid to the constraints imposed on women's access by sociocultural, economic, legal and, in some cases, educational barriers, as well as financial institutions' limited understanding of rural communities and gender dynamics in particular.
51. Financing from the banking sector (including microfinance institutions or MFIs) for the whole agricultural sector in the targeted GGW countries and Côte d'Ivoire and Ghana accounts for roughly only two per cent of the total credit offered⁹¹. When credit is available to farmers, interest rates are often prohibitive for smallholders. While the agricultural banks in Burkina Faso, Côte d'Ivoire, Mali and Senegal have set interest rates⁹² at around 10 per cent, commercial banks and MFIs, including Ghanaian banks, charge around 20 per cent⁹³, which is much higher than international standards⁹⁴. High interest rates prevent smallholder farmers – especially youth and women as 'agripreneurs' – from investing in climate resilient, low emission agriculture and raising their productivity to respond to the high demand in the face of changing climate conditions⁹⁵. Furthermore, collateral requirements are high and repayment periods are short, making the loans unattractive and unaffordable for smallholder farmers⁹⁶. In general, green financing for agriculture projects is scarce in the region. The Comprehensive Africa Agriculture Development Programme (CAADP)⁹⁷ called on countries to increase public investment in sustainable agriculture and dedicate a minimum of 10 per cent of national budgets to such investments⁹⁸.

IGREENFIN I aims to promote access to highly concessional green lines of credit and technical assistance to foster climate resilient COVID-19 recovery.

⁸⁷ Ibid.

⁸⁸ Katharina Hopp (2020). Between customs and legislation: land rights in West Africa Konrad-Adenauer-Stiftung e. V, <https://www.kas.de/documents/252038/7938566/Between+customs+and+legislation+-+land+rights+in+West+Africa.pdf/3a385772-41ac-1ff3-77a3-a739003cec25?version=1.0&t=1600863380158>

⁸⁹ UN Women, 2019, Inclusive Rural Finance in West Africa

⁹⁰ Ibid.

⁹¹ Terfa Abraham (2019). Between customs and legislation.

⁹² See Tables 24 and 25 for the interest rates of the LNABs. Also see: BCEAO (2019), Rapport sur la situation de l'inclusion financière dans l'UEMOA au cours de l'année 2018, https://www.bceao.int/sites/default/files/2019-10/Rapport_Annuel_2018_Situation_Inclusion_Financiere.pdf

⁹³ Trading Economics (N.D.), Ghana Bank Lending Rate, <https://tradingeconomics.com/ghana/bank-lending-rate>; USAID/Crossboundary (2018), On the functioning of agricultural markets in Mali.

⁹⁴ World Bank (2021), Financial Inclusion Overview.

⁹⁵ IFAD (2021), IFAD Inclusive Financial Services Portfolio Stocktaking.

⁹⁶ For more details on the barriers to credit for farmers, see Annex 2 Feasibility Study.

⁹⁷ CAADP: The Comprehensive Africa Agriculture Development Programme (CAADP) is Africa's policy framework for agricultural transformation, wealth creation, food security and nutrition, economic growth and prosperity for all.

⁹⁸ AUDA-NEPAD (N.D.), CAADP Overview, <https://www.nepad.org/caadp/overview>

52. **Renewable energy:** Access to energy has been acknowledged as being key for reducing poverty and supporting social and economic development given the limited access to energy across the Sahel⁹⁹. According to data from the Africa Energy Portal (AEP) database, in Burkina Faso, Côte d'Ivoire, Ghana, Mali and Senegal, there are 38.8 million people living in rural areas without access to electricity or approximately 33 per cent of the total population. In the five countries, 64 per cent of people living in the rural areas do not have access to electricity, yet the five countries heavily depend on their rural populations to produce livelihood. These countries struggle to respond to the growing demand for energy spurred on by rapid population growth and urbanization, and their energy sectors are still heavily reliant on fossil fuels and wood, leading to several environmental and social problems at the local, regional and global levels, including the depletion of non-renewable resources (even the ozone), acidification and global warming. As a result, the energy sector is still a major contributor to emissions: its share of overall emissions increased from 66 per cent in 1990 to 73 per cent in 2016¹⁰⁰. Aware of the need to transition towards more sustainable energy sources and their country's great potential for solar power generation, the target countries are increasing their investment in renewable energy sources. Off-grid renewable energy systems and technologies are an effective alternative for supplying sustainable energy for lighting, heating, cooking and powering all segments of the agricultural value chains (production, processing and storage), especially to poor smallholder farmers in areas far from the grid or cities. According to the IFAD renewable energy for smallholder agriculture approach (RESA), renewable energy in smallholder agriculture will eliminate some of the drudgery associated with production, raise incomes, reduce GHG emissions and boost the adaptive capacity of smallholder farmers, while producing various benefits across thematic areas such as, gender empowerment, nutrition and youth employment¹⁰¹. IGREENFIN I will thus support the promotion of renewable energy sources and technologies to address different energy needs in the target areas. These additional investments will help close the energy gap in smallholder farming and enable smallholder households and businesses to engage in value-added activities (processing, drying, and storage) and increase their productivity, incomes and food security. Access to energy, particularly RETs, can trigger rural transformation and the transition from subsistence living to vibrant rural economies, if their full potential is leveraged. Table 6 below shows the most common applications of RETs for the agriculture sector in the region.

Table 6: Common applications of RETs in agriculture

RET source	Examples of applications
Solar (PV and thermal)	• Pumping for potable water and irrigation. • Milling, hulling, pressing, heating, drying, cooling of products (fish, meat, dairy, fruits). • Communications (mobile phones, radios, TVs, etc.) • Fish ponds and shrimp farms, lighting for security, poultry incubators. • Domestic uses
Wind energy	• Water pumping, grinding.
Micro-hydro energy	• Grinding, milling, etc. • Domestic uses.
Biomass energy	• Anaerobic digesters: biogas for cooking, lighting and for engines. • Briquettes from biomass residues (such as rice husk, coconut shells, olive seeds and pulp and by-products of oil extraction) for steam engines.

Source: IFAD renewable for smallholder agriculture approach (RESA), 2019

53. **Climate change and conflict:** According to the Global Peace Index 2021 report, sub-Saharan Africa experienced a slight fall in peacefulness, with the average country score declining 0.5 per cent. Changes in the regional climate are fuelling conflicts related to the availability of natural resources and food insecurity, which can lead to migration, more conflict or a combination of the two¹⁰². Climate change affects livelihoods that are directly dependent on natural resources the most - for example, by causing agricultural yields to decrease, gradually contributing to the unsuitability of traditional grazing grounds or drying up important water bodies¹⁰³. Furthermore, incomplete institutionalization of land tenure and governance have allowed many natural resource conflicts to go on unabated. In some cases, this absence of clear, commonly accepted rules

⁹⁹ Julia Terrapon-Pfaff and Marie-Christine Gröne, Carmen Dienst, Willington Ortiz (2018), "Impact pathways of small-scale energy projects in the global south – Findings from a systematic evaluation", Renewable and Sustainable Energy Reviews 95 (November), 84-94, <https://www.sciencedirect.com/science/article/pii/S1364032118304787>

¹⁰⁰ AFDB (2020), Drivers of Greenhouse Gas Emissions in Africa: Focus on agriculture, forestry and other land use, <https://blogs.afdb.org/climate-change-africa/drivers-greenhouse-gas-emissions-africa-focus-agriculture-forestry-and-other>

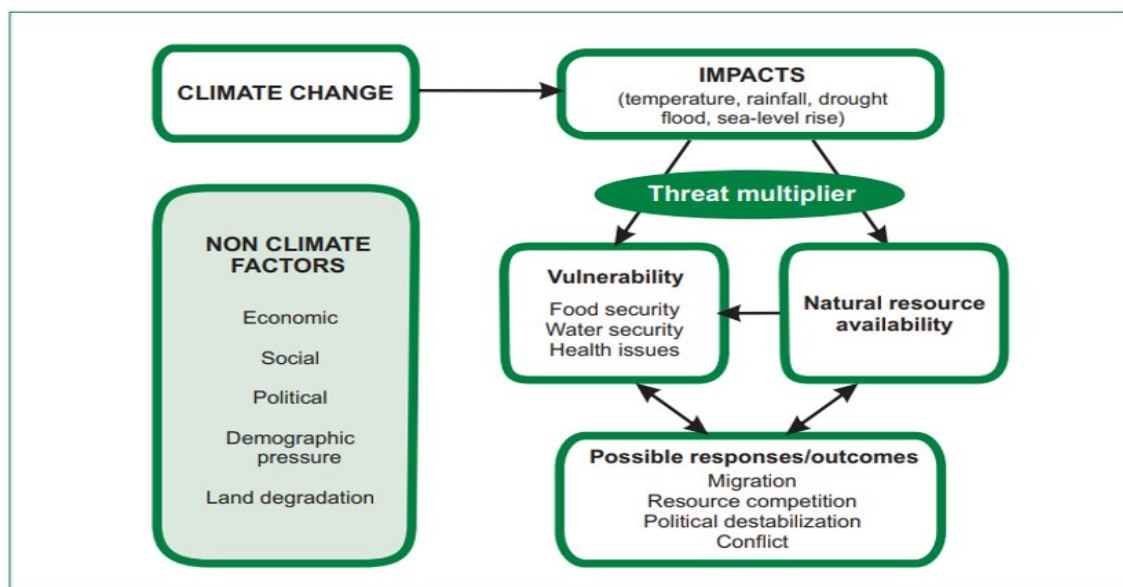
¹⁰¹ IFAD (2019), Renewable Energy for Smallholder Agriculture (RESA), <https://www.ifad.org/documents/38714170/41937394/resa.pdf/715e1a75-35df-bafc-f491-7effde867517>

¹⁰² UNEP (2019), Climate change and security risks.

¹⁰³ FAO (2017), The future of food and agriculture.

has added to the confusion over who owns and has access to these limited, but essential, natural resources¹⁰⁴. Demographic and climatic pressures are pushing the frontier of crop cultivation progressively northward in Burkina Faso, Côte d'Ivoire and Ghana, where it is increasingly encroaching on traditional pastoralist zones and transit routes. Pastoralists and their animals, who are also affected by changes in climate, are forced to travel through these areas of cultivation, damaging crops and possibly triggering conflict. In Burkina Faso, climate-linked internal migration has flowed from the central plateau to the more economically dynamic south-southwest and the more land-abundant east. It is probable that conflicts over scarce natural resources will continue to develop in these areas, especially those involving access to water and arable land¹⁰⁵. According to FAO, in northern Burkina Faso and Mali, the effects of climate change, combined with unresolved grievances of the pastoralist Tuareg population, have the potential to erupt into more intense violence and conflicts¹⁰⁶.

Figure 11: Climate change as a threat multiplier - conceptual framework



Source: UNEP (2011), Livelihood security, Climate Change, Migration and Conflict in the Sahel.

Table 7: Summary climate trends and impacts

Climate Trends			Impacts			
Increased temperature	Shifts in precipitation patterns	Extreme weather events	Agricultural food security and	Water	Financial system	Energy resources
Temperatures are expected to increase by between 1 and 1.72°C for 2031-2050 compared to 1986-2005. Temperature increases higher than 2°C are projected to decrease millet and sorghum	Precipitation is projected to decline by as much as 30% by the end of the century (RCP8.5). RCP 2.6; RCP 4.5, RCP 6.0 indicated a general trend of decline in precipitation between 30 to 40% by 2100 ¹⁰⁸ . Greater frequency and intensity of heavy	Higher temperatures will increase the frequency of floods, droughts, heatwaves, locust outbreaks, desertification and sandstorms ¹¹⁰	One in 6 trees in the region has died since the 1950s and 1/5 of all species has disappeared locally due to rising temperatures and lower rainfall. At some sites in the Sahel, temperatures rose 0.8 °C and rainfall decreased 48%. Trees shifted	Increase in temperature and intensification of hydrological cycles is expected to result in water stress, affecting supply of ground and surface water. Further, climate change is expected to impact water	Risks for financial institutions are higher, as climate change impacts can increase the number of defaults on loans; reluctance of farmers and	Hotter temperatures are likely to increase energy demand for cooling and food storage, consequently increase GHG emission and

¹⁰⁴ FAO (2020), Protecting land and natural resources tenure rights in the era of COVID-19 by heeding the lessons from the past, <https://reliefweb.int/sites/reliefweb.int/files/resources/CB0706EN.pdf>

¹⁰⁵ Julie Snorek, Jeffrey Stark and Katsuaki Terasawa (2014), Climate Change and Conflict in the Sahel: A Policy Brief on Findings from Niger and Burkina Faso. USAID, https://www.researchgate.net/publication/291331260_Climate_Change_and_Conflict_in_the_Sahel_A_Policy_Brief_on_Findings_from_Niger_and_Burkina_Faso.

¹⁰⁶ FAO (2017), The future of food and agriculture.

¹⁰⁸ IPCC 6th Report, 2021

¹¹⁰ S. Godfrey and F.A. Tunhuma (2020), The Climate Crisis: Climate Change Impacts, Trends and Vulnerabilities of Children in Sub Sahara Africa, UNICEF Eastern and Southern Africa Regional Office, Nairobi, September, https://reliefweb.int/sites/reliefweb.int/files/resources/73800_theclimatecrisisreportesawcarsep20.pdf

yields by 15-25% by 2080 ¹⁰⁷	rains that produce floods. In northern Senegal, Mali and Burkina Faso, decline in intensity and frequency of heavy precipitation events is accompanied by an increase in dry spell length (CDD) ¹⁰⁹		southward towards wetter areas ¹¹¹ contributing to desertification. CC projected to reduce yields and livestock growth.	quality and increase rainfall variability	MSMEs to borrow ¹¹²	use of fossil fuel ¹¹³
---	--	--	--	---	--------------------------------	-----------------------------------

54. **Non-climate stressors and impacts.** Livelihood vulnerability in the selected countries is also linked to many non-climatic factors, such as unequal land distribution, insecure land tenure, poorly developed markets, existing trade barriers and inadequate infrastructure¹¹⁴. Political instability, urbanization and pollution contribute to the degradation of land, forests and water and the destruction of biodiversity; these processes are likely to become more marked as climate change progresses. Tensions from increased competition over diminishing natural capital add fuel to existing conflicts, which exacerbate poverty and hunger, undermine human rights and fuel migration, especially of rural youth. These negative impacts are exacerbated when smallholders have low adaptive capacity and limited financial resources to withstand disasters and losses. According to UNEP, migration, competition over natural resources and other coping responses of households and communities faced with climate-related threats could increase the risk of domestic conflict and have regional and international repercussions¹¹⁵.
55. Since the **COVID-19** pandemic began, governments have had to face even greater budget constraints, as they were forced to direct record amounts of spending on national social, economic and humanitarian support plans and social safety net programmes to help their most vulnerable populations cope and recover from the impacts of COVID-19. As a result, countries now need even more support to ensure an inclusive and climate resilient recovery, which involves the participation of the private sector, and progress towards the achievement of the GGWI's objectives in the Sahel region and beyond. IGREENFIN I will play a key role in providing timely green financial support for investments in the climate resilient, low emission recovery and the development of the smallholder agricultural sector.

Figure 12: Projected hotspots of internal climate in- and out-migration in the Sahel in 2050

¹⁰⁷ USAID (2017), Climate Change Risk in West Africa Sahel.

¹⁰⁹ Ibid.

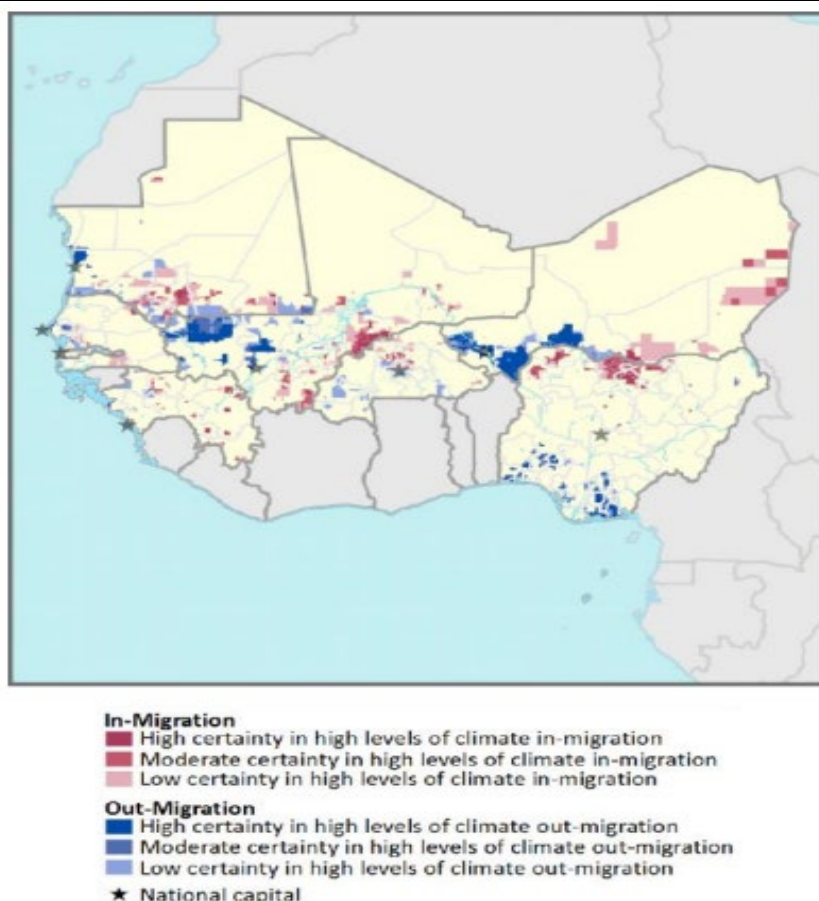
¹¹¹ P. Gonzalez et al. (2012), "Tree density and species decline in the African Sahel attributable to climate", *Journal of Arid Environments* 78 (March).

¹¹² IFAD (2021), IFAD Inclusive Financial Services Portfolio Stocktaking.

¹¹³ O. Hoegh-Guldberg et al. (2018), "Impacts of 1.5°C Global Warming on Natural and Human Systems", in Global Warming of 1.5°C. An IPCC Special Report on the impacts of global warming of 1.5°C above pre-industrial levels and related global greenhouse gas emission pathways, <https://www.ipcc.ch/sr15/chapter/chapter-3/>

¹¹⁴ IPCC, 2014, Chapter 13 Livelihoods and Poverty. https://archive.ipcc.ch/pdf/assessment-report/ar5/wg2/WGIIAR5-Chap13_FINAL.pdf

¹¹⁵ UNEP (2011), Livelihood security, Climate Change, Migration and Conflict in the Sahel.



Source: Kanta Kumari Rigaud et al. (2021), Groundswell Africa: Internal Climate Migration in West African Countries. Washington, DC: The World Bank.

56. In their efforts to develop their response to the impacts of climate change and variability (mainly temperatures and rainfall) affecting forests and agricultural production¹¹⁶ and to help smallholders move away from current unsustainable agricultural practices and energy use towards climate resilient, low emission technologies and practices, the five countries of IGREENFIN I (Burkina Faso, Côte d'Ivoire, Ghana, Mali and Senegal) elaborated Climate Smart Agriculture Investment Plans¹¹⁷. In these plans, they identified key value chains in their agricultural sector and for each one, climate smart agricultural practices that are effective in boosting yields and climate resilience and reducing GHG emissions (as a co-benefit) as investment priorities. These investment priorities are in alignment with their national adaptation and mitigation plans. Together, they constitute a broad range of green businesses to be prioritized for development with support from both the public and private sector.
57. Based on the findings of the Climate Smart and Adaptation Investment Plans (CSAIPs) and IFAD COSOPs, ten agricultural value chains have been pre-identified as priorities for investment in the IGREENFIN I programme, while taking into account the risks and vulnerabilities of each one, as well as the benefits of mainstreaming more sustainable practices. This will increase the region's resilience to future climate-related impacts which can no longer be avoided. The summary of the proposed intervention models for each value chain is presented in Table 8 below.

¹¹⁶ FAO (2017), The future of food and agriculture.

¹¹⁷ See, for example: [Cote d'Ivoire Climate-Smart Agriculture Investment Plan \(worldbank.org\)](https://www.worldbank.org/)

Table 8: Proposed initial intervention model for selected value chains for IGREENFIN I

Selected value chains		Climate risks	Evidence of impacts	Adaptation and mitigation interventions	Adaptation and mitigation benefits
Sustainable livestock production	Cattle	Projected increases in temperatures, rainfall variability and frequency and length of droughts across the Sahel expected to reduce food and water availability for cattle, contribute to disease/pest outbreaks, affect reproductive cycle (reduce calving rates from 60-70% to 25-30%) and growth, and degrade pastureland further.	Livestock disease kills 20% of ruminants in developing countries. Desertification and land degradation have resulted in loss of pastureland. Higher temperatures found to reduce feed intake, milk production, fertility and lead to energy deficits. During major droughts in the past, farmers reported losing entire herds.	<u>Adaptation:</u> Investment in fodder banks; crop-livestock integration; improvements to grazing corridors; efficient watering technologies; vaccination, pest monitoring and control systems, intensification of cultivated pastures; resilient breeds. <u>Mitigation:</u> Frequent removal of manure for composting. Use of RETs to store vaccines.	<u>Adaptation:</u> Improved access to water, feed and disease control will contribute to greater resilience and productivity of livestock, which is key for millions of small livestock producers in target countries. <u>Mitigation:</u> emissions from cattle production reduced through improved management of manure and displacement of fossil fuel in the storage of vaccines.
	Chicken	Projected increases in temperatures, high humidity and sunshine intensity have negative effects on poultry production, as they contribute to disease/pest outbreaks and affect production costs, the number of birds needed to raise to produce eggs and meat on the farm and feed grain availability (because high temperatures and low rainfall levels hinder grain harvest, their supply to the market and ultimately cost of poultry production)	Higher temperatures and humidity have been found to increase incidence of disease and reduce feed intake, affecting egg production and feed grain availability. Increases in temperature raise the mortality rate of chickens raised for meat by 8.4%, hens for eggs by 0.84%, and indigenous chickens by 0.32%.	<u>Adaptation:</u> Climate-smart livestock genotype; cross breeding with exotic birds; heat/drought/flood resilient and other climate resilient farming practices; Conserve & utilize climate resilient native poultry species; proper feed formulation; heat resilient housing; chicken waste processing into manure; extensive poultry management; vaccination, pest monitoring and control systems <u>Mitigation:</u> Frequent removal of manure for composting. Use of RETs to store vaccines.	<u>Adaptation:</u> Improved housing, health, feed and disease control will contribute to greater resilience and productivity, a key protein and income source for millions of producers and their families in target countries. <u>Mitigation:</u> emissions from chicken production reduced through improved management of manure and displacement of fossil fuel in the storage of vaccines.

Sustainable horticulture and vegetable production	Tomato	High temperatures, excess water or drought reduce productivity levels. Heat also impacts farmers' ability to market and conserve tomatoes, which are highly perishable. Climate variability increases incidence of disease.	Studies show significant declines in yields due to heat and water stress. High temperatures affect plant growth and reproduction, with increased incidence of abortion of fruit. Lack of water results in formation of fewer flowers and affects pollen. Farmers reported poor yields due to climate change.	Adaptation: Drip irrigation; soil management techniques to address low fertility and salinity (organic fertilizers, mulch); higher-yield, disease resistant varieties. Mitigation: promotion of RETS for storage and processing facilities and efficient irrigation systems to displace fossil fuel use.	Adaptation: Increased production levels and plant resilience, new job opportunities and reduction in post-harvest losses; all contribute to raising smallholders' income and resilience. Better use and conservation of water resources. Mitigation: Adoption of RETs reduces use of fossil fuels and wood as energy source and thus, emissions
	Sustainable Mango (agroforestry model 1)	Negative impacts of water stress and extreme temperatures on flower induction, fruit set and pollen survival. CC scenarios predict a rapid decline in the number of induction days by end of century. Shorter flowering periods reduce productivity. Rainfall variability – especially if there is rain during flowering period – may also reduce yields.	Influence of climate change on yield, time of maturation and quality	Adaptation: Drip irrigation, organic fertilizer, improved pruning practices, integration of mango trees in agroforestry systems; intercropping with cereals Mitigation: Mango trees have a significant capacity for carbon sequestration, tend to use up more carbon than they emit. RETs to power mango processing equipment to reduce post-harvest losses.	Adaptation: Mango trees serve as ecological barrier against CC, contribute to soil and water conservation. Better production and harvesting techniques increase yields and farmers' incomes and contribute to job creation. Growing mangos in agroforestry systems with native trees can increase yields three-fold and prevent soil erosion Mitigation: Mango trees contribute to carbon sequestration. Improved soil management through agroforestry contributes to carbon sequestration, and the adoption of RETs reduces use of fossil fuels and wood as energy source and thus, emissions.
Restoring degraded land and agroforestry	Cashew production with native species (agroforestry model 2)	Increased temperature, rainfall and humidity can lead to higher incidence of pests and disease, two major constraints to cashew production. Irregular rainfall, violent winds and prolonged drought can impact yields and quality. Though highly tolerant to drought, low rainfall during vegetative stage can prevent trees	Higher temperatures, erratic rainfall and increases in humidity increase the incidence of pests and disease. Irregular rainfall and temperature changes have been found to affect production.	Adaptation: Mulching to retain soil moisture, prevent weed growth and enhance soil structure; organic fertilizer; inclusion of cashew trees in agroforestry systems; drip irrigation. Investment in storage, conservation and processing techniques and infrastructure for reduced post-harvest losses and value addition. Mitigation: Cashew introduced to control soil erosion and for reforestation; ideal for carbon sequestration.	Adaptation: Improved productivity increase smallholders' incomes and create jobs. Cashew trees contribute to soil and moisture conservation, prevention of erosion and slowing desertification. Mitigation: Improved soil management and agroforestry systems that combine cashew trees with native species contribute to carbon sequestration. Adoption of RETs reduces use of fossil fuels and wood as energy source and thus, emissions.

		from obtaining enough water to sustain flowering.			
	Sustainable shea production (community forest model)	Favourable habitats of shea trees will decline by 13% (RCP8.5) by 2070 due to climate change. Higher risk of losing flowers and fruit due to heavy rainfall and winds; drought, flooding and wildfires increasing tree mortality and inhibit harvesting; decrease in water supply affecting growth and limiting processing; more frequent and prolonged droughts leading to the abortion of fruit. Deforestation biggest threat.	Higher temperatures and changes in rainfall patterns are already causing the trees to shift southward, with impacts on the trees' growth, regeneration and production levels.	<u>Adaptation</u> : climate resilient practices such as assisted natural regeneration and Zai, sustainable forest management techniques <u>Mitigation</u> : Agroforestry landscape in which shea is naturally grown acts like a carbon sink, controls soil erosion and good for reforestation. Adoption of RETs to power shea processing and storage facilities.	<u>Adaptation</u> : Agroforestry parklands conserve natural resource base, improve soil fertility, enhance resilience and allow farmers to diversify production. Investments in forest management and shea processing capacity improve incomes and create jobs, especially for women. <u>Mitigation</u> : shea value chain fixes 1.5 million tons of CO ₂ every year and every ton of shea kernel produced has a negative carbon footprint of 1.04 tons of CO ₂ (FAO-GSA, 2020). Valorisation of shea helps reduce deforestation and thus avoid emissions. Improved soil through combination of mixed native species with shea trees contributes to carbon sequestration. Adoption of RETs reduces use of fossil fuels and wood as energy source and thus, emissions.
Sustainable cereals-staple	Sustainable rice production	Increasing climate variability (rainfall) and high temperatures disrupt growing seasons. Greater risk of prolonged drought, heat waves, floods, water shortages, strong winds and storms, pests and diseases, causing substantial yield reductions or crop failure, which are higher for rainfed production. SLR increases risk of floods and salinization	Estimated reductions in rice yield range from 5-25% and up to 80%, in irrigated rice systems in the hot-dry season and around 40% in rainy season.	<u>Adaptation</u> : SRI; improved short-cycle rice varieties; efficient irrigation; management of soil salinity (namely drainage, flooding and organic matter addition). <u>Mitigation</u> : adoption of RETs to power processing facilities and irrigation; SRI involves alternate wetting and drying with minimal water application, thus reducing GHG emissions from flooded rice fields	<u>Adaptation</u> : More resilient and higher yielding rice production improves farmers' income and contributes to food security. More efficient water use. <u>Mitigation</u> : Reduced emissions from flooded rice fields thanks to SRI and adoption of RETs.
	Sustainable cassava production	Though resilient to climate change stressors, cassava production is vulnerable to pests and diseases.	Over a third of attainable cassava yield is lost every year to pests and disease, which could be exacerbated by	<u>Adaptation</u> : Efficient irrigation systems, intercropping with cover crops and trees (fruit and other), composting/organic fertilizer and pest control;	<u>Adaptation</u> : Improved productivity through more effective water, soil and pest management increases farmers' incomes and resilience

crops production			higher temperatures (e.g., whitefly).	sustainable land management techniques. <u>Mitigation:</u> Composting using biogasifiers, minimal tilling and adoption of RETs to power processing, storage and irrigation.	and contributes to food security. <u>Mitigation:</u> emission reductions due to biogasifiers and use of RETs.
	Sustainable groundnut production	Though moderately drought resistant, excessive heat and prolonged drought limit yields. Rainfall variability is an important risk as germination is poor in drier soils, whereas excessive moisture increases plant mortality and incidence of disease. Average production expected to decrease 11% over next 20 years.	Temperatures above 33°C found to decrease growth by up to 84%. Higher temperatures will have most dramatic impact on yields, which are traditionally low due to unreliable rain, poor soils, pests and disease.	<u>Adaptation:</u> improved short-cycle varieties, organic fertilizer, intercropping, stone bunds, water efficient irrigation systems, storage and conservation techniques. <u>Mitigation:</u> As a nitrogen-fixing crop, groundnuts reduce GHG emission intensity and use of chemical fertilizers. Solar energy instead of fossil fuels to power drip irrigation systems and other equipment.	<u>Adaptation:</u> More resilient production and higher yields improve farmers' incomes and food security and create jobs in value chain. More efficient resource management (soil and water) contributes to conservation. <u>Mitigation:</u> As a nitrogen-fixing crop, has mitigation benefits, as it maintains soil health. Reduced emissions from use of RETs.
	Sustainable millet production	Though well-adapted to hot areas, temperature increases higher than 2°C are projected to decrease millet yields by 10% over next 20 years and 15-25% by 2080. Rain variability contributes to lower yields, primarily excess rain during flowering, which can cause crop failure.	Production predicted to decrease on average by 10% over next 20 years.	<u>Adaptation:</u> Climate resilient practices such as inter-cropping; soil fertility (compost) and water retention (Zai and half-moons) techniques; climate information systems. <u>Mitigation:</u> Use of millet crop leftovers to maintain and increase soil carbon levels; adoption of RETs for processing to reduce post-harvest losses and displace diesel to power drip irrigation.	<u>Adaptation:</u> More resilient production and higher yields increase farmers' incomes and contribute to food security. Improved soil and water management. <u>Mitigation:</u> Crop leftovers have a high carbon content, making them particularly significant for increasing soil carbon levels. Emissions avoided due to adoption of RETs instead of fossil fuels.

58. Institutional and regulatory frameworks for green businesses: The five selected countries for IGREENFIN I have adopted various laws and regulations in the agricultural and renewable energy sectors, as well as policy frameworks that cover some areas relative to green business, but they do not have dedicated legal frameworks governing green business in the agricultural sector.¹¹⁸ There is thus a need for further policy dialogue between private and public sector stakeholders (governments, central, national and local banks, farmers' organizations, etc.) to develop adequate policy frameworks that could further increase attractiveness of these nascent, climate-focused agricultural sectors and support the creation of green businesses. The financial system currently faces institutional and regulatory challenges that reduce their readiness, such as limited climate risk management knowledge and practices at project level (standards), zero or few green lending products and the lack of instruments and tools such as ESG benchmarks, environmental data and awareness required for decision-making on investments and policies in agriculture¹¹⁹. Addressing these challenges requires innovative financing schemes

¹¹⁸ AfDB/FAO (2015), Agricultural Growth in West Africa: Market and Policy Drivers, Rome,

https://www.afdb.org/fileadmin/uploads/afdb/Documents/Publications/Agricultural_Growth_in_West_Africa_-_Market_and_policy_drivers_-_OSAN.pdf

¹¹⁹ UNEP FI (2005), CEO Briefing: Sustainability Banking in Africa, http://www.unepfi.org/fileadmin/documents/ceo_brief_sust_banking_in_africa_2005.pdf

and channels, such as blended finance with highly concessional loans, as well as the dissemination of regional/national public awareness materials on green finance and greening financial systems and capacity-building, particularly for public development banks and the central banks: the Central Bank of West African Monetary Union (WAMU) which covers Burkina Faso, Côte d'Ivoire, Mali and Senegal, and the Ghanaian Central Bank.

Regional Approach to Addressing Climate Change:

59. **The Great Green Wall Initiative:** To address climate change, restore and sustainably manage land and reduce poverty, the GGWI was launched as a Pan-African initiative with the endorsement of the African Union (AU). The GGWI is one of the earliest international land restoration initiatives that brings together African countries and international partners under the leadership of the AU and Pan-Africa Agency of the Great Green Wall (PAAGGW). A broad set of African and international partners are involved in the initiative through project implementation and development, or through the funding of several ongoing and future projects in all GGW countries. Combining the intervention zones reported by the eleven sub-Saharan GGW member states, the GGW initiative extends over a total area of 152 million ha, with the largest intervention zones located in Eritrea, Ethiopia, Mali, Niger and Nigeria¹²⁰. Several regional and transboundary programmes to support the implementation of the GGWI at the regional level have been implemented in recent years.
60. **The GGW Initiative objective:** The goal of GGWI is to restore 100 million hectares of degraded land, sequester 250 million tonnes of carbon and create 10 million green jobs in rural areas across the Sahel by 2030. With a span of 8,000 km (from Senegal in the west to Djibouti in the east) and a width of 15 km, the intervention zone is to become the home of a mosaic of trees, grasslands and other vegetation. The initiative currently involves more than 20 partner countries in Africa, of which the following 11 countries have been selected as intervention zones for IGREENFIN 0, I and II: Burkina Faso, Cameroon, Chad, Djibouti, Eritrea, Ethiopia, Ghana, Mali, Mauritania, Niger, Nigeria, Senegal and Sudan. These countries are among the world's most vulnerable and least resilient countries to climate change and the least ready to mobilize investments and convert them to adaptation actions, as evidenced by their low ranking on the ND-GAIN index¹²¹. Numerous initiatives have been implemented by GGW partners to contribute to the achievement of the GGW objectives, as described in the table below.

Table 9. Overview of regional initiatives supporting the GGW

Initiative	Implementing entity	Achievements
Action Against Desertification Program (2014-2019)	FAO	- Capacity-building for rural communities, government and NGO partners to create an enabling environment, large-scale restoration and SLM of degraded agro-silvo-pastoral lands, landscapes and forests. - Dissemination of good SLM practices - Creation of income-generating activities (IGA) and employment opportunities in rural areas, through the sustainable production, processing and marketing of agricultural products and forest goods and services - Farmer field schools and knowledge exchanges about the causes and the best ways to combat and prevent desertification
BRIDGES - Boosting Restoration, Income, Development, Generating Ecosystem Services (2017-2020)	FAO Turkey	- Restoration of 5,000 hectares of dryland forests and landscapes - Reinforcing value chains of non-timber forest products - Building GGW information and monitoring systems - Compiling, managing and sharing knowledge and good practices, promoting communications and visibility
The Sahel and West Africa Programme in support of the GGW (2013–2019)	World Bank	12 sustainable land and water management country projects - Training for farmers on SLM practices and improved agricultural technologies - One regional project on coordination, M&E and knowledge management

¹²⁰ Intervention areas in Burkina Faso and Niger are subject to confirmation by countries, as they do not coincide between the two sources used.

¹²¹ ND-GAIN Country Index rankings, <https://gain.nd.edu/our-work/country-index/>

Local Environmental Coalition for a Green Union (2014-2018)	UNCCD	• Micro-projects implemented in 23 communities in five countries (Mali, Niger, Burkina Faso, Chad, and Senegal) • Regional level capacity building activities on SLM and innovative financing complemented the projects on the ground.
Large-scale Assessment of Land Degradation to guide future investment in SLM in the GGW countries (2019-2024)	UNEP	• Assess available tools and methodologies for scientific measurement of the ecological impacts of land degradation and SLM practices in support of future investment decisions in the GGW region.
The IFAD-led GEF Integrated Approach Pilot on Food Security (2017- 2022)	IFAD	• Five GGW countries (Burkina Faso, Ethiopia, Niger, Nigeria and Senegal (out of 12 countries) • Projects to ensure the long-term sustainability and resilience of production systems, particularly in drylands. • Institutional and policy strengthening
Closing the Gaps in the Great Green Wall: Linking Sectors and Stakeholders for Increased Synergy and Scaling-up (2016-2019)	IUCN	• Improve the involvement of civil society organizations and vulnerable groups in the GGW Initiative and the national GGW agencies.

Challenges and barriers to the implementation of the Great Green Wall

61. The report 'The Great Green Wall: Implementation Status and Way Ahead to 2030'¹²² published by UNCCD in September 2020 provides an overview of the main challenges encountered by the countries and implementation partners in the execution of GGW activities, which include:
62. **Weak environmental governance and donor fragmentation:** An overarching issue appearing in most GGW countries is weak governance in the field of environmental change, which is a main barrier to implementation. Institutional challenges include the lack of high-level political support for the environmental policy agenda from the governments of the GGW member states and weak organizational structures and processes for the implementation of environmental projects, development initiatives or programmes. As a result of the lack of structures and processes for handling environmental matters and developing policy frameworks, national GGW agencies have not yet been established.
63. **Limited coordination of knowledge management:** The lack of managed knowledge, information sharing and coordination mechanisms at the national and regional level leads to insufficient coordination and collaboration between GGW countries, project developers and donors. This is especially important for taking advantage of lessons learned, innovations and success stories, whose replication and scaling up could accelerate and ensure an efficient expansion of the GGWI's activities.
64. **Unstructured monitoring and evaluation (M&E) systems:** A fundamental barrier is the lack of a system to monitor and report on GGW activities on the ground. The problems identified between 2011 and 2017 clearly indicate a need for better overall follow-up and monitoring. M&E expertise is absent in general, which hampers the establishment of proper M&E systems at the project and national levels and across the GGW Initiative as a whole. It also results in shortcomings in documentation and hinders the sharing of lessons learned, which is key to avoid negative developments and capitalize on positive results achieved by the projects under the GGW.
65. **Access to financing, particularly from the private sector:** An estimated US\$2.6-4.3 billion is needed to restore the 100 million hectares of land in Sahel and achieve the GGW objectives. Mobilizing and absorbing finance requires adequate structures, processes and capabilities to be in place within national governments, at the local level and among project developers/managers. Most GGW countries still face difficulties in establishing the required governance and project structures for attracting, accessing, managing and reporting on large(r) finance flows. Exploring "untraditional" finance mechanisms by involving the private sector (IUCN Conference on Mobilizing Private Investment in the GGW¹²³) in land use

¹²² UNCCD (2020), The Great Green Wall: Implementation Status and Way Ahead to 2030, <https://www.unccd.int/publications/great-green-wall-implementation-status-and-way-ahead-2030>

¹²³ Mobilizing Private Investment in the GGW, Recommendations and Actions, IUCN Conference, 23-25 March 2021.

and land restoration activities under the GGW initiative could be a great opportunity to complement the somewhat limited funding that has been mobilized so far. However, this will be a very challenging undertaking.

66. There are various challenges in developing bankable projects in such an extreme environment when targeting the land use sector, such as scale, land rights/ownership, community involvement and low(er) profitability expectations, among others. Apart from a few larger, commercial forestry or agribusiness/food operations, most conventional restoration projects generate returns that are too low to attract private investors. Furthermore, it is difficult to ensure the continuity, long-term sustainability and replication of these projects in the context of the Sahel and thus, they are considered riskier than other investments.
67. Pilot initiatives have established a range of restoration and sustainable land management approaches that have great potential to go to scale. However, there is a significant shortfall in resources and particularly in private investment in the region. Agriculture in the Sahel is insufficiently financed in general, despite this sector's contribution to employment, sustainable growth and innovation. Renewable energy, such as solar power, also has significant potential in terms of investment, job creation and private sector involvement. However, long-term financing is insufficient, as is the technical assistance needed to strengthen the monitoring of social and environmental impacts and build support for cooperatives for value-addition (including processing and marketing). Conflicts and issues related to governance, land tenure, climate impacts and limited data are some of the challenges faced by companies willing to invest in the agricultural value chains in the GGW countries. Promising opportunities include investing in agri-SMEs, global value chains, high value crops and new markets (such as food, carbon and humanitarian relief markets).
68. Overcoming the barriers to investment in the GGW requires adequate information, awareness raising, implementation of relevant investment pilots, capacity-building (e.g., in mobilizing financing for the GGW) and addressing 'aggregators', such as land tenure and public policy.

Table 10. Summary of relevant Great Green Wall definitions, processes and key organizations

Term	Description
Great Green Wall Initiative	The Great Green Wall Initiative is a Pan-African initiative created under the auspice of the African Union with overall objective of contributing to the fight against desert encroachment and the development of Saharan-Sahelian areas through sustainable management of natural resources and the fight against poverty. Its specific objectives include (i) preserve and enhance biodiversity (ii) restore and conserve soil (iii) diversify productive systems (iv) satisfy households needs (v) increase revenue through the promotion of income-generating activities and the installation of basic social infrastructure and (vi) improve capacity for carbon sequestration in soils and vegetation cover ¹²⁴ .
African Union	The African Union launched the Great Green Wall Initiative in 2007 and is coordinating the Steering Committee of the Great Green Wall Initiative.
Pan Africa Agency for the Great Green Wall (PAAGGW)	Pan Africa Agency of the Great Green Wall is an intergovernmental organization that aims to achieve the establishment of the Great Green Wall Initiative. It is responsible for the coordination and monitoring of implementation activities of the Great Green Wall Initiative and the mobilization of resources.
Great Green Wall Accelerator	The Great Green Wall Accelerator aims to help all actors for the Great Green Wall Initiative to better coordinate, monitor and measure the impact of their actions. It will build on and support the existing Great Green Wall institutions, entities and processes. The overarching goal is to accelerate the realization of the GGW ambitions by 2030 by supporting GGW partners in accessing the full range of funding sources and options. The Accelerator is being established by the UNCCD.
Great Green Wall Country	A country where the GGW Initiative is being implemented. These countries are Burkina Faso, Chad, Djibouti, Ethiopia, Eritrea, Mali, Mauritania, Niger, Nigeria, Senegal and Sudan.
Great Green Wall Focal Point	The GGW Focal Points are ministries in each country in charge of coordinating and monitoring GGW activities. They act as liaison entities with the PAAGGW.
UNCCD	A historic partner of the Great Green Wall, the UNCCD has supported the GGW through the EU funded Local Environmental Coalition for a Green Union project (2014–2018), a public awareness campaign "Growing a World Wonder", and a landmark report on the status of the UNCCD. Since January 2021, the UNCCD has been working to establish the GGW Accelerator.

¹²⁴ Memorandum of Understanding between the Pan Africa Agency for the Great Green Wall and the African Forest Forum, <https://www.grandemuraillevierte.org/images/DocsOfficiels/MOUAFFAPGMVENGLISH.pdf>

Closing the Gaps in the Great Green Wall

The Great Green Wall Accelerator

69. The Pan Africa Agency for the GGW (PAAGWW) created in 2010 is responsible for the coordination and monitoring of the implementation of the GGW and for the mobilization of necessary resources in relation with the AU and the Member States. The nine Regional Structural Programmes (RSP) coordinated by the PAAGGW are structured around the five major strategic axes of the GGWI, implemented in all GGW countries and tailored to each country's specific needs. They all share a common thread focused on the restoration of degraded land, economic development, increasing adaptation and resilience to climate change and the fight against food insecurity and migration. At the national level, Member States have created National GGW Agencies or focal points to supervise and coordinate the implementation of national GGW priority actions. The PAAGGW plays a key role in coordinating country-level commitments to advancing the objectives of the GGW and is currently being supported by the UNCCD to implement the Great Green Wall Accelerator.
70. As indicated in the September 2021 UNCCD report on the status of the GGWI¹²⁵, if the current level of funding and pace of implementation continue, the GGWI's objectives and all the co-benefits of this flagship initiative will not be met. According to this report, it is estimated that a minimum investment of US\$33 billion from national and international sources will be required to support countries in achieving the GGW objectives by 2030. With the shift in focus of the GGWI from tree planting to more comprehensive rural development, including improving the production systems of smallholder farmers and pastoralist communities to increase food security, create jobs and boost the local economy through sustainable land management, countries are now tackling systemic challenges in an integrated and holistic manner to ensure long-term sustainability in the entire Sahel region. Stronger support, coordination and engagement by development partners, international financing institutions and the private sector are critical for harnessing emerging technologies and innovations and promoting policy options that unlock market opportunities and capital for scaling up interventions. Financial needs for impact at scale are estimated to average between US\$40-130 million per country annually. More resources will thus have to be mobilized from a range of domestic and international sources, including multilateral development banks, special financing mechanisms such as the GEF, the Least Developed Countries Fund, and the GCF, as well as bilateral partners, civil society organizations and the private sector¹²⁶.
71. **The Great Green Wall Accelerator:** In September 2020, at the online meeting of the GGW Ministers organized by the UNCCD, the GGW countries invited the GCF and the other international entities to boost their support for the GGW initiative¹²⁷. In response to this invitation, in January 2021, during the One Planet Submit, development partners committed to support the GGW objectives in the eleven GGW countries through the launch of a Great Green Wall Accelerator to facilitate and strengthen the coordination and collaboration of donors and stakeholders involved in the GGWI and better monitor and measure the impact of their actions. The GGW Accelerator will be coordinated through the PAAGGW, with initial support from UNCCD.
72. This Accelerator aims to address several of the barriers presented above by providing a common M&E framework to all donors who pledged additional support to the GGW objectives during the One Planet Submit. The Accelerator will also give greater visibility to the GGWI and mobilize additional financial resources from the private sector to accelerate the pace of implementation. The Accelerator will organize the reporting of donors along the following pillars:
- Investment in small and medium-sized farms and strengthening of value chains, local markets, organization of exports
 - Sustainable management of ecosystems and land restoration
 - Climate resilient infrastructures and access to renewable energy
 - Favourable economic and institutional framework for effective governance, sustainability, stability and security
 - Capacity-building
73. The Great Green Wall Accelerator will:
- Coordinate the efforts of all stakeholders through the harmonization of the impact measurement framework
 - Support the implementation of the GGW Initiative through a structured, multi-stakeholder approach based on the five pillars
 - Facilitate a more comprehensive mapping of available funding and projects
 - Connect stakeholders to scale up successful initiatives and promote the Sahel as a land of opportunity

¹²⁵ UNCCD (2020), The Great Green Wall: Implementation Status and Way Ahead to 2030.

¹²⁶ GEF/World Bank (2019), Great Green Wall Initiative: Supporting Resilient Livelihoods and Landscapes in the Sahel, https://www.thegef.org/sites/default/files/publications/gef_great_green_wall_initiative_august_2019_EN_0.pdf

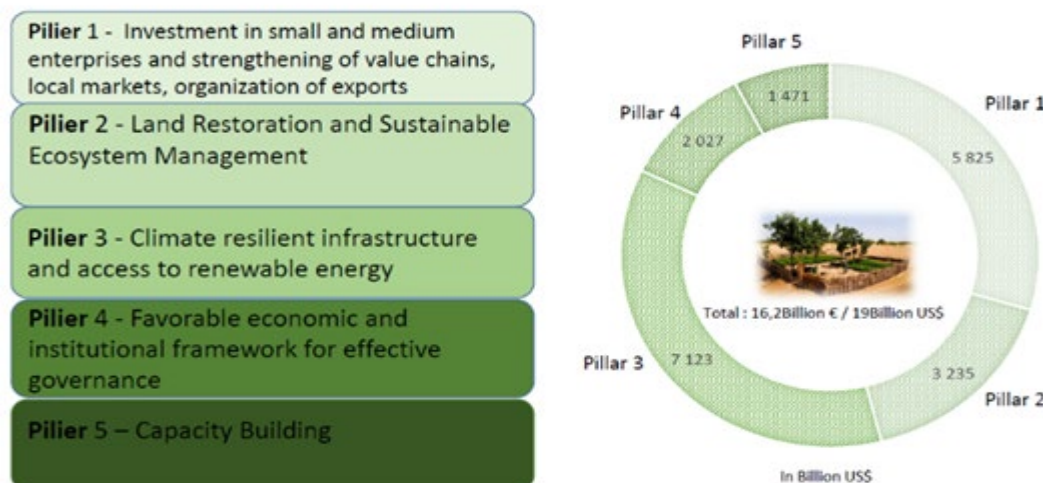
¹²⁷ See the Ministerial Declaration on the Great Green Wall from 7 September 2020: <https://www.unccd.int/news-events/ministerial-declaration-great-green-wall>

- Involve the private sector, civil society and research/innovation in GGW efforts.

74. Pledges amounting to 14.3 billion euros in funding for the GGWI until 2025 were made by several multilateral and bilateral organizations at the Summit. The reporting of these pledges will be facilitated by the Accelerator through a common reporting framework¹²⁸. The GCF has committed to addressing Africa's and LDCs' needs and the Sahel is one of the key areas in terms of climate vulnerability.

75. Figure 13 below presents the amounts pledged to the GGW Accelerator per pillar:

Figure 13: Pledges for the GGW Accelerator pillars



Source, UNCDD, GGW Accelerator 2021

The Global Environmental Facility investments in the GGW

76. The Global Environment Facility (GEF) has played a leading role in influencing a shift from the initial vision of a tree planting venture to one focused on integrated management of natural resources for improving livelihoods and landscapes. In response to demands of the countries in the region, the GEF has invested over US\$800 million in grants and leveraged an additional US\$6 billion from national governments, development partners and other multi-lateral donors for projects in the Sahel region. This financing has helped GGW member countries to boost their efforts to promote practices for improving crop and livestock productivity, restoring degraded parklands and strengthening resilience and adapting to climate change¹²⁹.

77. Building on the GEF's experience with programmes using an integrated landscape approach¹³⁰, the GEF 7 "Harnessing the GGWI for a Sustainable and Resilient Sahel" project is currently being implemented by UNEP to engage with all GGW stakeholders in fostering dialogue with countries and fleshing out a longer-term vision. It will serve as the vehicle to design a program with potential to mobilize larger investments in GEF-8. GEF-8 resources to support the GGW will be determined upon completion of the replenishment process and discussion with beneficiary countries on the use of their STAR allocations.

The Green Climate Fund Great Green Wall Regional Support Programme

78. In response to the call for pledges during the One Planet Summit in January 2021, the Green Climate Fund invited IFAD to develop a Great Green Wall Regional Support Programme (GGW RSP). The objective of the GGW RSP is to enhance the collective impact of individual GCF projects and programmes which may otherwise be implemented in silos by facilitating knowledge exchange and innovation. The GCF will continue to set the strategic direction of its investments and approve projects presented by Accredited Entities that contribute to the GGW Accelerator's five pillars. The GCF GGW Regional Support Programme will contribute to enhanced collaboration and efficiency, improved streamlining of actions and stronger

¹²⁸ The proposed reporting framework has been shared with donors and discussed with the 11 GGW countries but is not available to the public yet.

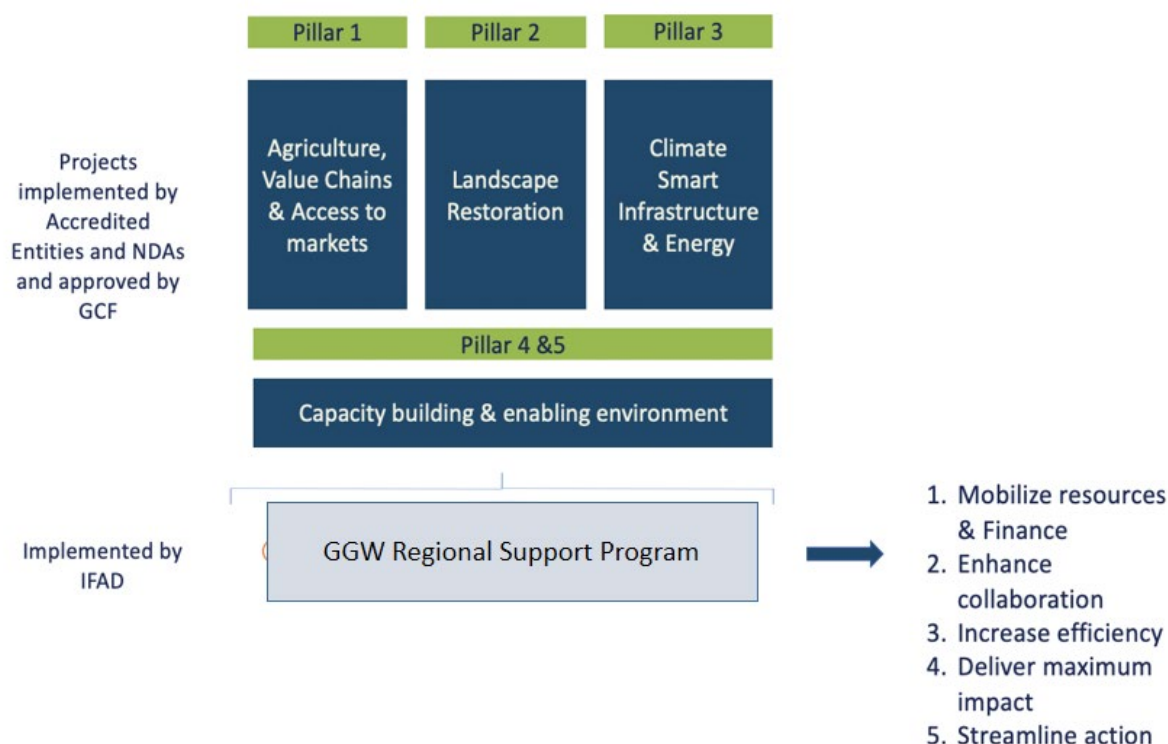
¹²⁹ For more information on GEF supported projects in the GGW from the GEF-4 through to the GEF-6 cycles, see: https://www.thegef.org/sites/default/files/publications/gef_great_green_wall_initiative_august_2019_EN_0.pdf

¹³⁰ Relevant GGW-related integrated approach programmes are: Terrafrica and the GEF-4 Strategic Investment Program (SIP); GEF-5 Sahel and West Africa Programme to Support the GGWI (SAWAP/BRICKS); GEF-6 IAP on Food Security (RFS); GEF-7 Food Systems, Land-Use and Restoration Impact Program (FOLUR); GEF-7 Dryland Sustainable Landscapes Impact Programme (DSL IP).

impacts across projects but will not play a coordination role in the preparation and implementation of projects, nor in the GCF's strategic investments in the GGW.

79. In addition to support to GCF projects on knowledge management and innovation, the GCF invited the Food and Agriculture Organization to develop a monitoring and evaluation framework to facilitate consolidated reporting of GCF projects across the GGW. The monitoring and evaluation framework will provide a multi-scale Monitoring and Evaluation of ecosystem services, low-emissions, climate-resilient pathways. The framework will focus on (i) GGW Monitoring and Reporting information integrated into the Framework for Ecosystem Restoration Monitoring (FERM) (ii) GGW baseline information prepared using innovative geospatial and technologies. The monitoring and evaluation framework for the GGW will be implemented under the project SURAGWA project currently under development.

Figure 14: Role of the GGW Regional Support Programme in support of GCF approved projects in the GGW



80. As of October 2021, 29 projects had been approved by the GCF and are being implemented in the 11 GGW countries¹³¹ by 17 different Accredited Entities. They are listed below in table 11 below. In addition to the already approved projects and taking into account the draft Entity Work Programmes submitted by some AEs, the Country Work Programmes submitted by some GGW countries and informal discussions that took place with GGW countries and AEs prior to the One Planet Submit, it is expected that GCF will provide approximately US\$1 billion to GGW countries. Additional information on the approved projects are found in Annex 2 Feasibility Study.

Table 11: Overview of the projects approved by the GCF in the GGW

¹³¹ Several of these are multi-countries projects and implemented in countries outside the GGW area, as shown in table 11.

Projects ID	Approved Reference	AE	Board Approval	Main GGF Pillar	Project Name	Countries List	Sector	Co-Financing	GCF
12660	FP003	CSE	B11	3	Increasing the resilience of ecosystems and communities through the restoration of the productive bases of salinized lands	Senegal	Public	546 000	7 610 000
13440	FP012	WB	B13	5	Africa Hydromet Program – Strengthening Climate Resilience in Sub-Saharan Africa: Mali Country Project	Mali	Public	8 250 000	22 750 000
13840	FP021	AFD	B14	3	Senegal Integrated Urban Flood Management Project	Senegal	Public	65 116 279	17 441 860
15320	FP027	Deutch	B14	3	Universal Green Energy Access Programme (UGEAP)	Ethiopia	Private	37 672 000	13 600 000
		Deutch	B14	3		Nigeria	Private	37 672 000	13 600 000
14920	FP049	WFP	B18	1	Building the climate resilience of food insecure smallholder farmers through integrated management of climate risk (R4)	Senegal	Public		9 983 521
17220	FP058	MOFEC	B18	1	Responding to the increasing risk of drought: building gender-responsive resilience of the most vulnerable communities	Ethiopia	Public	4 958 095	45 002 759
14130	FP074	WB	B19	5	Africa Hydromet Program – Strengthening Climate Resilience in Sub-Saharan Africa: Burkina Faso Country Project	Burkina Faso	Public	2 500 000	22 500 000
16280	FP078	ACUMEN	B19	1	Acumen Resilient Agriculture Fund (ARAF)	Nigeria	Private	7 500 000	6 500 000
17150	FP092	AFDB	B21	1	Programme for integrated development and adaptation to climate change in the Niger Basin (PIDACC/NB)	Burkina Faso	Public	14 558 171	6 942 091
		AFDB	B21	1		Chad	Public	15 253 177	7 273 506
		AFDB	B21	1		Mali	Public	19 309 510	9 207 776
		AFDB	B21	1		Niger (the)	Public	16 368 882	7 805 532
		AFDB	B21	1		Nigeria	Public	26 425 859	12 601 220
20220	FP093	AFDB	B21	3	Yeleen Rural Electrification Project in Burkina Faso	Burkina Faso	Public	33 488 372	28 255 814
17550	FP095	AFD	B21	1	Transforming Financial Systems for Climate	Burkina Faso	Private	6 291 047	3 655 814
		AFD	B21	1		Nigeria	Private	40 483 605	23 525 581
		AFD	B21	1		Senegal	Private	6 243 023	3 627 907
16110	FP099	FMO	B21	3	Climate Investor One	Djibouti	Private	40 086 540	5 556 000
		FMO	B21	3		Ethiopia	Private	40 086 540	5 556 000
		FMO	B21	3		Nigeria	Private	40 086 540	5 556 000
		FMO	B21	3		Senegal	Private	40 086 540	5 556 000
17750	FP102	BOAD	B22	3	Mali solar rural electrification project	Mali	Public	9 632 087	30 196 060
19770	FP103	GIZ	B22	3	Promotion of Climate-Friendly Cooking: Kenya and Senegal	Senegal	Public	4 181 192	9 598 256
20390	FP105	BOAD	B22	3	BOAD Climate Finance Facility to Scale Up Solar Energy Investments in Francophone West Africa LDCs	Burkina Faso	Private	11 821 942	11 821 942
		BOAD	B22	3		Mali	Private	11 821 942	11 821 942
		BOAD	B22	3		Niger (the)	Private	11 821 942	11 821 942
18950	FP128	MUFG	B25	2	Arbaro Fund – Sustainable Forestry Fund	Ethiopia	Private	25 007 500	3 572 500
19590	FP136	WB	B26	2	Resilient Landscapes and Livelihoods Project	Ethiopia	Public	132 000 000	165 237 592
22200	FP138	BOAD	B26	3	ASER Solar Rural Electrification Project	Senegal	Public	143 310 472	87 726 949
14630	FP139	UNDP	B26	1	Building resilience in the face of climate change within traditional rain fed agricultural and pastoral systems in Sudan	Sudan	Public	15 540 000	25 645 114

24500	FP148	ACUMEN	B27	3	Participation in Energy Access Relief Facility ("EARF")	Nigeria	Private	6 660 000	6 660 000	
		ACUMEN	B27	3		Senegal	Private	1 260 000	1 260 000	
24780	FP151	IUCN	B27	1	Global Subnational Climate Fund (SnCF Global) – Technical Assistance (TA) Facility	Burkina Faso	Private	226 195	440 485	
		IUCN	B27	1		Mali	Private	226 195	440 485	
		IUCN	B27	1		Mauritania	Private	226 195	440 485	
		IUCN	B27	1		Nigeria	Private	226 195	440 485	
		IUCN	B27	1		Senegal	Private	226 195	440 485	
18730	FP152	PCA	B27	1	Global Subnational Climate Fund (SnCF Global) – Equity	Burkina Faso	Private	14 286 000	3 571 500	
		PCA	B27	1		Mali	Private	14 286 000	3 571 500	
		PCA	B27	1		Mauritania	Private	14 286 000	3 571 500	
		PCA	B27	1		Nigeria	Private	14 286 000	3 571 500	
		PCA	B27	1		Senegal	Private	14 286 000	3 571 500	
21260	FP162	IFAD	B28	1	The Africa Integrated Climate Risk Management Programme: Building the resilience of smallholder farmers to climate change impacts in 7 Sahelian Countries of the Great Green Wall (GGW)	Burkina Faso	Public	8 639 744	11 835 937	
		IFAD	B28	1		Chad	Public	8 639 744	11 835 937	
		IFAD	B28	1		Mali	Public	8 639 744	11 835 937	
		IFAD	B28	1		Mauritania	Public	8 639 744	11 835 937	
		IFAD	B28	1		Niger (the)	Public	8 639 744	11 835 937	
		IFAD	B28	1		Senegal	Public	8 638 535	11 834 280	
24560	FP163	WB	B28	3	Sustainable Renewables Risk Mitigation Initiative (SRMI) Facility	Mali	Public	183 347 975	39 998 000	
23600	FP168	AFDB	B29	3	Leveraging Energy Access Finance (LEAF) Framework	Ethiopia	Private	149 910 000	32 471 000	
		AFDB	B29	3		Nigeria	Private	220 920 000	47 852 000	
23380	FP176	BOAD	B30	1	Hydro-agricultural development with smart agriculture practices resilient to climate change in Niger	Niger (the)	Public	17 912 143	35 045 084	
25180	FP178	AFDB	B30	3	Desert to Power G5 Sahel Facility	Burkina Faso	Private	192 663 399	35 385 000	
		AFDB	B30	3		Chad	Private	182 699 459	33 555 000	
		AFDB	B30	3		Mali	Private	201 728 951	37 050 000	
		AFDB	B30	3		Mauritania	Private	119 812 296	22 005 000	
		AFDB	B30	3		Niger (the)	Private	119 812 296	22 005 000	
21180	SAP012	IFAD	B24	1	Inclusive Green Financing for Climate Resilient and Low Emission Smallholder Agriculture	Niger (the)	Public	3 459 302	9 883 721	
20620	SAP019	FAO	B27	2	Gums for Adaptation and Mitigation in Sudan (GAMS): Enhancing adaptive capacity of local communities and restoring carbon sink potential of the Gum Arabic belt, expanding Africa's Great Green Wall	Sudan	Public		9 975 000	
Total								2 392 703 307	1 065 768 370	

Long-term vision for collaboration between GCF and GEF

81. The Long-Term Vision on Complementarity, Coherence, and Collaboration between the Green Climate Fund (GCF) and the Global Environment Facility (GEF)¹³² responds to the need to build on the experiences from ongoing collaboration between the GCF and the GEF and respond to developing countries' needs.

¹³² For more on the Long-Term Vision, see: https://www.thegef.org/sites/default/files/council-meeting-documents/EN_GEF_C.60_08_Long-Term%20Vision%20on%20Complementarity%20Coherence%20and%20Collaboration%20between%20the%20Green%20Climate%20Fund%20and%20the%20Global%20Environment%20Facility.pdf

Figure 15. Summary of the Long-Term Vision on Complementarity, Coherence, and Collaboration between the Green Climate Fund (GCF) and the Global Environment Facility (GEF)

The Long-Term Vision on Complementarity, Coherence, and Collaboration between the Green Climate Fund (GCF) and the Global Environment Facility (GEF)

1. **Coordinated support for major initiatives:** To provide support in line with each Fund's strengths, as well as exploring mainstreaming gender as such collaborations are developed.
 - 1.1. **Facilitation of national investment planning:** Support, through existing programming processes and communicating of the long-term vision on complementarity, country-owned national planning and priority setting for programming to utilize opportunities across the GEF and GCF.
 - 1.2. **Develop a list of activities or programs each fund will prioritize:** The two funds will aim for better coordination and greater efficiency of project and program preparation by streamlining consultation processes with countries, optimizing number of missions and consultants involved, and identifying potential duplication of funding, among others.
 - 1.3. **Support collaborating financing platforms:** As part of the coordinated support for major initiatives (see objective 1 above), the two funds could support the establishment and deployment of collaborating financing platforms.
2. **Sharing information, lessons learned and knowledge**
 - 2.1. **Information sharing, exchange and application of lessons learnt from portfolio to strengthen implementation of projects and programmes:** The Funds will review and apply relevant lessons learned across institutions in areas of project monitoring, indicators, and other relevant aspects of project implementation.
 - 2.2. **Collaborate on development of methodologies and guidance to maximize climate impacts through strong project design:** The two funds agree to produce and use a set of guidance products and methodologies for designing projects and measuring their impact, to maximize emissions reduction and adaptation impacts.
3. **Communication and outreach**
 - 3.1. **Communication and inclusion of long-term vision on complementarity in respective fund's Programming:** Relevant elements in this LTV will be included in the strategic documents in current and future replenishment and programming cycles.
 - 3.2. **Collaborate on communications, outreach, and sharing of lessons learned:** The two secretariats will enhance outreach and communication efforts, and on sharing of lessons learned from coordination and complementarity efforts.

82. Advancing the GGW vision will require an approach that is underpinned by holistic and systemic thinking to maximize potential for harnessing synergies at scale. In this context, the GGW Regional Support Programme will create an opportunity to implement this long-term vision and enhance complementarity and collaboration at the regional and country level through specific activities that foster the facilitation of national investment planning for the GGW countries and the sharing of information, lessons learned and knowledge (as described in section B.3).

Table 12. Complementarities between the GGW Regional Support Programme, the GEF/UNEP MSP and the GGW Accelerator

	GCF GGW Regional Support Programme	GEF MSP - Harnessing the GGWI for a Sustainable and Resilient Sahel	GGW Accelerator
Objectives	Increase the collective impact of GCF projects in the GGW	Long-term vision developed to contribute to the institutional strengthening of the GGWI and the mobilization of adequate investments for a resilient and sustainable Sahel	•Coordinate the efforts of all actors through the harmonization of the impact measurement framework •Support the implementation of the GGW Initiative through a structured, multi-stakeholder approach based on five pillars •Enable a more comprehensive mapping of available funding and projects •Connect actors to scale up successful initiatives and promote the Sahel as a land of opportunity •Involve the private sector, civil society and research/innovation in GGW efforts.
Target 'Beneficiaries' / Actors	GCF projects – (US\$1 billion of implementation and readiness projects over next 3-5 years) NDA / National Focal Points GCF, GGW and GEF and Accredited Entities	GGW countries entities (National GGW Agencies)	GGW countries, private sector, investors, project developers, donors
Thematic Area 1 - Knowledge Management and Communication			
Components/ Activities	• Establishing a GCF GGW KM and communication strategy • Creating a knowledge hub • Fostering knowledge sharing, learning through communities or practice and exchange workshops • Generating knowledge products based on GCF, GEF projects and other knowledge • Strengthening coordination at country level through national coordination committees	• Investment opportunities to transform key economic systems in the Sahel region identified and prioritized • A report documenting on innovative practices, monitoring and the Knowledge Management Platform is elaborated • Relevant stakeholders convened for the launch of a shared vision for the GGWI and Sahel region • An advocacy programme to help set up the coordination and partnership mechanism is developed and implementation started • A summary document on priority areas and potential flagship programmes for targeted donors' support prepared	Communications • Creation of a communication task force with PAAGGW, AU and Accelerator partners • Mobilization of GGW stakeholders • Publications and meetings (webinars, CSOs/private sector series, online trainings) • Project readiness: GGW success stories, pipeline of flagship projects
Thematic Area 2 - Monitoring and Evaluation			
	<i>To be developed by FAO</i> • GGW Monitoring and Reporting information integrated into the Framework on Ecosystem Restoration Monitoring • GGW baseline information prepared using innovative geospatial and technologies	• Support to the countries on software/training for M&E	• Core harmonized results management framework for the 5 pillars • GGW label • Accountability framework • Harmonized monitoring and evaluation system
Thematic Area 3 – Resource Mobilization and Private Sector Engagement			
	N/A	• Identifying and prioritizing investment opportunities to transform key economic systems in the Sahel • A report documenting targeted investments in existing initiatives	Platform • Matchmaking hub (projects/banks & investors) • Knowledge hub (partner agencies, researchers, etc.)

		as models for the integrated landscape approach is elaborated • A baseline report on emerging opportunities is prepared • Relevant stakeholders convened to launch a shared vision for the GGWI and Sahel region • A blueprint for mobilizing private sector engagement in the GGWI is elaborated and disseminated • A framework for countries to leverage resources for adaptation and resilience is established • The institutional and governance framework for GGWI coordination and partnership is strengthened • A resource mobilization and deployment strategy is developed	Communication • Preparation of a mobilization strategy for the GGW countries Mobilization of GGW stakeholders • creation of a US\$50 million-fund to prepare SMEs for investments
Thematic Area 4 Capacity-building and Governance			
	Knowledge management • Support to GCF NDAs, AEs and GEF, AF and GGW national focal points for the programming and design of climate and Environment fund projects	Strengthening the institutional and governance framework for GGWI coordination and partnership • Roadmap for a strengthened GGWI governance and institutional framework developed • A Capacity Development Plan for PAAGGW and Focal Points (GGW, Rio conventions) prepared	• Capacity-building in communication for the PAAGGW and the communication officers of the ministries of environment of the 11 countries • Online M&E training for non-state stakeholders to adopt the harmonized results-based management framework • Support to countries to strengthen their leadership in the GGWI and enhance governance tools and modalities.
Thematic Area 5 – Digital Innovation			
	• Building an ecosystem for innovation and digital solutions • Creating a marketplace of digital solutions and innovation focused on agricultural value chains, climate and landscape restoration • Identify new opportunities of innovation for GGW investments	• Identifying and prioritizing investment opportunities to transform key economic systems in the Sahel region • A report documenting innovative practices and monitoring is prepared and a Knowledge Management Platform is established	The GGW Accelerator multipurpose platform to be developed and released by February 2022 will include: • a data hub • a knowledge hub • a matchmaking hub – an online platform to collect monitoring data from countries and partners • publish online trainings and scientific articles • propose financial and networking opportunities (a crowdfunding functionality will be developed as well as 4 annual calls for applications for financial prizes)
Implementing entities	IFAD	UNEP Terrestrial Ecosystems Unit in collaboration with the following Executing Partners: National Great Green Wall Agencies/Coordination; Pan-African Agency of Great Green Wall (PAAGGW); UNCCD Secretariat-GM, UNEP Africa Office	UNCCD with PAAGGW
ME / Indicators	GCF Results Framework	GEF7 core indicators	New GGW Accelerator harmonized monitoring and evaluation system
Where Located – Key Base of Operations	TBD	Nouakchott	Initially Bonn (UNCCD), then Nouakchott (PAAGGW)
Resources	US\$14.9 million (GCF grant)	US\$2 million (GEF grant) with US\$19 million in co-financing	TBD
Reporting to ?	GGW countries (NDAs)	GGW countries (governments and populations)	GGW countries (governments and populations)

Timeline	2022-2028	2020-2021 for preparatory phase. Additional activities may be approved by GEF donors as part of GEF8 Impact Program on Ecosystems Restoration.	2021-2030
-----------------	-----------	--	-----------

Key lessons learned from past GGW projects/programmes

83. Regional projects, programmes and programmatic approaches offer many lessons learned. The GGW GCF Regional Programme will build on these lessons – particularly those from the World Bank SAWAP and the GEF7 Impact Program on Food Resilience - to maximize the overall impacts of individual projects.

Table 13. Lessons learned and recommendations from regional approaches to programmes

Lessons learned	Recommendations
- Limiting the number of implementing agencies of regional support programmes helps reduce the complexity of reporting and bureaucracy. - It is key to ensure enough resources for knowledge management and M&E, both financial and human resources in support units and at the country level. - When building M&E and knowledge management platforms, their sustainability and maintenance (hosted vs. managed by service provider) should be carefully considered. - When working across a region with various languages, there is a need to allocate sufficient resources for the translation of documents - Consistency of indicators across projects will allow for comparing/sharing.	- A consultative committee for regional multi-agency programmes is preferred to a steering committee to reduce the complexity of governance arrangements. - A common ground for defining knowledge management activities (languages, sub-region, thematic and size of projects, etc.) has proved to increase impact. - Countries are interested in designing an M&E system to harmonize indicators and methodologies and create a community of practice¹³³. - All projects should include geo-referencing, mapping and baseline data collection for all intervention sites¹³⁴.

Addressing climate change

84. To address climate change impacts, IGREENFIN I and the Regional Support Programme will use an approach that builds on the countries' climate adaptation and mitigation investment plans and on green financing models prepared for each country and compiled in Annex 2 - EFA. The green business projects to be supported by this programme are in line with the countries' NDCs on the reduction of GHG emissions and vulnerabilities to the effects of climate change (see Table 14 below). The selected West African countries have ratified the UN Convention on Biological Diversity (CBD), the Convention to Combat Desertification (CCD), the Framework Convention on Climate Change (UNFCCC) and the Kyoto Protocol and are part of the Great Green Wall initiative (Côte d'Ivoire and Ghana are not members of the GGWI). As IFAD has more than 40 years experiences working in countries affected by natural disasters, including all the countries of the GGW. It will leverage this experience to implement IGREENFIN I.

Table 14: NDCs of the target countries of IGREENFIN I

Countries	NDC Objectives
IGREENFIN I – Phase 1	
Burkina Faso (28/9/2015)	An unconditional pledge to reduce emissions by 6.6% below business-as-usual (BAU) levels by 2030, with a further 11.6% reduction conditional upon international support. Includes interim

¹³³ Bob Winterbottom et al. (2021), Sustainable Land Management in the Sahel: Lessons from the Sahel and West Africa Program in Support of the Great Green Wall (SAWAP), Washington, D.C.: World Bank Group, <https://documents.worldbank.org/en/publication/documents-reports/documentdetail/343311608752196338/sustainable-land-management-in-the-sahel-lessons-from-the-sahel-and-west-africa-program-in-support-of-the-great-green-wall-sawap>

¹³⁴ Ibid.

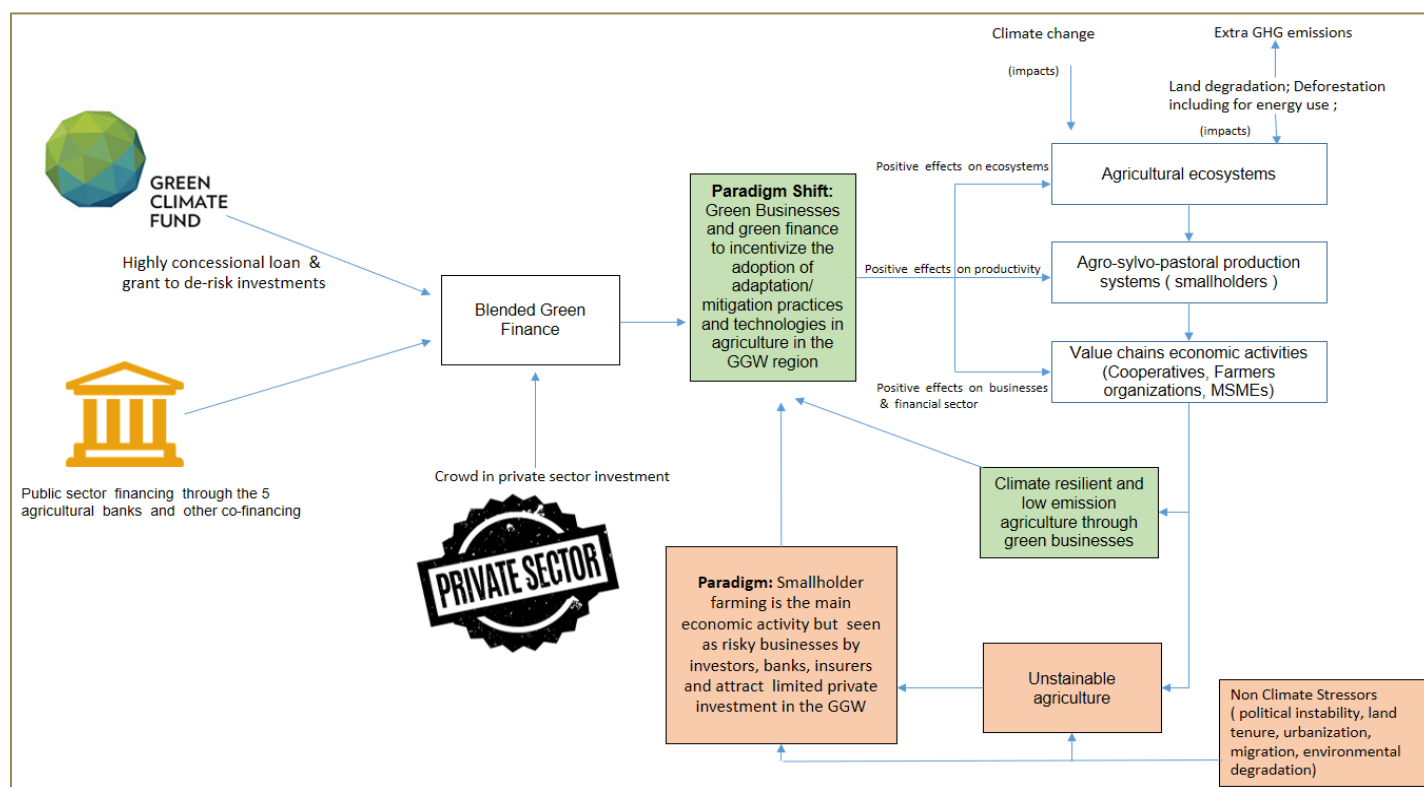
	pledges for 2020 and 2025. In the section on adaptation, actions proposed would reduce emissions by a further 36.95%, taking the total reductions up to a potential 55.15% below BAU levels.
Côte d'Ivoire (30/09/2015)	Côte d'Ivoire's goal is to unconditionally lower its GHG emissions by 28% relative to a BAU emissions scenario of 34.25 MtCO ₂ e.2 by 2030 in energy, water, agriculture, transport, waste and industry. Adaptation target sectors are water resources, agriculture, livestock and fisheries, coastal zones, energy, land use and forestry.
Ghana (11/09/2015)	Ghana's emission reduction goal is to unconditionally lower its GHG emissions by 15% relative to a BAU emissions scenario of 73.95 MtCO ₂ e.2 by 2030. Targeted sectors are energy, water, agriculture, transport, waste and industry.
Mali (29/09/2015)	Commits to reducing emissions by 29% for agriculture, 31% for energy and 21% for forests and land use, each by 2030 and in comparison to a BAU scenario. This is an average reduction of 27%. This is conditional upon international support, although around 40% of this can be met unconditionally. Includes a section on adaptation, though only for the period 2015-2020.
Senegal (26/09/2015)	An unconditional reduction in emissions of 5% by 2030, compared to BAU levels, with interim targets of 3% by 2020 and 5% by 2025. Accompanied by a conditional target subject to international financial support of 7% by 2020, 15% by 2025 and 21% by 2030, compared to BAU levels. Contains section on adaptation, as well as information on potential obstacles.

Source: <http://www4.unfccc.int/submissions/INDC/Published%20Documents>

B.2. Theory of change (max. 1000 words, approximately 2 pages plus diagram)

85. **The climate problems** that the proposed IGREENFIN I programme will address are: i) the vulnerability of the selected agricultural value chains to climatic shocks and natural hazards (floods, droughts, diseases, intense rain, wildfires and locust outbreaks) related to the target countries' geographic location and characteristics; ii) the impacts of climate change (primarily higher temperatures and reduced precipitation) and climate variability on agricultural production and productivity and water availability for agriculture, which lead to greater food insecurity, unemployment, migration and conflicts; iii) non-climate stressors (political instability, environmental and land degradation, urbanization, land tenure issues, etc.) that affect agricultural production and contribute to climate change; iv) recurrent climatic shocks, which increase the risks for financial institutions and reduce investments in key agricultural value chains, and v) the contributions of agriculture (livestock, forest and land use) and the energy sector to GHG emissions in the GGW region.
86. Moreover, there is an absence of a regional support programme and an overarching framework for GCF funded projects and partners and countries funded projects to help strengthen institutional and regulatory capacity and reporting, generate and share innovative solutions and knowledge and increase private sector financial participation in green business in the region.
87. **Shift in the development pathway.** Building the low-carbon, climate resilient market in agriculture in the GGW countries requires measures to encourage smallholder farmers to organize themselves as cooperatives, FOs and/or MSMEs so that they can collectively access concessional loans, thereby providing much needed access to financing to groups that have traditionally been financially excluded from markets. Smallholder farmers are perceived as risky and although agriculture is the backbone of the target countries' economies and social development, there is still need for well-designed incentives that can encourage the private sector to invest more in climate resilient, low emission agriculture and bring innovation to the sector all along the GGW.
88. The agricultural sector offers considerable opportunities for mobilizing private investment for green businesses. In this proposal, "green business" refers to investments in the selected agricultural value chains that promote climate resilient, low emission practices in line with the GGWI objectives and the target countries' climate change adaptation and mitigation agenda in the following seven sectors: sustainable livestock production, sustainable horticulture production (tomato), sustainable tree crop production (mango, cashew), NTFPs (shea), sustainable cereal production (rice, millet), sustainable cassava production and sustainable groundnut production.

Figure 16. Paradigm shift for green business interventions in the GGW region, Côte d'Ivoire and Ghana



89. Key assumptions for IGREENFIN I are summarized below:

90. **Assumption 1:** The creation of an innovative financing facility helps overcome growing insecurity heightened by the pandemic's restrictions and the disruptions it has caused by reducing financial market limitations (liquidity and interest rates) and promoting greater uptake of concessional loans by smallholder farmers mainly youth and women, FOs, MSMEs and cooperatives to finance their green business projects.
91. **Assumption 2:** The adoption of climate resilient, low emission agriculture (CRLEA) by rural communities and smallholder farmers (youth and women) increases the resilience of their health, well-being, food and water security, and the implementation of agroforestry, land restoration and other CRLEA techniques will be sufficiently widespread to generate a measurable positive impact on ecosystem functions and services.
92. **Assumption 3:** Smallholder farmers are willing to adopt alternative, sustainable water management technologies and adaptation practices for key crops and value chains, including efficient irrigation systems and RETs, which improve the availability of water for agricultural uses in rural areas and thus, their productivity levels.
93. **Assumption 4:** Rural communities and smallholder farmers mainly youth and women are willing to adopt agroforestry techniques, land use practices and RETs and are motivated to sustain them over the programme's duration (6 years) and lifetime (25 years). The use of these technologies and practices results in the avoidance of 5,6 million tCO₂ eq., as expressed in the five countries' NDCs.
94. **Assumption 5:** (IGREENFIN I & GGW Regional Support Program) IFAD and other financial and technical partners are willing to coordinate with the target countries and other stakeholders to build on GCF-funded projects to strengthen institutional capacity and reporting, generate and share innovative solutions and knowledge, and increase private sector financial participation in green business projects.
95. To address the above climate challenges, the five selected countries under IGREENFIN I have mapped a detailed list of green practices and technology-related green business needs. However, given that this is a nascent market, most of these technologies have not yet been made readily available and significant technology gaps remain - mainly due to the limited affordability and availability of financial instruments for this sector. Capacity, knowledge and policy gaps that hinder the full transition of agriculture to climate resilient, low carbon pathways also remain.

Table 15: Summary of key barriers and opportunities for green financing

Key barriers	Activity to address them
Limited access to green finance for green business projects in the GGW region and in Côte d'Ivoire and Ghana	Establish a Green Business Financing Facility within each of the selected banks to catalyse adequate finance for green business projects and key value chains through: Window 1 (adaptation) - Pillars 1, 2 and of the GGW Accelerator: will finance the adoption of CRLEA practices in the 10 selected value chains (listed in Table 8), including tree crops/NTFPs, which will contribute to the restoration of degraded land, Window 2 (mitigation) – Pillar 3 of the GGW Accelerator: solar power for agriculture (see Table 8). These windows will provide green loans to FOs, MSMEs, cooperatives and women and youth organizations. A technical assistance facility (Pillar 5) will be established to provide them support to build scalable and innovative green business models, thereby enhancing risk/return profiles and the bankability of the projects.
Capacity and knowledge gap hindering the transfer and adoption of effective climate adaptation and mitigation practices in green business projects	Provide the necessary technical assistance, training and knowledge transfer activities to strengthen capacity to perform feasibility studies and prepare and implement bankable projects for key stakeholders across the 10 selected agricultural value chains (Table 8). The programme will also support leadership activities for women and youth to increase their access to resources and participation. A technical assistance facility (Pillar 5) will be established to provide them support to build scalable and innovative green business models, thereby enhancing risk/return profiles and the bankability of the projects.
Limited policy and regulatory frameworks to increase the economic feasibility of potential green business projects in the GGW area	The five selected countries have adopted various laws and regulations in the area of green finance and renewable energy. Even so, there is a need for further policy dialogue between the private and public sector to develop the necessary frameworks for increasing the attractiveness of green business projects. The programme will provide capacity-building to empower smallholders to demand policy changes in favour of low-cost RETs (e.g., waiver of import duties on RETs) and technical and financial assistance to support the uptake of climate resilient, low emission agricultural techniques, thus generating more demand for RETs companies in rural areas. The technical assistance facility (Pillar 5) will help address limited policy and regulatory frameworks
Absence of a regional support programme and an overarching framework for GCF-funded projects to strengthen institutional capacity on reporting, generating and sharing innovative solutions and knowledge, and increase private sector financial participation in green business projects	The overall objective of the Regional Support Programme is to enhance the collective climate resilient impact of GCF-funded projects in the GGW area, as well as Côte d'Ivoire and Ghana by establishing a culture of knowledge sharing and collaboration between all these GCF projects and programmes and their respective AEs and for the benefit of the GCF NDAs, the GGW focal points and the AEs supporting these countries. If deemed appropriate by the national GEF and AF focal points, this intentional and coordinated effort to support knowledge creation and dissemination among projects and countries could also be extended to the GEF and AF projects. Specifically, the Regional Support Programme aims to improve coordination of knowledge management¹³⁵ and robust M&E systems¹³⁶ among GCF projects, which will result in increased project and portfolio impact. It will also foster innovation and strengthen the participation of the private sector. It will contribute to changing the narrative to one where the Sahel is a 'land of opportunities'¹³⁷ by increasing the collective impact of individual GCF projects that are focused on climate change adaptation and mitigation and offer multiple co-benefits.

¹³⁵ Paul Oluikpe, Mu Sohail and Frank Odhiambo (2011), "The Role of Knowledge Management in Development Projects, JIKM, 10, https://www.researchgate.net/publication/220588584_The_Role_of_Knowledge_Management_in_Development_Project

¹³⁶ Estelle Raimondo (2016), "What Difference Does Good Monitoring and Evaluation Make to World Bank Project Performance?", Policy Research Working Paper No. 7726, World Bank, <https://openknowledge.worldbank.org/handle/10986/24644>

¹³⁷ UNCCD (N/D), The Sahel: Land of Opportunities, Land with a Future, <https://knowledge.unccd.int/knowledge-products-and-pillars/unccd-e-library/sahel-land-opportunities-land-future>

96. **Theory of Change:** Against the climate impacts, the Theory of Change (ToC) below summarizes how a combination of the proposed interventions described under B3 are expected to yield maximum benefits in terms of transforming the target communities into more resilient ones through the proposed components and activities. In the rural communities of the selected countries, to improve agricultural value chains, there is a need to address climate change impacts to stimulate a green business market. This requires establishing green business financing to facilitate better access to concessional green finance, offering technical assistance and improving coordination to help smallholder farmers, FOs, cooperatives, MSMEs and agricultural banks access knowledge and technologies on climate resilient, low emission agriculture (CRLEA) previously out of their reach and build their technical capacity to design, invest in and implement viable and scalable green business plans. In the case of banks (including LNABs) and relevant government agencies, there is a need to establish adequate policy frameworks for green finance for climate resilient, low emission agriculture at scale in the region. There is also the need to adopt best agricultural and land use and forestry (LULUCF) practices, as well as sustainable energy to power profitable agricultural value chains. As a result, the programme is expected to i) reduce emissions by 5.6 million tCO₂e through increased low-emission energy access and power generation in the selected agricultural value chains including processing; ii) reduce vulnerability of smallholder farmers, particularly women and youth, their ecosystems and ecosystem services to climate change; iii) increase the resilience of 378,600 direct beneficiaries and 2,494,000 indirect beneficiaries, their health and well-being, and food and water security, ecosystems and ecosystem services; iv) identify financially viable green businesses, which will be financed by public and private investment; v) build the technical capacity of FOs, cooperatives, MSMEs, LNABs and IFIs, and vi) strengthen the institutional and regulatory framework for the adoption of climate resilient, low emission agricultural activities and mitigation.

Figure 17: Theory of Change for IGREENFIN I and the Regional Support Programme

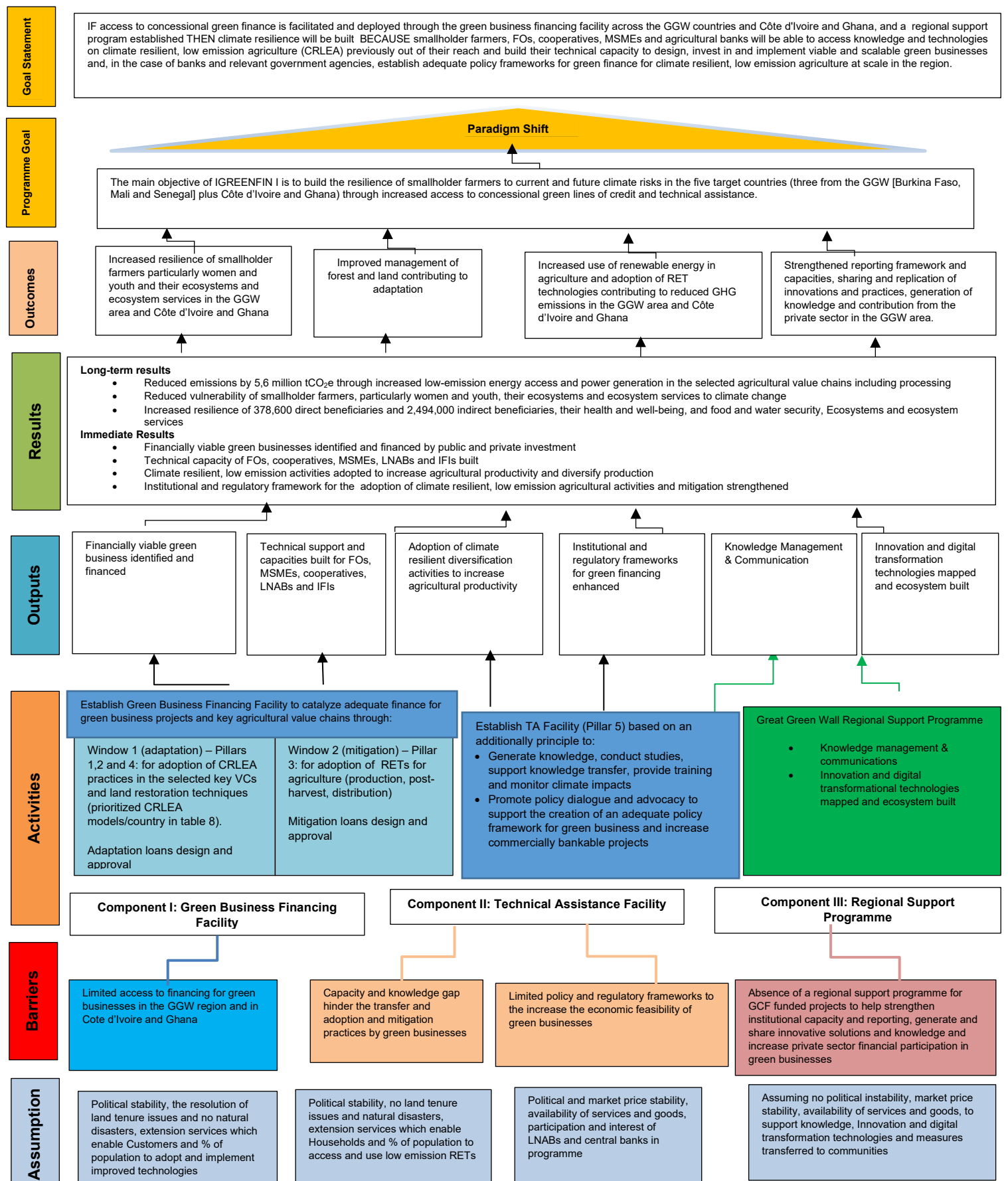


Table 16: Preliminary selected sustainable value chains

Main climate risks		- Cattle: increasing temperatures and rainfall variability reduce food and water for animals, increase incidence of disease/pests, degrade pastureland further, affecting overall growth and reproduction. - Horticulture (tomato): heat and water stress reduce yields, affect growth and increased incidence of fruit abortion and disease - Mango: water stress and extreme temperatures affect flower induction, fruit set and pollen survival, reducing yields - Cashew: Rising temperatures, irregular rainfall and drought negatively affect yields and quality; warmer and more humid conditions increase incidence of pests/disease, main constraint to cashew production - Rice: Erratic rainfall and high temperatures cause disruptions in growing season; greater risk of floods, water shortages, strong winds, pests and disease lead to lower yields or crop failure - Cassava: Vulnerability to pests and disease due to higher temperatures - Groundnut: excessive heat (above 33 C) and rainfall variability reduce yields; excessive moisture increases plant mortality and disease - Millet: temperature increases and rainfall variability reduce yields - Shea: Heavy rainfall and wind increase risk of loss of flowers and fruit; drought, floods and wildfires increase tree mortality; drought increases risk of fruit abortion. Risk of decline in favourable habitats.				
Selected value chains		Climate risks	Green businesses	Evidence of impacts	Adaptation and mitigation interventions	Adaptation and mitigation benefits
Sustainable livestock production	Cattle	Projected increases in temperatures, rainfall variability and frequency and length of droughts across the Sahel expected to reduce food and water availability for cattle, contribute to disease/pest outbreaks, affect reproductive cycle (reduce calving rates from 60-70% to 25-30%) and growth, and degrade pastureland further.	10,000 cattle green businesses	Livestock disease kills 20% of ruminants in developing countries. Desertification and land degradation have resulted in loss of pastureland. Higher temperatures found to reduce feed intake, milk production, fertility and lead to energy deficits. During major droughts in the past, farmers reported that they had lost entire herds.	<u>Adaptation</u>: Investment in fodder banks; crop-livestock integration; improvements to grazing corridors; efficient watering technologies; vaccination, pest monitoring and control systems, intensification of cultivated pastures; resilient breeds. <u>Mitigation</u>: Frequent removal of manure for composting and RETs to store vaccines.	<u>Adaptation</u>: Improved access to water, feed and disease control will contribute to greater resilience and productivity of livestock, which is key for millions of small livestock producers in target countries. <u>Mitigation</u>: emissions from cattle production reduced through improved management of manure and use of RETs to store vaccines.
	Chicken	Projected increases in temperatures, high humidity, sunshine intensity have negative effects on poultry production, contribute to disease/pest outbreaks and affect the cost of production, the number of birds needed to raise for eggs and meat and feed grain availability (high temperatures and low rainfall affect the grain harvest, the supply of grains to the market and ultimately the cost of poultry production).	15,000 poultry green businesses	Higher temperatures have been found to reduce feed intake, egg production, lead to reduced feed grain availability. The increase in temperature ultimately increases the mortality rate of chickens grown for meat by 8.4%, laying hens by 0.84%, and indigenous chickens by 0.32%.	<u>Adaptation</u>: Climate-smart livestock genotype; crossbreeding with exotic birds; heat/drought/flood/saline resilient climate resilient farming practices; conserve & utilize climate resilient native poultry; proper feed formulation; heat resilient housing; processing of chicken waste into fertilizer; extensive poultry management; vaccination, pest monitoring and control systems. <u>Mitigation</u>: Frequent removal of manure for composting. RETs to store vaccines.	<u>Adaptation</u>: Improved housing, health, feed and disease control will contribute to greater resilience and productivity chicken meat and egg, which is key for millions of small chicken producers in target countries. <u>Mitigation</u>: emissions from chicken production reduced through improved management of manure and the use of RETs to store vaccines instead of fossil fuels.

Sustainable horticulture and vegetable production	Tomato	High temperatures, excess water or drought reduce productivity levels. Heat also impacts farmers' ability to market and conserve tomatoes, which are highly perishable. Climate variability increases incidence of disease.	1,500 tomato green businesses	Studies show significant declines in yields due to heat and water stress. High temperatures affect plant growth and reproduction, with increased incidence of abortion of fruit. Lack of water results in formation of fewer flowers and affects pollen. Farmers reported poor yields due to climate change.	Adaptation: Drip irrigation; soil management techniques to address low fertility and salinity (organic fertilizers, mulch); higher-yield, disease resistant varieties. Mitigation: promotion of RETS for storage and processing facilities and efficient irrigation systems to displace fossil fuel use.	Adaptation: Increased production levels and plant resilience, new job opportunities and reduction in post-harvest losses; all contribute to raising smallholders' income and resilience. Better use and conservation of water resources. Mitigation: Adoption of RETs reduces use of fossil fuels and wood as energy source and thus, emissions.
	Sustainable Mango with native species (agroforestry model 1)	Negative impacts of water stress and extreme temperatures on flower induction, fruit set and pollen survival. CC scenarios predict a rapid decline in the number of induction days by end of century. Shorter flowering periods reduce productivity. Rainfall variability – especially if there is rain during flowering period – may also reduce yields.	6,000 mango green businesses	Influence of climate change on yield, time of maturation and quality	Adaptation: Drip irrigation, organic fertilizer, improved pruning practices, integration of mango trees in agroforestry systems; intercropping with cereals. Mitigation: Mango trees have a significant capacity for carbon sequestration, tend to use up more carbon than they emit. RETs to power mango processing equipment to reduce post-harvest losses.	Adaptation: Mango trees serve as ecological barrier against CC, contribute to soil and water conservation. Better production and harvesting techniques increase yields and farmers' incomes and contribute to job creation. Growing mangos in agroforestry systems with native trees can increase yields three-fold and prevent soil erosion. Mitigation: Improved soil management through agroforestry contribute to carbon sequestration, adoption of RETs reduces use of fossil fuels and wood as energy source and thus, emissions.
Restoring degraded land and agroforestry	Cashew production with native species (agroforestry model 2)	Increased temperature, rainfall and humidity can lead to higher incidence of pests and disease, two major constraints to cashew production. Irregular rainfall, violent winds and prolonged drought can impact yields and quality. Though highly tolerant to drought, low rainfall during vegetative stage can prevent trees from obtaining enough water to sustain flowering.	6,000 cashew green businesses	Higher temperatures, erratic rainfall and increases in humidity increase the incidence of pests and disease. Irregular rainfall and temperature changes have been found to affect production.	Adaptation: Mulching to retain soil moisture, prevent weed growth and enhance soil structure; organic fertilizer; inclusion of cashew trees in agroforestry systems; drip irrigation. Investment in storage, conservation and processing techniques and infrastructure for reduced post-harvest losses and value addition. Mitigation: Cashew introduced to control soil erosion and for reforestation; ideal for	Adaptation: Improved productivity increase smallholders' incomes and create jobs. Cashew trees contribute to soil and moisture conservation, prevention of erosion and slowing desertification. Mitigation: Improved soil management through agroforestry models using mixed native species with cashew trees contributes to carbon sequestration, adoption of RETs reduces use of fossil

					carbon sequestration. RETs to power cashew processing equipment.	fuels and wood as energy source and thus, emissions.
	Sustainable Shea (community forest model)	Favourable habitats of shea trees will decline by 13% (RCP8.5) by 2070 due to climate change. Higher risk of losing flowers and fruit due to heavy rainfall and winds; drought, flooding and wildfires increasing tree mortality and inhibit harvesting; decrease in water supply affecting growth and limiting processing; more frequent and prolonged droughts leading to the abortion of fruit. Deforestation biggest threat.	5,000 shea green businesses	Higher temperatures and changes in rainfall patterns are already causing the trees to shift southward, with impacts on the trees' growth, regeneration and production levels.	<u>Adaptation:</u> climate resilient practices such as assisted natural regeneration and Zaï, sustainable forest management techniques. <u>Mitigation:</u> Agroforestry landscape in which shea is naturally grown acts like a carbon sink, controls soil erosion; adoption of RETs to power shea processing and storage facilities.	<u>Adaptation:</u> Agroforestry parklands conserve natural resource base, improve soil fertility, enhance resilience and allow farmers to diversify production. Investments in forest management and shea processing capacity improve incomes and create jobs, especially for women. <u>Mitigation:</u> shea value chain fixes 1.5 million tons of CO₂ every year and every ton of shea kernels produced has a negative carbon footprint of 1.04 tons of CO₂ (FAO-GSA, 2020). Valorization of shea helps reduce deforestation and thus avoid emissions. Improved soil management through maintenance of shea trees contributes to carbon sequestration, adoption of RETs reduces use of fossil fuels and wood as energy source and thus, emissions.
	Sustainable Rice production	Increasing climate variability (rainfall) and high temperatures disrupt growing seasons. Greater risk of prolonged drought, heat waves, floods, water shortages, strong winds and storms, pests and diseases, causing substantial yield reductions or crop failure, which are higher for rainfed production. SLR increases risk of floods and salinization	20,000 rice green businesses	Estimated reductions in rice yields range from 5-25% and up to 80%, in irrigated rice systems in the hot-dry season and around 40% in rainy season.	<u>Adaptation:</u> SRI; improved short-cycle rice varieties; efficient irrigation; management of soil salinity (namely drainage, flooding and organic matter addition). <u>Mitigation:</u> adoption of RETs to power processing facilities and irrigation; SRI involves alternate wetting and drying with minimal water application, thus reducing GHG emissions from flooded rice fields.	<u>Adaptation:</u> More resilient and higher yielding rice production improves farmers' income and contributes to food security. More efficient water use. <u>Mitigation:</u> Reduced emissions from flooded rice fields thanks to SRI and adoption of RETs.
	Sustainable Cassava production	Though resilient to climate change stressors, cassava production will be vulnerable to pests and diseases.	18,000 cassava green businesses	Over a third of attainable cassava yield is lost every year to pests and disease, which could be exacerbated by higher	<u>Adaptation:</u> Efficient irrigation systems, intercropping with cover crops and trees (fruit and other), composting/organic fertilizer and pest	<u>Adaptation:</u> Improved productivity through more effective water, soil and pest management increases farmers' incomes and

Sustainable cereals-staple crops production				temperatures (e.g., whitefly).	control; sustainable land management techniques. <u>Mitigation:</u> Composting, minimal tilling and adoption of RETs to power processing, storage and irrigation.	resilience and contributes to food security. <u>Mitigation:</u> emission reductions due to use of RETs.
	Sustainable Groundnut production	Though moderately drought resistant, excessive heat and prolonged drought limit yields. Rainfall variability is in important risk as germination is poor in drier soils, whereas excessive moisture increases plant mortality and incidence of disease. Average production expected to decrease 11% over next 20 years.	35,000 groundnuts green businesses	Temperatures above 33°C found to decrease growth by up to 84%. Higher temperatures will have most dramatic impact on yields, which are traditionally low due to unreliable rain, poor soils, pests and disease.	<u>Adaptation:</u> improved short-cycle varieties, organic fertilizer, intercropping, stone bunds, water efficient irrigation systems, storage and conservation techniques. <u>Mitigation:</u> As a nitrogen-fixing crop, groundnuts reduce GHG emission intensity and use of chemical fertilizers. Solar energy instead of fossil fuels to power drip irrigation systems and other equipment.	<u>Adaptation:</u> More resilient production and higher yields improve farmers' incomes and food security and create jobs in value chain. More efficient resource management (soil and water) contributes to conservation. <u>Mitigation:</u> As a nitrogen-fixing crop, has mitigation benefits, as it maintains soil health. Reduced emissions from use of RETs.
	Sustainable Millet Production	Though well-adapted to hot areas, temperature increases higher than 2°C are projected to decrease millet yields by 10% over next 20 years and 15-25% by 2080. Rain variability contributes to lower yields, primarily excess rain during flowering, which can cause crop failure.	35,000 millet green businesses	Production predicted to decrease on average by 10% over next 20 years.	<u>Adaptation:</u> Climate resilient ag practices such as inter-cropping; soil fertility (compost) and water retention (Zai and half-moons) techniques; climate information systems. <u>Mitigation:</u> Use of millet crop leftovers to maintain and increase soil carbon levels; adoption of RETs	<u>Adaptation:</u> More resilient production with higher yields increases farmers' incomes and contributes to food security. Improved soil and water management. <u>Mitigation:</u> Crop leftovers have a high carbon content, making them particularly significant for increasing soil carbon levels. Remissions avoided due to adoption of RETs instead of fossil fuels.

97. The preliminary eligibility criteria for activities that can be financed by loan proceeds are presented in Table 17 below. While this list of eligible uses applies to all target countries, there will be country-specific needs, priorities and timeframes for implementation. Country-specific components will provide thorough justification for the selection of specific value chains and will be approved and implemented in accordance with the adaptation and mitigation list of activities in table 17.

Table 17: Preliminary eligibility criteria for activities that can be financed by loan proceeds

Sector and value chain	Eligibility criteria: Initial list of activities for each value chain eligible for financing from loan proceeds	
	Adaptation	Mitigation
Sustainable livestock production [Sustainable Cattle, goat, sheep production]	<u>Investment in:</u> - Grazing corridors for livestock to inhibit overgrazing - Vaccination and disease prevention measures - Monitoring and the production of knowledge and applied research systems on pests and diseases - Drought-resilient fodder cultivation - Improved animal feed and fodder banks - Native ruminant breeds - Sustainable agricultural practices - Equipment for eco-friendly production equipment for production, processing and marketing - Clean vehicles for transportation - Fodder production with resilient native species - Water harvesting, efficient irrigation infrastructure, checking dams, flood management and drainage, agricultural insurance/EWS - EWS equipment and infrastructure - Index-based insurance	<u>Investment in:</u> Installation of solar mini grids (refrigeration of vaccine storage where other sources of reliable electricity rely on fossil fuel use); drip irrigation and post-harvest processing
Sustainable livestock production [Sustainable Chicken Production]	<u>Investment in:</u> - Climate-smart livestock genotype - Cross breeding with exotic birds - Heat/drought/flood/saline resilient farming practices - Proper feed formulation - Heat resilient housing - Investment on chicken waste processing - Adapting extensive poultry management - Water supply - EWS equipment and infrastructure - Index-based insurance - Promote vermi compost - Processing chicken waste into compost	<u>Investment in:</u> RETs for building, storage and processing facilities and efficient lighting of the chicken farms
Sustainable horticulture and vegetable production [tomato]	<u>Investment in:</u> - Changes in cropping patterns - Heat-tolerant, higher yield and disease resistant varieties and species - Improved use of seasonal forecasts to inform planting time - Precision farming - Improved soil management practices to retain soil moisture and techniques to address low fertility and salinity (organic fertilizers, mulching); - Use of recycled wastewater for irrigation - Use of dry processing and packaging methods - More efficient water management policies and water use regulations - Improved early warning system dissemination - Establishment of soil conservation practices to reduce erosion	<u>Investment in:</u> RETs for storage and processing facilities and efficient irrigation systems.

	- Climate resilient infrastructure for storage and processing - Improve water storage capacity - Drip irrigation techniques et technologies - Greenhouse technologies and systems - EWS equipment and infrastructure - Sustainable land use - Crops that sequester GHGs	
Restoring degraded land and agroforestry [Sustainable Mango production with native species]	<u>Investment in:</u> - Selection and reproduction native genetic materials and indigenous species - Nurseries of improved seedlings and varieties - Organic fertilizers - Eco-friendly technologies and equipment for production, irrigation and harvesting - Conservation planting and storage that meet environmental norms - Best agricultural practices such as agroforestry - Low emission technologies and practices for efficient water use, such as drip irrigation - Biodegradable packaging - Improved pruning practices, integration of mango trees in agroforestry systems - Intercropping with cereals - Firebreaks around forested areas - Stakeholder engagement and awareness and agroforestry monitoring and protection activities - Promote woodlots for climate-resilient production of fuelwood to avoid deforestation. - Restoration of degraded farmland through soil and water conservation techniques (improved Zai, agricultural half-moons, assisted natural regeneration) - Agroforestry techniques with other natives specifies - Crops that sequester GHGs	<u>Investment in:</u> Solar mini grids equipment for production, processing and storage
Restoring degraded land and agroforestry [Sustainable Cashew production with native species]	<u>Investment in:</u> - Selection and reproduction of native genetic materials - Nurseries of improved seedlings and varieties - Mulching to retain soil moisture, prevent weed growth and enhance soil structure - Forest restoration using trees that provide NTFP combined with cashew and integration of cashew trees in agroforestry systems - Organic fertilizer - Drip irrigation - Investment in storage, conservation and processing techniques and infrastructure for reduced post-harvest losses and value addition - Promote woodlots for climate-resilient production of fuelwood to avoid deforestation - Firebreaks around forested areas - M&E awareness and protection activities - Restoration of degraded agricultural land through soil and water conservation techniques (improved Zai, agricultural half-moons, assisted natural regeneration) - Planting cashew trees to control soil erosion and for reforestation and carbon sequestration	<u>Investment in:</u> - Solar mini grids to power irrigation and processing and storage .

	•	
Restoring degraded land and agroforestry [Sustainable Shea production]	<u>Investment in:</u> - Climate resilient soil and water conservation practices (assisted natural regeneration, Zaï, sustainable forest management) - Nurseries of improved seedlings and varieties - Firebreaks around forested areas - Agroforestry - M&E awareness and protection activities - Storage of seeds and harvesting tools - Water conservation and efficiency measures - Index-based insurance - Indigenous species integrated with other tree species and seeds adapted to future climate change conditions (e.g., drought tolerant trees) - Forest restoration with other trees that provided NTFPs (Nere, Acacia Seyal, Acacia Nilotica, Acacia Radinia, Commifora Sp, Acacia Albida, Balanites Egyptiaca, Bauhina Rufescens, etc.) - Promote woodlots for climate-resilient production of fuelwood to avoid deforestation. - Preservation of shea in agroforestry parklands, which serves as a carbon sink	<u>Investment in:</u> Adoption of RETs to power shea processing and storage facilities.
Sustainable cereals-staple crops production [Sustainable Rice production]	<u>Investment in:</u> - Sustainable rice intensification (SRI) techniques - Improved short-cycle/heat and drought tolerant rice varieties - Efficient irrigation and water-saving technologies - Management of soil salinity (drainage, flooding and organic matter addition). - Production techniques such as water harvesting, infrastructure, checking dams, flood management and drainage - Organic fertilizers - Precision farming - No tillage practices - Integrated rice-based farming & cultivation system - Integrated rice crop management system - Weather-proof farming equipment and post-harvest facilities - Climate forecasting and monitoring system using remote sensing - Equipment and infrastructure to collect information for EWS projects and the meteorological agency and index-based insurance - SRI for its contribution to reducing GHG emissions from flooded rice fields - Short-cycle production - Use of rice cultivars with low methane emission potential - Management of crop residue - Application of methanotrophs	<u>Investment in:</u> RET-powered processing facilities, processing and irrigation
Sustainable cereals-staple	<u>Investment in:</u> Selection and reproduction of native genetic materials	<u>Investment in:</u> Solar mini grids equipment for production, processing, storage and irrigation

crops production [Sustainable Cassava production]	- Nurseries of improved varieties - Efficient irrigation systems - Intercropping with cover crops and trees (fruit and other), - Composting/organic fertilizer and pest control - Sustainable land management techniques - Agroforestry - Composting - Minimal tilling	
Sustainable cereals-staple crops production [Sustainable Groundnut production]	<u>Investment in:</u> - Digital solutions on climate change and agriculture - Crop rotation, intercropping, cover cropping and agroforestry - Organic composting, Zai, contour bunding and vegetative barriers - Bio-pesticides and organic fertilizer - Reimbursable technical assistance and training on climate resilient agriculture - Integrated pest management (e.g., push-pull methods) - Water harvesting techniques and infrastructure, - Water conservation and efficiency techniques - Efficient irrigation infrastructure, checking dams, flood management and drainage - Storage of seeds and tools - Rehabilitation of degraded land - Agricultural insurance - Equipment and infrastructure to collect information for EWS projects and the meteorological agency and index-based insurance	<u>Investment in:</u> Solar mini grids to power irrigation and processing equipment.
Sustainable Millet Production	<u>Investment in:</u> - Inter-cropping - Soil fertility (compost) and water retention (Zai and half-moons) techniques - Climate information systems - Use and storage of climate resilient seeds - Water harvesting and infrastructure - Tools - Rehabilitation of degraded lands - Equipment and infrastructure to collect information for EWS projects and the meteorological agency - Index-based insurance - Use of millet crop leftovers to maintain and increase soil carbon levels; - Sustainable land use practices	<u>Investment in:</u> Adoption of RETS

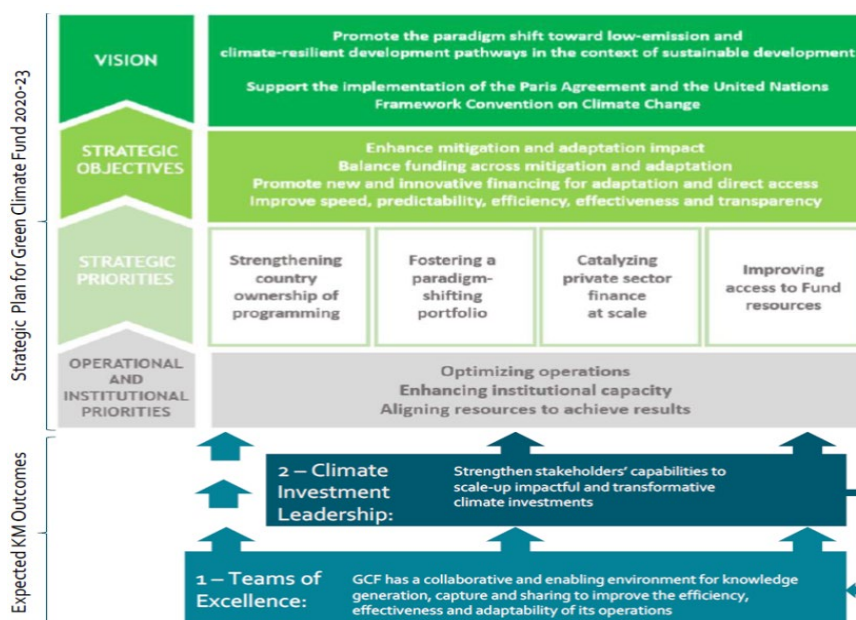
Theory of Change of Regional Support Programme

98. Several GCF projects and programmes are already being implemented in the GGW countries and new projects are being elaborated by Direct and International Accredited Entities. These projects contribute to achieving the GGWI's objectives by implementing specific activities focused on one or more of the five pillars mentioned in Section B.1 (value chain development, landscape restoration, climate-smart infrastructure, capacity building and institutional development). However, in the absence on an overarching framework, these GCF projects and programmes are being implemented in isolation, resulting in missed opportunities for improved delivery of the GCF mandate. The GGW Regional Support Programme will address:

- Lack of analysis and sharing of lessons and experiences across GCF projects in GGW countries.
- Duplication of activities by AEs due to limited awareness of AEs and project staff of similar topics or activities being implemented in different countries by GCF and other projects including those supported by the GEF.
- Limited evidence-based knowledge generated and tailored to specific audience to inform GCF project development.

- Increased costs for AEs, countries and the GCF due to limited exchanges of knowledge on GCF project implementation and available studies (e.g., climate analysis, gender assessments, etc.)
 - Limited coordination at the country level to strategically plan the allocation of GCF, GEF and other resources that can contribute to the achievement of the GGW objectives resulting in a duplication of projects and activities. NDAs are not necessarily aware of the lessons learnt, best practices or innovation from other countries or other GCF projects, which could help them to better plan their respective Country Work Plan to the GCF, projects, iteration of their NDCs or NAPs.
 - Innovative solutions, in particular digital solutions, to addressing climate change in value chains and landscape restoration developed by GCF, GEF and other stakeholders' projects are not widely available and spread across the GCF project portfolio.
99. While innovation and digital solutions are increasingly recognized as a strategic path for addressing climate change impacts, the ecosystem to support innovation and digital solutions developers is in its very early stage of development, meaning missed opportunities for replication and scaling up.
100. The benefits from the implementation of the knowledge management and innovation activities will contribute to several outcomes of the GCF Updated Strategic Plan (see Figure 18 below) by creating a transparent and collaborative environment for knowledge generation at the country level and across GCF funded projects. Knowledge generated by GCF projects will be captured, analysed and shared with specific audiences to improve the efficiency and effectiveness of GCF projects, as described above. Similarly, the collaborative environment created by the GGW RSP to foster innovation will increase stakeholders' capacities across the region to adopt and scale up innovative and digital solutions through coalitions. Increased access of GCF NDAs, GGW, GEF and AF focal points, AEs and DAEs to knowledge and innovation will increase their capacities to replicate, streamline and scale up climate activities and investments that have proven effective in similar contexts.

Figure 18. GCF expected contribution of KM and coalitions to the Strategic Plan for Green Climate 2020-23

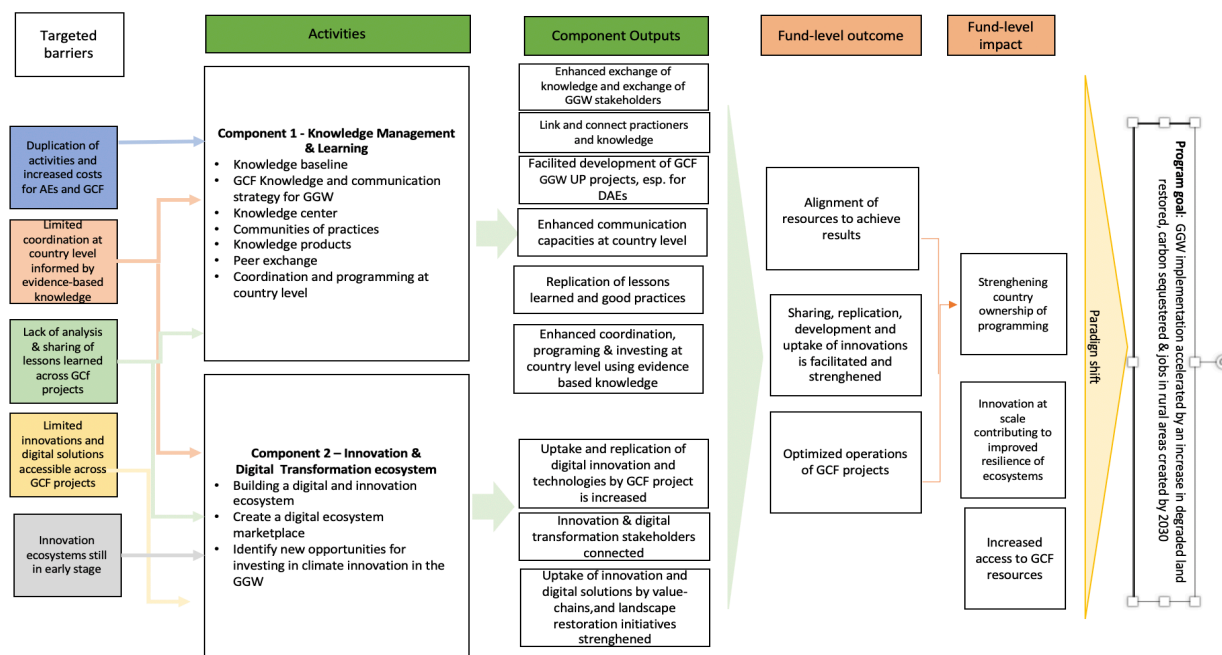


101. To address the challenges mentioned above, the Regional Support Programme (RSP) will improve access to best practices, encourage learning and foster innovation and digital transformation across Great Green Wall countries so as to increase individual GCF projects' collective impact. The RSP's added contribution is that its entire structure and all its actions are geared towards building the knowledge and systems necessary to identify transformational pathways for the GGW area. The RSP will not replace the knowledge management and innovation activities of each GCF project, nor take over the responsibilities of the corresponding AEs. It aims to create bridges through a structured framework to ensure that project specific information, knowledge and learning is properly captured and disseminated across GCF projects and AEs, but also GEF projects and other entities contributing to tackling climate change in the GGW. The GGW RSP will be uniquely placed as a facilitator between individual GCF projects to promote the scaling up of existing innovation in the GGW¹³⁸ and to increase the uptake of innovations and digital solutions developed by other stakeholders in current and future GCF projects.

¹³⁸ GCF (2021), Innovation is imperative, GCF tells Leaders Climate Summit, 23 April, Press Release, <https://www.greenclimate.fund/news/innovation-imperative-gcf-tells-leaders-climate-summit>

The GGW RSP will promote and amplify action on the ground to bring it to scale and generate spillover effects that positively affect other GGW initiatives and undertakings.

Figure 19: Theory of Change of GGW Regional Support Programme



B.3. Project/programme description (max. 2000 words, approximately 4 pages)

102. **The overall objective** of this regional flagship programme (IGREENFIN I and the Regional Support Programme) is to build the resilience of smallholder farmers to current and future climate risks in the five target countries (three from the GGW [Burkina Faso, Mali and Senegal] plus Côte d'Ivoire and Ghana) by: i) providing support to selected local national agricultural banks (LNABs) for the creation of green credit lines to finance the green business projects of smallholder farmers organized in FOs, cooperatives, women and youth-led organizations and MSMEs; ii) addressing capacity-building, knowledge and policy gaps that hinder the transfer and adoption of adaptation and mitigation practices by green businesses, and iii) enhancing the collective impact of all GCF funded projects, including IGREENFIN I, through the establishment of a coordination framework for knowledge generation and sharing, project implementation and reporting, mobilizing private investment and promoting innovation at scale to contribute to the improved resilience of ecosystems. The flagship programme focuses on creating an enabling environment to increase the flow of funds to green businesses engaging in climate resilient, low emission agricultural practices in the GGW area and Côte d'Ivoire and Ghana. It will further contribute with a potential total GHG emission abatement of **5,637,184** million tCO₂eq from the use of renewable energy technologies (RETs) and sustainable forest and land management practices. This mitigation potential represents 3.8 per cent of the GGW' emission reduction target (250 MtCO₂).
103. The Regional Support Programme covers all GCF projects and programmes being implemented by Direct and International Access Entities only in the 11 countries of the GGW (Burkina Faso, Chad, Djibouti, Eritrea, Ethiopia, Mali, Mauritania, Niger, Nigeria, Senegal and Sudan).
104. **Programme beneficiaries:** The programme will directly benefit 378,600 smallholder farmers who must be organized in FOs, cooperatives, women and youth organization and MSMEs and over 2,494,000 people indirectly, of which 50 per cent will be women and 50 per cent youth by increasing their climate resilience through access to affordable, long-term green loans. Other beneficiaries include: 1,500 MSMEs and 2,500 FOs, cooperatives or women- or youth-led organizations, two commercial banks (Attijariwafa Bank and the Islamic Bank of Senegal) and potential LNABs partners, such as MFIs. Furthermore, 50 technologies and innovative solutions will be transferred or licensed to support climate resilient, low-emission development during the implementation of the programme.
105. The **direct beneficiaries of the Regional Support Programme** include: GCF National Designated Authorities, GGW National Focal Points and regional entities, GEF national focal points, Accredited Entities, country project teams and key

country stakeholders implementing GCF country projects under the GGW Regional Support Programme. Indirect beneficiaries include the total number of beneficiaries reached by each of the GCF projects funded under the GGW RSP.

106. The programme has three components: (I) **Green Business Financing Facility** (Loan); (II) **Technical Assistance Facility** (Grant) and (III) the **Regional Support Programme** (Grant). This programme is in line with each of the five selected countries' Nationally Determined Contributions (NDC), as well as local climate policy initiatives as outlined under Section D.5.
107. **Component 1 – Green Business Financing Facility (Pillars 1, 2, 3 and 4 of the GGW Accelerator)** will be established in each of the five selected LNABs to provide highly concessional loans to FOs, women and youth-led organizations, cooperatives and MSMEs (including agribusiness dealers and photovoltaic operators) to finance the adoption of best adaptation and mitigation practices in the selected agricultural value chains in each target country and to address key climate hazards such floods, water scarcity, wildfire and drought, among others. These include, for instance, efficient irrigation techniques for both rainy and dry seasons, sustainable rice intensification, ecosystem-based adaptation (EbA) or nature-based solutions and mitigation measures such as solar energy to power the selected value chains, among others (refer to Table 8). Of the EUR73, 971 million from the GCF loan, EUR49, 314 million will be provided as a loan at highly concessional rates to promote adaptation practices, and the remaining EUR24, 657 million of GCF resources will be allocated to mitigation practices (renewable energy use in the 10 selected agriculture value chains).
108. The beneficiaries of the loans (Component 1) are FOs, cooperatives and MSMEs. The sub-borrowers include also Attijariwafa Bank which operates only in Burkina Faso, Côte d'Ivoire, Mali and Senegal; the Islamic Bank of Senegal identified during the design and other Intermediary Financial Institutions (IFIs) to be identified at implementation stage. Eligibility criteria for Attijariwafa Bank, the Islamic Bank of Senegal and other IFIs to access loan proceeds are described in Annex 21 Operations Manual and in the project description.

Criteria for the selection of sub-borrowers under each financing window of Component 1:

109. **Definition of smallholder farmers:** The definition of smallholder agriculture varies from one country to another. For this programme, the mostly widely used definition of smallholder farmers was adopted, which is farmers who farm less than 2.0 hectares of land¹³⁹.
110. **Definition of Farmer Organizations (FOs)**
- These organizations shall have: (i) a defined membership, (ii) a purpose for assembling, and (iii) an organizational structure in place to support members pursuing individual and collective interests.
- They may differ in their: (i) scale of membership (e.g., from national federations to local associations), (ii) function (e.g., policy, economic, technical and/or local development or advocacy), and (iii) legal designation (e.g., registered as cooperatives, clubs, groups or associations), among others. In the host country and the targeted regions in general, they combine their forces to improve bargaining power, particularly on pricing, achieve economies of scale in operations and/or improve quality control and access to financing and markets.
111. **Definition of cooperative:** An autonomous association of people who come together voluntarily to meet common economic, social and cultural aspirations and needs within a collective enterprise where power is exercised democratically. For this project, it is an enterprise which seeks a fair balance between profit and meeting the needs and interests of its members and their communities. Agricultural cooperatives allow their members to seize new economic opportunities by offering them a range of services in the agricultural sector.
112. **General definition of MSMEs**
- The definition for MSMEs is described in Tables 18 and 19 below.

Table 18: Variables for defining MSMEs in the host country (Burkina, Côte d'Ivoire, Mali and Senegal - WAEMU countries)

¹³⁹ HLPE (2013), Investing in smallholder agriculture for food security. A report by the High Level Panel of Experts on Food Security and Nutrition of the Committee on World Food Security, Rome, <https://www.fao.org/3/i2953e/i2953e.pdf>

Category	Quantitative Aspects
Medium enterprise	Employs between 10 and 100 people; subscription to the chamber of commerce; internal accounting system in place; annual profit above 50 million XOF, but not exceeding 500 million XOF ¹⁴⁰ ; minimum investment of 10 million XOF
Small enterprise	Employs between four and 10 people; subscription to the chamber of commerce; internal accounting system in place; annual profit above 10 million XOF, but not exceeding 50 million XOF; minimum investment of 1 million XOF
Very small and micro enterprises	Employs between one and three people; subscription to the chamber of commerce; internal accounting system in place; annual profit not exceeding 10 million XOF

Table 19: Quantitative variables for defining MSMEs in Ghana

Category	Quantitative aspects
Medium enterprise	Employs between 30 and 100 people; has a total annual turnover in the range of GHS138,000 to GHS 352,000; and with fixed assets not exceeding \$1,000,000.
Small enterprise	Employs between 10 to 29 people; has a total annual turnover in the range of GHS 15,000 to below GHS 138,000; and with fixed assets not exceeding more than \$100,000.
Very small and micro enterprises	Employs between one to nine people; has a total annual turnover between GHS2,400 and GHS15,000; and with fixed assets not exceeding \$10,000.

Source: Adopted and modified from Afful-Koomson et al. (2014) and AWFAFA (2019).

113. This programme uses a combination of the definitions above to determine the eligibility of MSMEs for support: MSMEs eligible for this programme will be all those that employ from one to 100 employees and have financial needs of up to 500 million XOF. In addition, given that the programme targets youth- and women-led MSMEs, the following additional criteria will apply and help identify them as priorities for support in all five countries. These criteria will be further refined during the first year of implementation, as specified in the OM:

i) For women-led MSMEs:

- Must be an officially registered agribusiness enterprise, with more than 51 per cent of its assets, holding or shares owned by women. This will be assessed and validated with firm registration information from the relevant public registry and their respective share of profits
- At least 30 per cent of the board of director members or individuals in senior management positions are women. This will be assessed and validated based on firm registration information from the relevant public registry
- Minimum firm size of three employees, of which 60 percent are women
- Activities are mainly carried out along the agricultural value chain
- Track record of loan repayments
- Bank account must be operational for more than two years
- Records and bookkeeping systems in place
- Agriculture land holding of between two and 10 hectares, and
- Any other characteristics deemed necessary by the LNABs, subject to the Executing Entity's confirmation.

ii) For youth-led MSMEs:

- A formally registered agribusiness enterprise with more than 51 per cent of its assets, holding or shares owned by youth, defined as individuals between 18 and 35 years of age. This will be assessed and validated based on firm registration information from the relevant public registry and their respective share of profits
- At least 30 per cent of board of director members or individuals in senior management positions are youth. This will be assessed and validated based on firm registration information from the relevant public registry
- Minimum firm size of three employees, of which 60 per cent are youth
- Activities are mainly carried out along the agricultural value chain

¹⁴⁰ XOF: a common currency, the official currency of the eight member states of the West African Economic and Monetary Union (WAEMU): Benin, Burkina Faso, Côte d'Ivoire, Guinea-Bissau, Mali, Niger, Senegal and Togo. 655.957 XOF = 1 EUR

- Track record of loan repayments
- Bank account must be operational for more than two years
- Records and bookkeeping systems in place
- Agriculture land holding of between two and 10 hectares, and
- Any other characteristics deemed necessary by LNABs, subject to the Executing Entity's confirmation.
- Eligible MSMEs include solar operators and farm-based associations, such as FOs and cooperatives, provided they meet the requirements described below.

For the sub-loans granted under the Green Business Financing Facility

114. Definitions:

- All destinations of the planned investments for which a loan is requested must be justified within the framework of IGREENFIN I and have the final goal of increasing resilience and reducing risk exposure of the MSME, cooperatives and FOs to the negative impacts of climate change and climate variability.
- Working capital: working capital will be understood as the available cash flow necessary for the administrative, financial and operative management of the MSMEs, cooperatives and FOs.
- Fixed asset: any tangible investment that the MSME will use for a considerable period that is not acquired for the purpose of selling them and that is acquired, constructed or used to adopt adaptation measures to increase the climate resilience of the MSME, cooperatives and FOs and enhance ecosystem services.
- Productive unit: the place where the business of the final beneficiary is located.
- Commercial unit: the place where the final beneficiary stores or sells his or her products.

115. Further conditions

- Instalments: The instalments that the final beneficiary will pay to the LNABs will be established in accordance with the IGREENFIN I conditions.
- Capital amortizations of IGREENFIN I disbursements will be scheduled quarterly, semi-annually or annually, according to the negotiations between the NEEs and the LNABs. Amortization of loans granted to MSMEs, cooperatives and FOs will be established during the negotiations between the NEEs and the LNABs.

116. The eligibility criteria to access credit are presented in Table 20 below.

Table 20: Eligibility criteria for access to credit

Current situation	GCF lines of credit	
Financial sector	Revolving lending facility (Window 1) Adaptation	Revolving lending facility (Window 2) Mitigation
Eligibility criteria: - Own and/or lease farmland or engage in sharecropping - Have the required experience and qualifications - Able to prove sufficient repayment capacity as FOs, cooperatives, MSMEs	Eligibility criteria: - Be an officially registered FO, women or youth organisation, cooperative, MSMEs - Have basic accounting and financial reporting systems in place or demonstrate capacity to put this kind of system in place within a short time - Interventions need to be linked to climate change impacts on the agricultural sector - Compliance with smart credit scoring tools including climate, environmental and social safeguards to be developed under this programme - FOs, cooperatives and MSMEs must submit an adaptation business plan	Eligibility criteria: - Be a formal FO, cooperative, MSMEs solar importers, wholesalers, retailers, installers or solar electricity service providers - Have basic accounting and financial reporting systems in place or demonstrate capacity to establish this kind of system within a short time - Interventions need to be linked to climate change impacts on the agricultural sector - Compliance with smart credit scoring tools including climate, environmental and social safeguards to be developed under this programme - FOs, cooperatives and MSMEs must submit an adaptation business plan

	- Targeted beneficiaries organized as FOs, cooperatives and MSMEs must show that they are vulnerable to climate change	- Targeted beneficiaries must show that they are vulnerable to climate change
Amount threshold to Sub-Borrowers	The amount of a Sub-Loan shall be: (a) in the range of EUR1,000 – EUR 150,000 (one thousand euros to eighty thousand euros) for FOs, MSMEs and/or cooperatives; and (b) up to seven hundred thousand Euro (700,000 EUR) for MFIs and commercial banks; pursuant to the details included in the Budget for Component 1.	

117. A summary of the preliminary investment criteria for green business projects to obtain financing through windows 1 and 2 is presented in Table 21 below.

Table 21: Investment windows and lending terms

BASELINE (Financial Sector-BANKS)	GCF LINE OF CREDIT – ADAPTATION WINDOW 1	GCF LINE OF CREDIT – MITIGATION WINDOW 2
Purpose: Finance the agricultural sector and provide green financial products to farmers, agricultural enterprises, traders and processors	Purpose: Provide financial support to smallholder farmers organized in FOs, MSMEs or cooperatives for the uptake of CRLEA practices and techniques (resilient seeds and livestock breeds, composting, drip irrigation, crop rotation, etc. - see Table 17 for list) working in selected VCs to strengthen their resilience and adaptive capacity, while improving productivity, income generation, investment and risk management and food security. Also provide financial support for sustainable land restoration, forest and other natural resource management systems.	Purpose: Strengthen, develop and scale up innovative rural electrification models that use off-grid, stand-alone solar systems (mini grids) and related equipment (for multi-purpose water pumps, mills for grinding, hulling and pressing etc.) to power agricultural value chains by LNABs'
Target Group: Smallholder farmers, FOs, MSMEs, traders and processors of agricultural products	Target Group: FOs including, women and youth organizations, MSMEs and cooperatives	Target Group: FOs including women and youth organizations, cooperatives, MSMEs, solar power companies, traders and importers
Target Areas: all countries and selected areas	Target Areas: all countries and selected areas of IGREENFIN I	Target Areas: all countries and selected areas of IGREENFIN I
Sectors: Agriculture, livestock, poultry, fishery, forestry, agricultural products processing, commerce, vet, irrigation, extension services, land restoration	Sectors: Climate resilient, low emission agriculture, including organic farmers, market gardens, climate resilient seed and animal breed growers; agroforestry and NFTP; EbA/nature-based solutions; climate information systems and knowledge platforms; suppliers of efficient irrigation equipment and climate resilient infrastructures, etc. Biodiversity conservation and land restoration.	Sectors: renewable energy for agriculture (off-grid, stand-alone solar micro and mini grids), enterprise financing (working capital or direct investment) for solar importers, wholesalers, retailers, installers and solar electricity service providers to power agricultural value chains

Type of loans: 1. Farm operating loans: loans for production cycle (short term, renewed annually through a simple request) 2. Investment loans: For setting up and developing a business or restructuring an agricultural business Tenor: Up to three (3) years starting from the execution of the sub-loan agreement between the selected banks and a sub-borrower.	Type of loans: 1. Direct investment loan for concrete eligible EbA measures that lead to income generation and high return on investment 2. Working capital loans for FOs, cooperatives, MSMEs, MFIs working on EbA and other adaptation measures 3. Loans for land restoration for viable green projects 5. Loans to invest in NTFPs	Type of loans: 1. Loans provided to FOs, cooperatives and MSMEs including importers, wholesalers, retailers, installers and solar electricity service providers to invest in solar panel equipment and systems for irrigation, processing, etc. 2. Working capital loans for solar importers, wholesalers, retailers, installers and solar electricity service providers
Lending terms: based on each project and assessment of the costs. Repayment begins in less than one year with an interest rate of more than 10 %	Lending terms: 0% interest and loan to be repaid within less than 2 years Floor loan: EUR 821,9 and variable ceiling	Lending terms: 1% interest and a loan repayment period of less than 2 years Floor loan: EUR 821,9 and variable ceiling

118. The eligible subprojects for Window 1 – Adaptation (i.e., measures/activities/equipment to be financed through sub-loans and, if applicable, sub grants) are listed in table 17 and are also available in Annex 21 Operations Manual.

119. In regard to Window 2 – Mitigation, a sustainable business model for the use of the GCF loan to unlock the potential of RETs to power agricultural value chains follows below. This model will be tailored to each country context (Figure 20).

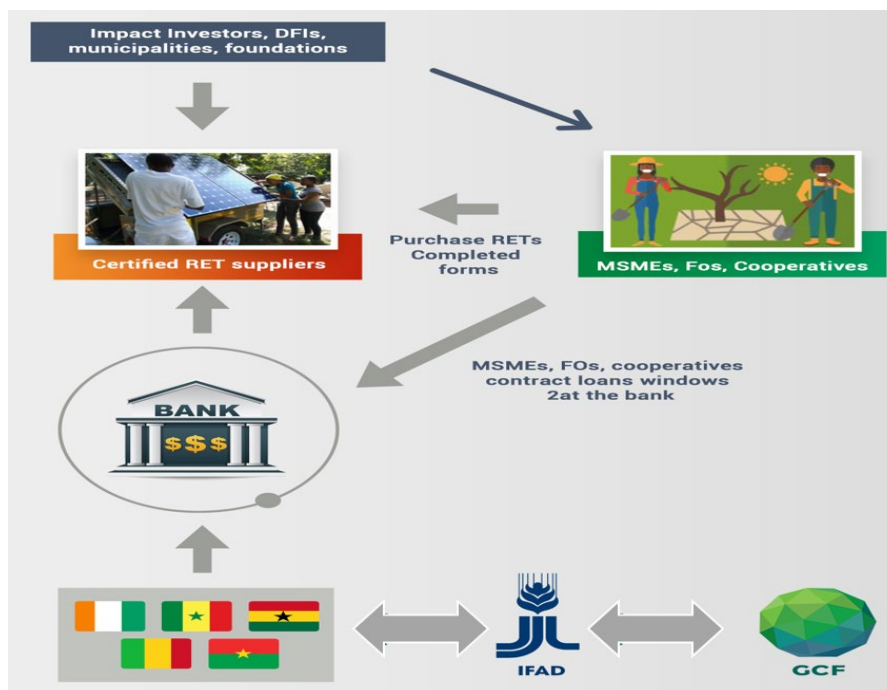


Figure 20: Proposed RET business model

120. The IGREENFIN I programme will finance the installation of 14.4 MWp of solar mini grids across the five target countries. A total of 1,800 modular systems with an average capacity of 8 kWp per system will be installed to reduce the use of diesel in agricultural value chains and approximately 640,000 tCO₂eq emissions over the 25-year lifetime of the mini grids. Mini grid assets will be owned by the borrowers who shall be either cooperatives or private sector operators depending on the specific regulatory framework and arrangement in each country. The following table summarizes mini grid assets ownership across the five target countries.

Table 22: Mini grid assets ownership – IGREENFIN I selected countries

Country	Lender	Asset Owner	Regulatory Arrangement
Burkina Faso	Agricultural Bank of Burkina Faso	COOPEL Private sector mini grid operator	COOPEL license from ARSE Concession Agreement from ARSE
Mali	National Agricultural Bank of Mali	Community-based organization Private sector operator	PPP with AER-Mali License from AMADER
Senegal	Agricultural Bank of Senegal	Farmer cooperatives in the area of undertaking	ERIL Framework
Côte d'Ivoire	National Investment Bank of Côte d'Ivoire	Private sector Single farmer	Tender of licensing by Ministry of Energy for serving agricultural areas with solar mini grids Smallholder farmers do not sell electricity to third parties and only own for their own use.
Ghana	ARB Apex Bank Limited	PAYGO operator with transfer to farmers associations Single farmer	System financed and installed by a PAYGO operator and transferred to FOs based on payments in instalments. System is maintained by operator during the initial phase. Smallholder farmers do not sell electricity to third parties and owns for own use.

121. A summary of the key barriers to the implementation of RETs in the five countries is presented in table 23 below.

Table 23: key barriers to the implementation of RETs – IGREENFIN I selected countries

Country	Barrier	Mitigation Strategy
Burkina Faso, Mali & Senegal	Long processing timelines for obtaining the operator's license for private sector owned mini grids which limits the private sector's appetite for owning/operating mini grids in Burkina Faso, Senegal and Mali.	Introduction of streamlined permitting processes to reduce the administrative burden. This will be achieved thanks to this programme's approach.
Burkina Faso, Côte d'Ivoire, Ghana, Senegal & Mali	Lack of government tariff subsidies that would lead to uniform electricity pricing	IGREENFIN I will work with the LNABs to provide concessional finance to smallholder farmer cooperatives and/or private sector for owning and operating mini grids. Concessional funds will significantly reduce the need for subsidies.
Burkina Faso	Limited technical capabilities of COOPELs, which increases reliance on external technical providers (so called "fermiers" in French) to manage the distribution network.	Knowledge management and information sharing activities will also target COOPELs.
Côte d'Ivoire	Low levels of consumer awareness of solar solutions.	Knowledge management and information sharing activities will target farmers' associations.
Ghana	The energy policy gives priority to national grid extension.	Options already in place that define what will happen when the grid arrives, including compensation to mini grid owners to enable them to exit the mini grid business.

Eligible local national agricultural banks (LNABs)

122. Eligible banks for the implementation of the GCF loan proceeds (Component 1) that have been subject to capacity assessments and criteria for selection are:

- The Agricultural Bank of Burkina Faso,
- The National Investment Bank of Côte d'Ivoire,
- The ARB Apex Bank Limited of Ghana,
- The Agricultural Bank of Mali,
- The Agricultural Bank of Senegal.

They will be the primary recipients of the GCF loan proceeds.

Eligible Intermediary Financial Institutions (IFIs)

123. **IFIs** will be selected following the Accredited Entity's procedures, and criteria and loans will be provided to such IFIs in accordance with all relevant requirements stipulated in the AMA and the FAA. The same conditions used for the LNABs will be applied to IFIs.

Commercial banks and microfinance institutions

124. **Definition of commercial bank:** A commercial bank is a financial institution whose basic activities are oriented mainly towards individuals (deposits, investments, savings and credit), businesses or public authorities. They can be physical establishments (branches, distributors) or virtual establishments (online or remote presence only). Their capital is held by shareholders who are not necessarily clients. This programme targets commercial banks working in the agricultural sector and smallholder farming in the host countries.

125. **Definition of microfinance institutions (MFI):** A microfinance institution is a local structure (NGO, association, cooperative, etc.) that delivers financial services to populations without access to the traditional banking system. MFIs mainly offer microcredit, but some also offer other types of services such as savings, micro-insurance, support and training services to deliver both social and economic benefits, since MFIs must be sustainable and financially balanced.

126. **The initial commercial banks eligible to implement the loan proceeds (Component 1) are:** the Islamic Bank for Senegal and the Attijariwafa Bank.

Other Intermediary Financial Institutions (IFIs)

127. IFIs may be private or public banks or MFIs that will act as intermediaries between the eligible LNABs under section II and the final beneficiaries and will ensure that financial resources are channelled between them in accordance with the same GCF conditions and the financial terms and conditions applied by the LNAB in each country. Eligible **LNABs** will enter into partnership with IFIs only in areas where a LNAB does not have any representation in targeted rural areas.

128. The **eligibility criteria for selection of IFIs** are as follow:

- Have extensive experience in deploying financing towards economic sectors that could potentially include adaptation activities
- Have a demonstrable track record of working with MSMEs, cooperatives, FOs and smallholders in the targeted countries
- Have satisfactory experience working as retailer with the eligible LNABs
- Must have a portfolio of programmes dedicated to agriculture (agriculture, livestock, agroforestry), food processing and energy use
- Must undertake financial services in the programme's target areas where selected LNABs are not represented
- Must have a monitoring system in place to track the flow of funds between LNABs and the beneficiaries and operational track record of loan repayments
- Must be an IFI authorized by the Central Bank of West African States (BCEAO) to operate in Burkina Faso, Côte d'Ivoire, Mali and Senegal and by the Central Bank of Ghana
- Must apply the same GCF conditions and lending terms that the LNABs apply to the end users
- Any other characteristics deemed necessary by LNABs and aligned with the GCF.

129. The analysis for an IFI's eligibility to the green lines of credit will be carried out during the preparation and analysis stage of each LNAB's project cycle. In this stage, the IFI's credit worthiness will be analysed according to the type of institution it is

using a credit risk methodology to assign it a credit quota which it may later use via the different intermediated credit programmes.

130. Each one of the above credit risk methodologies analyses areas such as the institution's capital, the quality of its assets, efficiency, rate of return and liquidity. For each area, financial indicators are used to determine the IFI's credit risk rating. Among the most important indicators there is financial vulnerability and delinquency, which reflect the quality of the institution's assets; the health and performance of the IFI is measured based on such indicators.
131. Also, relevant documents on environmental and social aspects of the IFI's operations will be reviewed to study and mitigate environmental and social risks. Such analysis takes into account two types of impacts: direct and indirect. Direct impacts derive from the institution's own activities while indirect ones stem from the projects included in the IFI's loan portfolio; the latter defines the level of the institution's environmental and social risk.
132. An environmental and social risk analysis will be carried out for each financial institution. It will review the size and probability of direct and indirect impacts of the IFI, assign it to a category of environmental and social risk and identify existing controls to mitigate the risks, as well as any additional measures that will need to be established to minimize, mitigate or compensate for risks. This process will inform the development of an Environmental Impact Study and Plan (EISP), which stipulates the measures that the IFI shall take to manage the social and environmental risks and/or impacts stemming from its operations and those of its clients. To determine if the IFI is eligible for financing through the programme's green credit lines, the EISP and its credit rating is submitted to the LNAB's credit committee, which is comprised by different technical areas of the selected LNAB. After approval by the credit committee, this analysis is submitted to the LNAB for final approval.
133. The LNAB is responsible for managing the direct relationship with the client and the underlying financing operations – from the identification and preparation stage until the loan is repaid in full – in order to stay adequately and timely informed of any relevant, quantitative and/or qualitative event that could affect the client's business or operations, credit rating and compliance with their contractual obligations to the LNAB. The PIU of the LNABs could recommend to the NEEs to turn down the applications of clients deemed financially non-viable after assessment. The National Steering Committee chaired by the NEEs will provide an opinion on the financially non-viable clients but the final decision will be taken by the NEEs.
134. The LNAB's follow-up credit area will make periodic visits to the IFI at least once a year. The visits must be duly documented in the client's credit file. The IFI's obligations to the LNAB, derived from the credit agreement, will have equal rank with other liabilities of borrowers of the same gender, nature or type. In no case shall such credits, regardless of their nature, have a payment priority superior to that maintained with the LNAB. IFIs will be trained on climate change adaptation aspects as well as monitoring and evaluation of adaptation measures and the implementation arrangements at the beginning of the programme and will receive training regularly throughout the IGREENFIN I programme.
135. As the Accredited Entity, IFAD will provide support to the PMUs and PIUs during the first years to ensure that the IFIs in the targeted countries receive adequate training. To this end, the LNABs will make several departments available to support the IGREENFIN I programme (human resources, communications, 5 specialists of the Environmental and Social Office, legal affairs and international relations, among others).
136. All IFIs willing to participate in the programme and disburse funds will have to participate in the training on climate change, appropriate adaptation measures supported by the IGREENFIN I programme and monitoring arrangements.
137. IFIs' activities will include the following:
 - Identifying, assessing, giving clearance and monitoring the sub-loans to FOs, MSMEs and cooperatives for adaptation /mitigation investments. The final approval of all cleared subprojects will be performed by the NEE represented by the Ministry of Finance
 - Coordinating with TAFs on the provision of capacity-building activities
 - Planning and monitoring the technical aspects of the programme, including regular field visits and periodic reporting to the LNAB
 - Actively participating in all relevant programme activities where appropriate.
 - If applicable, adopting during the programme the systems, programmes and tools developed by the programme to ensure sustainability of the programme's outcomes.
 - Playing an active role in coordinating with other stakeholders throughout the programme and maintaining a close relationship with their respective LNAB

- Preparing and submitting periodic progress reports and holding regular consultations with beneficiaries and contractors.

Table 27: Distribution of GCF funding as a whole

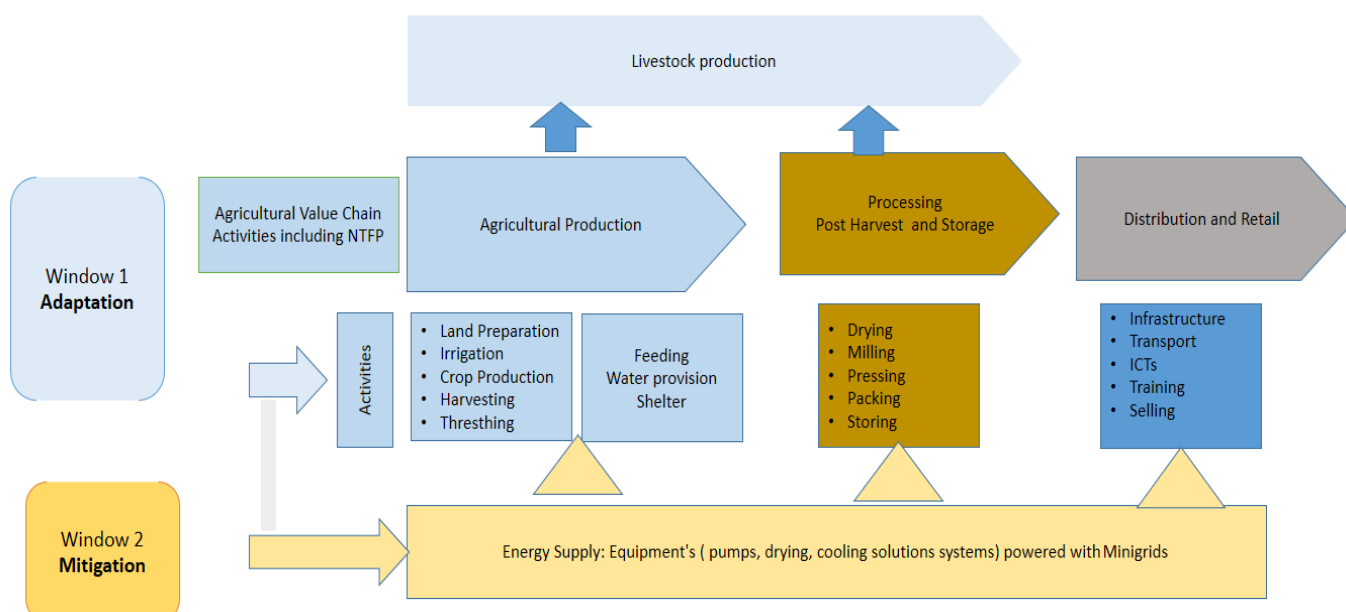
Credit lines	% of total funding	Example of eligible activities*	Excluded activities	Funding
Window 1: Adaptation	Climate resilient, low emission agriculture (CRLEA) practices (65%) Land restoration and agroforestry techniques (cross-cutting) (15%)	See Table 17: Initial list of activities for each value chain eligible for financing from the loan proceeds	GMOs, pesticides, herbicides (see ESMF Annex)	US\$60 million from GCF resources and co-financing
Window 2: Mitigation	Renewable energy equipment/ technologies for agriculture (20%)	See Table 17: Initial list of activities for each value chain eligible for financing from the loan proceeds	Invasive tree species (see ESMF Annex)	US\$30 million from GCF resources and co-financing

* More details are available in Annex 23 - Investment Criteria.

138. It is expected that a growing pipeline of green business projects will enable the LNABs to start aligning their portfolios with their respective country's NDCs while building their internal capacity to leverage more concessional investments to support the uptake of climate resiliencies, low emission agriculture in the region.
139. Key outputs of Component 1 are:
- **Output 1.1:** Viable green businesses run by FOs, youth and women led-MSMEs and cooperatives created, which attract funding, and commercialized RET technologies are adopted
 - **Output 1.2.** Technical support and capacities built for FOs, MSMEs, cooperatives, LNABs and IFIs
140. **Key activities under component I** are:
- **Activity 1.1.1.** Establish a Green Business Financing Facility (revolving lending facility) - **window 1** (adaptation, pillars 1, 2 and 4) to catalyse adequate finance for viable green businesses in key value chains (livestock – cattle, sheep, goats and poultry; horticulture – tomatoes; mango; cashews; shea; rice; millet; groundnuts and cassava) for investment in the adoption of CRLEA practices, including land restoration techniques.
 - **Activity 1.1.2.** Establish Green Business Financing Facility (revolving lending facility) - **window 2** (mitigation, pillars 3) to catalyse adequate finance for renewable energy equipment and technologies to power the selected value chains. IGREENFIN I will promote the provision of solar energy for the following activities: (i) drip irrigation systems to address declines in agricultural yields, (ii) processing techniques to reduce post-harvest losses, and (iii) storage of vaccines to address growing livestock diseases due to climate change. Solar mini grids that power efficient drip irrigation systems are needed to address water scarcity and rainfall variability, particularly during periods of drought. Additionally, solar mini grids that power the processing, storage and packaging of agricultural products will enable beneficiaries to seize new and existing marketing opportunities, and gradually lead to reductions in the use of fossil fuel and wood in the target countries. The demand for energy for agricultural processing and transformation within IGREENFIN I's scope of work in Burkina Faso, Mali and Senegal is estimated at 4,606 MWh per year, and 3,456 MWh per year in Côte d'Ivoire and Ghana.
 - **Activity 1.1.3:** Provide technical assistance to set up the Green Business Financing Facility with green credit line windows 1 and 2. This activity will support LNABs in setting up the facility and its two windows within each bank and developing a platform which will include all the eligibility criteria for assessing green business proposals, climate projections and priorities per country. Under this activity, the staff of the PIU will be regularly trained to better assess all green loan applications submitted to the LNABs.

141. To ensure the replenishment of the Green Business Financing Facility to guarantee resources for reinvestment in future green business plans, only financially viable climate change adaptation and mitigation activities proposed by FOs, MSMEs and cooperatives will be financed. The frequency of repayments of principal from the LNABS to the NEEs will start from the sixth year on given that the grace period for the NEE is 10 years. The Green Business Financing Facility's Window 2 will complement Window 1. Figure 21 below presents the links between the two windows and complementary.

Figure 21: Link between elements from Windows 1 and 2



142. **Component 2: Technical Assistance Facility (TAF) – Pillars 2, 3 & 5** (EUR 23,013 2million of the GCF Grant) will be created to address capacity, knowledge and policy gaps hindering the uptake of green business projects in the selected countries. It will target LNABs, central banks and clients (FOs, MSMEs and cooperatives), while placing a special focus on women and youth. The TAF will provide technical support to Direct Access Entities such as i) the Agricultural Bank of Senegal to build its internal capacity to mobilize additional concessional climate financing during the implementation of IGREENFIN I, and ii) the Centre de Suivi Ecologique (CSE), a strategic IGREENFIN I partner, to enhance its capacity to better monitor land use and forest management activities in all selected countries. The TAF will also support strategic partnerships between the LNABs and Attijariwafa Bank (a GCF accredited commercial bank) to crowd in additional private sector financing into the programme in the following countries: Burkina Faso, Côte d'Ivoire, Mali and Senegal. The TAF will provide the necessary technical support to the Central Bank of West African States (BCEAO) for Burkina Faso, Côte d'Ivoire, Mali and Senegal, and the Central Bank of Ghana for the development of strategic studies and knowledge to address policy and regulatory gaps and create an enabling environment for the uptake of green business projects. This component will contribute to Pillar 2 (sustainable management of ecosystems and land restoration), Pillar 3 (climate resilient infrastructures and access to renewable energy) and Pillar 5 (capacity-building) of the Great Green Wall Accelerator. The TAF will be implemented by the respective ministries of agriculture in each country through an agreement with the NEEs, as described in Section A.20 above.
143. **Eligible beneficiaries for capacity-building activities:** This component will provide technical support to build capacities on both the demand side (MSMEs, cooperatives, FOs and women and youth organizations) and the supply side (governments, LNABs, MFIs, commercial banks). The expected outcome is enhanced capacity of LNABs and other financial institutions, national institutions and recipients to finance and implement climate resilient, low emission agriculture. As the programme adopts a regional approach, capacity-building exchanges between countries will be undertaken between the financial institutions and between MSMEs, FOs, and cooperatives from the different participating countries (events, visits, exchanges, etc.).
144. Technical support and capacity-building will be provided to: (i) 378,600 beneficiaries (189,000 male and 189,000 female) from 1,500 MSMEs and 2,500 FOs or cooperatives; (ii) 100 pipelines of bankable projects; (iii) personnel of 30 commercial banks and MFIs on ESMS systems and green banking (50% women) and (iv) 2-4 green products and policies developed per country (v) LNABs and two central banks.

145. **Eligible Technical Assistance Services Providers (TASPs) contracted by each IFAD baseline investment:** In each country, IGREENFIN I will identify organizations in the country and the region that are qualified to provide technical assistance to the LNABs, MFIs, commercial banks and MSMEs, FOs and cooperatives, including solar operators. Their training activities will be focused on the capacity-building needs identified during consultations and will provide a detailed introduction of all the important aspects of the programme. It should be noted that these organizations have not yet been selected. Consultants from the identified network of TASPs will be hired depending on the specific needs of the activity and/or beneficiaries.
146. TASPs should be qualified to give the following training content:
- Climate issues (risks, forecast technologies)
 - Climate resilient, low emission agriculture (adaptation and mitigation measures, green technologies, soil and water management, etc.)
 - Financial literacy, business plans and management
 - Transition towards green finance
 - Institutional and regulatory framework
 - Promotion of knowledge management, South-South cooperation, exchanges
147. **Applicable rules for technical assistance and capacity-building activities:**
- All technical assistance processes shall start with the collection of data on the needs of eligible recipients from the recipients themselves.
 - When submitting documentation for approval, the original copy of all documents must be presented. The time and date of the submission must be recorded on all documentation.
 - No approval process will be started unless all requirements are fulfilled.
 - A checklist will be provided to ensure compliance with all requirements specified in the programme's OM (Annex 21).
 - Only forms filled out using a computer or typing machine will be accepted.
 - All technical assistance must be assessed by the Technical Assistance Committee appointed by the NEEs in coordination with the National Steering Committee. Any decisions or action adopted by the TAC will be subject to a favourable vote or non-objection by the NEEs. An official record must be written and shall include a description of all evaluation variables listed in the OM. The signatures of all reviewers or committee members must be included in this official record.
 - The Technical Assistance Committee shall be representative of all stakeholders involved in the process.
 - The majority of consultants contracted to provide technical assistance must be nationals from the country in question, with a quota of at least 50% of women.

Component 2 has two main outputs:

Output 2.1 – Generate knowledge, conduct studies, support knowledge transfer, provide training and monitor climate impacts

148. The green business market has yet to be created in the GGW region and many technologies and practices required for climate resilient, low emission agriculture are still at a nascent stage, meaning that significant capacity and knowledge gaps remain. In addition, to make green business projects commercially viable, a better understanding is required of both market opportunities and the technology needs of FOs, women and youth-led organizations, cooperatives and MSMEs. The activities under this output will contribute to incremental capacity and knowledge building so as to accelerate the adoption of climate adaptation and mitigation practices and technologies. By increasing the capacities of FOs, women and youth-led organizations, cooperatives and MSMEs, this output strives to contribute to the restoration of 200,000 ha of degraded areas and increase smallholder farmers' yields for the following crops: millet, cassava, rice and groundnut. It will promote the adoption of best climate resilient agricultural and land use practices, such as intercropping cereals with legumes and fruit and other trees in agroforestry systems, best practices in fruit and NTFP chains (shea butter, cashews and mango, etc.), efficient irrigation and soil conservation techniques, etc. Technical assistance (TA) will also be provided to FOs, women and youth-led organizations, cooperatives and MSMEs to prepare viable green business plans and tried and tested practices will be compiled in a catalogue of the best adaptation and mitigation solutions available in each country and across the region.
149. At the level of the LNABs, training will be offered to the banks' officials, directors and staff to strengthen their capacity to identify potential clients, assess potential beneficiaries' green businesses against the eligibility criteria and identify viable bankable green business plans and pipelines. The IFAD baseline PMUs (under the ministries of agriculture) will use the grant portion in coordination with the LNABs' PIU to hire the relevant consultants and TASPs to support LNABs and selected

partners in the area of green lending and climate risk management. FOs, cooperatives and MSMEs will be recipients of the TA proceeds. Training will be provided on how to conduct market assessments, data collection and analysis, the selection of target clients, preliminary eligibility and creditworthiness assessments of potentially viable business plans proposed by FOs, cooperatives and MSMEs, including ones focused on the acquisition and use of renewable energy equipment required along the agricultural value chains. Additionally, the programme will support the installation of solar mini grids for food processing, storage and packaging to enable smallholders to seize marketing opportunities, which will gradually lead to reduced use of fossil fuel and wood for energy in the target countries. The TAF will assist with the creation of a package of tools and instruments to better identify investment risks and increase awareness and training in climate resilient, low emission agriculture (CRLEA) practices. The programme will support LNABs and partners (MFIs and commercial banks) in developing and updating their environmental and risk management systems (ESMS) and skills, environmental and climate strategies and related plans. Activities for this output include (for activities disaggregated by country, see the appendix “Activities breakdown per country”):

- **Activity 2.1.1.** Provide capacity-building to FOs, women and youth-led organizations, cooperatives and MSMEs on understanding climate risks and climate resilient, low emission practices and integrating them into green business plans
- **Activity 2.1.2.** Offer capacity-building on the development of bankable green business plans
- **Activity 2.1.3.** Provide training to FOs, MSMEs and cooperatives on how to implement financed green business plans for results
- **Activity 2.1.4.** Support women and youth-led FOs and MSMEs to increase their participation, improve their financial literacy, natural resources management capacity, involvement in decision-making processes, ability to use labour-saving and cost-competitive CRLEA technologies and support for gender-related studies to guide decision-making.
- **Activity 2.1.5.** Provide technical assistance on green lending and climate risk management to LNABs
- **Activity 2.1.6** Provide technical assistance to enhance governance and management of loan products, fill the green credit lines policy gap and ensure use in each LNAB and IFI.
- **Activity 2.1.7.** Develop a package of tools and instruments, including ESMS, for LNABs to assess green business plans
- **Activity 2.1.8.** Conduct awareness-raising and training on climate resilient, low emission agriculture within LNABs.

Output 2.2: Improve policy dialogue, government technical and institutional capacity, advocacy, training, knowledge management, information dissemination and stakeholder management.

150. This output will promote a transformative approach and a feedback loop for scaling up investments in green business through enhanced public and private sector engagement and partnerships and the provision of capacity-building and training for national institutions on the economic development objectives associated with the two financing windows. Furthermore, it will strengthen knowledge management, information dissemination and stakeholder engagement through the creation of solution-oriented platforms, including a databank on innovative projects, that will be used to inform and facilitate a high-level policy dialogue between FOs, LNABs, MFIs and government authorities (national and local). With regards to renewable energy in agriculture, institutional and regulatory frameworks and planning on renewable energy will be improved to better support knowledge sharing, capitalization and dissemination of lessons learned. The programme’s impact, institutionalization and replicability will be enhanced through the involvement of the relevant sector ministries (energy, environment and agriculture, economy and finance) responsible for national and sector policies, planning and strategies. The programme will also organize high-level and technical events and develop a platform for exchange between value chain actors and financial institutions. To support stakeholder engagement and capacity-building on green lending, a stakeholder engagement plan will be developed. To promote the uptake of technologies and adaptation at scale, the programme will promote South-South and Triangular cooperation with similar programmes in Latin America and Asia financed by the GCF (e.g., AfD-GCF project on financing systems, the Productive Investment Initiative for Adaptation to Climate Change, or CAMBio II, in Central America, the Niger Inclusive Green Financing Project). Other activities include: the production of 40 policy briefs and five publications and catalogues for decision makers. With the support of the Regional Support Programme (Component 3), IGREENFIN I will promote the continued use of green finance worldwide and connect the selected LNABs with similar initiatives around the world.

151. To ensure sustainability beyond the programme’s life cycle, technical assistance will be provided to the target countries’ central banks on conducting strategic studies, designing and/or reviewing policies and regulations, developing climate risk management practices at the programme level (standards), developing and harmonizing green lending products, instruments and tools such as ESG benchmarks; generating and disseminating knowledge on green finance within the financial system

and climate allocations from both the private sector and government to the agricultural sector, as well as mechanisms for sharing best practices. This output will also support the dissemination of regional/national public awareness materials on green finance and greening financial systems, particularly for public development banks and the Central Bank of the West African Monetary Union (WAMU), which covers Burkina Faso, Côte d'Ivoire, Mali and Senegal, and the Ghanaian Central Bank. It includes outreach activities to support key actors of the financial system involved in the agricultural sector (banks, regulators, insurance companies, investors, agricultural and development banks, MFIs), policymakers, senior government officials and private actors on green financing and CRLEA. IGREENFIN I will support the establishment of a platform of green development banks/central banks regulators, public and private sector in the region to advance the policy dialogue and coordination among governments and partners on green finance geared towards the achievement of SDGs, NDCs and the goals of the Paris Climate Agreement.

152. To incentivize the sector, the TAF will launch a Sahel & GGW Award. The Sahel & GGW Award process consists of: i) the granting of an Annual Sahel Award incentive in the form of debt cancellation to promote sustainability in smallholder agriculture, and ii) the holding of an institutionalized annual award ceremony. The Award will be handed out to the five best projects in each country submitted during the year under each funding window described in Component 1 (10 projects per year per country). For each winner, a maximum of EUR 6986, 15 of debt on top of the loan principal will be cancelled. The prize will be administered as follows: EUR 3287, 6 will be cancelled at the time of selection; EUR 2465, 7 at the time of the midterm review, and EUR 1232, 85 once adaptation/mitigation measures have been successfully implemented and monitored. As the administrator of the award, the recipient government (NEE) will hold the right to revoke the debt forgiveness to recipients who at the time of the midterm review have not complied with original terms. The Sahel & GGW Award trophies will be given to the selected beneficiaries and the award will be considered collateral when applying for additional loans from banks. The recipient country will administer these resources through its Ministry of Finance. IFAD and selected banks will review applications and make recommendations to the government based on their expert opinion. The award selection committee will be further detailed in the implementation manual, but decisions will remain with the governments. Measures will also be taken to ensure that among the winners of the award, 50 per cent are MFIs, FOs, cooperatives and MSMEs led by women and 50 per cent are led by youth. The Annual Sahel & GGW Award Ceremony will be organized to honour and give greater visibility to changes in attitudes and behavioural changes and the best adaptation and mitigation projects and initiatives developed at the national level by FOs, cooperatives, women and youth organizations and MSMEs. A Sahel & GGW Award Trophy will also be designed. Also, under this output, women and youth will be incentivized to implement climate adaptation and mitigation measures and RET in agricultural value chains. Targeting will be carried out according to the definition of women- and youth-led MSMEs, FOs and cooperatives in each country (see above).

153. Activities include (for activities disaggregated by country, see the appendix "Activities breakdown per country"):

- **Activity 2.2.1.** Provide technical support to Central Banks through the production of studies and the development of policies to improve the institutional and regulatory framework, standards and ESG benchmarks on climate risk management and green finance for agriculture.
- **Activity 2.2.2.** Organize high-level round table events and exchanges on green finance
- **Activity 2.2.3.** Organize South-South and triangular cooperation visits and tours for experience sharing
- **Activity 2.2.4.** Enhance knowledge management and information sharing
- **Activity 2.2.5.** Organize annual Sahel & GGW Award event
- **Activity 2.2.6.** Organize events to replenish the Sahel & GGW Award
- **Activity 2.2.7.** Develop application protocols and framework
- **Activity 2.2.8.** Enrol women and youth entrepreneurs to compete for the Sahel & GGW Award.

154. Knowledge and innovations generated and systematized under Component 3 will be taken into account when implementing the two outputs of Component 2.

Component 3. GGW Regional Support Programme (GGW RSP)

Output 3.1: Enhanced knowledge management and exchanges accelerating the uptake of best practices and learning and informing policy and investments across projects of the GCF and others

155. In this proposal, knowledge management is the collection of methods for creating, sharing, using and managing knowledge and information with the goal of generating a continuous learning loop and improving processes. Lessons learned from

regional programmes that aim to strengthen the impacts of individual projects show that these projects give relatively little priority to knowledge management, particularly in their initial phases. This often results in delays in implementation due to loss of information or the repetition of common errors from previous projects implemented by other partners or in other countries. Projects usually do not have dedicated knowledge management experts on staff and M&E officers are often not well equipped to conduct knowledge management processes. KM is almost never part of the TORs of country staff, nor are there clear KM indicators and targets for the activities¹⁴¹. An overarching global knowledge management approach will not only strengthen exchanges among projects and countries but also make project and programme implementation more efficient and effective by ensuring that existing knowledge is appropriated from the early stages on.

156. The GCF GGW Regional Support Programme managed by IFAD will contribute to the effective management of the knowledge generated by GCF and GEF projects by providing faster and more effective access to relevant internal and external knowledge to maximize impact in the GGW. The GGW RSP Knowledge Management component will foster an enhanced culture of knowledge sharing whose benefits include:

- **Easy access to information and resources** for GCF NDAs; GGW, GEF and AF focal points; AEs and especially DAEs. With information more readily available to them, these actors and entities will be better equipped to support countries in implementing their NDCs and NAPS through present and future GCF projects.
- **Increased pace of learning uptake.** Well-targeted learning events such as South-South exchanges have proven to be an effective means of facilitating peer-to-peer learning among experts from different countries¹⁴². Sharing experiences from countries where project implementation is more advanced will help teams accelerate the progress of less mature projects, which will be able to take advantage of existing best practices and effective approaches.
- **Knowledge for reuse:** Reusing ideas, documents and expertise minimizes the need to reinvent the wheel, avoids repeating problems and mistakes, saves time and accelerates progress, as it enables teams to adapt successful experiences and good practices to the context they are working in. For instance, some of the background assessments elaborated by AEs and reviewed by the GCF Secretariat and iTAP for a specific FP could be reused or adapted by another AE.
- **Opportunities to take advantage of existing expertise and experience:** The more complementary the expertise of the team members, the greater the power of the team.
- **Identification and dissemination of methods, tools, templates, techniques, and examples that can serve as the building blocks for supporting processes and procedures.** Using these consistently streamlines work, improves quality and enhances overall efficiency.
- **Data for evidence-based decision-making** by NDAs, DAEs and international AEs, and GGW, GEF and AF focal points: Using knowledge generated from the analysis of data, especially when it is directed to specific audiences, allows teams to make decisions based on actual experiences and lessons learned.
- **Effective KM will promote operational excellence, technical innovation and efficiency gains** for NDAs, GCF projects under implementation and AEs. Innovations are shared systematically and timely among projects, countries and entities, allowing for faster replication and scaling up.

Knowledge management structure

157. GCF projects have KM strategies and activities in place and have allocated financial and human resources to achieve the proposed results. The support for knowledge management provided by the Regional Support Programme will not replace the existing KM strategies of each GCF project or the responsibilities of the corresponding AEs. On the contrary, it aims to build bridges between all these project-specific KM strategies and resources and create a structured framework to ensure that information and knowledge is properly captured and shared systematically with other GCF projects, entities and GGW countries. The Regional Support Programme will provide support at two levels:

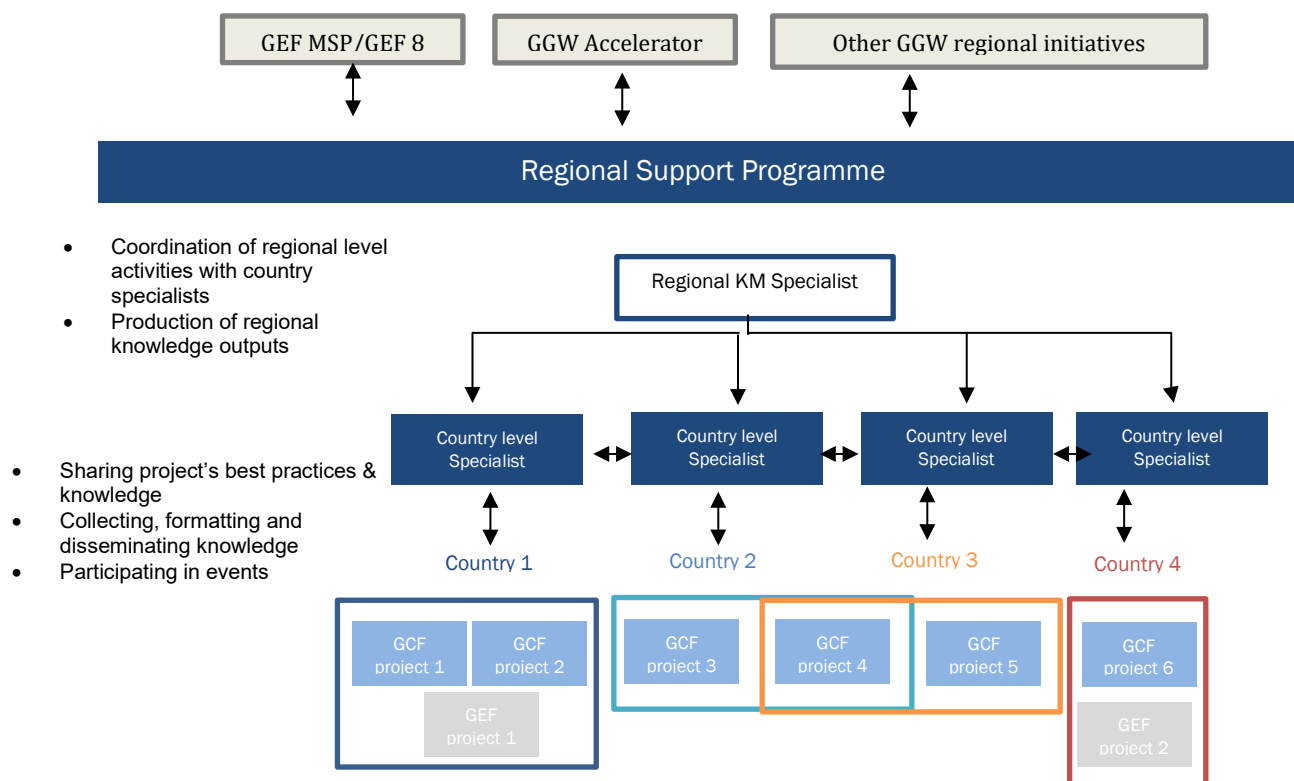
- **Regional level:** structure, streamline, analyse and disseminate knowledge generated by GCF projects across the GGW countries through a KM strategy, a knowledge centre, specific knowledge and communication products and the organization of events. In addition, synergies with the GGW Accelerator, GEF and AF will be created at the regional level for specific activities described below. The Regional Support Programme Unit will hire a Regional Knowledge Management Specialist who will be in charge of coordinating activities at the regional level.
- **Country level:** facilitate and support GCF projects in sharing the knowledge generated by each project and fostering linkages and learning based on past and present projects in the country. The RSP will arrange for a Knowledge Management Specialist to be hosted by the NDA in each country, as a way to address the issue of the limited human

¹⁴¹ IAP Midterm review report draft

¹⁴² GEF Secretariat (2021), GEF-6 Commodities IAP Program: Emerging Lessons from the Good Growth Partnership, <http://assembly.thegef.org/publications/gef-6-commodities-iap-program-emerging-lessons-good-growth-partnership>

resources that each project assigns to knowledge management and help anchor knowledge in the appropriate institution¹⁴³. The Country Knowledge Management Specialist will be support the AEs in systematically formatting and sharing knowledge through the different channels developed by the programme (knowledge centre, events, knowledge products).

Figure 22: Regional Support Programme



158. Key activities for Output 3.1 are:

- **Activity 3.1.1 Establish a knowledge baseline.** The rationale goal here is to map all relevant information on climate financing within GGW countries and determine where relevant knowledge is missing.
- **Activity 3.1.2. Develop a Knowledge and Communication Strategy.** Based on the knowledge baseline assessment, the RSP Support Unit will work with all AEs implementing projects in the GGW to build a knowledge management and communication strategy¹⁴⁴. The strategy will cover the role of KM and communication for the GGW RSP (external communication, internal communication, gathering, analysing, documenting, translating information, adapting information), strategic objectives, target audiences, coordination of products to be defined, timeline for reporting knowledge, templates, KM specific indicators, and responsibilities for individual projects. This activity will respond to the challenges faced by most regional programmes (IAP, BRICKS), namely the limited resources allocated to KM in each project, the implementation of projects in an isolated manner and variations in institutional standards.
- **Activity 3.1.3 Create a knowledge centre.** A knowledge centre will be created to execute the knowledge management strategy. It will make knowledge and information on key thematic areas and country project activities readily available to multiple stakeholders and easy to disseminate. It will organize knowledge into the GGW RSP's five pillars, and by thematic areas (M&E, communication), project or country, etc. New and existing knowledge disseminated through this centre will include good practices, improved indigenous knowledge, CRLEA, land restoration, value chain development, climate-resilient infrastructures, water management, etc. It will collect existing gender assessments, climate assessments and legal analyses prepared by AEs for GCF Funding Proposals and GEF project proposals¹⁴⁵ and make them available to all AEs, especially DAEs, in the GGW countries. The knowledge centre will function mainly in English

¹⁴³ IAP Lessons learned report on knowledge management shows the need for additional support at country level to ensure coordination of KM activities and formatting of knowledge products in a consistent manner.

¹⁴⁴ An example of a regional programme KM strategy that brings together knowledge generated by various projects is the IAP knowledge management strategy.

¹⁴⁵ In line with the proposed activities of the long-term vision for collaboration between GCF and GEF (see section B.1).

and French, be open access and be designed primarily to address country project's needs and to allow country projects to easily upload and request resources. The knowledge centre will be conceived as a collaborative platform to facilitate interaction among projects and countries, with options that allow them to contribute to blogs, news, online discussions and online communities of practice and to share and comment on content easily. The knowledge centre will be hosted by an existing website or platform from an institution that covers the region and that is very likely to continue to engage in the same area of work beyond the programme's life cycle. It will be connected to and coordinate with the GGW Accelerator knowledge hub currently under development to facilitate exchange and avoid duplication of information and efforts. The Regional Knowledge Management Specialist will be in charge of the development and implementation of the knowledge centre, while the Country Knowledge Specialists will support AEs in uploading their knowledge products, reviewing the knowledge submitted to the centre and promoting the centre's use. The centre's platform will include an interactive helpdesk to facilitate its use and ensure that all users can maximize their use of the knowledge available in different formats and across different sectors. IFAD will select and procure the "institution" that will host this platform.

- **Activity 3.1.4. Create communities of practice.** Communities of practices (CoP) will help connect practitioners from the projects/programmes of GCF, GEF, AF and other AEs aligned with the GGW Accelerator's five pillars so that they may learn from experiences and projects and incentivize cross-regional knowledge exchanges. Based on IFAD's experience in this area¹⁴⁶, CoPs are effective in identifying best solutions and lessons learned based on evidence and in disseminating information on them, which help to scale up and ensure faster and more efficient delivery of GCF projects. The RSP's regional and national KM experts will support the creation and facilitation of CoPs to generate a constructive environment of mutual support and trust among the experts and other stakeholders. The CoPs will be set up: i) **at the country level** to connect the KM experts of GCF and GEF projects¹⁴⁷ and programmes implemented in the country by different AEs who will structure and systematize the exchange of information, good practices, lessons learned and innovation among those projects, and draft policy recommendations with the GCF NDA, the GEF and AF national focal points, and the GGW focal point, and ii) **at the regional level**, mainly to enhance exchanges between countries in the region and organize South-South exchanges.
- **Activity 3.1.5 Generate knowledge products to disseminate lessons learned and scalable good practices and inform evidence-based investment and policymaking.** The RSP will bring added value to the existing knowledge from individual GCF and GEF projects¹⁴⁸ by creating knowledge products that facilitate the analysis of experiences in different countries and projects and the dissemination of the knowledge across the 11 countries. The result of this analysis will be a set of good practices that can be easily adapted to and adopted in similar contexts and scaled up across the region. In addition, data gathered across the 11 countries will be used to develop specific knowledge products that target investors and policymakers to inform evidence-based decision-making. The specific knowledge products such as policy briefs, social media pages, monthly newsletters, internal bulletins and tool kits to be developed under this component will be defined in the Knowledge Management and Communication Strategy (see activity 3.1.2). Through regular interaction with the RSP KM specialists in each of the 11 GGW countries and with all GCF and GEF projects in the GGW countries, as well as desk reviews, portfolio reviews and existing academic and public/private sector research, this activity will methodically gather, capture and share lessons learned and best practices from all the key result areas of the GCF and relevant GEF projects which are contributing to the objectives of the GGW. The Country Knowledge Management Specialist will be in charge of drafting country level knowledge products and the Regional Knowledge Management Specialist will be coordinating the creation of regional products¹⁴⁹.
- **Activity 3.1.6 Organize peer-based knowledge exchanges and knowledge sharing activities.** Knowledge exchanges targeting specific audiences will be organized at different levels. The exchanges and sharing activities will be part of a broader process, and not stand-alone activities, to guarantee uptake of the knowledge and engagement in continuous learning:
 - **Country level workshops** will focus on: (i) raising the awareness of stakeholders to improve the collective understanding of the relationship between KM, M&E, communications, advocacy, staffing plans and overall strategies of GCF projects, and (ii) offering training on thematic issues such as the application of analytical tools for regional-scale monitoring, knowledge platforms for learning and the synthesis and dissemination of

¹⁴⁶ For more information on CoPs in IFAD, see <https://ifadkmcentre.weebly.com/cops-in-ifad.html>

¹⁴⁷ In line with the proposed activities of the long-term vision for collaboration between GCF and GEF (see section B1)

¹⁴⁸ Idem.

¹⁴⁹ The IAP midterm review showed the importance of regional programmes playing an active role in supporting implementing entities in defining and designing their knowledge products, trainings and approaches.

innovations. These workshops will also provide an opportunity to connect projects, including the ones funded by GEF or AF, with similar intervention plans and objectives¹⁵⁰.

- **Annual Regional Knowledge Share Fairs** to bring together the knowledge and lessons learned from other GGW partners such as the Permanent Interstate Committee for Drought Control in the Sahel (CILSS) and independent units such as Agrhymet, the Sahara and Sahel Observatory (OSS), IUCN, the Economic Community of West African States (ECOWAS), FAO, the EU and other international partners, UNCCD, UNEP, World Agroforestry Centre (ICRAF), as well as knowledge providers such as the World Overview of Conservation Approaches and Technologies (WOCAT) and the Africa Regreening Initiative, among others. Knowledge Share Fairs are facilitated events designed to share knowledge in an interactive way. The key to a Share Fair's success lies in the linkages and connections people make at the event. Share Fairs bridge social, hierarchical and technological gaps through various participatory formats. The participatory aspect of the events also contributes to the generation of a community around the themes discussed.
- **South-South Exchanges** will be organized to share the best scalable good practices from other regions. These events will include training and periodic study tours of GCF, GEF and AF projects.
- **Activity 3.1.7. Enhance coordination and programming among GCF NDAs, AEs and GEF, AF and GGW national focal points:** At the regional level, the Regional KM Specialist will seek to build synergies between the Regional Support Programme and the GEF/UNEP regional project "Harnessing the Great Green Wall Initiative for a Sustainable and Resilient Sahel" (and the future GEF8 programme when defined) through ongoing dialogue to ensure the complementarity of activities and the mobilization of financing. At the country level, coordination committees will be established by building on relevant existing committees and bringing together NDAs, GGW, GEF and AF focal points, relevant sectoral ministries and AEs. The coordination committees will aim to streamline the development of GCF Country Work Plans, GEF STAR Allocations and the countries' NDCs and contribute to the design of impactful climate-related projects by sharing work programmes, information, knowledge and updates on portfolios at both the design and implementation phases. The Country Knowledge Management Specialists will coordinate the meetings and facilitate the activities of the Coordination Committees.

Output 3.2 Innovation and digital transformation technologies mapped and ecosystem built

159. In this proposal, innovation is defined as finding and applying new approaches to address existing problems or serve unmet needs. From a development perspective, an innovation is a new solution with the transformative ability to accelerate impact. Innovation can be fuelled by science and technology, entail improved ways of working with new and diverse partners or involve new social and business models or policy, creative financing mechanisms or path-breaking improvements in delivering essential services and products. Innovation has been and will be pivotal for reaching sustained, scalable solutions to the world's most complex problems, including climate change¹⁵¹.
160. Adopted in 2015, the Paris Agreement stipulates that accelerating, encouraging and enabling innovation is critical for an effective, long-term global response to climate change and promoting economic growth and sustainable development. The growing innovations associated with digital technologies create new opportunities for information-sharing and alternative forms of connectivity, financing models and governance arrangements¹⁵². According to UN and other experts, artificial intelligence and digital technology could bring a 10-20 per cent reduction in global carbon dioxide emissions by 2030. Disruptive climate solutions for technology, policy, finance, business models, leadership and capacity-building are key to accelerating climate resilient, low emission resilient development¹⁵³.
161. Innovation is closely linked to digital transformation, as they both break down silos and create transparent and unified data for objective decision-making. Information and communication technologies offer solutions to monitor, mitigate and adapt to the impacts of climate change, which range from using remote sensing observation technologies to track deforestation to developing smart grids to accelerate the energy transition and strengthen early warning systems against the rising number of extreme weather events. In the context of the GGW, innovation and digital solutions can contribute to the transformation of practices and bring new services at scale to sustainably increase food productivity, build resilience to climate change and protect natural resources. Examples of emerging innovations include:

¹⁵⁰ These two types of workshops at the country level have proven to respond to the needs for training on specific issues while also supporting country level projects to understand the role of knowledge management and its contribution to M&E and policymaking (GEF Secretariat, 2021).

¹⁵¹ This definition was published in the call for innovation in international development by USAID, Gates Foundation, DFID, Canada, Sweden, UNICEF, Australia, Rockefeller Foundation, Results for Development and the World Bank Group: <https://www.usaid.gov/ffdc3/innovation>

¹⁵² Bennett and Segerberg (2012); Karpouzoglou et al. (2016); Soma et al. (2016).

¹⁵³ UN Climate Change News (2021), UN Climate Change Boosts Innovation for Climate Action, 24 March, <https://unfccc.int/news/un-climate-change-boosts-innovation-for-climate-action>

- **Social enterprise innovations** that bridge service delivery gaps: For instance, the firm Hello Tractor¹⁵⁴ in Nigeria connects producers with providers of agricultural services (e.g., tractor owners) using Internet of Things (IoT) platforms similar to the one Uber uses to connect drivers with passengers. These solutions not only provide farmers with access to machinery, but also connect them to an ecosystem of partners who use the tractor's data to provide additional services such as financing and crop-specific weather information.
- **Climate technologies** such as early warning systems that provide farmers information and advice on when to plant or sell their livestock. The FAO Global Information and Early Warning System provides countries tools to monitor crops and food prices and produce analyses and briefs in real time.
- **Digital finance services** that can give smallholder farmers who do not have access to traditional financial services access to loans. In Kenya, Apollo¹⁵⁵ provides optimized bundled services for farmers that include financing, farming inputs, advisory services and insurance. Apollo's vertically integrated digital approach leverages machine learning and satellite data to service customers throughout the entire lifecycle, from acquisition to decisions on credit and finally, loan payments.
- **Open data and geo-mapping** for improving nature resource planning and monitoring in countries. Tools such as Open Foris, SEPAL¹⁵⁶ and trends.earth¹⁵⁷ provide digital technology and satellite imagery to support countries in assessing, monitoring and planning land use, including the monitoring of deforestation and desertification. The technology offers wide coverage, is faster than ground methods and facilitates long-term monitoring of land coverage and usage.
- **Clean technologies** (e.g., solar, wind) that generate sustainable energy. The World Bank Lighting Africa programme¹⁵⁸ provides electricity through off-grid solar products by working with local distributorships in sub-Saharan Africa.
- **Financial innovations** (e.g., crowdfunding, social impact bonds, mobile money). FSD Africa has developed an innovative approach to financing mechanisms that drive sustainability in conservation areas in Kenya¹⁵⁹.
- **Innovative adaptation approaches** that provide climate smart services to villages. CGIAR climate-smart villages¹⁶⁰ bring a holistic approach to village development by combining agroforestry, precision application of fertilizers and energy-efficient machinery with other climate-smart services, such as tailored weather forecasts to help farmer's better plan planting, harvesting and other activities on the farm. These services are delivered through phones which farmers can also use to buy index-based insurance that gives them some protection in the event of extreme weather.

162. In the context of the GGW RSP and IGREENFIN I, specific attention will be given to support accredited entities in leveraging opportunities to integrate precision farming technologies in projects in view of the potential they have to boost value chain resilience and contribute to landscape restoration. Precision farming technologies can be defined as a whole-farm management approach that uses information technology, satellite positioning data, remote sensing and proximal data collection for more efficient use of certain inputs and water resources. In the GGW, these technologies will be introduced with the goal of optimizing input usage and reducing environmental impacts, including GHG emissions¹⁶¹. Broadly speaking, there are three main components to technology applications: (i) hardware and sensors (which are involved in using or collecting data); (ii) data analysis and decision support systems (the software and/or cloud systems and machine learning algorithms which gather and process data to turn it into something meaningful), and (iii) commodity and whole-farm focus (the process whereby decision support systems and models are made commercially and utilized across whole farms, and not just individual fields). Alongside the economic benefits of precision farming, there is also much discussion on the vital role these technologies play in building a climate resilient and sustainable agricultural sector. Several studies note that precision farming enables farmers to maintain or increase productivity/yields while reducing overall inputs (including fuel, feed, nutrients or water, etc). As such, they should contribute to the reduction in GHG emissions directly or indirectly. While there is some direct evidence of emission reductions, to date, few studies have been performed on precision farming technologies to determine their direct contribution to GHG emissions mitigation¹⁶².

163. Recent decades have seen considerable improvements in the performance and diffusion of several flagship adaptation and low carbon innovations. Despite the numerous success stories and encouraging trends, overall progress towards innovative solutions has been modest in the GGW. Additional support needs to be provided for: (i) the development or strengthening of innovation systems and improved understanding of the specific needs of the countries where transformational innovation

¹⁵⁴ For more information, consult the company's website: <http://www.hellotractor.com/>

¹⁵⁵ <https://www.apolloagriculture.com>

¹⁵⁶ <https://www.fao.org/national-forest-monitoring/tools/en/>

¹⁵⁷ <https://trends.earth/docs/en/>

¹⁵⁸ <https://www.lightingafrica.org/about/our-impact/>

¹⁵⁹ <https://www.fsdafrica.org/programme/world-wildlife-fund-social-impact-bonds/>

¹⁶⁰ <https://ccafs.cgiar.org/climate-smart-villages>

¹⁶¹ [https://www.europarl.europa.eu/RegData/etudes/note/join/2014/529049/IPOL-AGRI_NT\(2014\)529049_EN.pdf](https://www.europarl.europa.eu/RegData/etudes/note/join/2014/529049/IPOL-AGRI_NT(2014)529049_EN.pdf)

¹⁶² <https://businesswales.gov.wales/farmingconnect/news-and-events/technical-articles/can-precision-farming-help-mitigate-climate-change>

efforts are to be based; (ii) piloting, testing or expanding technologies in new environments and adapting climate technologies to local contexts; (iii) the establishment of coordination mechanisms to effectively identify and prioritize climate technology solutions; (iv) feasibility assessments of selected climate technologies for mitigation and adaptation, and (v) market preparation, business planning and financing for the deployment and scaling-up of climate technology solutions¹⁶³. Output 3.2 focuses on digital innovations and precision farming and will be directly linked to the 10 selected value chains described in Table 16 above. During the programme's inception and implementation, linkages will be created with regional and national public and private research and development institutions in the relevant sectors (including the CILSS, OSS and others), and other innovation platforms such as the UNFCCC Global Innovation Hub Launched for Transformative Climate Solutions¹⁶⁴, launched during COP26. When providing TA under Component 2 and green credit lines under Component 1, IFAD will ensure that the innovations generated and systematized under Output 3.2 will be taken into account and replicated whenever appropriate.

164. Key activities for Output 3.2 include:

- **Activity 3.2.1 Build a digital and innovation ecosystem.** A digital and innovation ecosystem is the evolving set of actors, activities, artifacts, institutions and relations (including complementary and substitute relations) that are important for the adoption of digital technologies that facilitate multi-actor interaction and contribute to the development of innovative climate adaptability solutions that can be deployed at scale. Such an ecosystem fosters economic stability and promotes resource sharing for the development of innovation and technologies and the creation of new market opportunities. Considering the challenges in the GGW, the RSP will act as a broker of the digital ecosystem by promoting collaboration among different actors, including government(s), towards the achievement of key objectives of the GGW. It will do so by:
 - **Mapping existing services and digital platforms that provide extension services, financial services, farming precision technologies and digital technologies that help monitor and plan the use of natural resources, among other services.** This mapping will include key players in the region and existing platforms and services, including those promoted or implemented by GCF and GEF projects, as well as technologies and platforms developed by MSMEs, entrepreneurs, academia and other players in the region. This mapping will be done at the time of the programme's inception to allow the RSP to define the best strategy for developing and hosting the platform in the GGW context, which is evolving very rapidly since the One Planet Summit of January 2021. The mapping will also include recommendations for opportunities to leverage these technologies across the pillars of the GGW Accelerator and the promising solutions that can be scaled up – for instance, through GCF support for climate technology incubators and accelerators¹⁶⁵.
 - **Providing technical support** for brokering the relationship between solutions providers participating in the marketplace and GCF, GEF and other partners at country level.
 - **Facilitating a series of online events** to showcase specific innovations and share expertise across the region, which will be accessible to GCF, GEF and other partners at the country and regional level.
 - **Organizing annual fairs** based on the model of SahellInnov Expo¹⁶⁶ where climate technologies in specific areas (e.g. farm precision technologies, digital solutions for monitoring and planning restoration, access to financing, etc.) in the GGW region and beyond will be showcased and sessions will be organized to discuss and address specific climate challenges. The fairs' events will facilitate the interaction of different skills, resources, knowledge or needs by matching interests or creating standardized technological interfaces and communication forums. The events will be complementary to the marketplace by providing opportunities for real-time demonstrations and networking.
- **Activity 3.2.2 Create a digital marketplace with all services provided by platforms and other providers.** Based on the mapping from activity 3.2.1, an online marketplace will be created to bring together financial and non-financial institutions, civil society organizations, AgTech and FinTech agencies, value chain players, MSMEs that provide services for value chain development, climate solutions and landscape restoration as a one-stop-shop open to everyone across the region. In many productive sectors of the economy, platforms have already become a preferred method of collaboration and value co-creation¹⁶⁷. The marketplace will enhance accessibility, promote economies of scale and foster new market opportunities. The marketplace will feature platforms and services providers that provide extension

¹⁶³ https://www.greenclimate.fund/sites/default/files/document/gcf-brief-support-technology_0.pdf

¹⁶⁴ <https://unfccc.int/news/cop26-global-innovation-hub-launched-for-transformative-climate-solutions>

¹⁶⁵ https://www.greenclimate.fund/sites/default/files/document/gcf-brief-support-technology_0.pdf

¹⁶⁶ <https://blogs.worldbank.org/psd/start-scratch-how-entrepreneurship-can-generate-sustainable-development-and-inclusion-sahel>

¹⁶⁷ Smedlund, Lindblom, and Mitronen, 2018

services, financial services, farming precision technologies, digital technologies, organized by area of service or expertise and country and will provide a feature to directly enter in contact.

- **Activity 3.2.3 Identify new opportunities for investing in climate innovation in the GGW.** While some innovations are already being implemented in other regions (e.g. carbon financing, PES schemes, etc.) or on a small scale, their potential for being upscaled in the GGW still needs to be assessed. A series of knowledge products and feasibility studies for specific technologies will be conducted to identify new opportunities for implementing innovative solutions in the region with the support of future GCF investments in the GGW, based on the results of Activity 3.2.1.

Output 3.3 GGW RSP Support Unit and governance structure established:

- **Activity 3.3.1 Establish Support Unit** to support the implementation of all activities under Component 3
- **Activity 3.3.2 Ensure governance of RSP is operational** with the recruited staff
- **Activity 3.3.3 Establish the M&E System** to assess the progress against the targets & objectives
- **Activity 3.3.4 Conduct impact evaluations** that can be attributed to the interventions of the program

B.4. Implementation arrangements (max. 1500 words, approximately 3 pages plus diagrams)

Organization and staffing of IGREENFIN I

IFAD Regional Coordination Units (RCU):

165. In its capacity as AE for IGREENFIN I, IFAD will create a Regional Coordination Unit (RCU) which will oversee, compile, consolidate and control the work of the EEs. The EEs will implement the programme through IFAD baseline investment PMUs (grant proceeds) and the LNABs' PIUs. The RCU finance team will be established and located in IFAD's West and Central Africa Hub based in Dakar, Senegal. The RCU's financial management unit will be composed of one employee: the Finance Officer. IFAD PMU staff will undergo an annual performance evaluation in line with IFAD performance assessment policy and procedures. Refer to Section 1 of Annex 22 for the duties and responsibilities for each AE and EE.
166. **National Executing Entities (NEEs):** The NEEs for IGREENFIN I are the Republic of Burkina Faso, the Republic of Côte d'Ivoire, the Republic of Ghana, the Republic of Mali and the Republic of Senegal, represented by their respective ministries of economy and finance. The AE will inform each NEE of the relevant obligations and requirements established in the Accreditation Master Agreement (AMA) and Funded Activity Agreement (FAA). The NEEs will be recipients of the GCF financing and guarantee the repayment of the GCF loan portion. The NEEs have a proven track record in similar projects and have been working with IFAD and other financial institutions on rural finance.
167. The selected LNABs responsible for the implementation of green credit lines with the GCF loan proceeds (Component 1) are: the Agricultural Bank of Burkina Faso, the National Investment Bank of Côte d'Ivoire, the ARB Apex Bank Limited of Ghana, the Agricultural Bank of Mali and the Agricultural Bank of Senegal. The LNABs shall submit all applications for financing under the green credit lines (windows 1 and 2), together with their preliminary assessment and recommendations based on independent expert opinion, to the NEEs. LNABs shall not be entitled or authorized to give the final clearance for a loan or sub-grant application; this decision will be made exclusively by the NEEs. Accordingly, the NEEs will verify whether the application meets the eligibility criteria and decide whether to endorse the LNABs' recommendations, but the final approval or rejection of a subproject lies with the NEE. For the implementation of the GCF grant proceeds (Component 2), the NEEs will make arrangement with the Ministries of Agriculture of the respective countries who will act through IFAD PMUs. The executing entities of the GCF grant proceeds for IGREENFIN I are: the Ministry of Agriculture, Hydro-Agricultural Development and Mechanization of Burkina Faso; the Ministry of Agriculture and Rural Development of Côte d'Ivoire; the Ministry of Agriculture of Mali; the Ministry of Agriculture of Ghana, and the Ministry of Agriculture and Rural Equipment of Senegal. IFAD has conducted a capacity assessment of the LNABs and the Ministries of Agriculture of the five targeted countries.
168. **Regional Steering Committee (RSC):** Given the geographical and thematic scope of the different components, a joint inter-ministerial decision will establish the Regional Steering Committee (RSC) and specify its composition, mandate and functioning. The nomination of members to the RSC will be done by each NEE in compliance with the GCF Gender Policy. The RSC will provide oversight, direction and guidance for the programme's implementation and approve the programme's AWPBs and its periodic progress reports. It will designate the key stakeholders that are to be part of the National Steering Committees. These includes the NEEs, their respective ministries of agriculture and forestry and representatives of MFIs, FOs, cooperatives and MSMEs, relevant regional institutions, private sector entities and civil society organizations (CSO). Any recommendations or measures adopted by the RSC are subject to final approval by the NEE, represented by its ministry

of economy and finance. IFAD PMUs will be supported by the IFAD Country Office, which will be able to provide or call upon expertise in institutional development, if necessary.

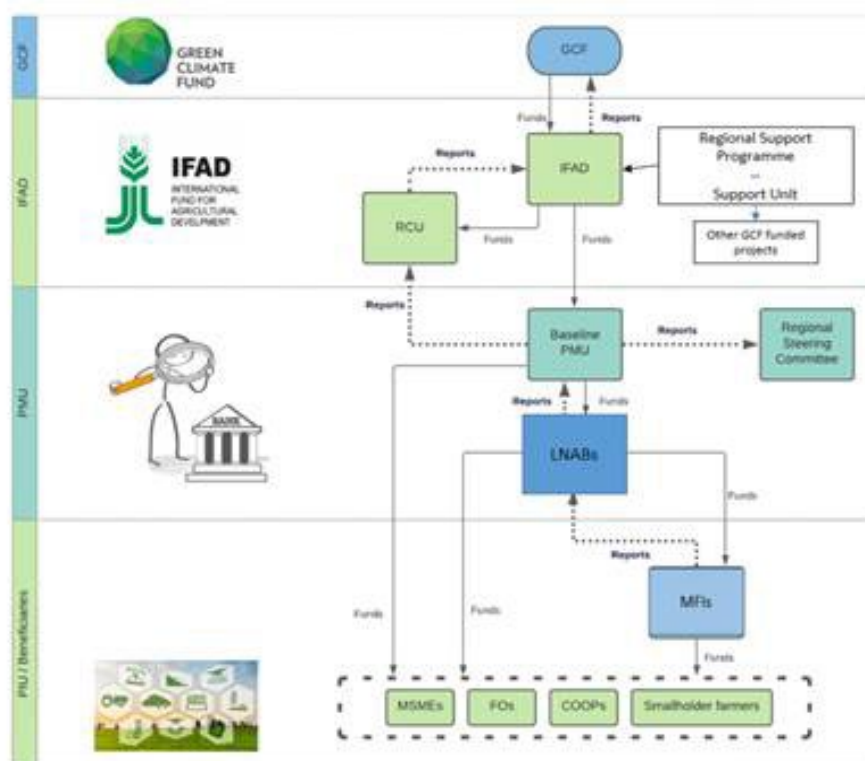
IFAD Baseline Project Management Units (PMUs):

169. In each of the selected countries, the grant proceeds will be anchored within each IFAD baseline PMU. Each baseline PMU will budget for two additional headcounts to manage IGREENFIN I. Baseline PMU staff working on IGREENFIN I will undergo an annual performance evaluation in line with baseline PMU's performance assessment policy and procedures.

LNABs Project Investment Units (PIUs):

170. In each country, the selected LNABs will implement the IGREENFIN I loan proceeds. Each LNAB will set up a dedicated Project Investment Unit (PIU) whose primary role will be to receive the GCF proceeds. The funds will be disbursed to the LNAB through the IFAD baseline PMUs. The LNABs will on-lend to beneficiaries or other IFIs. Financial management activities will be managed using the LNAB's existing financial human resources and systems in place. Each LNAB will also name a focal point within its structure that will be responsible for all financial management and accounting issues related to the management of GCF funds within the bank. The PIU will work with rural finance institutions (RFIs) to gain better access to target beneficiaries. It is the NEEs' responsibility to determine how they will work with RFIs. The PIU will also be responsible for conducting a due diligence review of the partner institutions that will be involved in the programme and ensure that mitigating actions are put in place to deal with the key risks identified and ensure that they can respond to all operational and reporting requirements. PIUs will be required to obtain a non-objection letter from IFAD regarding their choice of partner institutions.

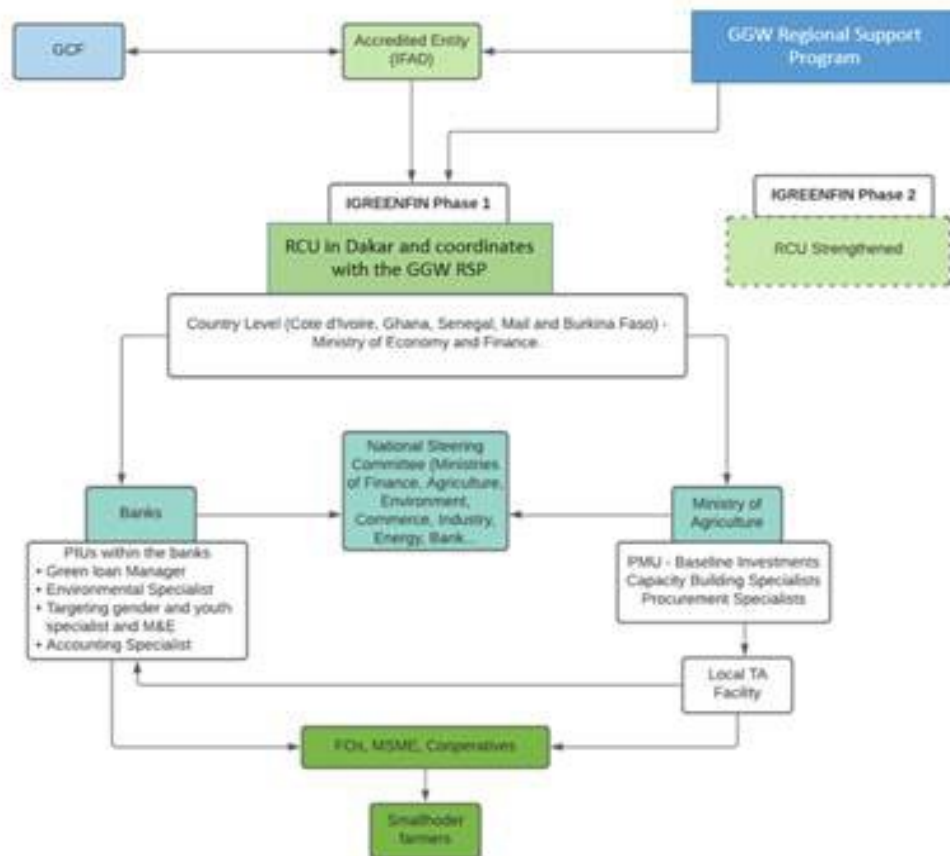
Figure 23: IGREENFIN I Organizational Structure



Disbursement arrangements and flow of funds

171. Refer to the flow chart below for the proposed flow of funds

Figure 24: IGREENFIN I flow of funds



IFAD RCU:

172. The IFAD RCU will carry out an ex-post review of a risk-based sample of Withdrawal Applications (WAs) As but will not be directly involved when Baseline PMUs make WA or Direct Payment requests to IFAD. The ex-post review is to support improvements in the quality of financial management. Any discrepancies noted will be communicated to IFAD and the baseline PMU and recommended actions included in the assurance review tracker for monitoring of implementation.

IFAD Baseline PMU in the flow of funds:

173. Baseline PMUs will open two new bank accounts to use exclusively to manage IGREENFIN I proceeds, namely a Grant Designated Account and a Loan Designated Account. The Grant Designated Account will be used for payment of project operating costs relating to IGREENFIN I incurred by the baseline project and for the replenishment of Bank PIU overhead costs. IGREENFIN I loan proceeds will be deposited in the baseline project's Loan Designated Account and transferred to the bank against appropriate reporting using the revolving fund mechanism.

PIU within the LNABs:

174. Disbursement of IGREENFIN I loan proceeds to the LNABs will be based on a revolving funds mechanism. The LNABs will open two IGREENFIN I operating accounts within the bank; one account for the IGREENFIN I loan proceeds and the other account for the IGREENFIN I grant proceeds. The amount of the IGREENFIN I proceeds allocated to each LNABs for the entire life of the programme will be stipulated in the IFAD Financing Agreement. The LNABs PIU will receive an initial deposit amounting to the budget for the first six (6) months of operations for both the loan and grant proceeds, as stipulated in the AWPB.

175. The LNAB's PIU will then transmit further disbursement requests to the Baseline PMU for approval and disbursements on a quarterly basis, justifying the expenditures related to the previous tranches of payment made in the grant and loan proceeds accounts. Apart from the initial deposit, subsequent disbursements will be preceded by the submission of Loan Disbursement/ Repayment Report and supporting documents, which will serve as justification for the previous instalments.

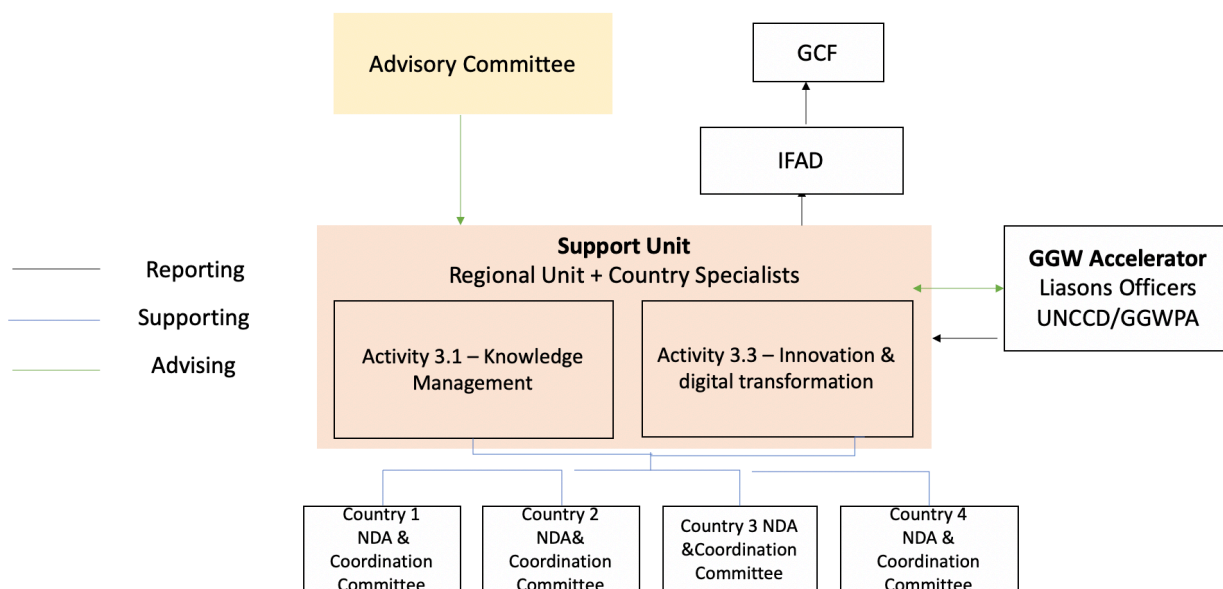
INTERNAL CONTROLS

176. All stakeholders involved in IGREENFIN I must pass the KYBP test, which consists of filling the KYBP Form, providing the list of documents on the KYBP checklist and having these reviewed and signed off by the Finance Manager / Controller of the EEs prior to commencement of business with the business partner in question.

External Audit:

177. IFAD RCU will be in charge of reviewing each of the five baseline PMU's annual audit reports and preparing a consolidated tracker of all IGREENFIN I assurance review recommendations. It will ensure that IFAD non-objection is granted each year on the external auditor's terms of reference by verifying that GCF requirements regarding external audit are fully captured in the EE external auditors TORs. The IFAD RCU will ensure a timely transmission of each country's external audit report to GCF.
178. Each LNAB and the respective Ministry of Agriculture will each undergo an annual external audit as mandated by the IFAD Financing Agreement and will be in charge of the planning, preparation and coordination of its annual external audit. Each external auditor's ToR will include a standalone audit of IGREENFIN I, which will provide a specific opinion on IGREENFIN I, and as part of the annual external audit, a separate disclosure on IGREENFIN I within the final accounts, the notes to the accounts and management letter. Such ToRs will be submitted to IFAD for non-objection at least one month prior to the commencement of the audit. The external auditors of the Baseline PMU will rely on the work of the LNAB's external auditor to give an audit opinion on the use of IGREENFIN I funds managed by the LNAB and on the related financial statements.
179. At the regional level, an external audit will be undertaken on the RCU of IGREENFIN I and the Regional Support Programme's activities annually.
180. The management and oversight structure proposed below intends to take into account factors that ensure an efficient structure while still involving stakeholders with deep knowledge and experience in the GGW and national and regional actors.

Figure 25: GGW Regional Support Programme governance



181. **Advisory Committee:** provides overall advice on the Regional Support Programme components. The Advisory Committee will meet twice a year to review and make recommendations on: knowledge management and innovation, GGW RSP direction and achievements through GCF projects, linkages with relevant initiatives and processes such as the GGW Accelerator. Foreseen composition: IFAD, UNCCD, GEF, African Union, the PAAGGW, two rotating GGW countries representing the different geographical regions (East and West Africa) (e.g., on a yearly basis).

182. **Support Unit:** The Support Unit will be responsible for implementing activities under Component 3 (RSP). It will be composed of a regional unit based in Dakar, Senegal and of one country-level specialist per country who will support the NDAs. It will be composed of the following regional and country-level specialists:
- Program Coordinator: overall coordination of Component 3
 - Knowledge Management Specialist: responsible for Output 3.1
 - Innovation Specialist: responsible for Output 3.2
 - Finance Specialist
 - Administrative Assistant
- Country level specialists: For each country, IFAD will hire a Knowledge Management Specialist to support the NDAs and GGW Focal Point with the implementation of the Regional Support Programme's activities. Country specialists will report to the Regional Support Unit's KM Specialist and will share their time between the NDAs office and the IFAD office to coordination of activities. Country KM Specialists will be supported by the regional knowledge management and policy specialist. They will have a coordinating role for knowledge management, communication and policy in support of the NDAs.
183. **GGW RSP Liaison Officers:** The GGW Accelerator will host a GGW Regional Support Programme liaison officer to ensure close alignment and collaboration with the GGW Accelerator. These liaison functions will include:
- a. Ensure the collation of results and impacts by aligning the RSP's knowledge management and innovation activities with those of the GGW Accelerator
 - b. Coordinating the reporting of GGW RSP activities and progress to the GGW Accelerator to increase the collective impact of the RSP and Accelerator projects and programmes
 - c. Ensure alignment and regular feedback between the RSP knowledge hub and the Accelerator GGW knowledge hub on best practices and lessons learned
 - d. Liaise with GGW structures, processes and institutions (PAAGGW, AU, and GGW Steering Committee) to ensure they are well informed of the progress of the RSP.
184. A second liaison officer will be co-funded with the UNCCD and hosted by the Great Green Wall Pan-African Agency to ensure coordination with the regional entity's activities.
185. **Country Coordination Committees** will be composed of NDAs, GGW Focal Points, GEF Focal Points, Adaptation Fund Focal Points, UNCCD, CBD and UNFCCC Focal Points, relevant sectoral ministries and AEs. Building on relevant platforms wherever possible, committees are responsible for coordinating the alignment of GCF work programmes, GEF STAR allocations and prioritizing GGW investments at the country level. The Country Coordination Committees will also be responsible for the coordination of the activities reported to the GGW Accelerator to ensure transparent and comprehensive exchange of information and reporting. Countries may decide to invite additional stakeholders to take part in the Country Coordination Committee.

Technical and Financial Reporting

186. The Support Unit will be responsible for preparing the following reports for submission to the IGREENFIN PMU Coordinator:
- M&E indicators
 - Lessons learned, best practices
 - Financial report
 - Annual work plans
 - Annual progress reports
 - Bi-annual progress updates
187. The progress reports should present the main achievements, issues and constraints of the reporting period, information on financial and physical achievements in comparison with targets set in the annual work plans as well as possible impacts and outreach. The reports should highlight the implementation strategy and indicate challenges and the reasons for them. Specific reference should be made to recommendations by supervision missions.

Technical and Financial Reporting

188. The Executing Entities will provide the following reporting outputs to the Coordinator of the Support Unit for review and approval:

- M&E indicators
- Lessons learned, best practices
- Financial report
- Annual work plans
- Annual progress reports
- Bi-annual progress updates

189. The progress reports should present the main achievements, issues and constraints of the reporting period, information on financial and physical achievements in comparison with targets set in the annual work plans, as well as possible impacts and outreach. The reports should highlight the implementation strategy and indicate challenges and the reasons for them. Specific reference should be made to recommendations by supervision missions.

190. The Support Unit will report to IFAD and will submit the following reporting outputs for review and approval:

- Annual work plans
- Annual progress reports
- Annual financial reports

191. **Loans repayment:** Beneficiaries (FOs, cooperatives, MSMEs, commercial banks, MFIs) will repay their loans to the selected national agricultural banks (Agricultural Bank of Burkina Faso, the National Investment Bank of Côte d'Ivoire, the ARB Apex Bank Limited of Ghana, the Agricultural Development Bank of Mali and the Agricultural Bank of Senegal) back to the EEs (The Republic of Burkina Faso, Côte d'Ivoire, Ghana, Mali and Senegal, represented by their respective ministries of economy and finance, the only National Executing Entities). Each Host Country shall repay the loans that they will receive from IFAD (as Accredited Entity of the GCF), irrespective of the performance of the on-lending to the respective LNABs, and therefore IFAD will only assume the credit risk of the sovereign borrowers

192. **Grant proceeds** will be used to support the financial system in each country in: (i) complying with regulations and overcoming barriers to attracting investments; (ii) enhancing coordination at the financial industry level, and (iii) financing and offering advisory services to financial institutions and beneficiaries. The Agricultural Bank of Burkina Faso, the National Investment Bank of Côte d'Ivoire, the ARB Apex Bank Limited of Ghana, the Agricultural Development Bank of Mali and the Agricultural Bank of Senegal will disburse GCF and co-financing loans to beneficiaries (FOs, cooperatives, MSMEs, etc.) in a manner that is consistent with the mandate of the Programme. The IFAD baseline PMUs will provide technical assistance and capacity-building through selected consultants and service providers to help LNABs conduct marketing activities, develop internal procedures to evaluate eligible technologies and/or projects, check compliance with the mandate and put in place monitoring systems. IFAD will report to the GCF based on the conditions established in the FAA and AMA. In addition, contractual obligations will be established for both the LNABs and beneficiaries (FOs, cooperatives, MSMEs,). The AEs will contractually ensure that the conditions required by the GCF are transferred through the Credit Line Agreements to end users. No business plans and subprojects may be financed with GCF resources if they do not meet the GCF's conditions stipulated in the FAA.

B.5. Justification for GCF funding request (max. 1000 words, approximately 2 pages)

193. In the GGW region where markets are still small; the cost of capital, high; credit terms, unfavourable, and the upfront costs of adopting adaptation and mitigation measures are high, GCF financing can help to attract the private sector to participate in green business investment and support the achievement of the GGW's objectives by 2030. The main barriers hindering investment in green business projects in the GGW region and Côte d'Ivoire and Ghana are limited local financial market capacity and appetite of international institutional investors for green business initiatives. This is mainly due to the perception of green technologies/business models along the targeted agricultural value chains as high-risk investments, but also concerns with the macroeconomic environment, limited credit worthiness of smallholder farmers, country risks, policy support and limited liquidity of national agricultural banks. To close the financing gap, countries need public concessional finance that will leverage additional private sector investment to create a viable green business market in the region. Covid-19 has further impacted investment activity in the five selected countries, as institutional investors - particularly those interested in smallholder agriculture - become all the more risk averse. The West Africa region's growth prospects were much less favourable in the second and third quarters of 2020, which are expected to lead to negative annual growth. According to AfDB, West African countries were expected to enter recession in 2020, with a 3.6 per cent contraction of the regional economy. In 2020, forecasts predicted a sharp deterioration in the current account deficit for ECOWAS as a whole, which was expected to stand at 4.3 per cent, against 2 per cent in 2019. Forecasts suggested on average an

increase in ECOWAS public debt as a proportion of GDP, which would stand at 41 per cent in 2020 and 42 per cent in 2021, against 35 per cent in 2019.

194. Therefore, IFAD has approached the GCF to seek the required “capital protection” to incentivize and de-risk public and private institutional investment and create local and regional green business markets which will attract private sector investments. The public sector in the targeted countries can play an important role in green business but limited public finances and fragility due to the impacts of COVID-19 impede the countries’ ability to promote climate resilient, low emission agriculture. Because of COVID-19, these economies are predicted to go into recession and therefore, all development gains are under threat if concessional financing is not mobilized to address the needs of the most vulnerable communities. As the countries’ fiscal space has been greatly narrowed and the financial resources that they do have are being directed toward the health crisis, more funding is needed to prevent a widespread food crisis from emerging under a changing climate. To attract private investment, the GGW countries, Côte d’Ivoire and Ghana are leveraging highly concessional climate finance from the GCF to provide incentives so that private funds flow more easily to green businesses to scale up and accelerate the achievement of the GGW objectives.
195. For the targeted beneficiaries of this programme, which are the smallholder farmers organized in cooperatives, MSMEs and FOs defined as the left behind and excluded from the financial sector, IFAD concessional loans and grants require additional concessional climate finance to attract private funds to address market failures and support the development of nascent green business market. In the current context, without concessional funds from the GCF, the prospects of attracting public and private sector investment for green business projects in the GGW remain limited.
196. The justification of the use of GCF concessional funds is based on the fact most sub-Saharan African countries have some of the lowest capacity to mobilize financing to address climate impacts and shocks in local markets. When resources are available, lending terms are ill adapted: interest rates are high (as seen in tables 24 and 25) and repayment periods are short. This limits investment in climate resilient, low emission agriculture. GCF concessional finance will de-risk investments, cover the incremental costs for green businesses, and incentivize MSMEs, FOs, cooperatives and local public investors such as the LNABs to engage in or finance green businesses.
197. The GCF Grant instrument under Component 2 will support the LNABs in transitioning towards green banking (green portfolio, governance and policies) and attracting new market opportunities. IGREENFIN I will help the selected banks build their capacity and access to knowledge and skills to mobilize additional concessional finance from development partners and the private sector beyond the programme’s life cycle to respond to the demand of a new promising market. All LNABs are committed to embrace green financing. GCF grant resources will be used to support them to kick off their first green business pipelines, improve their environmental and climate governance procedures and with co-financing from IGREENFIN I, align part of their investment portfolios with their NDCs and the Paris Climate Agreement. IGREENFIN I will support policy dialogue on green finance in agriculture with central banks and other local commercial and MFIs. More specifically, the added value of the GCF funding will be to help operationalize these countries’ commitments under the Paris Agreement and through their NDCs, as well as the involvement of private actors. The grant proceeds will also support capacity-building for the demand side (FOs, MSMEs, cooperatives) to build their capacity to develop bankable green businesses on high return on investment value chains
198. Due to the target countries’ huge solar potential, the programme will work closely with cooperatives, FOs and private sector energy services suppliers to produce electricity for powering agricultural activities. Modalities to build, operate and transfer (BOT) will be promoted, whenever possible, to ensure the sustainability of the energy infrastructures.
199. Regarding the loan proceeds: The reduced interest rates triggered by zero percent GCF concessional resources will provide MSMEs, FOs, youth- and women-led cooperatives and final beneficiaries the security they need to invest in and try innovative practices. It is only with zero percent interest rate that public banks would be able to define an exit blended interest rate that is attractive to FOs, MSMEs and cooperatives. The investments modelled in the 10 selected value chains would result in an IRR presented in Annex 2 EFA for each crop with longer repayment periods. The GCF loan at zero per cent interest rate will reduce the cost of finance and increase its tenor making the amortization profile more suitable to any potential green businesses. De-risking support can catalyse positive lending decisions by enhancing the collateral profile and/or reducing expected losses for financiers and, in turn, improving profitability. With this programme, IFAD is nurturing a potential brand on green finance, which will contribute to greening sub-Saharan agricultural development banks and financial sectors.
200. This programme will also benefit from the wider Country Support Programme to be coordinated by IFAD. This will result in closer collaboration and coordination of each country project, thus facilitating the identification of complimentary actions and opportunities to scale up adaptation and mitigation co-benefits across the GGW countries. These synergies will result in positive feedback loops that will enhance the programme’s impact given the common development challenges related to green finance and rural development. The future GGW Regional Support Programme will also provide a unique platform for

strengthened and targeted knowledge management to share valuable lessons learned across the region and foster more robust implementation and management outcomes, further enhancing the impact of GCF funding.

201. The GCF and IFAD are piloting an innovative approach aimed at enhancing the collective impact of all GCF projects and programmes implemented in the GGW countries and maximizing their contributions to the GGW objectives. This approach consists mainly of financing activities related to KM and to GIS, which are not expected to generate return on investments, and are typically financed through grants.
202. The implementation of all mini grids will be anchored on existing frameworks. Implementation arrangements are provided in Table 22 and will take into account the specifics of each country.

B.6. Exit strategy and sustainability (max. 500 words, approximately 1 page)

203. When it comes to market investment particularity in sub-Saharan Africa, investor confidence is very important. The Green Business Financing Facility and the TAF will help generate that confidence during the programme lifecycle. With the number of green businesses to be developed and capacity built both from the supply (financial sector) and demand side (FOs, MSMEs and cooperatives); a green agricultural market will be built on the most suitable value chains which will help, in turn, to crowd in more private investment.
204. IGREENFIN I will establish the TAF to build the capacity of the LNABs and the central banks to develop feasibility studies, policy documents and knowledge to address all challenges hindering the uptake of green business projects in the selected countries. Through the work on central bank-related activities such institutional and regulatory framework support to develop climate risk management practices at the programme level (standards), develop and harmonize green lending products, instruments and tools such as ESG benchmarks; strategic studies and knowledge on green finance within the financial system and climate investments from both private sector and government in the agricultural sector and mechanisms for sharing best practices, the programme will set the pathway for sustainability in the region. IGREENFIN I will support the establishment of a platform for green development banks/central banks regulators in the region to advance the policy dialogue and coordination with governments and partners on green finance for the SDGs; the NDCs and the Paris Climate Agreements, which can be sustained of supported by other partners after the programme ends. IGREENFIN I is building a network of banks with the highest standards of ESG and setting up an open-source platform (ESMF underwriting procedures), which eventually will help attract and guide other prospective players in the market. The TAF will support a better alignment of policy and regularly frameworks with the goal of embedding green financing in the agriculture sector plans and all climate investments. The programme will engage in all strategic discussions and policy work at the country level.
205. In addition to the mechanisms described above, policy dialogue between the public and private sector, including producers, will contribute to the development of a regulatory framework to sustain programme interventions beyond its duration. Further, throughout the programme, support from the TAF will help banks increase their capacity readiness so they are better able to obtain accreditation with the green funds (Adaptation Fund, Global Environment Fund), thus aiding countries to become champions of climate finance in the region and make the transition towards a green economy.
206. The establishment of the green financing windows with the loan proceeds in each country will provide incentives for the financial sector. The adoption a regulatory framework on interest rates and extension services will also help raise awareness on the extent and significance of climate risks and support better access to concessional finance. The banking systems' sustainability strategy will ensure that farmers can access financing from local, regional and national markets at affordable credit levels. Building on the impact of reduced interest rates and increased lending, the institutionalization of the Sahel Award will further raise the public profile of the RETs market and climate resilient, low emission agricultural techniques in the region. Successful green business projects will attract mainstream financing to the sector. It is expected that a critical mass of expertise and skills will develop to the point where they are able to catalyse investments from the private sector (national banking associations, insurance companies, etc.). Impact assessments will be conducted in each country to inform the exit strategy throughout the life of the investment and monitor the market to optimize the exit. Most renewable and resource efficiency (water and wastewater management) green businesses projects have a life span of 25 years and have long-term offtake agreements. Hence, the continuity and impact of the projects even after IGREENFIN I exits and other investors take over.
207. By creating a self-sustaining loan portfolio that ensures institutionalization of green credit lines within the banks in the long term, GCF bank partners (commercial banks, MFIs) are expected to lend at the same interest rate and thus, influence market rates. Subsequently, interest rates are likely to decrease, gradually making banks loans become more attractive once the IGREENFIN I programme has ended, stimulating more loan applications and investment in resilient agriculture and RETs. GCF funds will allow concessional sub-loans to be extended to MSMEs including the RET operators through

the Green Business Financing Facilities operated by the LNABs. Financing from the mitigation window will support public and private partnerships with solar operators under various modalities, including build, operate and transfer (BOT). Through these partnerships, the programme will stimulate the private sector to invest in this strategic sector linked to the modernization of agriculture through clean energy services. Such partnerships will ensure energy infrastructures are maintained beyond the project.

208. Training and capacity-building for secondary FIs (including ones from the private sector) offered in Component 2 will stimulate the creation of a green business market. The creation of a green business market will be supported through: i) training for bank staff, (ii) training for suppliers, vendors and other service providers in the sector on green finance, (iii) outsourcing of development/operating of adaptation and mitigation activities in certain markets/regions to local players in order to expand quickly and cost-effectively, and (iv) partnerships with technical experts from overseas to bring in much needed expertise on digital innovation in agriculture and banking to support the inclusion of youth and women in rural areas using mobile banking, GIS, remote sensing, big data and digital finance.
209. The programme will enable smallholder farmers, MSMEs and FOs currently excluded from the financial market to access to formal market and its services (financial literacy, bank accounts and transactions), which will enable them to improve their production and income and thus compete in local markets. Capacity-building and training for FOs, women and youth organizations, MSME, MFIs and solar company loan recipients on business plan development will ensure that appropriate measures are taken to guarantee maintenance and the continued operation of installed hybridized solar mini grids systems for agricultural value chains beyond the programme's completion.
210. As a future investment pillar of the GCF GGW programmatic approach based on the GCF Integrated Result Management Framework and the GCF sector guidance on agriculture, forests and land use, ecosystems and ecosystem services and water, IGREENFIN I will use the RSP framework to scale up the strategy to sustain the initiative in the region. IGREENFIN I will align its programming and planning on GCF results management with two main objectives: (i) to facilitate more consistent and credible reporting of climate results of GCF-funded projects/programmes, which can be aggregated and reported as part of the interventions of the GGWI, and (ii) allow the GCF to assess the contribution of its investments in the promotion of a paradigm shift towards climate resilient, low emission development pathways and in supporting the implementation of the UNFCCC and Paris Agreement in the context of the GGW initiative.
211. The GGW RSP is a very innovative endeavour which differs from the typical project-based approach of the GCF. This effort is expected to harness the collective strength of all the AEs and entities involved in supporting the GGW objectives. It is also expected that this approach will foster complementarity between the GCF, the GEF and the AF at the country and regional levels, and will enhance private sector engagement. The benefits and potential drawbacks of this approach will be assessed during the implementation of the project, and an impact assessment will be carried over before project closure. It will provide vital information to the GCF, the GEF, the AF, IFAD, the NDAs and the AEs to better assess if and how such an approach should be extended in the future, and how to make it more effective. If proven effective, this new modality could become a new window within the GCF, complementary to the standard FP approach. In the future, Direct Access Entities, especially the regional ones, could take the lead of this type of Regional Support Programme.
212. In the unlikely event of the national grid arriving to rural or hard to reach areas that mini grids will serve under the IGREENFIN programme various scenarios are possible including an indemnity paid to mini-grid owners by national grid operator for exit, integration of the mini-grid to the national grid with a power purchase agreement, or national grid serving surrounding communities while mini-grid continues to serve the agricultural asset.

C. FINANCING INFORMATION							
C.1. Total financing							
(a) Requested GCF funding (i + ii + iii + iv + v + vi + vii)		Total amount		Currency			
		104,415,189		million euro (€)			
GCF financial instrument		Amount	Tenor	Grace period	Pricing		
(i)	Senior loans	73,971,000	40 years (Burkina, Senegal, Mali) and 25 years (cote d'Ivoire and Ghana)	10 years	(proposed 0% interest rate)		
(ii)	Subordinated loans	Enter amount	Enter years	Enter years			
(iii)	Equity	Enter amount					
(iv)	Guarantees	Enter amount	Enter years				
(v)	Reimbursable grants	Enter amount					
(vi)	Grants	30,444,189					
(vii)	Results-based payments	Enter amount					
(b) Co-financing information		Total amount		Currency			
		73,971,000		million euro (€)			
Name of institution		Financial instrument	Amount	Currency	Tenor & grace	Pricing	Seniority
IFAD		GrantGrant	26,218,790	million euro (€)			Options
IFAD		Loans	11,506,000	million euro (€)	For Burkina, Mali, Senegal : 40 years, 10 years For Côte d'Ivoire, Ghana : 25 years, 5 years	0,75% 0,9%	
AfDB		GrantGrant	14,794,200	million euro (€)			Options
ISDB		Senior LoansSenior Loans	12,328,500	million euro (€)	25 years 5 years	7%	Options
WAEMU Banks (Senegal, Mali, Cote d'Ivoire, Burkina)		Senior LoansSenior Loans	6,164,250	million euro (€)	20 years 5years	12,5%	Options
Ghana		Loans	2,054,750	million euro (€)	20 years 5 years	20%	
(c) Total financing (c) = (a)+(b)		Amount		Currency			
		177,481,678		million euro (€)			
(d) Other financing arrangements and contributions (max. 250		The GCF Programme will be complementary to the overall IFAD baseline investments in the selected countries. Current contributions from IFAD in form of grants amount to EUR 37, 724, 790 However, IFAD PMUs' contributions provided through input of human resources, office structures and technical input will be counted as co-financing. These co-					

words, approximately 0.5 page)	financing resources foster the financial sustainability of the programme beyond the timeframe of the execution of GCF funds. Other contributions are also expected from the 5 selected banks EUR 8,219,000, AfDB, EUR 14 794 200; ISDB, EUR 12,328,500 and IFAD EUR 37, 724, 790. AfDB will provide resources through the Agri-SMEs programme in the selected countries while for ISDB, co-financing is provided through a loan from the Islamic Bank of Senegal. The financial terms will apply to the GCF loans to be provided by IFAD to the NEEs which will be passed on to the 5 banks. The minimum co financing ratio will be applied at sub-loan level. Government contributions are in form of tax exemptions and presented in the IFAD baseline investments
---------------------------------------	--

C.2. Financing by component (EUR)

Component	Output	Indicative cost Options	GCF financing		Co-financing		
			Amount Options	Financial Instrument	Amount Options	Financial Instrument	Name of Institutions
Component 1: Green Business Financing Facility	1.1. Establish Green Business Financing Facility (revolving lending facility) - window 1 (adaptation) Pillars 1,2 & 4	65, 135, 575	49,314,000	Loans	4,109,500	Loans	5 LNABs
					5,958,775	Loans	IsDB
					5,753,300	loan	IFAD
	1.2. Establish Green Business Financing Facility (revolving lending facility) window 2 (mitigation) – Pillar 3	40,478,575	24,657,000	Loans	4,109,500	Loans	5 LNABS
					5,958,775	Loans	IsDB
					5,753,3000	loan	IFAD
	1.3: Technical assistance to set up the Green Financing Facility with the (windows 1 and 2)	410,950	410,950	Grants			
Component 2 - Technical Assistance Facility (TAF) (Pillar 5)	2.1: Generate knowledge, conduct studies, support knowledge transfer, provide training and	42,804,972.8	9,107,072.	Grants	19,725,600	Grant	IFAD
					13,561,350	Grant	AFDB

	monitor climate impacts				410,950	Grant	IsDB
	2.2. Improved policy dialogue, government technical and institutional capacity, advocacy, training, knowledge management, information dissemination and stakeholder management	13,895,481.1	7,188,776	Grants	4,720,841	grants	IFAD
					1,232,850	grants	AfDB
Component 3. Regional Support Programme	3.1: Enhanced knowledge management and exchanges.	6,617,939	6,617,939	Grants			
	3.2. Innovation and digital transformation technologies mapped and ecosystem built	682,177	682,177	Grants			
	3.3 Support Unit & Governance	4,922,852	4,922,852	Grants			
PMC	3,286,171	1,514,423	1,514,423		1,771,748	Grants	IFAD
Indicative total cost (USD)		177,481,678	+ 73,971,000 Other co-financing being confirmed with selected banks and partners (AfDB, ISDB, IFAD) at the design stage of each country				

C.3 Capacity building and technology development/transfer (max. 250 words, approximately 0.5 page)

C.3.1 Does GCF funding finance capacity building activities?

Yes ☒ No ☐

C.3.2. Does GCF funding finance technology development/transfer?

Yes ☒ No ☐

The investment of the GCF funds will enable about 50 technologies in climate resilient, low emission agriculture, including RETs to power agricultural value chains, to be transferred. These are local or imported technologies and a capacity-building programme will help sustain the transfer of technologies. The South-South and Triangulation cooperation will facilitate the transfer of technologies between the selected countries and other regions. The transfer of knowledge and innovation on green business through the GGW Regional Support Programme will sustain the programme's results and impacts beyond its lifecycle. Awareness-raising and dialogue will ensure the mainstreaming of

the objectives of IGREENFIN I in the financial sector (central banks, LNABs, MFIs) and maximize the impact of the program through replication with IGREENFIN II

D. EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

This section refers to the performance of the project/programme against the investment criteria as set out in the GCF's Initial Investment Framework.

D.1. Impact potential (max. 500 words, approximately 1 page)

213. IGREENFIN I will support the greening of the financial system in the five selected countries by building the capacity of five local national agricultural banks to offer green financing for climate resilient, low emission agriculture projects (green governance and lending products, green credit indicator system for management, long-term loan periods and unlocking barriers to innovative climate finance). It will also support 1,500 MSMEs and 2,500 FOs and cooperatives, as well as the transfer or licensing of 200 technologies and innovative solutions for climate resilient, low-emission agricultural development. Its interventions will contribute to reducing the targeted countries' GHG emissions by 5.6 million tCO₂ eq over the lifetime of the programme. The adaptation benefits of this programme will reach 378,600 direct and 2,494,000 indirect beneficiaries, of which 50 per cent will be women, 50 per cent men (50 per cent youth and 50 per cent adult males). It will build the capacity of the Agricultural Bank of Senegal and the Centre de Suivi Ecologique (CSE), and other four LNABs' readiness and ability to mobilize additional concessional finance from both the private and public sector.
214. Innovative green business models including climate resistant breeds and crop varieties, sustainable land restoration techniques, solar energy to power drip irrigation and processing, packaging and storage equipment, among others, for the ten selected value chains provide an indication of how EUR 177,481,678 million in funds for IGREENFIN I (i.e., GCF resources plus additional resources mobilized) can be used to produce positive impacts and ensure both financial and economic returns on investment. The green business could be scaled up by the financial sector in each of the targeted countries and in the GGW region.
215. IGREENFIN I will contribute to the implementation of the GGWI, which aims to restore 100 million hectares of currently degraded land, sequester 250 million tons of carbon and create 10 million jobs in rural areas by 2030. Spanning the entire width of Africa – 7,775 km from Senegal to Djibouti – the “Wall” is a mosaic of projects implemented by countries in the region with the support of development partners as a pan-African sustainable landscape programme to address land degradation and desertification, boost food, environmental and economic security and support the adaptation of communities to climate change. It aims to do much more than just grow trees: its goals are to i) foster climate resilience for local communities in a region set to be hit by a 3 to 5°C temperature increase by 2050; ii) safeguard food security for the 20 million people in the Sahel that go hungry every year; iii) support the creation of “green” jobs to provide stable incomes for African women and youth; iv) build a global response to the migration crisis, with EUR 49,314 million people set to migrate to Europe from degraded parts of Africa by 2021, and v) grow peace and security in the Sahel, where conflict has already displaced 2.8 million people so far in 2021. Through IGREENFIN I, IGREENFIN II and the Regional Support Programme, this programme will serve as the investment pillar of the future GCF-GGW RSP, which is supported by UNCCD and the 3S Initiative, two major African-led initiatives set up with the bold ambition to restore the productivity and vitality of the Sahel region, while ‘growing solutions’ to the continent’s most urgent development challenges. IGREENFIN I Phase 1 will channel resources to beneficiaries and provide crucial funds to help accelerate the implementation of the GGW, which began in 2007. It will also contribute to advances on the African Union’s Agenda 2063 in the region.
216. Climate impacts will be monitored at the programme level using the core GCF mitigation/adaptation indicators, consistent with the GCF results management and performance measurement frameworks. The tool to track these indicators will be defined in cooperation with the LNABs, IFAD baseline investments and in accordance with the green business plans that will be developed by FOs, MSMEs, cooperatives and targeted sectors through the credit line. The final impacts will be measured by an ex-post impact assessment with evidence-based proof and will depend on the portfolio of eligible projects that the LNABs will approve in line with the set of eligibility criteria proposed. The TAF will oversee the measuring and controlling of these impacts.

D.2. Paradigm shift potential (max. 500 words, approximately 1 page)

217. **Potential for scale up and replication:** By promoting the scaling up of the Niger GCF SAP business model, IFAD is driving the transition to climate resilient agriculture in the West and Central Africa (WCA) and beyond. This is key to achieving a paradigm shift towards more climate resilient, low emission agriculture and sustainable bank portfolios and sectors not only in the region but throughout the developing world since the market barriers the

programme is designed to address are not unique to GGW countries. This regional approach, IGREENFIN I and the Regional Support Programme will be implemented using a phased approach:

- **Phase 0:** The Simplified Approval Process IGREENFIN I project in Niger (SAP012) approved by the GCF Board in November 2019.
- **Phase 1** will be implemented in the first set of three GGW countries (Burkina Faso, Mali and Senegal) plus Côte d'Ivoire and Ghana
- **Phase 2** will be implemented in the second set of seven GGW countries (Chad, Djibouti, Eritrea, Ethiopia, Mauritania, Nigeria and Sudan)

218. IGREENFIN I aims to engage national agricultural banks, national banking associations, banking regulators, green standards settings bodies, local universities and business schools on green finance in agriculture. Through the mainstreaming work, these institutions will be able to sustain the GCF grant and concessionality beyond the programme's duration. The programme will support the production of studies and policy papers on climate risk management to define the standards, develop and harmonize green lending products, instruments and tools such as ESG benchmarks. It will also build knowledge on green finance invested in the agricultural sector by both the private sector and governments. IGREENFIN I will work closely with the national agricultural banks to develop to improve their ESG, international capacity and better align part of their lending portfolio on the NDCs. By building the banks' capacity to offer affordable green finance for smallholder agriculture, the programme is starting to fill the gap in financing for agriculture and alter the perception smallholder from a risky business to a vast market with great potential and opportunities

219. Green credit lines and awareness campaigns will support the development of demand for green products, particularly for youth and women, and concrete viable pipelines of green business projects, reduce the financial sector's perception of smallholders as high risk and hence, the development of new green markets in sectors that have been traditionally difficult to reach (for example, the MSME sector). Beneficiaries such as MSMEs will benefit from training activities and affordable finance (available from local financial institutions) and techniques that help them reduce their operating costs while increasing their productivity and incomes. The early stages of the programme's operations are expected to demonstrate to end users the effectiveness and value of the solutions proposed, which will stimulate demand beyond the target financial institutions and target countries to other market sectors and areas in the GGW after the programme ends. This programme will facilitate the transition towards granted funded rural financial projects to sustainable long term markets rural finance projects

220. **Potential for knowledge sharing and learning:** Markets for climate change adaptation and mitigation solutions, particularly for small and medium-sized projects in the GGW countries and the Sahel, remain underdeveloped and thus require support to develop at the pace needed to achieve national emission reduction targets. IGREENFIN I is designed to foster the development of new markets for green finance and climate resilient, low emission agriculture in the GGW areas, contribute to overcoming barriers and create a long-term enabling environment for investments in these sectors. IGREENFIN I will put in place long-term robust structures for green lending operated by five LNABs in the five countries of the GGW during the programme's first phase. The green credit lines to be created under component 1, when combined with the capacity-building on the development of green business plans offered by the TAF to FOs, MSMEs, cooperatives and women and youth-led organisations, will help reduce the risks of both lending and borrowing in the sector, thus contributing to even more robust long-term impacts and sustainable market growth while aligning part of the LNABs portfolio on the country's NDCs.

221. **Contribution to the regulatory framework and policies:** In addition to inadequate legal frameworks to systematically drive investment in climate change projects, the element that is missing is accessible and affordable financing, which this programme is designed to address through the establishment of green finance market. As outlined above, the programme is well-aligned with national emission reduction targets and priorities. Successful performance of the programme and the creation of an enabling market environment will encourage national and local authorities to prioritize strong policies in these sectors. To address any remaining barriers in the market, IGREENFIN I will continue to carry out policy dialogue activities through the central banks, LNABs and other partners. This programme will also build upon and complement nascent agreements focusing on greening the financial sector in the region. In addition to all the knowledge built, studies, trainings and monitoring of climate impacts carried out throughout the programme cycle by the TAF and the support it will offer for aligning national banks' portfolios with the NDCs, IGREENFIN I will also support the implementation of countries' Climate Investment Plan and NAPs.

222. The GCF GGW RSP introduces a new approach piloted by IFAD and the GCF, which aims to catalyse impact beyond a one-off project or programme investment by creating bridges for knowledge sharing and learning between GCF projects that would otherwise be implemented in silos, and among AEs, GCF NDAs, GGW and GEF focal points. This innovative approach has potential for scale up in Africa beyond the GGW countries and could also be replicated in other large landscapes and ecosystems such as the Amazon or in countries with relatively similar needs, such as some SIDS. This UP will also contribute to strengthening collaboration at the country level between the GCF, the GEF and the AF, and bring to life a concrete example of complementarity and collaboration between the UNFCCC and the UNCCD.

D.3. Sustainable development (max. 500 words, approximately 1 page)

223. IGREENFIN I and the Regional Support Programme have identified and adopted the following UN SDGs and mapped each of the target sectors to them: No poverty (SDG 1); Zero hunger (SDG 2); Gender equality (SDG 5); Clean water and sanitation (SDG 6); Affordable and clean energy (SDG 7); Decent work and economic growth (SDG 8); Climate action (SDG 13); Life on land (SDG 15), and Partnerships for the Goals (SDG 17).

224. **Environmental co-benefits:** The programme is designed to help secure multiple environmental benefits for water, air, land and nature by promoting the adoption of best adaptation and mitigation practices and technology, which will enhance the sustainable management of natural resources, protect threatened dryland biodiversity against erosion and desertification and increase the potential for carbon sequestration. These benefits could also contribute to the increased resilience of the region's ecosystems and human livelihoods to climate change and variability and the GGW's overall objectives (land restoration, job creation and CO₂ emission reduction). The programme intends to reduce the use of energy from biomass, including firewood and agricultural residues, imported diesel fuel and charcoal, which are the main sources of electricity in the five targeted countries in IGREENFIN I, by providing greater access to renewable energy. It is expected that CO₂ emissions (5.6 million tCO₂ eq) will be reduced during the programme's lifecycle thanks to agroforestry and SLM practices (this amount will further increase once the avoided emission from RET is integrated) and limits will be placed on the use of biomass and diesel for energy consumption through the adoption of RET. With the promotion of EbA and agroforestry techniques, more than 100,000 ha of degraded land will be restored and sustainably managed with improved ecosystem goods and services. The programme will build up each link in the selected agricultural value chains (including livestock and NTFPs) from production to markets, increasing the productivity for key commodities and farmers' incomes while safeguarding the environment. Through the adoption of climate resilient agriculture, the programme will reduce soil erosion and soil fertility loss, enhance soil moisture retention and maintain soil temperatures suitable for crop production.

225. **Social co-benefits including health impacts:** The activities proposed under this programme will generate multiple job opportunities for rural people, particularly for youth and women, promote inclusion and support coherent investment through FOs, cooperatives and MSMEs. By promoting local climate resilient, low emission options, the programme will contribute to cost-saving health benefits thanks to cleaner air for local communities and access to clean energy, water and food, thus reducing food insecurity. As described in the gender action analysis and action plan, the programme will support women's participation in decision-making, increase their access to financing and assets and improve their livelihoods. The targeted areas considered the most vulnerable and selected through IFAD baseline investments will be positively impacted by the programme.

226. **Economic co-benefits:** The economic benefits of this programme are linked to its high potential to reduce poverty through increased productivity of key crops and livestock production, as well as income generation activities (IGA), such as the production of and value addition to different fruit and NTFPs, including tamarind, Arabic gum, baobab leaves as well as fodder and seedlings sales. The GGW implementation status report indicated that revenues from IGA since 2007 have amounted to approximately EUR73,971 million across all 11 GGWI countries, with Burkina Faso being more successful in creating IGA opportunities than the others. Revenues reported per country are as follow: Burkina Faso, US\$6.8 million, and US\$50,000 for Mali and Senegal each. A stable inflow of revenues from natural products and services is important in terms of exit strategies that contribute to the sustainability of the restoration projects over time. By promoting IGA and opportunities for livelihood improvement, the programme will help better the living conditions in the five participating countries that are among the poorest and the most vulnerable to climate

change in the world. The programme will contribute to create 703,600 jobs¹⁶⁸. By doing so, the programme provides a unique opportunity to put an end to poverty traps.

227. Gender-sensitive development impacts: Women are increasingly and disproportionately affected by climate change. In the five participating countries, women play a central role in feeding their families and ensuring other basic needs (water, fuel for firewood, etc.). This responsibility has become harder to assume because of declining crop yields due to climate change (primarily higher temperatures and unpredictable rainfall patterns). The programme will increase opportunities for livelihood improvement and provide concrete benefits to smallholder farmers and pastoralists, both men and women. The implementation of targeted activities can potentially offer an array of advantages to local communities, such as greater yields, improved soil fertility, fodder availability, as well as shorter wood and water collection time for women, freeing up their time so they may engage in other productive tasks. Through its focus on gender mainstreaming and women's economic empowerment, this programme will support the reduction of entrenched gender inequalities in rights to land resources and access to technology and information that hamper women's capacity to manage current climate risks and adapt their livelihoods to long-term climate change trends. The programme has set targets: 50 per cent of loans are to be granted to women-led MSMEs, cooperatives and FOs. The other 50 per cent will be dedicated to Men of which 50% will be youth-led MSMEs, cooperatives and FOs in which young women participate. Specific actions will be developed to strengthen the technical and managerial capacities of women aimed at providing them with appropriate tools for identifying and developing bankable business plans and improving their level of financial education. As stated in the gender action plan, gender-disaggregated data will be assessed against the appropriate indicator to measure women's enhanced access to loans. The project intends to close the gender gap, as women represent 60 to 70 per cent of the work force and do not have access to productive assets, finance and knowledge. At least 1/3 of the women will be included into the National Steering Committee to ensure that they influence the main strategic decisions.

D.4. Needs of recipients (max. 500 words, approximately 1 page)

228. Participating countries in this programme are highly vulnerable to drought, flooding, rising temperatures and variability in rainfall, as well as other climate change hazards (wildfires, locust infestations, etc.). Their exposure and vulnerability to climate hazards is projected to increase in the future. Climate projections suggest that the West Sahel and northern parts of Côte d'Ivoire and Ghana will be hotter and drier with more frequent extreme events. In most scenarios, temperature rise will accelerate, meaning that the continent on average could be between 2 and 6°C warmer by end of this century (ACMAD, 2020).

229. Climate impacts have negative impacts on agriculture. They reduce productivity, food security and GDP, cause endemic diseases (malaria) and increase rural poverty. Given the five countries' high poverty rate (see feasibility study for more details) and the impacts of climate change, the country has few financial institutions that want to risk investing in smallholder farming despite the fact that agriculture remains the main source of economic growth.

230. The five countries' financial system remains weak and has a limited number of instruments for financing agriculture. LNABs have limited possibilities of mobilizing highly concessional loans in the capital markets. FOs and MSMEs, cooperatives, youth and women-led organizations active in the agricultural sector often lack access to long-term financing with affordable conditions high(interest rates, tenors and grace periods) due to small loan sizes and relatively high transaction costs. Key barriers are limited adaptation capacity to climate change across the country, lack of skills and knowledge on climate finance and climate resilient, low emission agriculture, the nascent regulatory framework for these sectors and the need to de-risk the sector.

231. Technical assistance and funding for the establishment of a Green Financing Facility that will provide concessional loans and climate smart credit risk tools will enhance access to credit, contribute to de-risking the sector and foster greater knowledge on climate finance and adaptation across the selected countries. They will directly help overcome the key barriers that LNABs face in rural finance and enable them to engage in green financing.

D.5. Country ownership (max. 500 words, approximately 1 page)

¹⁶⁸ The number of job created is calculated by adding the direct number of jobs created for each of the direct beneficiaries plus the average number of jobs created for the 1500 MSMEs and 2500 FOs or Cooperatives assuming that one MSME has on average 50 people and FOs about 70 people

232. The participating countries are parties to the UNFCCC and have signed and ratified the Kyoto Protocol. By ratifying the UNFCCC, these countries have committed to implementing measures to adapt to climate change and reporting on their NDCs. This programme will contribute to the implementation of objectives of the three Rio conventions ratified by all countries, including the UNFCCC, as well as the Paris Climate Agreement, the SDGs and the Sendai Framework for Disaster Risk Reduction.

233. This programme is fully aligned with the countries' national development plans and their national commitments on climate mitigation and adaptation included in their NAPAs, National Climate Change Policies and Strategies, NDCs, National Communications (NCs), SDGs and National Strategies for Disaster Risk Reduction.

234. Table 28 below presents the priorities of the five countries' NDCs. As the table reveals, IGREENFIN I has high potential to contribute to targets in two main sectors: Agriculture, forestry and other land use (AFOLU) and Energy.

Table 28: Priorities of Target Countries' NDCs relevant to IGREENFIN I

Countries	NDC targets relevant to IGREENFIN I
Burkina Faso (28/9/2015)	An unconditional pledge to reduce emissions by 6.6% below business-as-usual levels by 2030, with a further 11.6% reduction conditional upon international support. Includes interim pledges for 2020 and 2025. In the section on adaptation, actions proposed would reduce emissions by a further 36.95%, taking the total reductions up to a potential 55.15% below business-as-usual levels. Key sectors targeted water resources, agriculture, livestock and fisheries, transport, energy, land use and forestry
Côte d'Ivoire (22/09/2015)	28% reduction from business-as-usual (BAU) by 2030 in energy, transport, waste, agriculture and land use. Adaptation sectors targeted water resources, agriculture, livestock and fisheries, coastal zones, energy, land use and forestry.
Ghana (23/09/2015)	Unconditional pledge to lower its GHG emissions by 15% relative to a BAU scenario emission of 73.95MtCO ₂ e by 2030. An additional 30% emission reduction is attainable on condition that external support is made available to Ghana to cover the full cost of implementing mitigation action (finance, technology transfer, capacity-building). With this external support, a total emission reduction of 45% below the BAU emission levels can be achieved by 2030, with energy and AFOLU offering the largest sources of emission reduction potential.
Mali (29/9/15)	Commits to reducing emissions by 29% for agriculture, 31% for energy and 21% for forests and land use, each by 2030, and in comparison, to a BAU scenario. This is an average reduction of 27%. This is conditional upon international support, although around 40% of this can be met unconditionally. Includes a section on adaptation, though only for the period 2015-2020.
Senegal (26/9/2015)	An unconditional reduction in emissions of 5% by 2030, compared to BAU levels, with interim targets of 3% by 2020 and 5% by 2025. Accompanied by a conditional target, subject to international financial support, of 7% by 2020, 15% by 2025 and 21% by 2030, compared to BAU levels. Includes a section on adaptation, as well as information on potential obstacles. Sectors are, Transport, water resources, agriculture, livestock and fisheries, coastal zones, energy, land use and forestry.

235. This programme was designed at the request of the GCF national designated authorities (NDAs) of the respective countries following the approval of the Inclusive Green Financing for Climate Resilient and Low Emission Smallholder Agriculture project for Niger in late 2019. The Agricultural Bank of Niger (BAGRI) was the first financial institution in the region to take courageous new steps towards a greener approach to financing climate resilient smallholder agriculture, which was quick to spark interest in other parts of the region. An event held in Dakar in February 2020 brought together the NDAs, the CEOs of the agriculture and investment banks and government representatives from Burkina Faso, Côte d'Ivoire, Ghana, Mali and Senegal who requested IFAD's support to pursue similar projects with the GCF. This meeting also led to the formation of a coalition of local state-owned banks (agricultural and investment banks) that are ready to embrace green financing for climate resilient, low emission smallholder agriculture. IFAD then engaged with the NDAs of the five target countries through face-to-face meetings, presentations and email correspondence to discuss the programme. Once the consultations with all stakeholders, including civil society organizations, had been conducted, the GCF National Focal Points from the participating countries endorsed the proposals. All NOLs have been provided.

236. Feasibility studies and market analyses were conducted on agricultural value chains and sectors relevant to this programme to assess both demand and supply and the need to stimulate and create a new market for green business projects (see Annex 2 Feasibility Study, Market Assessment, Renewable energy report).
237. In 2020, IFAD was proposed to lead the Regional Support Programme of the GGF. IFAD will work in partnership with other agencies including UNCCD, FAO, GEF and WB on the GCF GGW RSP. IGREENFIN I will be one of the financing programmes under the future GGW RSP. Coordination with the GGW will help leverage additional resources and capacities.
238. The GGW RSP has been developed in response to a call from the GGW countries for the GCF and the international community to increase the level of support to accelerate the pace of the implementation of the activities contributing to the achievement of the GGW's objectives. It is fully aligned with the principle of country ownership and it respects, and builds on, the institutional structure already in place to support the implementation of the GGW Initiative. The GGW RSP is aligned with GGW Accelerator, which is led by the GGW Agency with the temporary support of the UNCCD. At the country level, the GGW RSP aims to enhance the capacities of the GCF NDAs, as well as of the GEF and AF focal points, by hiring GIS and KM specialists to help them during the planning of future GCF, GEF and AF projects and their implementation through enhanced KM exchanges.

D.6. Efficiency and effectiveness (max. 500 words, approximately 1 page)

239. In the GGW region, green businesses are by and large financially underserved in terms of the availability of both highly concessional loans from LNABs, commercial banks and MFIs and technical assistance. Section B3 describes how GCF resources will be used efficiently under interlinked activities to help overcome the barriers identified, including the approach to ensure minimum concessionality. Activities under IGREENFIN I aim to fill a financing gap by providing financial and non-financial support. In doing so, the programme will jump start a nascent market that is currently far from reaching its full potential. As this market develops over time, opportunities will be created to attract other public and private sector investors in these LDCs (Burkina Faso, Mali and Senegal) and the two LMICs (Côte d'Ivoire and Ghana) that have all the characteristics of LDCs, namely high levels of poverty and vulnerability to climate change.
240. IGREENFIN I will nurture and scale up all the green financing products and green business models it generates to other relevant projects underway in the targeted countries and regions. By building synergies and leveraging IGREENFIN I resources, the programme and other entities will adequately reach the maximum beneficiaries. Partnerships with LNABs and EEs will be promoted through the provision of green loans and technical assistance to de-risk all potential subprojects. GCF financing will be essential to overcome existing barriers to access to financial and non-financial services that hinder the adoption and implementation of best climate change adaptation and mitigation measures (including solar energy for agriculture, among others). Additionally, even though LNABs have been performing quite well over the past years with various loan products, their loans are still not concessional enough to stimulate greater adherence to climate resilient, low emission agriculture and generate a strong green business pipeline. In the energy sector, several bilateral donors have put in place concessional financing schemes that combine grants with credit to promote agricultural and rural finance; they work mainly through MFIs and state-linked financial institutions, such as the targeted LNABs. Early results from these programmes are highly encouraging, although stronger oversight and technical assistance (TA) are still needed. This new approach to transform the LNABs' internal governance and set up a facility with green finance windows is considered innovative and transformational in the context of climate change. From an effectiveness point of view, IGREENFIN I is projected to achieve a leverage ratio of 1:0.7 (i.e., EU0.57 of additional public and private funding leveraged for every EUR 0, 82 of GCF investment); thus, a EUR104, 134 million investment from the GCF is expected to leverage about EUR 73,971 million in additional public and private contributions (including co-financing as per Section C.1.b).
241. Given the historic market demand for such loans, IGREENFIN I believes the total proposed GCF facility size (EUR 104,134) equivalent will unlock the market and improve both demand and supply. It will stimulate the market and encourage bilateral lending organizations and private sector investors to adopt competitive rates to reduce the risk of being crowded out of the market. The NEEs, through their respective LNABs, intend to blend their internal resources with GCF funds to support rural transformation through the adoption and scaling up of climate resilient agriculture in all targeted countries. Through IGREENFIN I, countries are expected to develop a conducive regulatory framework which will sustain this programme beyond its duration.

242. Regarding adaptation, 10 value chains for each country were analyzed with estimated the economic and financial performance of the investment for different types of local stakeholders. The IRR for the various VCs is as follows: VC1 (20%), VC2 (40%), VC3 (55%), VC4 (70%), VC5 (10%), VC6 (60%), VC7 (80%), VC8 (40%), VC9 (30%), VC10 (50%) with payback time in the range of one to five years. When we consider country performance, the IRR ranges from 37 per cent for Mali to 51 per cent for Côte d'Ivoire. Finally, when considering different beneficiaries, the highest IRR is found for MSMEs in Burkina Faso and Ghana (over 80 per cent), for cooperatives in Côte d'Ivoire (47 per cent), Mali (68 per cent) and Senegal (52 per cent). The IRR for farmer organizations is generally the lowest in all countries, ranging from 24 to 35 per cent.
243. The GCF contribution will be very important to both improve the financial performance of the investment (i.e., by lowering the interest payment) and create a buffer for less productive years (e.g., sensitivity analysis shows that a 20 per cent decline in revenues or a similar increase in production costs will result in negative IRR and NPV for several VCs).
244. When taking a societal approach that considers the economic valuation of externalities (e.g., carbon and other ecosystem services), including those that do not generate a cash flow, the IRR increases to 100 per cent for adaptation. This highlights the value of the GCF contribution beyond the direct economic impacts of the investments proposed, resulting in a net positive contribution to society.
245. Regarding mitigation, the IRR and NPV are positive, and the debt coverage ratio (DCRE) is above one for energy investors in mini grids when considering the price of \$0.61 - \$0.87/kWh. This shows that mini grids are more financially viable than bringing centralized power generation to rural areas and using fully decentralized diesel power generators. It is worth noting that with any price lower than \$0.8 - \$1/kWh, farmers would also benefit from cost savings, thus making the investment advantageous for all economic actors. The IRR is 25 per cent (with a price of \$0.87/kWh) and declines to 14 per cent (with a price of \$0.61/kWh). The GCF contribution makes the DCR more desirable, increasing from 1.9 to 2.57 (with a price of \$0.87/kWh) or increasing from 1.39 to 1.83 (with a price of \$0.61/kWh).
246. With regard to farmers as investors/beneficiaries: If a farming cooperative were to invest in a mini grid, it would enable them to have access to affordable electricity and minimize spending on energy. The price that would allow the mini grid investment to be economically viable is \$0.4/kWh. This is the minimum price that would justify investing in a mini grid (i.e., generating a positive IRR and NPV). Furthermore, this price would allow consumers (farmers and those working in the food value chain) to save \$0.5/kWh consumed when compared to using diesel generators. If we compare the investment to cost savings, the IRR reaches 11 per cent and the NPV is positive for all countries. In other words, the savings are large enough to reduce the IRR to a value that is comparable to the interest rates requested by local national agricultural banks, and twice the rate requested by commercial banks. In this respect, the contribution of IGREENFIN I and GCF is essential. In fact, while the IRR and NPV are positive, the debt coverage ratio (DCR) is smaller than 1 when considering bank rates. When using the preferential rates guaranteed by the IGREENFIN I instead, the DCR is above 1, indicating economic viability for the programme considering all IRR, NPV and DCR. Further, if we carry out a more systemic analysis and consider indirect benefits and reduction of externalities, the IRR grows to a range of 20 to 40 per cent, depending on the scenario.
247. Overall, we find that with the implementation of the programme, investment in mini grids is economically viable for all economic actors interested in investing to replace reliance on diesel generators, which would generate both positive returns for investors and savings for consumers.

E. LOGICAL FRAMEWORK

This section refers to the project/programme's logical framework in accordance with the GCF's Performance Measurement Frameworks under the Results Management Framework to which the project/programme contributes as a whole, including in respect of any co-financing.

E.1. Paradigm shift objectives

Please select the appropriated expected result. For cross-cutting proposals, tick both.

- ☒ Shift to low-emission sustainable development pathways
- ☒ Increased climate resilient sustainable development

E.2. Core indicator targets

Provide specific numerical values for the GCF core indicators to be achieved by the project/programme. Methodologies for the calculations should be provided. This should be consistent with the information provided in section A.

E.2.1. Expected tonnes of carbon dioxide equivalent (t CO ₂ eq) to be reduced or avoided (mitigation and cross-cutting only)	Annual	225,487 t CO ₂ eq
	Lifetime	5, 637 184 million t CO ₂ eq
E.2.2. Estimated cost per t CO ₂ eq, defined as total investment cost / expected lifetime emission reductions (mitigation and cross-cutting only)	(a) Total project financing	<u>177,481,678</u> Choose an item
	(b) Requested GCF amount	<u>104,415,189</u> Choose an item.
	(c) Expected lifetime emission reductions	5, 637, 184 t CO ₂ eq
	(d) Estimated cost per t CO₂eq (d = a / c)	<u>31,76</u> / t CO ₂ eq
	(e) Estimated GCF cost per t CO₂eq removed (e = b / c)	<u>18,5</u> / t CO ₂ eq
E.2.3. Expected volume of finance to be leveraged by the proposed project/programme as a result of the Fund's financing, disaggregated by public and private sources (mitigation and cross-cutting only)	(f) Total finance leveraged	<u>73,971,000</u> Euros
	(g) Public source co-financed	<u>65,752,000</u> Euros
	(h) Private source finance leveraged	<u>8,219,000</u> Euros
	(i) Total Leverage ratio (i = f / b)	0,70
	(j) Public source co-financing ratio (j = g / b)	<u>0,63</u>
	(k) Private source leverage ratio (k = h / b)	<u>0,078</u>
E.2.4. Expected total number of direct and indirect beneficiaries, (disaggregated by sex)	Direct	378,600 (50% women; 50% men,) 50 % of the men beneficiaries and 50% of women beneficiaries will be youth.
	Indirect	2,494,000 (50% women; 50% men)
		378,600 people of which 189,000 will be women and 189,000 men
		2,494,000 people of which 1,247,000 women, 1,247,000 men. 50% of the men and 50% of women are youth as provided under Annex 17 The calculation Methodologies and assumptions is included in Annex 17
E.2.5. Number of beneficiaries relative to total population (disaggregated by sex)	Direct	0.25% of which 50% are women
	Indirect	1.74% of which 50% are women
		For a multi-country proposal, leave blank and provide the data per country in annex 17

E.3. Fund-level impacts

Expected Results	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term	Final	
M1.0 Reduced emissions through increased low-emission energy access and power generation	M1.1 Tonnes of carbon dioxide equivalent (t CO ₂ eq) reduced or avoided - gender-sensitive energy access power generation	Semester and annual banks reports on the MWh of power generated by installed mini	0	Tonnes CO ₂ eq reduced from solar mini grids and carbon sequestrati	272,048 Tonnes CO ₂ eq: reduced from solar mini grids and	544,097 Renewable energy-based rural electrification replacing the continuous, alternative use

		grids financed by RET loans Audit RCU reports and energy audit report from other partners		on from power generation as a result of Fund funded project	carbon sequestration from trees or avoided as a result of Fund funded project 1,800 mini grids will be installed across the 5 counties: 400 mini grids in Burkina Faso 400 mini grids in Mali 400 mini grids in Senegal 300 mini grids in Ghana 300 mini grids in Côte d'Ivoire	of fossil fuel-based solutions (CDM AMS-III.BL baseline methodology) in households, commercial facilities such as shops, public services/buildings and small, medium and micro enterprises (SMMEs) These targets are estimated based on the CDM methodology AMS-I. A v17. Average yearly emissions abatement is estimated ex-ante at 544,097 tCO₂eq (assumption is made that 40% of mini grids would have been commissioned at mid-term) Solar systems annual performance loss 1% We estimated the Total emission reductions from solar systems over 25 years is estimated at 525,450 tCO₂eq
--	--	---	--	--	--	---

<i>M4.0 Reduced emissions from land use, reforestation, reduced deforestation, and through sustainable forest management and conservation and enhancement of forest carbon stocks</i>	<i>M4.1 Tonnes of carbon dioxide equivalent (t CO2 eq) reduced or avoided (including increased removals) - forest and land use</i>	Programme reports, annual mitigation outcomes verification	TBC at inception phase	2,49 million Tonnes CO2 eq: reduced or avoided as a result of funded green projects	4,998, tonnes million Tonnes CO2 eq: reduced or avoided as a result of funded green projects	Adaptation co benefits from AFOLU builds on extensive community mobilization, and capacity building for Communities to sustainably manage land.
<i>A1.0 Increased resilience and enhanced livelihoods of the most vulnerable people, communities and regions</i>	<i>A1.2 Number of males and females benefiting from the adoption of diversified, climate resilient livelihood options (including fisheries, agriculture, tourism, etc.)</i>	Programme reports, stakeholder surveys including interviews to assess increased resilience	TBC at inception stage	1,247,000 people of which 623,500 females 623,500 males	2,494,000 people of which 1,247,400 females 1,247,000 males	Political stability, no major climate disasters and market price stability to support Climate resilient agricultural solutions financially viable, leading to their adoption by FOs, MSMEs, cooperatives, youth and women-led organizations
<i>A2.0 Increased resilience of health and well-being, and food and water security</i>	<i>A2.2 Number of food secure households (in areas/periods at risk of climate change impacts)</i>	Programme reports, stakeholder surveys, other ministries reports (ministry of health, water, local development)	0	189,000 Household of which 94,500 females 94,500 males	378,600 Households of which 189,000 females 189,000 males	Political stability, no land tenure issues and natural disasters, extension services to support FOs, MSMEs, cooperatives, MFIs and communities to adopt climate resilient agricultural solutions and increase the resilience of their health, well-being, food

						and water security
A4.0 Improved resilience of ecosystems and ecosystem services	A4.1 Coverage/scale of ecosystems protected and strengthened in response to climate variability and change	Programme - level field surveys GIS maps of project intervention sites.	TBC at inception phase	125,000 ha	250,000 ha	Political stability, no land tenure issues and natural disasters, extension services to support Target communities and authorities to adopt and implement the proposed transformative knowledge and measures transferred through the project to protect ecosystems and strengthen response to climate change, agroforestry activities will have a significant effect on ecosystem functioning

.4. Fund-level outcomes						
Expected Outcomes	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term)	Final	
	Number of technologies and innovative solutions transferred or licensed to support low-emission development as a result of fund support (by type and number)	Programme M&E Reports	0	25 technologies	50 technologies	Political stability, no land tenure issues and natural disasters, extension services which enable customers and % of population to adopt and implement

						improved technologies
M6.0 Increased number of small, medium and large low-emission power suppliers <i>Choose expected outcome</i>	<i>M6.2 Number of households and individuals (males and females) with improved access to low-emission energy sources</i>	Programme reports, national and regional statistic	0	189,000 Household and 1,247,000 individuals of which 623,500 females 623,500 males	378,600 Households And 2,494,000 individuals of which 1,247,400 females 1,247,000 males	Political stability, no land tenure issues and natural disasters, extension services which enable households and % of population to access and use low emission RETs 1 direct beneficiary is equivalent to 1 household. On average each household has 6 people which corresponds to indirect beneficiaries
	<i>M6.3 MWs of low-emission energy capacity installed, generated and/or rehabilitated as a result of GCF support</i>	National energy statistics	TBC at inception phase	5,7 MW	14.4 MW	Political stability, no land tenure issues and natural disasters, extension services, price market which enable Communities and stakeholders to adopt RETs
M9.0 Improved management of land or forest areas contributing to emissions reductions	<i>M9.1 Hectares of land or forests under improved and effective management that contributes</i>	national and regional statistics every two years, programme M&E reports	0 ha	125,000 ha	250,000 ha	Mitigation and adaptation will happen in selected areas and target communities when there is

	to CO2 emission reductions					Political stability, no land tenure issues and natural disasters, extension services, price market knowledge and measures transferred through the project to protect ecosystems and strengthen response to climate change variability
A7.0 Strengthened adaptive capacity and reduced exposure to climate risks	A7.1 Use by vulnerable, communities, businesses, and public sector services of fund-supported tools, instruments, strategies and activities to respond to climate change and variability	Programme and bank reports as well as stakeholder and surveys	TBC at inception phase	7 MFIs, 5 commercial banks, 750 MSMEs and 1250 FOs, cooperatives or women- or youth-led organizations use one or more tool, instrument, strategy or product, worthiness tool, safeguard) to respond to climate risks in their lending portfolio	15 MFIs, 10 commercial banks, 1,500 MSMEs and 2,500 FOs, cooperatives or women- or youth-led organizations use one or more tool, instrument, strategy or product, worthiness tool, safeguard) to respond to climate risks in their lending portfolio	Identification of bankable and de-risked projects, credit worthiness tool, safeguard instruments, strategies and activities developed by the project will improve adaptive capacity
A8.0 Strengthened awareness of climate threats and risk reduction processes	A8.1 Number of males and females made aware of climate threats and	Programme reports, surveys	0 males 0 females	189,000 beneficiaries (94,500 male and	378,600 beneficiaries (189,000 male and 189,000	Assuming 100% of targeted rural communities are willing to

E.5. Project/programme performance indicators

Expected Results	Indicator	Means of Verification (MoV)	Baseline	Target		Assumptions
				Mid-term	Final	
1.1. Financially viable green business identified and financed	a. Total project financing volume	Banks reports And IGREENF IN I RCU reports	0	EUR 73,9 (of which EUR 49,3 million for Adaptation mitigation and 24,6 EUR million for Mitigation)	EUR 73,9 million (of which half for mitigation and half for adaptation)	Assuming interest of LFIs in financing mitigation projects. See annex 3 (Financial Model)
	b. Number of FOs, MSMEs, cooperatives benefited			2 agricultural banks 5 commercial banks, 7 MFIs, 750 MSMEs, 1250 FOs or cooperatives.	5 agricultural banks 10 commercial banks, 15 MFIs, 1500 MSMEs, 2500 FOs or cooperatives	
2.1 Technical support and capacities built for FOs, MSMEs, cooperatives, LNABs and IFIs	Technical support and capacities built for FOs, MSMEs, cooperatives, personnel trained in Project Assessment. Pipeline of bankable projects	Banks reports and IGREENF IN I PMUs and RCU reports	0	At least 189,000 trained (94,500 male and 94,500 female) from 750 MSMEs, 1250 FOs or cooperatives 50 pipelines of bankable projects	378,600 people beneficiaries (189,000 male and 189,000 female) from 1500 MSMEs, 2500 FOs or cooperatives. 100 pipelines of bankable projects	Assuming participation and interest of FOs, MSMEs, cooperatives in the Programme See Annex 3 (Financial Model)
	Technical support and capacities built for banks, MFIs personnel trained in Green Business plan Assessment,			7, commercial banks, MFIs personnel trained on ESMS systems and green banking (50% women)	30 commercial banks, MFIs personnel trained on ESMS systems	

	tools , green lending				and green banking (50% women)	
2.2. Adoption of climate resilient diversification activities to increase agricultural productivity	People participating in income-generating activities supported by the project disaggregated by gender	Based on information collected as part of stakeholder / beneficiary survey and project reporting	0	750 MSMEs, 1250 F0s or cooperatives (50% women-led) trained	1500 MSMEs, 2500 F0s or cooperatives trained	The target value reflects a household adoption rate of 40%. Female-headed households (approx. 15% of all households) are targeted at a rate higher than 50%.
2.3. Institutional and regulatory frameworks for green financing enhanced	Number of Central Banks and LBDPs with green finance policies and number of LNABs with green products	Banks reports And IGREENFIN I PMUs and RCU reports	TBC at inception phase	At least 1 to 2 green finance policies developed in coordination with the central bank per country and at least 2green products per banks per country	2-4 green policies developed per country and at least 4 green products LNABs per country	Assuming no political instability, market price stability, availability of services and goods, participation and interest of banks, Central banks in programme. See annex 3 (Financial Model)
3.1 Knowledge Management & Communication	Technical support and capacities built for GCF projects (and other projects, including GEF) to share knowledge, increase uptake of good practices and inform decision-making	Support Unit progress report	0	At least 20 GCF projects participating in knowledge sharing activities and staff trained At least 3 new projects approved by GCF using knowledge generated by the RSP	At least 29 GCF projects participating in knowledge sharing activities and staff trained At least 7 new projects approved by GCF using knowledge generated by the RSP	Assuming participation and interest of GCF projects, NDAs, GEF projects and other stakeholders to participation in knowledge sharing activities

Output 3.2 Innovation and digital transformation technologies mapped and ecosystem built	Technical support to build an innovation and digital solutions ecosystems in the GGW	Support Unit progress report	0	At least 20 innovators participating actively in the ecosystem At least 3 new projects approved by GCF using taking on innovation and digital solutions promoted through the ecosystem	At least 50 innovators participating actively in the ecosystem At least 6 new projects approved by GCF using taking on innovation and digital solutions promoted through the ecosystem	Assuming participation and interest of innovators and other stakeholders to participation in ecosystem
<i>All project activities should be listed here with a description and sub-activities. Significant deliverables should be reflected in the implementation timetable. Add rows as needed.</i>						
Activity	Description	Sub-activities		Deliverables		
Activity 1.1.1. Establish Green Financing Facility (revolving lending facility) - window 1 (adaptation)	This activity consists of establishing the green line of credit for adaptation within each selected LNAB	- Preparation of the ToRs - Hired consultants - Set up the line of credit		- Line of credit for adaptation established and functional		
Activity 1.1.2. Establish Green Financing Facility (revolving lending facility) - window 2 (mitigation)	This activity consists of establishing the green line of credit for mitigation within each selected LNAB	- Preparation of the ToRs - Hired consultants - Set up the line of credit		- Line of credit for mitigation established and functional		
Activity 1.1.3: Provide technical assistance to set up the Green Financing Facility with green credit line windows 1 and 2	This activity will provide technical support to set up each of the two lines of credit (windows 1 and 2)	- Preparation of the ToRs - Hired consultants - Set up the line of credit .1.1.1 and 1.1.2		- The two green lines of credit established		
Activity 2.1.1. Provide capacity building on understanding climate risks and climate resilient, low emission practices and integrating them into green business plans	The programme will support capacity-building on product on understanding climate risks and integrating them into green business plans	- ToRs prepared - Consultants hired and - Training material prepared - Provide training on understanding climate risks and integrating them into green business plans		- Smallholder farmers who are members of 1500 MSMEs and 2500 FOs, cooperatives, youth and women organizations trained on understanding climate risks and integrating them into green business plans in each country (Burkina Faso, Côte d'Ivoire, Mali, Ghana, Senegal,)		

Activity 2.1.2. Offer capacity building on bankable green business plan development	The programme will support capacity-building on green bankable business plan development	- ToRs prepared - Consultants hired and - Training material prepared - Provide training on green bankable business plan development	- Smallholder farmers members of 1500 MSMEs, 2500 FOs, cooperatives, youth and women organizations trained on bankable green business plan development in each of the country (Burkina Faso, Côte d'Ivoire, Mali, Ghana, Senegal,)
Activity 2.1.3. Provide training to FOs, MSMEs and cooperatives on how to implement financed green business plans for results	The programme will train FOs, MSMEs and cooperatives to implement financed business plans for results	- ToRs prepared - Consultants hired and - Training material prepared - Provide training on the implementation financed business plans for results	- Smallholder farmers members of 1500 MSMEs and 2500 FOs, cooperatives, youth and women organizations trained to implement financed green business plans for results in each of the country (Burkina Faso, Côte d'Ivoire, Ghana, Mali and Senegal,)
Activity 2.1.4. Support FOs and MSMEs led by women and youth to increase their participation, improve their financial literacy, natural resources management capacity, involvement in decision-making processes, ability to use labour-saving and cost-competitive CRLEA technologies and support for gender-related studies to guide decision-making	The programme will support FOs, MSMEs and cooperatives led by women and youth for increased participation, improved financial literacy, natural resources management, decision-making, labour saving and cost-competitive technologies for developing gardens.	- ToRs prepared - Consultants hired and - Training material prepared - Provide training on financial literacy, natural resources management, decision-making, labour saving and cost-competitive technologies for developing vegetable gardens	- Smallholder farmers members of 1500 MSMEs and 2500 FOs, cooperatives, youth and women organizations trained on financial literacy, natural resources management, decision-making, labour saving and cost-competitive technologies in each country (Burkina Faso, Côte d'Ivoire, Mali, Ghana, Senegal)
Activity 2.1.5. Provide technical assistance on green lending and climate risk management to LNABs	The programme will support technically green lending and climate risk management	- ToRs prepared - Consultants hired and - Training material prepared - Provide training on green lending and climate risk management	- Banks and financial institutions trained on green lending and climate risk management
Activity 2.1.6. Provide technical assistance to enhance governance and management of loan products and green credit lines policy gap and ensure use in each LNAB and IFI	The programme will provide technical support to enhance governance and management of loan products and green lines policy gap	- ToRs prepared - Consultants hired and - Training material prepared - Provide training on governance and management of loan products and green lines policy gap	- Banks and financial institutions trained and have access to new policy products and policy
Activity 2.1.7. Develop a package of tools and instruments, including ESMS, for LNABs to assess green business plans	The programme will support the development of tools and instruments	- ToRs prepared - Consultants/ firms hired and - Training package of tools and instruments developed	- Various tools and instruments developed to support greening of banks

Activity 2.1.8: Conduct awareness-raising and training on climate resilient, low emission agriculture	The programme will support the development of tools and instruments	- ToRs prepared - Consultants hired and - Training of banks/ financial institutions/ MFIs	- All, FOs, Cooperatives, MSMEs MFIs and banks trained
Activity 2.2.1 Provide technical support to Central Banks through the production of studies and the development of policies to improve the institutional and regulatory framework, standards and ESG benchmarks on climate risk management and green finance for agriculture.		-	-
Activity 2.2.2 Organize high-level round table events and exchanges on green finance	The programme will support the round table and events	- ToRs prepared - Consultants hired and - Events	- Events organized
Activity 2.2.3: Organize South-South and triangular cooperation visits and tours for experience sharing	The programme will support the organization of the South-South and triangular cooperation visit tour for experience sharing.	- ToRs prepared - Consultants hired and - Visits & Tours	- Visits tour exchanged
Activity 2.2.4: Enhance knowledge management and information sharing	The programme will support the strengthening of knowledge management and information sharing	- ToRs prepared - Consultants hired and - KM materials	- KM materials produced
Activity 2.2.5 Organize annual Sahel & GGW Award event	The programme will organize an annual Sahel Award event to celebrate the best projects	- standard tender document for procurements - Call for tender - Selection of best service providers	- 1 Sahel award event organized in each country in each country (Burkina Faso, Côte d'Ivoire, Ghana, Mali, Senegal)
Activity 2.2.6: Organize events to replenish the SAHEL & GGW Award	The programme will organize events to replenish the Sahel Award	- ToRs prepared - Resource mobilization meeting - Events	- At least 2 events organized
Activity 2.2.7: Develop application protocols and framework and organize events to replenish the Sahel and GGW Award	The programme will develop application protocols and framework and organize events to replenish the Sahel Award	- ToRs prepared - Consultants hired and - Training material prepared - application protocols and framework developed and the event organized	- 1 protocol and 1 framework developed for the replenishment event organized in each country (Burkina Faso, Côte d'Ivoire, Ghana, Mali, Senegal)

Activity 2.2.8: Enrol women and youth entrepreneurs to compete for the Sahel & GGW Award.	The programme will support the mobilization of women and youth entrepreneurs to compete for the Sahel Award.	- ToRs developed - Hiring of consultants - Women and youth sensitized and mobilized	- Events organized and 50% of women and 50 % youth mobilized under the Sahel Award
Activity 3.1.1: Establish a knowledge baseline	Stock taking of existing information and information gaps on climate interventions	- ToRs developed - Hiring of consultant -	- GGW knowledge baseline established
Activity 3.1.2 Develop a knowledge and communication strategy and plan	Stock taking of knowledge and communication needs of GCF projects and drafting of strategy and plan	- ToRs developed - Hiring of consultant - Consultation workshops in GCF projects and NDAs	- Communication strategy and plan established
Activity 3.1.3 Create a knowledge centre	Creation of an online knowledge centre to host GCF GGW RSP centre, repositories of knowledge products, blogs, group's etc. The Regional Support Programme will establish a structured and participatory learning programme in the region which will provide opportunities for knowledge exchanges among the 11 countries under the GCF Great Green Wall Umbrella Program throughout the programme's life cycle.	- ToRs developed - Hiring of vendor to design, build and implement knowledge centre - Establishing a regional web portal to support decision-making. Identifying and disseminating best practices in the region Holding regular South-South learning events, training, and periodic study tours - Supporting countries in mapping GCF projects and assessing their contributions towards NDC-NAP.	- Operational knowledge inside and outside the GCF Great Green Wall Umbrella Programme portfolio is regularly exchanged through a regional learning hub that network institutions and stakeholders are implementing 11 country investment operations. Operational knowledge centre
Activity 3.1.4. Create communities of practice	This activity will contribute to sharing information and experiences and raise awareness on the successful practices and pilots to be scaled up, for instance in agriculture and value chain development, as well as on respective needs from private sector investors and land	- - Workshops of Communities of practices at national and regional level	- Community of practice functional at country and regional level
Activity 3.1.5. Generate knowledge products to disseminate lessons learned and scalable good practices and inform evidence-based investment and policymaking	Take stock of best practices and lessons learned across GCF and GEF projects and prepare publications of best practices, lessons learned, policy publications, guidelines	- ToRs prepared - Consultants hired and - Report and publications prepared	- Lessons learnt, experiences, policy publications and guidelines produced and disseminated across the 11 countries
Activity 3.1.6. Organize peer-based knowledge	The programme will support the organization of: 1. Country level workshops for (i) awareness raising on KM, M&E, communications and	- Events & exchanges organized	- Peer-based knowledge exchanges organized

exchanges and knowledge sharing activities	advocacy, staffing plans and overall strategies of GCF projects (ii) training on thematic issues; 2. Regional knowledge fairs, and 3. South/South exchanges		
Activity 3.1.7. Enhance coordination and programming among GCF NDAs, AEs and GEF, AF and GGW national focal points	The programme will support coordination & programming between GCF NDAs, AEs and GEF, AF and GGW national focal points	- Meetings of Country coordination committees	- Work programmes coordinated at country level
Activity 3.2.1 Build a digital and innovation ecosystem	The programme will support the creation of an innovation and digital solutions ecosystems through stock taking, providing TA and organizing annual fairs to bring innovators together,	- ToRs prepared - Consultants hired and - Report and publications prepared - Events organized	- Ecosystem built and functional
Activity 3.2.2 Create a digital marketplace	A marketplace created as a one-stop-shop to find all innovation and digital solutions that can help to advance climate mitigation and adaptation, and in particular value chains	- ToRs developed - Hiring of vendor to design, build and implement the marketplace	- Marketplace available to all GCF projects as well as all stakeholders identified as key players in the ecosystem
Activity 3.2.3 Identify new opportunities for investing in climate innovation in the GGW	Mapping of new investment opportunities in innovation and digital transformation.	- ToRs developed - Consultant hired	- Opportunities for investing taken up by GCF projects

E.7. Monitoring, reporting and evaluation arrangements (max. 500 words, approximately 1 page)

248. **Reporting from IFAD to GCF:** Monitoring, reporting and evaluation arrangements will comply with IFAD's Accreditation Master Agreement and GCF policies, as well as the Funded Activity Agreement (FAA). IFAD through the RCU will provide annual progress reports on the status of the funded activity throughout the relevant reporting period, based on the above-described logical framework and reporting from IFAD baseline investments to IFAD RCU. The RCU will consolidate above described annual, mid-term and final reports from each country IGREENFIN I project submitted by IFAD baseline investments and send them to the GCF within additional thirty (30) days of the above-mentioned reporting periods by the IFAD. IFAD will deliver numbers in a log frame format: a) annually within sixty (60) calendar days after the end of each year of programme implementation; b) at mid-term within thirty (30) months from the effective date of the loan contract or when 50 per cent of the programme resources have been disbursed, whichever occurs first, and c) at programme end up to six (6) months after the day of the last disbursement to a participating country and its banks.

249. **IGREENFIN I monitoring and reporting:** IGREENFIN I will apply the standard procedures established by IFAD for monitoring and evaluating investment operations, including co-financing (IFAD, ISDB, and AfDB). The IFAD RCU will compile and maintain all information, indicators and parameters necessary for the preparation of programme reports, including annual reports, midterm review and final evaluation to be submitted to the GCF. The agricultural banks and selected partners (commercial banks, MFIs) will be responsible for assessing beneficiaries' business plans in accordance with the programme's eligibility criteria, as defined in the Annex 23: GCF Investment Criteria for sub loans. Business plan approved by the selected financial institutions must be properly identified in the banks' accounting systems and be in compliance with what is stated in the specific loan agreements. These records should allow for identifying financial conditions of each transaction (e.g., currency, maturity and interest rates), the value of the contract, loan proceeds and eventual use of proceeds for monitoring purposes, programme funds balances and default rates, if necessary. The monitoring process intends to follow up on the execution of the programme in order to identify the intermediate milestones achieved in each phase and evaluate its outcomes and fulfilment of proposed targets. The indicators to be monitored are included in the log frame in section E.

250. **ESG reporting:** The IFAD RCU will receive an annual report on the ESG performance of the national agricultural banks. The PIU teams will take inputs from relevant teams and compile the annual report for each applicable reporting period. Annual climate impact reporting will be performed by the banks, while the independent mid-term and final year verification of climate impact will be undertaken by an independent provider.

251. **GHG reporting:** ICRAF, a strategic partner of IGREENFIN I, and the Regional Support Programme will be in charge of the reporting on GHG emissions in line with the methodology in the EXCAT Annex.

Regional Support Programme impact assessments:

252. Impact evaluations are a rigorous tool to assess the contribution and impact of programmes and to improve the next-generation GCF programmes that can benefit from the best evidence-based design options available. The impact assessment of the Regional Support Programme will focus on the 11 countries. It will take an ex-ante/ex-post approach to assess the contribution of the Regional Support Programme's activities to the paradigm shift. It will entail the development of a questionnaire to define the outcome variables before the Regional Support Programme starts. A baseline will be established at the beginning of the implementation of the Regional Support Programme and the same questionnaire will be submitted upon completion to the beneficiaries, namely the 11 countries' NDAs and GGW Focal Points as well as the Accredited Entities.

Monitoring and evaluation in the context of the GCF GGW RSP

253. The Regional Support Programme will not substitute the AEs during the M&E process. Each AE remains accountable for the elaboration and submission of the Annual Performance Reports of its own projects to the GCF Secretariat. FAO will provide additional support through the SURAGWA project to the AEs and the NDAs to help them understand which means of verifications will be used by the GGW Accelerator, as they can also be used by the individual GCF projects directly.

F. ISK ASSESSMENT AND MANAGEMENT

F.1. Risk factors and mitigations measures (max. 3 pages)

Please describe financial, technical, operational, macroeconomic/political, money laundering/terrorist financing (ML/TF), sanctions, prohibited practices, and other risks that might prevent the project/programme objectives from being achieved. Also describe the proposed risk mitigation measures. Insert additional rows if necessary.

254. The programme may face risks in several categories: macroeconomic, implementation, financial, technical, environmental and social. The levels of these risks are considered moderate to low and expected to be mitigated by IFAD and partners' operational tools and control mechanisms. IFAD will regularly evaluate the overall potential risks related to the implementation of the programme and provide recommendations for procedures to mitigate and address the risks.

255. It is foreseen that the programme will be affected by the COVID-19 context, which could cause delays in the start-up and generate operational risks during implementation if there is third wave of COVID-19 in the target countries in 2021. IFAD has prepared a comprehensive COVID-19 risks management approach, which includes the setup of the COVID-19 Rural Poor Stimulus Facility to support countries in addressing the impact of COVID-19; guidelines for remote and tele-supervision, design and midterm review; repurposing baseline investments with a set of activities such as training on safe labour practices and transport; access to protective equipment such as masks and gloves, restrictions on workers in producers' field, use of drones and other digital extension tools for labour and input-saving practices; shared mechanization; digital marketing platforms and logistics, sanitary and phytosanitary controls; good practices in relation to the gender dimension of the COVID-19 pandemic to reduce women's exposure to violence; increased access to finance, agri-service centres for inputs; distribution of inputs (seed, fertilizer, forage/fodder saplings, fingerlings, vaccines, medicines); increasing allocations to debt redemption fund for fishers and farmers; outsourcing the affected activities to local institutions and NGOs, including executing partners such as FAO and international NGOs.

Selected Risk Factor 1

Category	Probability	Impact
CreditCreditCreditCredit	MediumMediumMediumMedium	MediumMediumMediumMedium
Description		

256. Credit default and end beneficiaries' failure to repay the loans and meet their contractual obligations. Credits risks are inherent to banking activities and can pose a challenge to banks performance.

Mitigation Measure(s)

257. The credits are administrated by credible financial institutions with excellent knowledge of their customers and the market within which they operate. IFAD has assessed each bank's financial management capacities. The selected banks internal system will ensure a proper assessment of loans based on investment criteria, including creditworthiness and other qualitative criteria deemed necessary. The bank policy and criteria for credit assessments and measures will safeguard and limit the risks. In addition, the GCF proceeds to support the loans programme are all guaranteed by the governments of the five participating countries that will act as a borrower of last resort.

Selected Risk Factor 2

Category	Probability	Impact
ForexForexForexForex	LowLowLowLow	LowLowLowLow

Description

258. Recipient countries experience currency devaluation or currency fluctuation linked to external shocks. Foreign exchange (FX) risk (or currency risk) exists in financial transactions under Component 1, as IFAD loans are denominated in Euros for the WAEMU zone and loans to final borrowers are likely to be disbursed in a currency other than USD (i.e., the local currency in each country), especially in the case of smaller firms for which transactions occur mainly in local currency. Thus, FX risk arises from the possibility of unfavourable fluctuations between the local currency and the Euros before the loan repayment is completed, as the associated value may decrease due to changes in the relative value of the involved currencies. The XOF currency is pegged to Euros and there won't be any loss. However, for Ghana which has its own currency called Ghana Cedis, the country is generally prone to undergo sporadic unstable economic conditions, but NEEs generally have the capacity to hedge this risk (i.e., place Euros in the market and manage profits to increase their capacity to offer concessional terms in local currency).

Mitigation Measure(s)

259. For Burkina Faso, Côte d'Ivoire, Mali and Senegal: These countries are part of the West African Economic and Monetary Union (WAEMU) which uses the FCFA as their currency (Ghana is the exception and will require more specific measures, as explained below). Owing to the FCFA's fixed exchange rate to the euro, which is currently established at EUR 1 = CFA 655.957, there is no immediate (short-term) foreign exchange risks for these four countries as the GCF proceeds will be denominated in euros. The latest devaluation occurred in 1994, which illustrates the stability of the FCFA, which is also supported by the good performance of WAEMU member's states. Devaluation or re-evaluation of the FCFA in the long term is a possibility in view of the ongoing discussion on the reform of the FCFA. If such risks should materialize, the WAEMU zone is well-known as a strong monetary zone with its members' countries positive GDP growth projections, which have, however, been affected by the COVID-19 pandemic. WAEMU monetary performance is supported by its strong monetary policy, liquidity buffer and continuous fiscal and macroeconomic consolidation efforts by member countries.

260. Ghana: The grant proceeds will be allocated to IFAD which has strong credit risk management, including hedging strategies to reduce exchange rate risks. In addition, Ghana is hedged against forex risk. The Bank of Ghana regulates the conduct of the interbank foreign exchange business in the Ghanaian financial market and establishes and regularly updates standards of practice expected of market participants to help ensure an efficient and effective FX market with the view to continue to develop and deepen the foreign exchange market in Ghana. The Bank of Ghana publish in 2019 its latest interbank forex market conduct and guidelines. As Green business exposed to FX risk, hedging strategies to mitigate it usually involve mechanisms such as forward contracts, options and others, which can protect them from exchange rate fluctuations. Capacity building and proper implementation of such mechanisms will be part of the TC activities under Component

Selected Risk Factor 3

Category	Probability	Impact
GovernanceGovernanceGovernanceGovernance	MediumMediumMediumMedium	MediumMediumMediumMedium

Description		
261. Governance systems remain weak and can pose a key challenge in all the countries included in this proposal.		
Mitigation Measure(s)		
262. IFAD's presence and engagement at the country level, the use of IFAD PMU and risk management framework, policy dialogue and partnerships will help countries manage this risk. The programme is designed grounded on a robust management arrangement using existing IFAD and countries project governance management systems. Additionally, the programme will establish a regional coordination programme to support its overall implementation and good performance. During its lifecycle, measures will be taken to ensure that the outcomes are sustained beyond the programme lifecycle. With regards to corruption, the programme will implement an absolute zero tolerance to corruption in accordance with IFAD's internal policy; regular audits and supervision missions will be undertaken.		
Selected Risk Factor 4		
Category	Probability	Impact
Technical and operational Technical and operational Technical and operational and operational	MediumMediumMediumMedium	MediumMediumMediumMedium
Description		
263. Sustainable finance practices and integration of climate risks management into banking operations is new for all the countries involved in this programme. Moreover, neither the banks, their clients nor regulators are accustomed to the new products to be developed under this programme.		
Mitigation Measure(s)		
264. The programme results from a demand-driven process. IFAD has initiated this programme upon confirmation of the interest and willingness of national development banks, smallholder farmers and governments during the design phase. The consultation processes occurred as part of the design to understand the need of the beneficiaries. Further consultations and policy dialogue interventions are planned during the programme implementation phase. The implementation of a participatory and inclusive approach will facilitate social mobilization and help ensure the adherence of the beneficiaries. Policy dialogue between public and private actors will be promoted to mitigate the risks.		
265. Proper due diligence process led by IFAD and the national executing entities, the local FIs' financial experts, technical experts from the ministries of agriculture and the oversight of the respective ministries of finance and regulators (the two central banks – Western African Countries Central Bank and the Central Bank of Ghana) will all contribute to ensuring robust engineering, procurement, and governance systems. Additionally, the agreements between the recipient governments and their respective relevant sector ministries are to ensure that they better manage the loans during and beyond the project lifecycle. Programme operation risk will be mitigated by a thorough due diligence process led by IFAD technical and financial experts and by ensuring a robust M&E system is set up and fully operational. TA agreements and procurement procedures will be established at each EE level to support and ensure the high delivery quality outcomes.		
266. The IFAD grievance mechanism will be used for the reporting of complaints or allegations of wrongdoing within the projects or activities (whistle blower programme). The description of the IFAD grievance mechanism is presented in the ESMF. The programme will establish a community engagement process and provide access to information on a regular basis. The objective of the IFAD Complaints Procedure is to ensure that appropriate mechanisms are in place to allow individuals and communities to contact IFAD directly and file a complaint if they believe they are or might be adversely affected by an IFAD-funded project/programme that does not comply with IFAD's Social and Environmental Policies and mandatory aspects of SECAP. Complaints must concern environmental, social and climate issues only and should not be accusations of fraudulent or corrupt activities in relation to programme implementation; these are dealt with by IFAD's Office of Audit and Oversight.		
267. Eligibility criteria according to IFAD's grievance mechanism: To file a complaint for alleged non-compliance with IFAD's social and environmental policies and mandatory aspects of its SECAP, IFAD will consider only complaints meeting the following criteria:		

- The complainants claim that IFAD has failed to apply its social and environmental policies and/or the mandatory provisions set out in SECAP and Safeguards.
- The complainants claim that they have been or will be adversely affected by IFAD's failure to apply these policies.
- Complaints must be put forward by at least two people who are both nationals of the country concerned and/or living in the programme area. Complaints from foreign locations or anonymous complaints will not be taken into account.
- Complaints must concern projects/programmes currently under design or implementation. Complaints concerning closed projects, or those that are more than 95 per cent disbursed, will not be considered.

268. The process according to IFAD's grievance mechanism:

Complainants should first bring the matter to the attention of the government or non-governmental organisation responsible for planning or executing the project or programme (ministry of agriculture implementing agency and the ministry of economy and finance and the environmental protection agencies involved in this programme that have with the responsibility for overseeing the work on the field. If the Implementing Agency does not adequately respond, then the matter may be brought to the attention of IFAD. The issue may be brought straight to IFAD if the complainants feel they might be subject to retaliation if they went to the Lead Agency directly.

How to submit a complaint:

A complaint relating to non-compliance with IFAD's Social and Environmental Policies and mandatory aspects of its SECAP can be submitted in any of the following ways:

- Download the complaints form (Word) through IFAD website: <https://www.ifad.org/en/accountability-and-complaints-procedures>
- Send an email to SECAPcomplaints@ifad.org

In addition, GCF sub projects will as much as possible utilize every available grievance redress mechanism including associations (including farmers' associations/organizations), traditional council (Paramount Chiefs and elders), village square engagement (consisting of representatives of men, women, and social groups), village general assembly, the project NCPU, etc.

Selected Risk Factor 5

Category	Probability	Impact
OtherOther	MediumMedium	MediumMedium
Description		

269. Political conflict and insecurity strongly affect some countries in this proposal.

Mitigation Measure(s)

270. Countries facing conflicts are Mali and Burkina Faso. The regions selected for the programme are secured sites in the countries concerned. However, each country will propose an action plan for the safety of the intervention sites that will be considered during the evaluation mission. Mali is undergoing a political transition process.

Selected Risk Factor 6

Category	Probability	Impact
OtherOther	MediumMedium	MediumMedium
Description		

271. COVID-19 effects on countries socioeconomic situation and outlook limit their fiscal space further and may hinder their ability to repay the debt.

Mitigation Measure(s)

272. All countries are affected by COVID-19 and have put into place some specific measures to address the impacts of the pandemic. These measures include economic recovery plans, repurposing existing loan investments and requesting more concessional resources such as grants, among others.

Selected Risk Factor 7

Category	Probability	Impact
OtherOtherOtherOther	MediumMedium	MediumMedium
Description		
273. Regulatory risks, including risks of changes in regulations or legislation, could affect the programmes or private actors' engagement in the programme in various ways, including with contracts.		
Mitigation Measure(s)		
274. In the selected countries, regulatory risks are low to medium and the programme proposes a series of activities under component 1 and component 3 to address potential regulatory risks. Part of GCF and IFAD financing will be in the form of a grant to the selected countries. The EEs will use their legal frameworks and internal policies to ensure that the activities they are in charge of are implemented within the programme timeline while supporting countries to enhance their regulatory frameworks to support proper programme implementation.		

Selected Risk Factor 8

Category	Probability	Impact
ML/FTML/FT	LowLow	MediumMedium
Description		
275. Fraud and other deceptive practices may be committed by people in senior management positions at NEEs or LNABs portfolio. Banks and clients may undermine due diligence efforts when such fraud is discovered, negatively affecting the programme's implementation and evaluation. National financial systems are subject to country compliance with the international standards of the Anti-Money Laundering and Combating the Financing of Terrorism policy, embedded in national policymaking of all target countries. As lending operations being considered in the context of this Programme will be part of financial intermediation schemes in local and international markets, a very low risk is foreseen on having unidentified financing of these practices		
Mitigation Measure(s)		
276. Policies and procedures on compliance, AML, etc. are in place to avoid these risks. Strong risk management and governance structures are followed by the banks, which are the LNABs and FIs All banks are regulated by the Central banks: the Central Bank of the WAEMU for Burkina Faso, Côte d'Ivoire, Mali and Senegal and for Ghana, the Central Bank of Ghana. Because all financial operations under Programme activities are carried out in the context of the financial and capital markets, subject to national regulation, this risk is considered low. Nonetheless, the IFAD will not only establish specific rules and eligibility criteria for all financial transactions in each target country (as part of IFAD project-cycle), but all portfolios financed will be subject to IFAD's non-objection through NOTUS online tool		

G.1. Environmental and social risk assessment (max. 750 words, approximately 1.5 pages)

277. IFAD has a strong working relationship with LNABs across Africa and works together with them and project developers to overcome key barriers to financing sustainable investments. IFAD SECAP is the key tool used to promote and enhance the management of environmental and social impacts in the financial sector, transfer knowledge and promote good practices in environmental and social governance (ESG) to strengthen the adoption of ESG principles by rural institutions. Such experience has been instrumental for the design and elaboration of the proposed programme and its ESMF, ensuring that appropriate and robust procedures and safeguards are in place for implementing climate change projects in the energy and land and forest use sectors. The ESMF and screening of subprojects by the LNABs will ensure that comprehensive and robust environmental and social screening and management systems are in place, and programme activities and investments and their social and environmental impacts are closely monitored, evaluated and reported on.
278. During the due diligence phase of the LNABs, IFAD will assess the ESMS of each institution and ensure that it is aligned with the programme's ESMF, including the Environmental and Social Safeguards of the banks and the GCF. Participating LNABs will receive support to develop effective environmental and social policies, assess environmental and social risks and categorize operations while evaluating the environmental and social risk of their portfolios. The measures to be applied are:
- Verify that the operations being financed are not prohibited by law.
 - Verify that the operations being financed are not included in the programme's List of exclusions in the ESMF.
 - Enforce compliance with national legislation related to permits and environmental licenses, use of natural resources and activities with dangerous substances.
 - Categorize and evaluate the environmental and social risk of the operations to be financed.
 - Verify that the amounts requested by their clients include the implementation of the measures to minimize the environmental and social risks of their operations.
 - Present reports on the environmental and social management of its operations according to the frequency established in the credit agreement, which will be fixed according to the risks of its portfolio and the magnitude of the credit granted by the LNABs or its partners.
279. When ESMS systems are either not in place or are weak, IGREENFIN I's TAF will provide technical assistance to help establish or improve such systems and, where possible, support LNABs in improving their ESMS to be able to participate in the programme. All sub loans will be screened by LNABs for their environmental and social risks and impacts prior to credit approval. During this screening phase, projects will be revised according to the LNABs Environmental and Social Risk Analysis and Management System to ensure they adhere to local regulations and laws, including ESIA requirements. The programme will not finance category A or high-risk projects, nor any investments described in the programme exclusion list in annex 23 of this proposal. The programme will be limited to category B projects. As such, it is qualified as I-2. Where necessary, environmental, and social management plans (ESMPs) will be either in place or developed at the final loan beneficiary level to manage identified risks and adverse impacts.

G.2. Gender assessment and action plan (max. 500 words, approximately 1 page)

280. IGREENFIN I will promote gender equality and women's empowerment both at the fund- and project-level. As presented in the Gender Analysis and Action Plan, IGREENFIN I will support the use of gender-disaggregated indicator systems, requiring participating LNABs (agricultural banks), commercial banks and MFIs to collect and report on gender-disaggregated data. IGREENFIN I proposes an action plan for all banks which help them integrate or strengthen their M&E systems to ensure that such gender-disaggregated data is collected. Of the loans granted by the Green Financing Facility, 50 per cent are to go to women-led and youth-led MSMEs, FOs and cooperatives. Under Output 2.3, IGREENFIN I will incentive and promote the participation of women-led MSMEs and FOs in capacity-building and technical assistance activity. Furthermore, IGREENFIN I will support awareness-raising campaigns and set clear gender-sensitive targeting strategies. IGREENFIN I will tackle many barriers to women's access to financial services and credit, including gendered division of labour, property rights and control over assets; lack or no awareness of financing opportunities and new instruments like green financing; lack of access to knowledge, particularly reliable climate information, and biased perception of the financial sector. Key activities to be financed by the green sub loans are concrete measures to reduce women's workloads, agri-preneurship support for women's organizations, etc. Working with men is critical, as they are often the gatekeepers of customary practices that limit women's access to resources or public spaces. Highly participatory approaches and cultural sensitivity will be promoted. Finally, IGREENFIN I will support financial education through the design and promotion of training programmes that contain specific strategies aimed at increasing the participation of female personnel and/or women-led MSMEs, cooperatives and FOs within the programme.

G.3. Financial management and procurement (max. 500 words, approximately 1 page) FMD

BUDGETING

281. The countries involved in IGREENFIN I, represented by their NEEs, will be recipients of the IGREENFIN I loan and grant proceeds and guarantee the repayment of the GCF loan portion. The NEEs will sign the relevant on lending agreements with each of the LNABs to implement the loan and grant proceeds.

IFAD RCU:

282. Each year, the IFAD RCU will communicate the budget timetable and templates to be used for submission to the Baseline PMUs, which must comply with GCF guidelines. The IFAD RCU will emphasize that the IGREENFIN I overhead costs should be disclosed as a separate component of budget and will be funded through the IGREENFIN I grant proceeds.

283. Upon receipt of the country's IGREENFIN I budget from each Baseline PMU, the IFAD RCU will review and ensure that each budget line item includes a note / description of underlying assumptions, harmonize the budgets and coordinate the obtention of IFAD non-objection letter for each country's IGREENFIN I budget. As soon as the non-objection letters are obtained from IFAD, the IFAD RCU will consolidate them into a single IGREENFIN I budget, ensuring that the IFAD RCU's grant budget is incorporated prior to submission to the GCF.

Baseline PMU:

284. The Baseline PMU staff will closely collaborate with the bank PIUs throughout the AWPB preparation process, from inception to reception of IFAD's non-objection letter. This is to ensure that IFAD/GCF budget guidelines are followed. Once the final draft budget of the bank PIU is submitted to the Baseline PMU, the Baseline PMU will check that it is multi-dimensional and complies with both budget guidelines and that the appropriate budget templates have been used in the budget submitted. Once the Baseline PMU is satisfied with the content and presentation of the Bank's IGREENFIN I budget, it will incorporate its portion of the grant budget into the budget and submit the draft budget to the RCU for comments and review before convening the NSC for review and approval, following its normal AWPB timetable. Once the NSC approves the budget, it is officially submitted to IFAD for non-objection.

BANK PIU:

285. The bank PIU will prepare the IGREENFIN I budget while paying attention to the milestones in the timetable communicated by the Baseline PMU. Most importantly, they must submit the final draft to the baseline PMU no later than 1 October every year and use prescribe templates. The budget will be broken down by donor, component / subcomponent / activity, category / sub-category of expenditure, zone / subzone, etc. The loan component of the budget will be drawn up from the forecast activities of target beneficiaries and RFIs while the grant component is drawn from the indirect costs required to operate the scheme.

ACCOUNTING STANDARDS AND FINANCIAL REPORTING:

286. Each EE will use its own accounting standards though its external auditors. ToRs must clearly state that they will prepare IGREENFIN I financial statements in accordance with IFRS. Refer to Section 6 of annex 22 for list of Financial Reports that the IFAD RCU will submit to the GCF, each Baseline PMU to the IFAD RCU, and each Bank PIU to their respective Baseline PMU. The role of the IFAD RCU Financial Unit is to harmonize, compile and consolidate country financial reports for submission to GCF. The baseline project will review and ensure that the format and content of Bank PIU financial reports comply with reporting guidelines prior to submission to IFAD RCU. Bank PIUs will ensure that they can generate the mandated financial reports in the format required by the Baseline PMU using their Financial Reporting System. All financial information related to IGREENFIN I must be expressed in the currency stipulated in the IFAD Financing Agreement, namely US dollars.

287. **Procurement:** Procurement for the programme will be carried out in accordance with IFAD applicable rules and practices, as well as the IFAD's Programme Implementation Manual (PIM), the programme specific PIM and loan administration arrangements established through the Letter to the Borrower and the Financing Agreement between IFAD and each National Borrower/Recipient.

G.4. Disclosure of funding proposal

Indicate below whether or not the funding proposal includes confidential information.

☒ **No confidential information:** The accredited entity confirms that the funding proposal, including its annexes, may be disclosed in full by the GCF, as no information is being provided in confidence.

☐ With confidential information: The accredited entity declares that the funding proposal, including its annexes, may not be disclosed in full by the GCF, as certain information is being provided in confidence. Accordingly, the accredited entity is providing to the Secretariat the following two copies of the funding proposal, including all annexes:

- ☐ full copy for internal use of the GCF in which the confidential portions are marked accordingly, together with an explanatory note regarding the said portions and the corresponding reason for confidentiality under the accredited entity's disclosure policy, and
- ☐ redacted copy for disclosure on the GCF website.

The funding proposal can only be processed upon receipt of the two copies above, if containing confidential information.

H. ANNEXES

H.1. Mandatory annexes

- ☐ Annex 1 NDA no-objection letter(s) **(template provided)**
- ☐ Annex 2 Feasibility study - and a market study, if applicable
- ☐ Annex 3 Economic and/or financial analyses in spreadsheet format
- ☐ Annex 4 Detailed budget plan **(template provided)**
- ☐ Annex 5 Implementation timetable including key project/programme milestones **(template provided)**
- ☐ Annex 6 E&S document corresponding to the E&S category (A, B or C; or I1, I2 or I3):
(ESS disclosure form provided)
 - ☐ Environmental and Social Impact Assessment (ESIA) or
 - ☐ Environmental and Social Management Plan (ESMP) or
 - ☐ Environmental and Social Management System (ESMS)
 - ☐ Others (please specify – e.g. Resettlement Action Plan, Resettlement Policy Framework, Indigenous People's Plan, Land Acquisition Plan, etc.)
- ☐ Annex 7 Summary of consultations and stakeholder engagement plan
- ☐ Annex 8 Gender assessment and project/programme-level action plan **(template provided)**
- ☐ Annex 9 Legal due diligence (regulation, taxation, and insurance)
- ☐ Annex 10 Procurement plan **(template provided)**
- ☐ Annex 11 Monitoring and evaluation plan **(template provided)**
- ☐ Annex 12 AE fee request **(template provided)**
- ☐ Annex 13 Co-financing commitment letter, if applicable **(template provided)**
- ☐ Annex 14 Term sheet including a detailed disbursement schedule and, if applicable, repayment schedule

H.2. Other annexes as applicable

- ☐ Annex 15 Evidence of internal approval **(template provided)**
- ☐ Annex 16 Map(s) indicating the location of proposed interventions
- ☐ Annex 17 Multi-country project/programme information **(template provided)**
- ☐ Annex 18 Appraisal, due diligence, or evaluation report for proposals based on up-scaling or replicating a pilot project
- ☐ Annex 19 Procedures for controlling procurement by third parties or executing entities undertaking projects financed by the entity
- ☐ Annex 20 First level AML/CFT (KYC) assessment
- ☐ Annex 21 Operations manual (Operations and maintenance)
- ☐ Annex x Other references

** Please note that a funding proposal will be considered complete only upon receipt of all the applicable supporting documents.*

No-objection letter template for projects



DIRECTION GÉNÉRALE DE
L'ENVIRONNEMENT

DIRECTION DE LA LUTTE CONTRE LES
CHANGEMENTS CLIMATIQUES



N° 00617 MINEDD/CAB1/DGE/DLCC/kkp

To: The Green Climate Fund ("GCF")

Abidjan, 17 May 2021

Re: Funding proposal for the GCF by The International Fund For Agricultural Development (IFAD) regarding "Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I"

Dear Yannick G Lemareck,

Executive Director of the Green Climate Fund,

We refer to **"Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I"** in Côte d'Ivoire as included in the funding proposal submitted by IFAD to us on 29 April 2020.

The undersigned is the duly authorized representative of the National Designated Authority/Focal point of the republic of Côte d'Ivoire.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to **Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I** as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of The Republic of Senegal has no-objection to **"Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in 13 countries of the Green Great Wall (GGW). Phase I"** as included in the funding proposal
- (b) The **"Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I"** as included in the funding proposal is in conformity with The Côte d'Ivoire's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, The **"Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial**

Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I” as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the **“Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I”** as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Mr. Kouadio Kumassi Philippe
National Designated Authority
Cote d'Ivoire





THE STATE OF ERITREA
Ministry of Land, Water & Environment
Department of Environment

ዕለት 23/06/2021
Date
ቁ. መዝገብ DO E/01/28/21
Ref. No.

التاريخ
رقم السجل

To: The Green Climate Fund ("GCF")

Re: Funding proposal for the GCF by International Fund for Agricultural Development (IFAD) regarding "Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission, Smallholder Agriculture in the Great Green Wall (GGW) countries. Phase I"

Dear Yannick Glemareck,
Executive Director of the Green Climate Fund,

We refer to the programme titled Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission, Smallholder Agriculture in the Great Green Wall (GGW) countries Phase I in Eritrea as included in the funding proposal submitted by IFAD to us on 25 May 2021.

The undersigned is the duly authorized representative of Ministry of Land, Water and Environment, the National Designated Authority of Eritrea.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

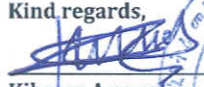
- The government of Eritrea has no-objection to the "Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in 13 countries of the Great Green Wall (GGW) Phase I" as included in the funding proposal;
- The "Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Great Green Wall (GGW) Phase I" as included in the funding proposal is in conformity with the national priorities, strategies and plans of Eritrea;
- In accordance with the GCF's environmental and social safeguards, the "Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Great Green Wall (GGW) Phase I" as included in the funding proposal is in conformity with relevant national laws and regulations.

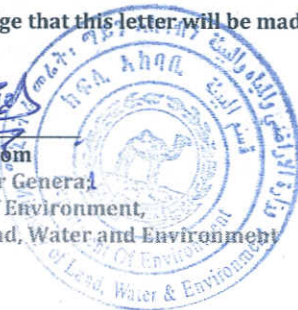
We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,


Kibrom Asmerom
Acting Director General
Department of Environment,
Ministry of Land, Water and Environment
Eritrea



05 JUL 2021



To: The Executive Director

Green Climate Fund ("GCF")

Re: Funding proposal for the GCF by The International Fund For Agricultural Development (IFAD) regarding "Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I"

Dear Madam, Sir,

We refer to **"Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I"** in Djibouti as included in the funding proposal submitted by IFAD to us on 25 May 2021.

The undersigned is the duly authorized representative of the National Designated Authority/Focal point of Djibouti.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to **Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I** as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Djibouti has no-objection to **"Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in 13 countries of the Green Great Wall (GGW). Phase I"** as included in the funding proposal;

- (b) The **"Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I"** as included in the funding proposal is in conformity with Djibouti's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, The **"Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I"** as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the **"Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I"** as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Mr. Dini Abdallah Omar

Secretary General, Ministry of Environment and Sustainable Development

Djibouti

GCF-Focal Point-Djibouti



PRIMATURE

SECRETARIAT GENERAL

**SECRETARIAT EXECUTIF DU FONDS VERT
POUR LE CLIMAT AU BURKINA FASO**

BURKINA FASO
Unité-Progrès-Justice

Ouagadougou, May 18, 2021

National Designed Authority

N° **2021-002** /PM/SG/SE-FVC/BF

To

***The Green Climate Fund
("GCF")***

Republic of Korea

Re: Funding proposal for the GCF by The International Fund for Agricultural Development (IFAD) regarding **"Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I"**

**Dear Yannick G Lemareck,
Executive Director of the Green Climate Fund,**

We refer to **"Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I"** in Burkina Faso as included in the funding proposal submitted by IFAD to us on 29 April 2020.

The undersigned is the duly authorized National Designated Authority of Burkina Faso.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to **"Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in**

the **Green Great Wall (GGW) countries. Phase I**” as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of The Republic of Burkina Faso has no-objection to **“Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I”** as included in the funding proposal;
- (b) The **“Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I”** as included in the funding proposal is in conformity with The Burkina Faso’s national priorities, strategies and plans;
- (c) In accordance with the GCF’s environmental and social safeguards, the **“Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I”** as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the **“Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I”** as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards.



Issaka OUEDRAOGO

*Advisor to the Prime Minister
Inspector of Environment*



REPUBLIC OF GHANA

MINISTRY OF FINANCE

*In case of reply, the
number and date of this
letter should be quoted*

Our Ref: **MOF/ESRD/NREC/GCF/06/21**

Your Ref:

Tel No:

16th
JUNE, 2021

Dear Mr. Yannick Glemarec

**RE: INCLUSIVE GREEN FINANCING INITIATIVE(IGREENFIN): GREENING
AGRICULTURAL BANKS & FINANCIAL SECTOR TO FOSTER CLIMATE RESILIENT,
LOW EMISSION SMALLHOLDER AGRICULTURE IN FIVE WEST AFRICAN
COUNTRIES OF THE GREEN GREAT WALL INITIATIVE (GGWI), PHASE I PROJECT**

We refer to the **"Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in five West African Countries of the Green Great Wall Initiative (GGWI), Phase I project"** as included in the funding proposal submitted by IFAD to the NDA, Ghana, on 14th May, 2021.

2. The undersigned is the duly authorized representative of the Ministry of Finance the National Designated Authority/focal point of the Republic of Ghana.

3. Pursuant to GCF decision B.08/10, the content of which we acknowledge to have been reviewed, we hereby communicate our no-objection to the IGREENFIN, Phase I project as included in the funding proposal.

4. By communicating our no-objection, it is implied that:

- (a) The government of the Republic of Ghana has no-objection to the IGREENFIN, Phase I project as included in the funding proposal;
- (b) The IGREENFIN, Phase I project as included in the funding proposal is in conformity with the Republic of Ghana's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, the IGREENFIN, Phase I project as included in the funding proposal is in conformity with relevant national laws and regulations.

5. In addition, we confirm that our national process for ascertaining no-objection to the IGREENFIN, Phase I project as included in the funding proposal has been duly followed.

6. We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.



7. We acknowledge that this letter will be made publicly available on the GCF website.



DR. ALHASSAN IDDRISU
DIRECTOR, ESRD/NDA, MOF
FOR: MINISTER

THE EXECUTIVE DIRECTOR
GREEN CLIMATE FUND ("GCF")
G-TOWER, 24-4 SONGDO-DONG,
YEONSU-GU INCHEON CITY,
REPUBLIC OF KOREA.

Cc: The Hon. Minister, MoF
The Senior Policy Advisors, MoF
The Chief Director, MoF
The Regional Director, W&CA,

IFAD

N'Djamena, le 23 juin 2021

N° 007/MEPDD/PNFFVC/2021

**The Focal Point****To****Mr Yannick Glemarec****Director Executive of Green Climate Fund, G-Tower, 24-4 Songdo dong, Yeonsu-gu, Incheon,
City Republic of Korea**

Re: Funding proposal for the GCF by The International Fund For Agricultural Development (IFAD) regarding "Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I"

Dear Yannick Glemarec,

We refer to "Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I" included in the funding proposal submitted by IFAD to us on 10 June 2021.

The undersigned is the duly authorized representative of the National Designated Authority/Focal point of the Republic of Chad

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I" as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of The Republic of Chad has no-objection to "Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I" as included in the funding proposal;
- (b) The "Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I" as included in the funding proposal is in conformity with The Chad's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, The "Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I" as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the "Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I" as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Mr. IBRAHIM CHARFADINE ABDELKERIM

FP/ National Designated Authority of Chad for the Green Climate Fund



وزارة البيئة والتنمية المستدامة
Ministère de l'Environnement
et du Développement Durable

مديرية المناخ والاقتصاد الأخضر
Direction climat et Economie Verte

Le Directeur

المدير

Nouakchott, le : 22 JUIN 2021
N° : 000010
انواكشوط، في
الرقم

To: The Green Climate Fund ("GCF")

Re: Funding proposal for the GCF by The International Fund For Agricultural Development (IFAD) regarding "Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I"

Dear Yannick GLemareck,

Executive Director of the Green Climate Fund,

We refer to "Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I" in Mauritania as included in the funding proposal submitted by IFAD to us on 10 June 2021.

The undersigned is the duly authorized representative of the National Designated Authority/Focal point of the Republic of Mauritania

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I as included in the funding proposal.

By communicating our no-objection, it is implied that:

- The government of The Republic of Mauritania has no-objection to "Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I" as included in the funding proposal;
- The "Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I" as included in the funding proposal is in conformity with The Mauritania's national priorities, strategies and plans;
- In accordance with the GCF's environmental and social safeguards, The "Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I" as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the "Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I" as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Mr. Sidi Mohamed EL WAVI

National Designated Authority of Mauritania for the Green Climate Fund

Director of Climate and Green Economy

Copie :

- MEDD





FEDERAL MINISTRY OF ENVIRONMENT

HEADQUARTERS, MABUSHI, ABUJA.

Ref. No. FMENV/DCC/030/68

Date. 15 June, 2021

The Green Climate Fund ("GCF")
Songdo International Business District,
175, Art Centre-daero,
Yeonsu-gu Incheon 406-840
Republic of Korea.

Re: Funding proposal for the GCF by The International Fund For Agricultural Development (IFAD) regarding "Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I"

Dear Yannick GLemareck,
Executive Director of the Green Climate Fund,

We refer to "Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I" in Nigeria as included in the funding proposal submitted by IFAD to us on 10 June 2021.

The undersigned is the duly authorized representative of the National Designated Authority/Focal point of the Republic of Nigeria

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of The Republic of Nigeria has no-objection to "Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I" as included in the funding proposal;
- (b) The "Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I" as included in the funding proposal is in conformity with The Nigeria's national priorities, strategies and plans;
- (c) In accordance with the GCF's environmental and social safeguards, The "Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster

Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I" as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the "Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I" as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,



Mrs. Halima Bawa-Bwari
Ag. Director, Department of Climate Change
Federal Ministry of Environment, Nigeria.

REPUBLIQUE DU NIGER



Fraternité - Travail - Progrès
CABINET DU PREMIER MINISTRE

=====

CONSEIL NATIONAL DE L'ENVIRONNEMENT
POUR UN DEVELOPPEMENT DURABLE

=====

SECRETARIAT EXECUTIF

Niamey, le 17 NOV 2021

LE SECRETAIRE EXECUTIF

0198

N°/

A

Monsieur Yannick Glemerac
Executive Director a.i.
Secretariat of the Green Climate Fund
175 Art center- daero
Yeonsu-gu, Incheon 406-840
Republic of Korea

Subject : Funding proposal for the GCF by The International Fund For Agricultural Development (IFAD) regarding "Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall Countries. Phase I"

Dear Mr. Glemerac,

We refer to "Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall Countries. Phase I" Niger as included in the funding proposal submitted by IFAD to us on 17 November 2021. The undersigned is the duly authorized representative of the National Designated Authority/Focal point of the Republic of Niger.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to "Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall Countries. Phase I" as included in the funding proposal.

C.N.E.D.D

BP : 10 193 Niamey

Tél : (227) 20 72 25 59/ 20 72 42 64

Email : biocnedd@intnet.ne

By communicating our no-objection, it is implied that :

- (a) The government of The Republic of Niger has no-objection to **"Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall Countries. Phase I"** as included in the funding proposal ;
- (b) The **"Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall Countries. Phase I"** as included in the funding proposal is in conformity with The Niger's national priorities, strategies and plans ;
- (c) In accordance with the GCF's environmental and social safeguards, The **"Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall Countries. Phase I"** as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the **"Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall Countries. Phase I"** as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,


Mr. Maazou Kamayé
National Designated Authority of Niger for the Green Climate Fund

We also confirm that our national process for ascertaining no-objection to the **"Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I"** as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,



Dr. Rachid Hassan Mekki

Secretary General, Higher Council for Environment and Natural Resources, Soudan





المجلس الأعلى للبيئة والموارد الطبيعية
The High Council for Environment & Natural Resources (HCENR)

الأمين العام
Secretary General

To: The Green Climate Fund ("GCF")

Khartoum, 27 May 2021

Re: Funding proposal for the GCF by The International Fund For Agricultural Development (IFAD) regarding "Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I"

Dear Yannick Glemareck,

Executive Director of the Green Climate Fund,

We refer to **"Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I"** in Sudan as included in the funding proposal submitted by IFAD to us on 25 May 2021.

The undersigned is the duly authorized representative of the National Designated Authority/Focal point of Sudan.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to **Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I** as included in the funding proposal.

By communicating our no-objection, it is implied that:

- The government of Sudan has no-objection to **"Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in 13 countries of the Green Great Wall (GGW). Phase I"** as included in the funding proposal;
- The **"Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I"** as included in the funding proposal is in conformity with Sudan's national priorities, strategies and plans;
- In accordance with the GCF's environmental and social safeguards, The **"Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries. Phase I"** as included in the funding proposal is in conformity with relevant national laws and regulations.

ቁጥር 2/12.1/2406/h
Ref. No.
ቀን 9/7/2021
Date

To: The Green Climate Fund ("GCF")

Re : Funding proposal for the GCF by International Fund for Agricultural Development (IFAD) regarding "Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission, Smallholder Agriculture in the Great Green Wall (GGW) countries Phase I"

Dear Yannick Glemareck,

We refer to the programme titled **Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission, Smallholder Agriculture in the Great Green Wall (GGW) countries Phase I** in Ethiopia as included in the funding proposal submitted by IFAD to us on 25 May 2021.

The undersigned is the duly authorized representative of Environment, Forest and Climate Change Commission, the National Designated Authority of Ethiopia.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.
By communicating our no-objection, it is implied that:

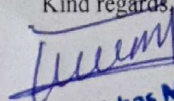
- The government of Ethiopia has no-objection to the **"Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in 13 countries of the Great Green Wall (GGW) Phase I"** as included in the funding proposal;
- The **"Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Great Green Wall (GGW) Phase I"** as included in the funding proposal is in conformity with the national priorities, strategies and plans of Ethiopia;
- In accordance with the GCF's environmental and social safeguards, the **"Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Great Green Wall (GGW) Phase I"** as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,


Tirhas Mebrahtu
Resource Mobilization Project
Monitoring & Evaluation
Directorate Director
His Excellency, Commissioner



☎ 251-(0)11-170-40 60
251-(0)11-170-40 39

TEEE ☎ 251-(0)11-170-4158/45

☎ 12760

Website: www.mefcc.gov.et

Agence de l'Environnement et du
Développement Durable
(AEDD)



Bamako, le **17.7.2021**...

*The Director General of the Environment
and Sustainable Development Agency*

To

**Mr. Yannick Glemarec, Executive Director a.i.
Secretariat of the Green Climate Fund
175 Art Center-daero
Yeonsu-gu, Incheon 406-840
Republic of Korea**

N° **0314** /MEADD/AEDD-DMRF

Re: Funding proposal for the GCF by The International Fund for Agricultural Development (IFAD) regarding **“Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Great Green Wall (GGW) countries. Phase I”**

Dear Mr. Glemarec,

We refer to the programme titled **“Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Great Green Wall (GGW) countries. Phase I”** in Mali as included in the funding proposal submitted by IFAD to us on 29 April 2020.

The undersigned is the duly authorized representative of the National Designated Authority of Mali.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

- (a) The government of Mali has no-objection to the programme as included in the funding proposal;
- (b) The programme as included in the funding proposal is in conformity with the national priorities, strategies and plans of Mali;
- (c) In accordance with the GCF's environmental and social safeguards, the programme as included in the funding proposal is in conformity with relevant national laws and regulations.

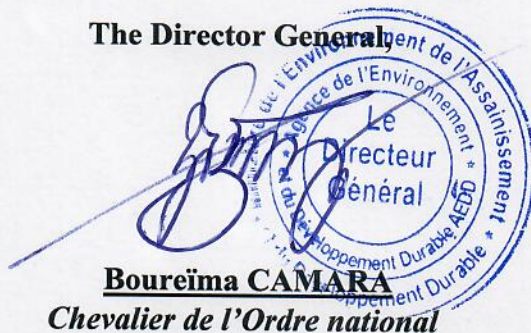
We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

The Director General



Boureïma CAMARA
Chevalier de l'Ordre national

The circular stamp contains the text: "Agence de l'Environnement et du Développement Durable", "Le Directeur Général", and "AEDD".



THE STATE OF ERITREA
Ministry of Land, Water & Environment
Department of Environment

ዕለት 23/06/2021
Date
ቁ. መዝገብ DO E/01/28/21
Ref. No.

التاريخ
رقم السجل

To: The Green Climate Fund ("GCF")

Re: Funding proposal for the GCF by International Fund for Agricultural Development (IFAD) regarding "Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission, Smallholder Agriculture in the Great Green Wall (GGW) countries. Phase I"

Dear Yannick Glemareck,
Executive Director of the Green Climate Fund,

We refer to the programme titled Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission, Smallholder Agriculture in the Great Green Wall (GGW) countries Phase I in Eritrea as included in the funding proposal submitted by IFAD to us on 25 May 2021.

The undersigned is the duly authorized representative of Ministry of Land, Water and Environment, the National Designated Authority of Eritrea.

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to the programme as included in the funding proposal.

By communicating our no-objection, it is implied that:

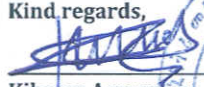
- The government of Eritrea has no-objection to the "Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in 13 countries of the Great Green Wall (GGW) Phase I" as included in the funding proposal;
- The "Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Great Green Wall (GGW) Phase I" as included in the funding proposal is in conformity with the national priorities, strategies and plans of Eritrea;
- In accordance with the GCF's environmental and social safeguards, the "Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Great Green Wall (GGW) Phase I" as included in the funding proposal is in conformity with relevant national laws and regulations.

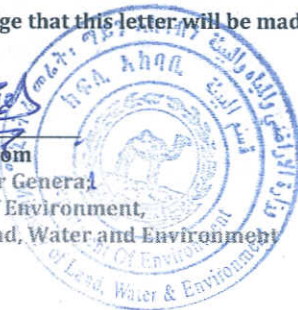
We also confirm that our national process for ascertaining no-objection to the programme as included in the funding proposal has been duly followed.

We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,


Kibrom Asmerom
Acting Director General
Department of Environment,
Ministry of Land, Water and Environment
Eritrea





N°.....17.....MEDD/DEEC/DCC

Dakar, le 21 Juin 2021

To: The Green Climate Fund ("GCF")

Mr. Yannick Glemarec

Executive Director

G-Tower, 24-4 Songdo-dong, Yeonsu-gu

Incheon City, Republic of Korea

Re: Funding proposal for the GCF by The International Fund For Agricultural Development (IFAD) regarding "Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries, Phase I"

Dear Yannick Glemarec,

Executive Director of the Green Climate Fund,

We refer to " **Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries, Phase I**" in Senegal as included in the funding proposal submitted by IFAD to us on 10 June 2021.

The undersigned is the duly authorized representative of the National Designated Authority/Focal point of the Republic of Senegal

Pursuant to GCF decision B.08/10, the content of which we acknowledge to have reviewed, we hereby communicate our no-objection to **Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries, Phase I**" as included in the funding proposal.

By communicating our no-objection, it is implied that:

- a) The government of The Republic of Senegal has no-objection to " **Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries, Phase I**" as included in the funding proposal;
- b) The " **Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall**

- a) **(GGW) countries, Phase I**” as included in the funding proposal is in conformity with The Senegal’s national priorities, strategies and plans;
- b) In accordance with the GCF’s environmental and social safeguards, the **“Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries, Phase I”** as included in the funding proposal is in conformity with relevant national laws and regulations.

We also confirm that our national process for ascertaining no-objection to the **“Inclusive Green Financing Initiative (IGREENFIN): Greening Agricultural Banks & Financial sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Green Great Wall (GGW) countries, Phase I”** as included in the funding proposal has been duly followed.

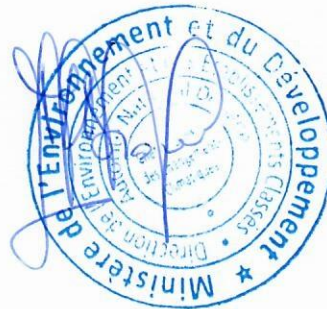
We also confirm that our no-objection applies to all projects or activities to be implemented within the scope of the programme.

We acknowledge that this letter will be made publicly available on the GCF website.

Kind regards,

Madeleine Diouf SARR

AND/SENEGAL



Environmental and social safeguards report form pursuant to para. 17 of the IDP

Basic project or programme information	
Project or programme title	Inclusive Green Financing Initiative (IGREENFINI): Greening Agricultural Banks & the Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Great Green Wall (GGW) countries - Phase I
Existence of subproject(s) to be identified after GCF Board approval	Yes
Sector (public or private)	Public
Accredited entity	International Fund for Agricultural Development (IFAD)
Environmental and social safeguards (ESS) category	Category I-2
Location – specific location(s) of project or target country or location(s) of programme	IGREENFIN I [Components 1 and 2]: Burkina Faso, Côte d'Ivoire, Ghana, Mali, and Senegal GCF GGW Regional Support Programme [Component 3]: Burkina Faso, Chad, Djibouti, Eritrea, Ethiopia, Mali, Mauritania, Niger, Nigeria, Senegal, and Sudan
Environmental and Social Impact Assessment (ESIA) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Environmental and Social Management Plan (ESMP) (if applicable)	
Date of disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Environmental and Social Management (ESMS) (if applicable)	
Date of disclosure on accredited entity's website	Wednesday, 5 January, 2022
Language(s) of disclosure	English and French
Explanation on language	English is the official language of Ghana. French is the official language of Burkina Faso, Cote d'Ivoire, Mali, and Senegal.
Link to disclosure	English: https://www.ifad.org/documents/38711624/40206666/IGREENFIN_esmf.pdf/338698a8-fda2-df85-dd14-e09dcc3f0dd4?t=1641392843760

	French: https://www.ifad.org/documents/38711624/40206666/IGREENFIN_summary_f.pdf/737219f0-e1c7-6aa8-7998-fc82ed544d8a?t=1641392925039
Other link(s)	https://www.ifad.org/en/secap-disclosed-documents https://www.badf.bf/wp-content/uploads/Annex-6-ESS-IFAD-IGREENFIN-I-ESMF-CLEAN-2022.pdf https://res.bnici.net/ http://www.labanqueagricole.sn/sites/default/files/2022-01/Annexe_IFAD%20IGREENFIN.pdf http://www.labanqueagricole.sn/sites/default/files/2022-01/Resume_Executif_CGES.pdf https://sites.google.com/view/fidafrique-ifadafrica/programme-r%C3%A9gional-sahel?authuser=0#h.8ss1exqyfrhc Mali: Résumé-Executif-CGES-IGREENFIN-1-FR.pdf (inclusifmali.org) Burkina Faso: https://www.badf.bf/wp-content/uploads/Resume-Executif-CGES-IGREENFIN-1R.pdf
Remarks	An ESMS consistent with the requirements for a Category I-2 project is contained in the “Environmental and Social Management Framework (ESMF)”. Environment and Social Management Plans consistent with the requirements for a Category I-2 project will be prepared at the sub-project levels.
Any other relevant ESS reports, e.g. Resettlement Action Plan (RAP), Resettlement Policy Framework (RPF), Indigenous Peoples Plan (IPP), IPP Framework (if applicable)	
Description of report/disclosure on accredited entity's website	N/A
Language(s) of disclosure	N/A
Explanation on language	N/A
Link to disclosure	N/A
Other link(s)	N/A
Remarks	N/A
Disclosure in locations convenient to affected peoples (stakeholders)	
Date	Wednesday, January 5, 2022
	Burkina Faso: Banque Agricole Du Faso; ZACA, Rue du commerce; 01 BP 4050 Ouagadougou 01. +226 25 32 99 00 Cote d'Ivoire: BNI, Immeuble SCIAM, Avenue Marchand Plateau, Abidjan, Tel: (225) 27 20 20 98 29

	Bureau FIDA, immeuble SAMA, Plateau Abidjan. Tel: +225 20 23 88 12 Ghana: IFAD Office N 69 Dr insert road, North Ridge 8th, Postal address Box CT 11187 Accra. Tel +233 302610945 Apex Bank: 5 GAmel Abdul Nasser Accra Mali: Bureau projet Inclusif, E4610 Magnambougou Faso Kano, Rue 30 porte 76, Bamako Tel: 00223 20 20 73 91 Sénégal: LBA, Place de d'indépendance, Dakar, BP 3890 Bureau regional FIDA, Bureau de Dakar / Villa 1427 Corniche Ouest - Almadies, Dakar, Sénégal Tel: + 221 33 868 05 87
Date of Board meeting in which the FP is intended to be considered	
Date of accredited entity's Board meeting	Thursday, February 10, 2022
Date of GCF's Board meeting	Monday, February 7, 2022

Note: This form was prepared by the accredited entity stated above.

Secretariat's assessment of FP183

Proposal name:	Inclusive Green Financing Initiative (IGREENFIN I): Greening Agricultural Banks & the Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Great Green Wall (GGW) Countries – Phase I
Accredited entity:	International Fund for Agricultural Development (IFAD)
Country/(ies):	IGREENFIN I [Components 1 and 2]: Burkina Faso, Côte d'Ivoire, Ghana, Mali and Senegal GCF GGW Regional Support Programme [Component 3]: Burkina Faso, Chad, Djibouti, Eritrea, Ethiopia, Mali, Mauritania, Niger, Nigeria, Senegal and Sudan
Project/programme size:	Medium

I. Overall assessment of the Secretariat

1. The funding proposal is presented to the Board for consideration with the following remarks:

Strengths	Points of caution
The programme will enhance access to credit and technical assistance for local farmers, farmers' organizations, cooperatives and micro and small sized enterprises to help them implement climate-resilient and low-emission agriculture/agroforestry and value chain processing and mini-grid supply linked to adaptive supply chains.	The use of GCF funds exclusively for climate-adaptive and mitigation practices need to be ensured.
The programme will incentivize the participation of the private sector by engaging with commercial banks and microfinance institutions.	The programme is subject to project-specific risks, such as macroeconomic, political, project implementation, and environmental and social risks.
The programme provides a framework for the current and future GCF development portfolio in the Sahel to share knowledge; scale up successful interventions in mitigation and adaptation; provide a platform for results monitoring and evaluation; and build a platform of digital and technical innovations to support farmer groups.	

2. The Board may wish to consider approving this funding proposal with the terms and conditions listed in the term sheet and addendum V, titled "List of proposed conditions and recommendations", respectively.

II. Summary of the Secretariat's assessment

2.1 Programme background

3. The 11 countries¹ of the Great Green Wall (GGW),² as well as Côte d'Ivoire and Ghana,³ are among the world's poorest and most vulnerable countries to climate change.⁴ They ranked at the bottom of the Human Development Index in 2020⁵ and a large majority of their population lacks access to employment, basic health care, education and natural resources, which contributes to food insecurity, youth migration and conflict.⁶ Agriculture, livestock and forestry activities are the foundations of their economies and more than 70 per cent of rural communities depend directly on rainfed agriculture (crops, livestock, fisheries and forestry) for nutrition and livelihoods.⁷ The agriculture sector is extremely vulnerable to both climate change and variability owing to the geographical location of these countries and their socioeconomic and technological characteristics.⁸ Over recent decades, climate change has increased the frequency, intensity and duration of droughts and floods, temperature variability, desertification, water stress and soil erosion, all of which reduce agricultural productivity and food security. Shocks from climate change and adverse weather conditions – resulting in the loss of assets, crops and livestock, disruptions in value chains and soaring food prices – in addition to the effects of the coronavirus disease 2019 (COVID-19) pandemic are pushing millions of smallholder farmers further into poverty.⁹

4. Climate projections indicate that precipitation levels will continue to decrease, while temperatures are expected to increase by between 1 and 1.72 °C during the period 2031–2050 compared with the 1986–2005 reference period.¹⁰ According to these projections, agricultural production is expected to decrease by at least 20 per cent in the region, which will reduce food availability and farmers' incomes, drive competition over diminishing natural capital and lead to greater instability and migration, especially among young people in rural areas.¹¹ Climate change also destabilizes local markets and curbs economic growth and investment, which in turn affects job creation, particularly for young people and women.¹² It will therefore make it more difficult to overcome the current lack of access to credit¹³ and technical capacity barriers, one of the primary barriers limiting the growth of rural smallholder farmers' organizations, cooperatives and micro, small and medium-sized enterprises (MSMEs) in the GGW region.

5. Increasing access to financing is key to improving on-farm productivity by providing working capital, allowing for investment in post-harvest value aggregation of products and

¹ Burkina Faso, Chad, Djibouti, Eritrea, Ethiopia, Mali, Mauritania, Niger, Nigeria, Senegal and Sudan.

² GGW is an African-led movement with an ambition to grow a natural wonder of the world measuring 8,000 km across the entire width of Africa. The goal is to restore 100 million hectares of degraded land, sequester 250 million tonnes of carbon dioxide equivalent and create 10 million green jobs in rural areas across the Sahel by 2030. For more information, see: <https://www.greatgreenwall.org/about-great-green-wall>.

³ Food and Agriculture Organization of the United Nations (FAO). 2020. *Twenty-second Session of the African Forestry and Wildlife Commission: Forests and Wildlife: Africa's Diversity for Shared Prosperity and Security*. Available at <http://www.fao.org/3/ca8011en/ca8011en.pdf>.

⁴ Intergovernmental Panel on Climate Change (IPCC). 2021. *AR6 Climate Change 2021: The Physical Science Basis*. Available at <https://www.ipcc.ch/report/ar6/wg1/>.

⁵ United Nations Development Programme. 2020. *Human Development Report*. Available at <http://hdr.undp.org/en/content/latest-human-development-index-ranking>.

⁶ Institute for Economics and Peace. 2020. *Global Peace Index Report 2020: Measuring Peace in a Complex World*. Available at <https://imctc.org/en/eLibrary/INTReports/Pages/Reports15012021.aspx>.

⁷ Organisation for Economic Co-operation and Development (OECD)/FAO. 2016. *Agriculture in Sub-Saharan Africa: Prospects and challenges for the next decade*. In: OECD and FAO. *OECD-FAO Agricultural Outlook 2016-2025*. OECD Publishing: Paris. Available at <http://www.fao.org/3/BO092E/BO092E.pdf>.

⁸ OECD. 2018. *Climate Change Profile: West African Sahel*. Available at <https://www.government.nl/binaries/government/documents/publications/2019/02/05/climate-change-profiles/West+African+Sahel.pdf>.

⁹ IPCC. 2019. *Food Security*.

¹⁰ Climate Analytics. 2019.

¹¹ Cordaid. 2020. *Saving Lives and Reducing Conflict in the Sahel Region*. Available at <https://reliefweb.int/report/burkina-faso/saving-lives-and-reducing-conflict-sahel-region>.

¹² International Monetary Fund. 2019. *Long-Term Macroeconomic Effects of Climate Change: A Cross-Country Analysis*.

¹³ IFAD. 2021. *IFAD Inclusive Financial Services: Portfolio Stocktaking*. Available at https://www.ifad.org/documents/38714170/43254162/IFS_Web.pdf/150d86c4-a444-26a3-2412-a386bbca3e3e?t=1623839327713.

processing systems and the acquisition of new equipment¹⁴ and, when combined with technical assistance, strengthening capacity and awareness of climate risks and their impacts on the production, trade and marketing of agricultural commodities.

2.2 Component-by-component analysis

6. The proposed IGREENFIN I programme has three components: component 1: the Green Business Financing Facility; component 2: the Technical Assistance Facility; and component 3: the GGW Regional Support Programme:

- (a) **Component 1:** Green Business Financing Facility – this will be established to provide concessional loans to foster best adaptation and mitigation practices for green businesses within the selected agricultural value chains in each target country. Operated by local national agricultural banks (LNABs), the facility will offer special lines of financing for green business projects developed by farmers' organizations, women and youth organizations, cooperatives and MSMEs (including agribusiness actors and operators of solar photovoltaic energy systems);
- (b) **Component 2:** Technical Assistance Facility (TAF) – this will be created to address capacity, knowledge and policy gaps hindering the uptake of green business projects in the selected countries. It will target LNABs, central banks and clients (farmers' organizations, MSMEs and cooperatives), with a special emphasis on women and youth; and
- (c) **Component 3:** GGW Regional Support Programme – to address the above-mentioned increasingly complex and interrelated challenges and support the transition towards climate-resilient, low-emission agriculture, the GGW initiative was launched in 2007 by the African Union. The initial objectives of the GGW initiative were to address land degradation, climate change adaptation and mitigation, and protect biodiversity and forests. There is a need to accelerate investment in the region to address these issues, especially since the onset of the COVID-19 pandemic, which has had severe impacts on the region's economic development. In a post-COVID-19 context, the ambition of the GGW initiative is to pursue a green growth recovery by restoring 100 million hectares of currently degraded land, sequestering 250 million tonnes of carbon dioxide equivalent (MtCO₂eq) and creating 10 million green jobs in rural areas by 2030.

7. The GGW Regional Support Programme component will increase the collective impacts both of individual GCF projects, as well as those undertaken in close collaboration with the Global Environment Facility (GEF), and programmes (including IGREENFIN I) through two outputs: (i) output 3.1 – enhanced knowledge management and exchange to accelerate the uptake of good practices, increase learning and inform policy and investments across GCF and other projects; and (ii) output 3.2 – mapping innovative and digital transformation technologies and building a digital and innovative ecosystem.

8. IGREENFIN I seeks EUR 73,971,000 in loans and EUR 30,444,189 in technical assistance grants from GCF. Additional co-financing is being mobilized from IFAD (USD 40 million), the Islamic Development Bank (USD 15 million) and the African Development Bank (USD 18 million). The GCF resources will be innovatively structured to stimulate and create a market for green business and will provide additional climate finance to complement IFAD baseline investments.¹⁵ The programme will support: (i) the acceleration of the transfer of climate-resilient, low-carbon technologies; (ii) the development of innovative green business models for

¹⁴ Magaja D and Agai OG. 2019. *Access to Finance for Inclusive Agri-business Development*. 2Scale. Available at https://www.2scale.org/upload/650938_2SCALE_paper11.pdf.

¹⁵ "IFAD baseline investments" refers to IFAD projects that have or are currently being designed to mobilize additional climate finance for IGREENFIN I to complement the work of IFAD.

agricultural value chain projects, pipeline planning and deployment; and (iii) the alignment of the portfolios and pipelines of the targeted LNABs with each country's nationally determined contribution (NDC) through capacity-building, policy dialogue and knowledge-building activities. This funding proposal covers only IGREENFIN I Phase I and the GGW Regional Support Programme.

9. IGREENFIN I intends to directly build the resilience and adaptive capacity of 378,600 smallholder farmers organized around 1,500 MSMEs and 2,500 farmers' organizations or cooperatives. The number of indirect beneficiaries of the programme is estimated at 2,494,000 in the five selected countries (Burkina Faso, Côte d'Ivoire, Ghana, Mali and Senegal), of which 50 per cent will be women and 50 per cent will be men (comprising 50 per cent youth and 50 per cent adult males), by increasing their climate resilience through access to affordable long-term green loans. The programme is expected to reduce 225,487 tCO₂eq annually (5,637,175 tCO₂eq over a 25-year lifespan of investments) from agriculture, forestry and land use, energy use and power generation.

III. Assessment of performance against investment criteria

3.1 Impact potential

Scale: Medium to high

10. IGREENFIN I will support the greening of the financial system in the five selected countries (Burkina Faso, Côte d'Ivoire, Ghana, Mali and Senegal) by building the capacity of five LNABs to offer green financing for climate-resilient, low-emission agriculture projects (green governance and lending products, green credit indicator systems, long-term loan periods and unlocking of barriers to innovative climate finance). It will also support 1,500 MSMEs and 2,500 farmers' organizations and cooperatives, as well as the transfer or licensing of 200 technologies and innovative solutions for climate-resilient, low-emission agricultural development. Its interventions will contribute to reducing the targeted countries' greenhouse gas emissions by 5,637,184 tCO₂eq over the 25-year lifetime of the programme. The adaptation benefits of this programme are estimated to reach 378,600 direct and 2,494,000 indirect beneficiaries, of which 50 per cent will be women and 50 per cent will be men (comprising 50 per cent youth and 50 per cent adult males). Innovative green business models, including climate-resistant breeds and crop varieties, sustainable land restoration techniques, solar energy to power drip irrigation, and processing, packaging and storage of equipment, for the 10 selected value chains (cattle, chicken, tomato, sustainable mango agro-forestry, cashew, shea, rice, groundnut, millet, cassava production), provide an indication of how the EUR 177,481,678 in requested funding for IGREENFIN I could be used to produce positive impacts and ensure both financial and economic returns on investment. The green business models could be scaled up by the financial sector in each of the targeted countries and in the GGW region.

11. IGREENFIN I will contribute to the implementation of the GGW initiative, which aims to restore 100 million hectares of currently degraded land, sequester 250 MtCO₂eq and create 10 million jobs in rural areas by 2030. Spanning the entire width of Africa – 8,000 km from Senegal to Djibouti – the "Wall" is a mosaic of projects implemented by countries in the region with the support of development partners as a pan-African sustainable landscape programme to address land degradation and desertification, boost food, environmental and economic security, and support the adaptation of communities to climate change. With the aim of achieving much more than just growing trees, its goals are to (i) foster climate resilience among local communities in a region expected to experience warming temperatures; (ii) safeguard food security for the 20 million people in the Sahel that lack sufficient food every year; (iii) support the creation of green jobs to provide stable incomes for African women and youth; (iv) build a global response to the migration crisis, with 60 million people set to migrate to Europe from degraded parts of Africa by 2021; and (v) foster peace and security in the Sahel, where conflict has already displaced 2.8

million people up to mid-2021. IGREENFIN I will channel resources to the programme beneficiaries and provide crucial funds to help accelerate the implementation of GGW, which was launched in 2007. It will also contribute to advancing the African Union Agenda 2063 in the region.

3.2 Paradigm shift potential

Scale: Medium to high

12. The proposal demonstrates the potential for a paradigm shift by shifting away from current unsustainable agricultural practices to low-emission and climate-resilient practices and a set of ecosystem-based adaptation (EbA) measures that will break the current model of degradation and lead to a sustainable agricultural model. The proposed climate-resilient agriculture interventions, such as crop rotation, intercropping, agroforestry and zai-planting techniques, are well-tested solutions in Africa and are aligned with good practices for achieving adaptation results. If implemented robustly and sustained after programme completion, such interventions can lead to transformational changes in the behaviours of farmers, as well as positive impacts on the livelihoods of the target beneficiaries. It is recommended that climate-resilient and low-emission technologies and solutions to transform agricultural practices in the medium and long term are explored and used.

13. IGREENFIN I aims to engage local national agricultural banks, national banking associations, banking regulators, green standard-setting bodies, local universities and business schools in green finance in the agriculture sector. Through the programme's mainstreaming work, these institutions will be able to sustain the GCF grant funded activities and concessionality beyond the programme's duration. The programme will support the development of studies and policy papers on climate risk management to define the relevant standards, and develop and harmonize green lending products, instruments and tools such as environmental and social governance benchmarks. It will also build knowledge on green finance invested in the agriculture sector by both the private sector and governments. IGREENFIN I will work closely with national agricultural banks to develop and improve their environmental and social governance and internal capacity, and better align part of their lending portfolio on NDCs. By building the banks' capacity to offer affordable green finance for smallholders in the agriculture sector, the programme will start to fill the gap in financing for agriculture and alter the perception of smallholders from a risky business to a vast market with great potential and opportunities.

14. Green credit lines and awareness campaigns will support the development of demand for green products, particularly among youth and women, support a viable pipeline of green business projects, reduce the financial sector's perception that smallholders present a high risk and hence promote the development of new green markets in sectors that have been traditionally difficult to reach (e.g. the MSME sector). Beneficiaries such as MSMEs will benefit from training activities and affordable finance (available from local financial institutions) and techniques that help them reduce their operating costs while increasing their productivity and income. The early stages of the programme's operations are expected to demonstrate to end users the effectiveness and value of the solutions proposed, which will stimulate demand beyond the target financial institutions and countries to other market sectors and areas covered by GGW after the programme ends. This programme will enable the transition away from grant-funded rural finance projects towards sustainable long-term markets for such projects.

15. The GGW Regional Support Programme introduces a new approach piloted by IFAD and GCF, which aims to catalyse impact beyond a one-off project or programme investment by creating bridges for knowledge-sharing and learning between GCF projects that would otherwise be implemented in silos, and among accredited entities (AEs), GCF national designated authorities (NDAs), and GGW and GEF focal points. This innovative approach has the potential to be scaled up in Africa beyond the GGW countries and could also be replicated in other large landscapes and ecosystems such as the Amazon or in countries with relatively

similar needs, such as some small island developing States. This regional support program will also contribute to strengthening collaboration at the country level between GCF, the GEF and the Adaptation Fund, and showcase a concrete example of complementarity and collaboration between the United Nations Framework Convention on Climate Change (UNFCCC) and the United Nations Convention to Combat Desertification.

16. The proposal demonstrates good potential for scaling up and replication at the country and regional levels. The grant component is essential as it will lead to tangible knowledge management for scaling up and replication of sustainable lending.

3.3 Sustainable development potential

Scale: High

17. The proposal demonstrates high potential for sustainable development. IGREENFIN I and the GGW Regional Support Programme have identified and adopted the objectives of the following Sustainable Development Goals (SDGs) and mapped each of the target sectors to them: No poverty (SDG 1); Zero hunger (SDG 2); Gender equality (SDG 5); Clean water and sanitation (SDG 6); Affordable and clean energy (SDG 7); Decent work and economic growth (SDG 8); Climate action (SDG 13); Life on land (SDG 15); and Partnerships for the Goals (SDG 17).

18. The programme is designed to help secure multiple environmental benefits for water, air, land and nature by promoting the adoption of best adaptation and mitigation practices and technology, which will enhance the sustainable management of natural resources, protect threatened dryland biodiversity against erosion and desertification, and increase the potential for carbon sequestration. These benefits could also contribute to the increased resilience of the region's ecosystems and human livelihoods to climate change and variability and the overall objectives of GGW (land restoration, job creation and carbon dioxide (CO₂) emission reduction). The programme intends to reduce the use of energy from biomass, including firewood and agricultural residues, imported diesel fuel and charcoal, which are the main sources of electricity in the five selected countries, by providing greater access to renewable energy. It is expected that CO₂ emissions will be reduced (total emission reductions of 5.6 MtCO₂eq) during the programme's lifecycle through improved agroforestry and sustainable land management practices (this emission reduction will further increase once the avoided emissions from renewable energy technologies (RET) are included) and limits will be placed on the use of biomass and diesel for energy consumption through the adoption of RET.

19. With the promotion of EbA and agroforestry techniques, more than 100,000 hectares of degraded land will be restored and sustainably managed with improved ecosystem goods and services. The programme will build up each link in the selected agricultural value chains (including livestock and non-timber forest products) from production to markets, increasing the productivity of key commodities and farmers' incomes while safeguarding the environment.

20. The activities proposed under this programme will generate multiple job opportunities for people in rural areas, particularly for youth and women, promote inclusion and support coherent investment through farmers' organizations, cooperatives and MSMEs. By promoting local low-emission and climate-resilient options, the programme will contribute to cost-saving health benefits due to cleaner air for local communities and access to clean energy, water and food, thus reducing food insecurity. As described in its gender analysis and action plan, the programme will support the participation of women in decision-making, facilitate their access to financing and assets and improve their livelihoods. The targeted areas considered the most vulnerable and selected through IFAD baseline investments will be positively impacted by the programme.

21. The economic benefits of this programme are linked to its high potential to reduce poverty through increased productivity of key crops and livestock production, as well as income-generation activities, such as the production of and value added to a variety of fruit and

non-timber forest products, including tamarind, Gum Arabic and baobab leaves, as well as fodder and seedling sales. The GGW Implementation Status Report indicates that revenues from income-generation activities since 2007 up to 2020 amount to approximately USD 90 million across all 11 GGW countries, with Burkina Faso being more successful than the other countries in creating income-generation opportunities.

22. The proposal seeks to reduce gender inequalities resulting from climate change impacts and promote equal participation of gender groups. Gender considerations have been adequately taken into account in the programme design. The programme has set a target of granting 50 per cent of loans to women-led MSMEs, cooperatives and farmers' organizations. The other 50 per cent will be granted to men, of which 50 per cent will be youth-led MSMEs, cooperatives and farmers' organizations in which young women also participate. Specific actions will be developed to strengthen the technical and managerial capacities of women aimed at providing them with appropriate tools for identifying and developing bankable green business plans and improving their level of financial education. As stated in the programme's gender action plan, gender-disaggregated data will be assessed against the appropriate indicators to measure women's enhanced access to loans.

3.4 Needs of the recipient

Scale: High

23. The vulnerability of the country and target populations and their adaptation needs are explained in the funding proposal and feasibility study, including a general analysis of the scale and intensity of exposure to climate change risks. The countries participating in this programme are highly vulnerable to drought, flooding, rising temperatures and variability of rainfall, as well as other climate change hazards (e.g. wildfires and locust infestations). Their exposure and vulnerability to climate hazards is projected to increase in the future. Climate projections suggest that the Western Sahel and northern parts of Côte d'Ivoire and Ghana will be hotter and drier with more frequent extreme events. Under most scenarios, temperature rise will accelerate, meaning that the continent of Africa could be between 2 and 6 °C warmer by the end of this century (African Center for Meteorological Applications for Development, 2020). Climate impacts have negative impacts on agriculture. They reduce productivity, food security and gross domestic product, cause endemic diseases (e.g. malaria) and increase rural poverty. Given the five countries' high poverty rate (see the feasibility study for further details) and the impacts of climate change, the countries have few financial institutions willing to risk investing in smallholder farming, despite the fact that agriculture remains the main source of economic growth.

24. The five countries' financial systems remain weak and have a limited number of instruments for financing agriculture. LNABs have limited possibilities of mobilizing highly concessional loans in capital markets. Farmers' organizations, MSMEs, cooperatives, and youth and women-led organizations active in the agriculture sector often lack access to long-term financing with affordable conditions (interest rates, tenors and grace periods) due to small loan sizes and relatively high transaction costs. Key barriers are limited adaptation capacity to climate change at the country level, lack of skills and knowledge of climate finance and climate-resilient, low-emission agriculture, the nascent regulatory frameworks and the need to de-risk the agriculture and banking sectors.

25. Technical assistance and funding for the establishment of a Green Business Financing Facility that will provide concessional loans and climate-smart credit risk tools will enhance access to credit, contribute to de-risking the sector and foster greater knowledge of climate finance and adaptation across the selected countries. This will directly help LNABs to overcome the key barriers they face in relation to providing access to finance to rural populations, and enable them to engage in green financing.

3.5 Country ownership

Scale: High

26. The proposal notes the programme's alignment with the participating countries' national climate change policies and strategies and NDCs. It is recommended that the programme's progress and results are monitored and quantified, and the extent to which the programme contributes to the countries' NDCs is measured.

27. The participating countries are Parties to the UNFCCC and have signed and ratified the Kyoto Protocol. By ratifying the UNFCCC, these countries have committed to implement measures to adapt to climate change and report on their NDCs. This programme will contribute to the implementation of the objectives of the three Rio Conventions ratified by all countries¹⁶, including the UNFCCC, as well as the Paris Agreement, the SDGs and the Sendai Framework for Disaster Risk Reduction 2015–2030.

28. This programme is fully aligned with the participating countries' national development plans and national commitments on climate mitigation and adaptation included in their national adaptation programmes of action, national climate change policies and strategies, NDCs, national communications and national strategies for disaster risk reduction, as well as the SDGs.

3.6 Efficiency and effectiveness

Scale: Medium to high

29. In the GGW region, green businesses are largely financially underserved, both in terms of availability of highly concessional loans from LNABs, commercial banks and microfinance institutions and in terms of technical assistance. Activities under IGREENFIN I aim to fill a financing gap through the provision of financial and non-financial support. In doing so, the programme will boost a nascent market that is currently far from tapping into its full potential. As this market develops over time, opportunities will be created to attract other public and private sector investors in these least developed countries (Burkina Faso, Mali and Senegal) and lower-middle income countries (Côte d'Ivoire and Ghana) that share the features of the least developed countries, namely a high level of poverty, given the size of the population and extreme vulnerability to climate change.

30. IGREENFIN I will promote and scale up green finance products and green business models in the targeted countries and region. By building synergies and leveraging IGREENFIN I resources, the programme will reach the maximum number of beneficiaries. Partnerships with LNABs and executing entities (EEs) will be promoted through the provision of green loans and technical assistance to de-risk all potential subprojects.

31. GCF financing will overcome the existing barriers to accessing financial and non-financial services that hinder the adoption and implementation of best climate change adaptation and mitigation measures (e.g. solar energy for agriculture). Additionally, even though LNABs have been performing quite well over recent years with various loan products, their loans are still not sufficiently concessional to stimulate greater adherence to low-emission and climate-resilient agriculture and generate a strong pipeline of green business projects. In the energy sector, several bilateral donors have put in place concessional financing schemes that combine grants with credit to promote agricultural and rural finance; they work mainly through microfinance institutions and State-linked financial institutions, such as the targeted LNABs. Early results from these programmes are highly encouraging, although strong oversight and technical assistance are necessary. IGREENFIN I is projected to achieve a leverage ratio of 1:0.7 (i.e. USD 0.7 of additional public and private funding leveraged per USD 1 of GCF investment); thus, the GCF investment is expected to leverage about USD 90 million in additional public and private contributions (including co-finance).

¹⁶ The Rio Conventions consist of the Convention on Biological Diversity, the United Nations Convention to Combat Desertification and the UNFCCC.

32. Given historic market demand for such loans, IGREENFIN I anticipates that the total proposed GCF facility with contribution of Eur 104,415,189 will unlock the green finance market and improve both demand and supply. This will stimulate the market and encourage bilateral lending organizations and private sector investors to adopt competitive rates to reduce the risk of crowding out the market. The national EEs, through their respective LNABs, intend to blend their internal resources with GCF funds to support rural transformation through the adoption and scaled-up implementation of climate-resilient agriculture in all targeted countries. Through IGREENFIN I, the participating countries are expected to develop a conducive regulatory framework which will sustain this programme beyond its lifetime. The programme is expected to bear an estimated GCF cost of EUR 5.3 per tCO₂eq, which is assessed to be highly cost-effective compared with similar investments in EbA, climate-resilient agriculture and renewable energy solutions in Africa.

33. The proposal aims to overcome existing barriers and bottlenecks for end-beneficiaries to access financial services for adopting and implementing best climate change mitigation and adaptation measures. The programme's proposed financing structure is assessed to be reasonable compared to business-as-usual financing, considering the additional investments needed to shift away from conventional unsustainable agricultural practices to climate-friendly investments.

IV. Assessment of consistency with GCF safeguards and policies

4.1 Environmental and social safeguards

34. The programme aims to support the building and scaling up of the resilience and adaptive capacity of farmers' organizations, cooperatives and MSMEs by increasing the beneficiaries' access to concessional credit lines for green agricultural investments that would result in low greenhouse gas emissions and climate resilient and sustainable agricultural practices in the target countries. The environmental co-benefits envisaged include enhancing the sustainability of natural resources, protecting threatened dryland biodiversity, and preventing erosion and desertification through the adoption of adaptation and mitigation practices and technologies. The social co-benefits for local communities include improved health due to cleaner air and increased access to clean energy, water and food, and enhanced food security.

35. Environmental and social risk category and safeguard instrument. The programme is categorized as medium level of intermediation (I-2), because the bulk of the GCF support is allocated to providing concessional loans to beneficiaries, through the LNABs, for adaptation and mitigation practices and businesses. The programme will not finance category A or high-risk subprojects, nor any investments listed in the programme exclusion list (e.g. activities that lead to or require involuntary resettlement or involve conversion or degradation of critical natural habitat, among others). An environmental and social management framework (ESMF) that outlines the potential risks and impacts of the proposed programme and the mitigation measures to address these impacts has been prepared.

36. Compliance with GCF's environmental and social safeguards standards. Paragraphs below describe how the programme proposes to address the requirements of the GCF environmental and social safeguards (ESS) standards.

37. **ESS 1: Assessment and management of environmental and social risks and impacts.** The AE has prepared an ESMF that provides a description of how environmental and social impacts and risks will be managed, mitigated and supervised at both the framework level and the programme operational level. The ESMF identifies the potential direct and indirect risks as well as both positive and negative risks and impacts of the activities, and provides plans and strategies on how the adverse risks and impacts will be avoided, minimized and/or mitigated.

38. **ESS 2: Labour and working conditions.** Given the generally poor working conditions and limited awareness of and non-compliance with health and safety standards in the agriculture and energy sectors in the rural areas in the target countries, risks related to labour and working conditions (e.g. child or forced labour) in financed subprojects may occur. Such risks may occur in programme activities that may require the participation of external service providers, such as during the installation of renewable energy technologies, the installation and maintenance of irrigation systems, or the drilling of boreholes, where underage workers may be engaged. The strategies that the programme intends to implement include screening and monitoring the service providers to ensure that any forms of child and forced labour in infrastructure construction related to the programme will not occur. The programme also includes training and capacity-building on existing labour regulations, as well as implementing a forced and child labour surveillance mechanism as part of the regular supervision visits of the programme during implementation. These actions are part of the AE's regular monitoring across its implementing partners. While most of the beneficiaries will be small-scale farmers who may invest in improving their agricultural practices, some beneficiaries (particularly the MSMEs) may employ workers, in which case the programme intends to ensure that they respect both the national standards and standards set by the International Labour Organization.

39. **ESS 3: Resource efficiency and pollution prevention.** The programme's activities include supporting "carbon sensitive processing and packaging", which could result in factories causing noise and air pollution, solid waste generation as a result of packaging waste, and water and soil pollution from factory effluents. To mitigate these risks, the programme requires participants to install pollution control technologies such as scrubbers, locate new factories in industrial zones or far from communities, promote the use of recyclable or reusable packaging materials, and develop waste and water management plans. To promote efficient use of resources, the programme will promote strategies such as using renewable energy sources as well as retrofitting with modern and efficient equipment. Regarding the potential release of agrochemicals such as fertilizers into water bodies during agricultural production, the programme will provide training to farmers on proper use of fertilizers and agrochemicals; and a pesticide management framework and plan is included in the ESMF. Considering there is environmental and health risk from potentially hazardous wastes in the RET installations (e.g. solar panels and batteries containing contaminants such as lithium and lead), the programme will develop a hazardous waste management framework/plan covering the operational and decommissioning phases of the installations to ensure that feasible actions for the proper management of the wastes is established. To ensure that borehole drilling and irrigation expansion activities will not result to overextraction, these activities will be required to comply with nationally and/or internationally established guidance and the programme will require a supply/demand assessment to be conducted prior to implementation to ensure that there is adequate supply for all users in the targeted area.

40. **ESS 4: Community health, safety and security.** The programme may have environmental health and safety related issues or impacts that may affect the communities as a result of the activities of the programme. These include health risks related to noise and air pollution, inadequate handling of RET wastes, release of agrochemicals into the surroundings and exposure to pesticides. The management measures planned to be implemented include training and capacity-building activities and development of hazardous waste and pesticide management plans. Potential risk of social conflicts, sexual exploitation, abuse and harassment (SEAH), and potential use of forced and/or child labour in the implementation of the programme's activities in the target countries will be mitigated through implementation of a "no tolerance" policy for SEAH in the programme's supported operations to ensure that appropriate precautionary and remedial measures for any occurrences of SEAH risks and/or complaints are incorporated in the programme design and are embodied in the AE's policy on SEAH. Regarding the potential for conflicts relating to the agriculture-based activities, the programme will ensure that prospective beneficiaries are made well aware of selection criteria in the selected project districts and regions.

41. **ESS 5: Land acquisition and involuntary resettlement.** The programme's activities will likely involve involuntary resettlement and land acquisition, particularly for activities that may require population resettlement (e.g. for agricultural production sites or agricultural produce processing and packaging facilities). These activities are not yet clearly defined so a resettlement policy framework, which will include provisions for the development and implementation of site-specific resettlement action plans, will be developed during programme implementation, taking into consideration the legal requirements in each of the countries. The action plans will then be developed as needed, if the activities under the programme lead to land acquisition and involuntary resettlement.
42. **ESS 6: Biodiversity conservation and sustainable management of living natural resources.** The activities under the programme will have a potential risk of habitat fragmentation through the development of production plots or infrastructure construction, as well as the potential risk of the introduction of alien or invasive species. To mitigate these impacts, the ESMF stipulates that critical biodiversity hotspots in target areas will be identified and mapped at the start of the programme for inclusion in the eligibility criteria. The mapping activity will be conducted by the programme's safeguards team in collaboration with the environmental protection agencies in the target countries so that the alien or invasive species identified can be included in the final project eligibility criteria. The programme as a whole will not support activities that will lead to conversion of wetlands, conversion of significant areas of natural forests or other wild lands into agricultural or pastureland; nor will it support activities that will lead to loss of natural habitats, biodiversity or decreased provision of ecosystem services in sensitive areas. The programme will provide capacity-building activities aimed at strengthening the capabilities of local and indigenous populations on natural resource management and safeguards issues.
43. **GCF Indigenous Peoples Policy and ESS 7: Indigenous peoples.** Pastoralists and other socially vulnerable groups are present in the countries where the program activities will be undertaken and therefore the requirements of the GCF Indigenous Peoples Policy will need to be considered and applied by the AE. The programme excludes activities that could lead to adverse social impacts to local communities, or that could result in displacement of at least 20 people. Within its ESMF framework, the AE provides the following entry points to ensure the consideration of indigenous peoples: an indigenous peoples plan outline that provides the framework for planning to exclude adverse impacts on indigenous peoples; and a rapid social analysis template that provides the basis to understand the baseline and social diversity at the subproject sites. The stakeholder engagement plan includes a distinct strategy for engaging with indigenous peoples. The AE notes that its free, prior and informed consent requirements apply not only to indigenous peoples and historically underserved local communities but also to other local rural communities whose land access and use rights may be affected. In terms of potential benefits for indigenous peoples, some targeted opportunities for indigenous peoples and local communities can be found in components 1, 2 and 3, and indicators for the number of beneficiary organizations composed principally of members of ethnic minority backgrounds as well as the engagement of vulnerable groups could be used to promote the participation of indigenous peoples. The programme will promote the benefits of indigenous knowledge by including the strengthening of indigenous climate risk management capabilities in its screening form that is used for evaluating the climate change vulnerability of applicants. A budget for training on indigenous peoples screening and development of an indigenous peoples plan is included under the budget for ESMF monitoring.
44. **ESS 8: Cultural heritage.** The programme will exclude activities that will lead to conversion and/or loss of physical cultural resources and this will be part of its screening criteria. Should instances where chance finds (of physical cultural resources that are not identified during screening) are discovered during construction and/or operation of the facilities, the programme will redirect the work into other sites to prevent any damage. Adequate "chance find" procedures in the operational guidelines of external service providers,

including the identification of chance finds and the designation of a point of contact with relevant state and/or archaeological actors, will also be developed.

45. **Implementation arrangements.** The AE will ensure coordination, implementation and monitoring of the environmental and social performance of the programme, and third-party sector-specific consultants may be brought on board to support the implementation of the environmental and social management plans. A regional coordination unit will be established and will have responsibility for the overall operational implementation of the ESMF. An ESS specialist will be engaged within the RCU to oversee the planning, delivery and monitoring of all safeguards-related activities. At the national level, a project management unit will be established within each bank, and the PMU will oversee the supervision and development of the subprojects, and ensure effective implementation of the ESMF guidelines among its contractors and sub-contractors. The banks will also be responsible for the environmental and social assessment, and management and monitoring of the subprojects under their respective portfolios. The AE's RCU will evaluate and monitor the supported banks on a continuous basis throughout the project cycle, including coordinating with the finance ministries in the target countries to monitor their environmental and social performance.

46. **Stakeholder engagement and grievance redress mechanism.** The programme has conducted various stakeholder consultations in each of the target countries during the baseline investment design stage. A stakeholder engagement strategy and engagement plan has also been developed to define programme context, principles of stakeholder engagement, applicable national and international regulations, analysis of programme stakeholders, approaches to stakeholder engagement, communication and information disclosure, grievance mechanism, and monitoring and reporting of stakeholder engagement activities. The programme will also require its borrowers to implement effective grievance mechanisms to receive and assist with the resolution of any concerns and grievances that stakeholders may raise in connection with the programme's environmental and social performance. This is in addition to the AE's institutional complaints procedure for receiving and resolving concerns and issues relating to non-compliance with its environmental and social policies and procedures

4.2 Gender policy

47. The AE has provided a gender assessment and gender action plan and therefore complies with the requirements of the Gender Policy of the GCF.

48. The gender assessment provides the context of the enabling environment in the five countries that are the focus of the program. It indicates that all five countries have ratified the Convention on the Elimination of All Forms of Discrimination against Women (CEDAW), as well as a number of other international conventions including the African Charter on Human and Peoples' Rights on the Rights of Women in Africa (Maputo Protocol). The constitutions of the respective countries also reflect an ambition for equality between women and men, and prohibition of discrimination based on gender. Each of the countries have established women's ministries/agencies with responsibilities of implementing, promoting and protecting the rights of women and other marginalized community members. Despite supportive institutional arrangements, legal frameworks and policies, IGREENFIN-targeted countries remain among the countries with the lowest rankings when it comes to gender equality and equity indicators. While Senegal ranks higher than the other focal countries, gender gaps remain high across all countries due to social and cultural norms; a lack of enforcement and a lack of implementation of laws and policies; and a lack of resources dedicated to reducing gender gaps.

49. The gender assessment was completed based on a desk review. The gender assessment found that gender inequalities persist in the agriculture and finance sectors of the five countries targeted by the IGREENFIN regional programme. This is particularly relevant when it comes to women's access to credit for agriculture, as well as gender inequalities for accessing other

productive resources. The gendered division of labour in households places the burden of securing water and fuel supplies, and caring for the children and the elderly largely on women, leaving them with very little time to engage in income-generating activities or to further their education. Women in the region have less access to skilled jobs, and are marginalized in the labour market in and outside the agricultural sector. They represent only 8 per cent of entrepreneurs. Gender-based inequalities exist in terms of participation in formal labour markets; and gender-based inequalities in agriculture also apply to access to and control over resources, which affect women's agricultural productivity. These gender inequalities exist in land ownership, access to inputs, access to financial resources, and access to information and training, among other things. Existing financial service providers and their products also generally fail to consider the particular needs of women and rarely conduct gender-based assessments when developing products. Cultural and societal norms continue to create barriers for women's engagement in leadership and decision-making. In addition, gender-based violence is prevalent throughout the five focal countries.

50. The AE has provided a gender action plan and therefore complies with the requirement of the Gender Policy of the GCF. The gender action plan provides activities that address the challenges faced by women and includes baseline, indicators, targets, timelines and budgets.

51. The gender action plan includes activities designed to increase access to finance for climate-resilient agricultural practices and enhanced utilization of RETs by women-led households, married women in households, youth, minority groups, farmers' organizations, cooperatives and MSMEs and increase women's participation in governance and decision-making in the energy finance sector. More specifically, the gender action plan includes activities to increase access to credit for farmers' organizations, enterprises and cooperatives managed or owned by women and young people. This includes activities designed to improve women's property rights and control over assets, and capacity-building for farmers' organizations, enterprises and cooperatives managed or owned by women and youth. The AE will build the capacity of banks and national ministries in gender mainstreaming and the development of institutional frameworks that facilitate women's engagement in decision-making processes. The action plan also includes the development of financial products that take into account the key barriers that women and women-run organizations traditionally face. Communication campaigns via visits, social media and radio, directed at raising awareness and understanding of renewable energy products and financial products will be undertaken, with women and women-run organizations as key target audiences. Award programmes that encourage the participation of women and youth run organizations implementing renewable energy projects are also part of the plan. The AE will hire a full-time gender specialist within each bank participating in the programme and ensure that gender is integrated throughout the program. This includes ensuring gender expertise in decision-making bodies. It will also develop a conflict resolution and grievance mechanism that will provide women with easy access to these mechanisms and the ability to report and address instances of gender-based violence.

52. As the gender assessment was based on a desk review, there are further studies to be conducted in order to ensure the engagement of stakeholders from different groups. The expectation is that these studies will support the fine-tuning of targets and indicators. The gender action plan includes the development of country-, sector- and site-specific gender assessments and further stakeholder consultations (including with women and women's groups) to better understand opportunities, barriers and challenges for women, engage marginalized groups, collect baseline data and refine targets for the gender action plan. The AE will submit a revised action plan before the end of the first year of the program.

4.3 Risks

4.3.1. Overall programme assessment (medium risk)

53. The programme aims to build the resilience of smallholder farmers to climate risks in Burkina Faso, Côte d'Ivoire, Ghana, Mali and Senegal. The programme intends to do so by: (i) providing support to selected LNABs to create green credit lines to finance green business projects from smallholder farmers' organizations, cooperatives, women and youth-led organizations and MSMEs; (ii) addressing capacity-building, knowledge and policy gaps that hinder the transfer and adoption of adaptation and mitigation practices in green agricultural projects; and (iii) enhancing the collective impact of all GCF-funded projects, by establishing a coordination framework for knowledge generation and sharing, implementation and reporting. The AE seeks EUR 74 million in loans and EUR 30.4 million in technical assistance grants from the GCF. Additional co-financing is to be mobilized from IFAD (EUR37.7 million), the Islamic Development Bank (EUR 12.3 million) and the African Development Bank (EUR 14.8 million).

4.3.2. Accredited entity/executing entity capability to execute the current programme (medium risk)

54. The AE, IFAD, has close to 40 years of experience in providing loans and grants to support governments in agriculture. IFAD will create a RCU that will oversee, compile, consolidate and control the work of the EEs. IFAD already has a presence in the region. Given the number of EEs involved in the programme, the AE will have an important coordinating role to ensure coherence among the activities.

55. The EEs for the reimbursable funds are Burkina Faso, Côte d'Ivoire, Ghana, Mali and Senegal, represented by their respective ministries of economy and finance. The EEs will be recipients of the GCF financing and guarantee the repayment of the GCF loan portion. The AE notes that the EEs have a track record in similar projects and have been working with the AE and other financial institutions on rural finance. The funds will be implemented by the LNABs. The AE has also performed capacity assessments on the LNABs in the programme.

56. The EEs of the GCF grant proceeds for the programme 1 are the Ministry of Agriculture, Hydro-Agricultural Development and Mechanization of Burkina Faso; the Ministry of Agriculture and Rural Development of Côte d'Ivoire; the Ministry of Agriculture of Mali; the Ministry of Agriculture of Ghana; and the Ministry of Agriculture and Rural Equipment of Senegal. IFAD has conducted a capacity assessment of the ministries of agriculture of the five targeted countries.

4.3.3. Programme-specific execution risks (medium risk)

57. Market demand: Low demand for agricultural financial products offered would result in lower impacts for the programme. Comfort is derived from confirmation received by the AE of the interest and willingness of national development banks, smallholder farmers and governments during the design phase of the programme.

58. Credit risk: GCF will be exposed to repayment risks from the governments of Burkina Faso (rated B by Standard & Poor's (S&P)), Côte d'Ivoire (rated BB-, S&P), Ghana (rated B-, S&P), Mali (Caa1, Moody's) and Senegal (B+, S&P). There is partial financial alignment with the AE because the AE will co-finance the programme to the amount of EUR 11.5m in reimbursable loans with equal tenor and grace period as GCF.

59. Political and security risk: Burkina Faso and Mali are exposed to conflicts. The regions selected for the programme are secured sites as stated in the assessment of the AE. Each country will propose an action plan for the safety of the intervention sites that will be considered during the evaluation mission.

60. Tenor: the long tenor of the programme (up to 40 years) could result in the proposed interventions becoming less additional or of lower impact over time as a result of changes in the operating environment. The financial terms and conditions offer a possibility to prepay the loans in case reinvestment in line with the eligibility criteria is no longer feasible.

61. Currency risk: GCF will provide the loans and grants in EUR. End beneficiaries are likely disbursed in local currency. For Burkina Faso, Côte d'Ivoire, Mali and Senegal, which are part of the West African Economic and Monetary Union and use the Central African franc (FCFA) as their currency, the short-term currency risk is limited because the FCFA is pegged to the Euro. However, over the long term a change in the peg could incur currency risk for the governments of these countries and affect the countries' ability to repay the loans. The currency of Ghana, the Ghanaian cedis, is not pegged to the Euro and therefore is also exposed to currency risk in the short term. Fluctuations in the currency could impact Ghana's ability to repay reimbursable funds. The AE notes that managing the foreign exchange risk is included as part of capacity-building activities.

62. Project viability and concessionality: GCF will provide both grants and low-concessional loans. The term sheet requires that the EEs be contractually obliged to ensure that concessionality is passed down to the final downstream recipients/borrowers. The interest rates for the sub-loans as presented in the Term Sheet are based on a near 1:1 co-financing ratio by the LNABs while co-financing requirements are 80:20 in the Term Sheet. It is advised that the margin the LNABs can charge be limited and that end-user pricing takes into account realized co-financing.

4.3.4. Compliance risk (medium risk)

63. Of the 13 country beneficiaries of this programme, at least 2 are subject to targeted United Nations Security Council resolutions: (1) [Resolution 2374 \(2017\)](#) concerning Mali; and (2) [Resolution 1591 \(2005\)](#) concerning the Sudan.

64. IFAD, as the AE, has confirmed that restrictive measures will be applied to Mali and the Sudan to ensure compliance as per all ongoing IFAD projects in those countries. The restrictive measures IFAD will apply are in line with the scope of United Nations Security Council sanctions regimes currently under implementation, as IFAD is also a United Nations entity. In crisis countries, IFAD operates under special measures and will reduce its operations if needed.

65. National financial systems are subject to country compliance with the international standards of anti-money-laundering and countering the financing of terrorism policies, which are embedded in the national policies of the target countries. As the lending operations being considered in the context of this programme will be part of financial intermediation schemes in local and international markets, a very low risk is foreseen of unidentified financing of these practices.

66. Furthermore, since all financial operations for the programme activities will be carried out on financial and capital markets that are subject to national regulations, this risk of money laundering and financing of terrorism is considered low. Nonetheless, IFAD will not only establish specific rules and eligibility criteria for all financial transactions in each target country (as part of the IFAD project cycle), but all portfolios financed will be subject to IFAD no objection through the IFAD No Objection Tracking Utility System (NOTUS) online tool.

67. IFAD included first-level due diligence in the financial management assessment questionnaire and in the financial management summary. IFAD confirmed that the questionnaire considered the risks and vulnerabilities of money-laundering, financing terrorism, prohibited practices and integrity issues in the programme and activities and among the counterparties.

68. The Office of Risk Management and Compliance (ORMC)/Compliance Team has reviewed the financial management assessment questionnaire and is satisfied that the preliminary risk assessment is satisfactory and that appropriate internal controls have been established to identify, monitor, mitigate and remedy any risks and incidents related to money-laundering, financing terrorism, prohibited practices and integrity issues that may arise.

69. IFAD confirmed that policies and procedures on compliance, anti-money-laundering and countering the financing of terrorism (AML/CTF) and other compliance-related topics are in place to avoid such risks. Strong risk management and governance structures are applied by the Local National Agricultural Banks, which are the national implementing entities (NIEs) and financial institutions (FIs).
70. IFAD confirmed that proper due diligence processes will be led by IFAD and the NIEs, financial experts from local FIs and technical experts from the respective ministries of agriculture. Oversight of the respective ministries of finance and regulators (i.e. the two central banks – the Central Bank of West African States and the Bank of Ghana) will contribute to ensuring robust engineering, procurement and governance systems. Additionally, agreements between recipient governments and relevant sector ministries will ensure better management of loans during and beyond the programme lifecycle.
71. IFAD confirmed that bank accounts have been opened with the two above-mentioned central banks for its projects or funded projects. The central banks have strict rules regarding AML / CFT and monitor all accounts, including those of the agricultural banks that are governed by the central banks. The AML/CFT policies of the central banks will apply to this programme and all IFAD-funded projects.
72. Programme operation risks will be mitigated by a comprehensive due diligence process led by IFAD technical and financial experts and by ensuring that a robust monitoring and evaluation system is set up and fully operational. Technical assistance agreements and procurement procedures will be established at each NIE level to support and ensure high-quality delivery outcomes.
73. IFAD confirmed that there will be no planned disbursements or distribution of cash, vouchers, commodities or other items of value either directly or indirectly to programme beneficiaries as part of any activities of this programme.
74. IFAD confirmed that procurement for the programme will be carried out in accordance with applicable IFAD rules and practices, as well as the IFAD Programme Implementation Manual (PIM), the programme-specific PIM and loan administration arrangements established through the letter to the borrower and the financing agreement between IFAD and each national borrower/recipient.
75. IFAD has procurement guidelines in place that require an IFAD no objection through the NOTUS platform for any procurement of goods and services. IFAD confirmed that it will ensure that procurement controls are safeguarded through this process.
76. IFAD confirmed that its grievance mechanism will be used for reporting complaints or allegations of wrongdoing related to the programme or associated activities (whistleblower programme). A description of the IFAD grievance mechanism is provided under the ESMF. The programme will establish a community engagement process and provide access to information on a regular basis.
77. Complainants should first bring any matter of concern to the attention of the government or non-governmental organization responsible for planning or executing the project or programme and overseeing the work in the field. If the EE does not adequately respond, the matter may then be brought to the attention of IFAD. The issue may be brought directly to the attention of IFAD if the complainants feel that they might be subject to retaliation if they submit their complaint to the EE directly.
78. The objective of the IFAD Complaints Procedure is to ensure that appropriate mechanisms are in place to allow individuals and communities to contact IFAD directly and file a complaint if they believe that they are, or might be, adversely affected by an IFAD-funded project or programme that does not comply with IFAD social and environmental policies and the mandatory aspects of IFAD Social, Environmental and Climate Assessment Procedures.

Complaints must concern environmental, social and climate issues only and should not involve accusations of fraudulent or corrupt activities in relation to programme implementation; these are dealt with by the IFAD Office of Audit and Oversight.

79. Additionally, IFAD confirmed that the programme will utilize (to the extent possible) every available grievance redress mechanism, including associations (e.g. farmers' associations/organizations), traditional councils (e.g. Paramount Chiefs and elders), village square engagement (consisting of representatives of men, women and social groups), village general assemblies and the programme's national programme coordination unit.

80. **Recommended risk rating:** The ORMC/Compliance Team has conducted a review of the programme in accordance with relevant GCF Board approved policies and does not find any material issue or deviation with respect to compliance issues. Based on available information for this funding proposal, the ORMC/Compliance Team has determined a risk rating of "medium" and has no objection to this request proceeding to the next steps for processing.

81. The ORMC/Compliance Team would like to remind IFAD, as the AE, of its continuing obligations and responsibilities with regard to monitoring and reporting any risks for money-laundering, financing terrorism, or prohibited practices among the intended counterparties, EEs, beneficiaries, or persons involved, or any of the proposed activities.

4.3.5. GCF portfolio concentration risk (low risk)

82. In the case of approval, the impact of this proposal on the GCF portfolio concentration in terms of result area and single proposal is not material.

4.3.6. Recommendation

83. It is recommended that the Board consider the above factors in its decision.

Summary risk assessment		Rationale
Overall programme	Medium	The overall programme risk is assessed as medium, as a result of the AE's capability and the number of EEs involved as well the operating environment, demand risks and risks related to the continued additionality of proceeds during the full lifetime of the programme.
Accredited entity/executing entity capability	Medium	
Programme-specific execution	Medium	
GCF portfolio concentration	Low	
Compliance	Medium	

4.4 Fiduciary

84. IFAD will serve as the AE for this programme. As the AE, IFAD will provide oversight and implementation support, ensure the quality of the programme, and be responsible for ensuring that programme reporting is provided in a timely manner.

85. The financial management of and procurement for the programme will be carried out in accordance with applicable IFAD rules and practices, as well as the IFAD PIM, the programme-specific PIM and loan administration arrangements established through the letter to the borrower. Financial statements will be prepared in accordance with internationally accepted standards.

86. Programme procurement plans will be implemented in accordance with the IFAD Procurement Handbook and procurement plan guidelines will be prepared annually and validated by the programme steering committee.

87. An audit will be conducted annually in accordance with the IFAD Handbook for Financial Reporting and Auditing of IFAD-Financed Projects.
88. IFAD will create an RCU which will oversee, compile, consolidate and control the work of the EEs. The EEs will implement the programme through IFAD baseline investment PMUs (grant proceeds) and the LNAB project investment units. The programme's national EEs are the five host countries represented by their respective ministries of economy and finance. The national EEs will be the recipients of the GCF financing and will guarantee the repayment of the GCF loan portion.
89. The national EEs will sign an on-lending agreement with each of the five LNABs (the Agricultural Bank of Burkina Faso, the National Investment Bank of Côte d'Ivoire, the ARB Apex Bank Limited of Ghana, the National Agricultural Development Bank of Mali and the Agricultural Bank of Senegal). The selected LNABs will be responsible for implementing green credit lines with the GCF loan proceeds under component 1. In addition, each LNAB will set up a dedicated project investment unit whose primary role will be to receive the GCF proceeds and on-lend to beneficiaries or other FIs.
90. To implement the GCF grant proceeds (component 2), the national EEs will make arrangements with the ministries of agriculture of the respective countries who will act through the IFAD PMUs. The EEs of the GCF grant proceeds for IGREENFIN I are the Ministry of Agriculture, Hydro-agricultural Development and Mechanization of Burkina Faso; the Ministry of Agriculture and Rural Development of Côte d'Ivoire; the Ministry of Agriculture of Mali; the Ministry of Food and Agriculture of Ghana; and the Ministry of Agriculture and Rural Equipment of Senegal. IFAD has conducted a capacity assessment of the LNABs and the ministries of agriculture of the five targeted countries.
91. The LNABs and respective ministries of agriculture will each undergo an annual external audit as mandated by the IFAD financing agreement. In addition, at the regional level, an external audit of the RCU of IGREENFIN I and the GGW Regional Support Programme's activities will be undertaken annually.
92. Procurement for the programme will be carried out in accordance with applicable IFAD rules and practices, as well as the IFAD PIM, the programme-specific PIM and loan administration arrangements established through the letter to the borrower and the financing agreement between IFAD and each national borrower/recipient.

4.5 Results monitoring and reporting

93. The cross-cutting programme covers 13 countries in Africa, namely Burkina Faso, Chad, Djibouti, Eritrea, Ethiopia, Mali, Mauritania, Niger, Nigeria, Senegal and the Sudan. The programme is part of the GGW Initiative and the second one within the umbrella programme supported by the GCF.
94. The thirteen countries of the programme are among the poorest and most climate-vulnerable countries in the world. Climate change, political instability and unsustainable agricultural practices have significantly impacted farmers throughout the region.
95. The program consists of three components. In component 1, a green financing facility will be established to provide concessional loans to foster adaptation and mitigation best practices for green businesses within the selected agricultural value chains in each target country. In component 2, a technical assistance facility will be created to address capacity, knowledge and policy gaps hindering the uptake of green business projects in the selected countries. In component 3, knowledge management, information exchange, digital transformation and other overarching activities will be supported.

96. The program is expected to have a cumulative mitigation impact of 5,637,184 tCo₂eq over the program life span and reach 378,600 direct and 2,494,000 indirect beneficiaries in the 13 countries. Program impacts were estimated based on transparent and credible methodologies and allow for consistent measurement of results during program implementation. The impact generated in each of the participating countries is described in Annex 17 of the funding proposal.
97. The Theory of Change clearly shows the logical pathways for generating results via program activities, and how those contribute to the outcomes and impacts of the program. Relevant assumptions have been also made and an explanation of the ways in which barriers are addressed has been provided.
98. The logical framework of the programme was designed in accordance with the GCF RMF/PMF and the programme Theory of Change. The logical framework has selected relevant impact, outcome and output indicators aligned with the result areas that the programme is contributing to, as well as the programmes specific activities.
99. The implementation table has been prepared in Annex 5 and the M&E plan in Annex 11 of the funding proposal.

4.6 Legal assessment

100. The AMA was signed with the AE on 24 September 2018, and it became effective on 9 November 2018.
101. The AE has not provided a legal opinion/certificate confirming that it has obtained all internal approvals and it has the capacity and authority to implement the programme. It is recommended that, prior to submission of the funding proposal to the Board (a) the AE has obtained all its internal approvals; and (b) GCF has received a certificate or legal opinion from the AE in form and substance satisfactory to GCF confirming that all final internal approvals by the AE have been obtained and that it has the authority and capacity to implement the programme.
102. The proposed programme will be implemented in 13 countries: Burkina Faso, Chad, Côte d'Ivoire, Djibouti, Eritrea, Ethiopia, Ghana, Mali, Mauritania, Niger, Nigeria, Senegal and Sudan. GCF is not provided with privileges and immunities in 12 of the 13 countries. This means that, among other things, GCF is not protected against litigation or expropriation in these countries, which risks need to be further assessed. The status of negotiation of bilateral agreements on privileges and immunities of GCF with the relevant countries is as follows:
- (a) Burkina Faso: An agreement on privileges and immunities was signed with Burkina Faso and became effective on 6 December 2021;
 - (b) Chad: The Secretariat sent a draft agreement together with a background note on privileges and immunities to the NDA in December 2015, and a meeting was held with the Chad delegation at the twenty-first session of the Conference of the Parties to the UNFCCC (COP 21) in December 2015. In response to a request from the NDA, the Secretariat sent the draft agreement and background note in both English and French in October 2016, and a reminder was sent in February 2017. However, no formal response has been received so far;
 - (c) Côte d'Ivoire: The Secretariat sent a draft agreement together with a background note on privileges and immunities to the NDA in November 2016 and a meeting was held during the Global NDA Conference in October 2018. However, no formal response or updates have been received so far;
 - (d) Djibouti: The Secretariat sent a draft agreement together with a background note on privileges and immunities to the NDA in December 2015 and since then emails

- containing comments on the draft agreement have been exchanged. A reminder email was sent to the NDA in August 2017; however, no formal response has been received so far;
- (e) Eritrea: Discussions with Eritrea on a draft agreement on privileges and immunities are yet to start;
 - (f) Ethiopia: The Secretariat sent a draft agreement together with a background note on privileges and immunities to the NDA in November 2015 and since then emails have been exchanged. A reminder email was sent in November 2017; however, no formal response has been received so far;
 - (g) Ghana: The Secretariat sent a draft agreement together with a background note on privileges and immunities to the NDA in November 2015. An in-person meeting was held with the delegation of Ghana during COP 22 in November 2016 and emails have since been exchanged. The last communication is an email received in February 2018 informing the Secretariat that the Ministry of Food and Agriculture “would ascertain whether they have any issues with it and get back to GCF”;
 - (h) Mali: The Secretariat sent a draft agreement together with a background note on privileges and immunities to the NDA in December 2015, and a meeting was held with NDA representatives during the Global NDA Conference in October 2018. However, no formal response on the text of the draft agreement has been received so far;
 - (i) Mauritania: The Secretariat sent a draft agreement together with a background note on privileges and immunities to the NDA in November 2016, followed by an official letter addressed to the Ministry of Environment and Sustainable Development mentioning the context of the draft agreement in February 2017. In August 2017, the Secretariat sent reminders to the NDA on the progress of the review and was informed that the draft agreement had been forwarded to the Ministry of Foreign Affairs. A reminder was sent in February 2018; however, no formal response has been received so far;
 - (j) Niger: The Secretariat sent a draft agreement together with a background note on privileges and immunities to the NDA in December 2015, and a meeting was held with the Niger delegation during COP 21 in December 2015. In February 2017, the draft agreement and background note were sent again in both English and French to the NDA; however, no formal response has been received so far;
 - (k) Nigeria: The Secretariat sent a draft agreement together with a background note on privileges and immunities to the NDA in September 2015, and a meeting was held with the delegation of Nigeria during COP 21 in December 2015. A reminder was sent to the NDA in November 2016; however, no formal response has been received so far;
 - (l) Senegal: The Secretariat sent a draft agreement together with a background note on privileges and immunities to the NDA in September 2015. Meetings were held with the delegation of Senegal at COP 21 in December 2015, at COP 22 in November 2016, and at the Senegal Permanent Mission to the United Nations in New York in August and October 2016, respectively. In February 2017, the NDA informed the Secretariat that all documents had been sent to the Ministry of Finance at the request of the Ministry of Foreign Affairs, and that it would inform GCF of the status of consideration of the documents by the Ministry of Foreign Affairs. A reminder email was sent in August 2017; however, no formal response has been received so far; and
 - (m) Sudan: A meeting was held with the Sudan delegation during COP 21 in December 2015, and the Secretariat sent a draft agreement together with a background note on privileges and immunities to the NDA in December 2015. A reminder was sent to the NDA in October 2018; however, no formal response has been received so far.

103. The Heads of the Independent Redress Mechanism and Independent Integrity Unit have both expressed that it would not be legally feasible to undertake their redress activities and/or investigations, as appropriate, in countries where GCF is not provided with relevant privileges and immunities. Therefore, it is recommended that disbursements by GCF are made only after GCF has obtained satisfactory protection against litigation and expropriation in the countries, or has been provided with appropriate privileges and immunities.

104. The AE requested from the Secretariat a deviation from the terms of the AMA to exclude the application of GCF step-in rights for this programme, indicating that since this is a public sector programme involving sovereign States as counterparties, it is not feasible for the AE to allow GCF or any other party to take over its contractual relationship with the EEs under the subsidiary agreements. Accordingly, and subject to, as a condition for execution of the funded activity agreement, the receipt by the Secretariat of relevant written confirmation from the AE that the NDAs of each host country have been informed of this deviation to the AMA and its implications, an express provision has been included in the term sheet to allow for such deviation from the AMA. This means that, subject to Board approval, in case the AE decides to discontinue programme implementation, GCF will not have recourse to take over the role of the AE or appoint another party to replace the AE, and the only alternative would be the cancellation of the programme, without prejudice to other remedies available to GCF under the AMA.

4.7 List of Proposed Conditions (including Legal)

105. In order to mitigate risk, it is recommended that any approval by the Board is made subject to the following conditions:

- (a) Submission by the AE to GCF of a certificate or legal opinion, in form and substance satisfactory to the GCF Secretariat, within 120 days after Board approval, confirming that the AE has obtained all final internal approvals needed by it and has the capacity and authority to implement the proposed programme. In case Board approval of the funding proposal is conditional on substantial modification of the funding proposal, in accordance with the terms of the AMA, the AE shall have a period of 180 days to obtain its final internal approvals and provide to GCF such certificate or opinion;
- (b) Signature of the funded activity agreement in a form and substance satisfactory to the GCF Secretariat within 180 days from the date of Board approval, or the date the AE has provided a certificate or legal opinion confirming that it has obtained all final internal approvals, whichever is later; and
- (c) Completion of the legal due diligence to the satisfaction of the GCF Secretariat.

Independent Technical Advisory Panel's assessment of FP183

Proposal name:	Inclusive Green Financing Initiative (IGREENFIN I): Greening Agricultural Banks & the Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Great Green Wall (GGW) Countries – Phase I
Accredited entity:	International Fund for Agricultural Development (IFAD)
Country/(ies):	IGREENFIN I [Components 1 and 2]: Burkina Faso, Côte d'Ivoire, Ghana, Mali and Senegal GCF GGW Regional Support Programme [Component 3]: Burkina Faso, Chad, Djibouti, Eritrea, Ethiopia, Mali, Mauritania, Niger, Nigeria, Senegal and Sudan
Programme size:	Medium

I. Assessment of the independent Technical Advisory Panel

1.1 Impact potential

Scale: High

1. This funding proposal is for a medium-sized, cross-cutting programme in environmental and social safeguards (ESS) Category B. The programme is submitted by 13 African States, 10 of which have least developed country (LDC) status. Five states will participate in Phase I of the IGREENFIN programme – three Great Green Wall (GGW) states, Burkina Faso, Mali and Senegal, plus Ghana and Côte d'Ivoire. Burkina Faso, Mali and Senegal, plus a further eight countries (Chad, Djibouti, Eritrea, Ethiopia, Mauritania, Niger, Nigeria and Sudan), will participate in a Regional Support Programme for the GCF Great Green Wall portfolio. The total cost of the funding proposal is USD 216,900,720, including a GCF loan contribution of USD 90,000,000 and a GCF grant contribution of USD 36,900,720, with the GCF funds forming 58.5 per cent of the total cost. The remaining USD 90 million, or 41.5 per cent of the total project cost, includes grant contributions for technical assistance from IFAD and the African Development Bank (AfDB) totalling USD 51 million. Concessional loan finance totalling USD 39 million, from IFAD, Islamic Development Bank and the five National Agricultural Banks of the Phase I countries, will be blended with highly concessional GCF loan funds to 'jumpstart' the market for loan facilities to organizations of small-scale farmers and agro-processing and renewable energy businesses that are investing in low emission, climate resilient practices.

2. The eleven countries of the GGW, and Côte d'Ivoire and Ghana, are among the world's poorest countries and among those most vulnerable to climate change. Agriculture, livestock and forestry activities are central to their economies and more than 70 per cent of rural communities depend directly on rainfed agriculture. The region's agriculture, forestry and other land use, and energy sectors are responsible for just under a quarter of anthropogenic greenhouse gas (GHG) emissions, generated mainly by deforestation (wood and clearing land for agriculture), livestock raising, soil and nutrient management and unsustainable rice production. Over recent decades, increasing annual average temperatures have combined with increasingly unpredictable rainfall patterns to put significant stress on production of staple and

other crops, and some parts of the region have experienced more frequent droughts and/or floods. Climate projections indicate that precipitation levels will increase in some areas, and decrease in others, with increasingly erratic timing of growing seasons, while temperatures are expected to increase between 1 and 1.72°C for the 2031–2050 period compared to the 1986–2005 reference period. These changes, combined with direct human pressures from unsustainable land management practices and, in the absence of adaptation interventions, are expected to cause a decline in agricultural productivity, reducing food availability and farmers' incomes and fuelling competition over diminishing natural capital, greater instability, conflict and migration, especially of rural youth.

3. The overall objective of the proposed GCF programme is to create an enabling environment to increase the flow of funds to green businesses engaging in climate resilient, low emission agricultural practices in the GGW area and Côte d'Ivoire and Ghana. IGREENFIN I (Components 1 and 2) aims to build the resilience of smallholder farmers to current and future climate risks in the five target countries by: (i) providing support to selected local national agricultural banks (LNABs) for the creation of 'green credit lines' to finance the business projects of smallholder farmers organized in farmer organizations, cooperatives, women and youth-led organizations, and micro, small and medium-sized enterprises (MSMEs) that will either result in lower emissions, or help adapt to changing climatic conditions, or both – such as solar photovoltaic-powered boreholes for drip irrigation; and (ii) addressing capacity-building, knowledge and policy gaps that hinder the transfer and adoption of adaptation and mitigation practices by green businesses. The Regional Support Programme (Component 3) aims to enhance the collective impact of all GCF-funded projects contributing to the GGW, including IGREENFIN I, through the establishment of a coordination framework for knowledge generation and sharing, project implementation and reporting, mobilizing private investment and promoting innovation at scale.

4. **Adaptation impact:** The programme will help agricultural communities in the three selected GGW countries under IGREENFIN I (Burkina Faso, Mali and Senegal) plus Côte d'Ivoire and Ghana, to reduce their vulnerability to a range of natural hazards, frequently worsened by climate change, including droughts, flooding, wildfires, water scarcity, sandstorms, extreme heat and desertification. IGREENFIN I intends to directly build the resilience and adaptive capacity of 378,600 smallholder farmers, 50 per cent female, and 50 per cent youth, indirectly benefitting the 2,494,000 people making up their households, their health and well-being, food and water security, as well as the health of the ecosystems on which their livelihoods depend. This will be achieved through the Green Business Financing Facility in the five selected countries – building the capacity of five local national agricultural banks to offer affordable financing for climate resilient, low emission agriculture interventions (green governance and lending products, green credit indicator system for management, long-term loan periods and unlocking barriers to innovative climate finance). It will support loans and technical assistance to an estimated 1,500 MSMEs and 2,500 producer organizations and cooperatives, as well as the transfer or licensing of 200 technologies and innovative solutions in agriculture and agro-processing.

5. In response to a question from the independent Technical Advisory Panel (TAP) about the narrow focus of eligibility criteria on 10 specific value chains, the AE highlighted the benefits of focusing the banks' loan portfolios in order to be able to provide end-to-end supply chain solutions, stating that "this approach has proven to be more effective in increasing access to financing and reducing agricultural costs and risks, than helping individual businesses in isolation". In discussion, the AE acknowledged that additional value chains with demonstrated demand for loans and with potential for climate resilience and profitability (on a sustainable basis) could be added, following the programme's mid-term review.

6. **Mitigation impact:** The cross-cutting project is expected to have a modest but significant impact in the five countries, through GHG emissions avoided as a result of the activities undertaken by the borrowing organizations using renewable energy technologies and

sustainable forest and land management practices (with mitigation benefits derived from 35 per cent of the programme's total costs). The major impact will come from activities undertaken in the farming/agroforestry sector, applying agroecological principles and using the farmer-managed natural regeneration and pasture restoration practices that are common in the region to switch from ongoing loss of above- and below-ground carbon through landscape degradation, to effective capture and storage of carbon at scale in healthy, productive agroecosystems. These activities, with ongoing technical assistance, are expected to result in better practices across 250,000 hectares, resulting in 4,998,515 tonnes less of CO₂eq emissions over the programme's 25-year lifetime, compared to a scenario without the project in which climate change combines with human actions to lead to ever-worsening land degradation and desertification, with concomitant loss of above- and below-ground carbon. In response to questions from the independent TAP about the model of rangeland management that lies behind the proposed interventions in the sustainable livestock production value chain, the AE provided additional information, but was unable to comment on the issue of reducing methane emissions from enteric fermentation, reductions of which have not been factored in to the emission reduction estimates.

7. The Regional Support Programme (RSP) is designed to reinforce the adaptation and mitigation benefits of IGREENFIN Phase 0 (the existing IFAD-Niger project) and Phase I, as well as 29 already approved GCF projects being implemented through 17 accredited entities (AEs) in 11 countries of the Great Green Wall. These countries are contributing to the GGW's African-led initiatives to restore the productivity and vitality of the highly climate change-vulnerable Sahel region, with the goals to restore 100 million hectares of degraded land, sequester 250 million tonnes of carbon, and create 10 million green jobs by 2030. The RSP aims to strengthen GGW countries' institutional capacity on reporting, generating and sharing innovative solutions and knowledge innovation on green business and accessible financing, and increase private sector financial participation in green business projects. The RSP has a technical capacity component that aims to enhance knowledge management and exchanges, accelerating the uptake of best practices and learning and informing policy and investments. However, the funding proposal is not explicit on the type of policy support envisaged, and detail is not provided on the concrete deliverables expected from the activities.

8. One of the aims of the RSP is "to facilitate more consistent and credible reporting of climate results of GCF-funded projects/programmes, which can be aggregated and reported as part of the interventions of the GGWI". The independent TAP foresees that it will be challenging to 'retrofit' new indicators for tracking and monitoring portfolio-wide GCF investments related to the GGW, given the diverse nature of the 29 projects already existing, and believes this process has the potential to cause confusion with the existing project-level monitoring and evaluation systems, but understands that it will be important to try and achieve this.

9. The independent TAP assesses the impact potential of this cross-cutting project as high, including both mitigation and adaptation aspects.

1.2 Paradigm shift potential

Scale: Medium

10. The programme aims to enable a paradigm shift to low emission and climate resilient agriculture and associated value chains by providing affordable loan finance in new green credit lines through the target countries' national agricultural banks. The intention, as expressed in the funding proposal, is that "Through the mainstreaming work, these institutions will be able to sustain the GCF grant and concessionality beyond the programme's duration...By creating a self-sustaining loan portfolio that ensures institutionalization of green credit lines within the banks in the long term, GCF bank partners (commercial banks, micro-finance institutions) are expected to lend at the same interest rate and thus influence market rates. Subsequently,

interest rates are likely to decrease, gradually making bank loans become more attractive once the IGREENFIN I programme has ended.”

11. In discussion with the AE, the independent TAP expressed the view that the programme’s capacity-building of the national agricultural banks and intermediary financial institutions, and the technical assistance to borrowers, would go some way to enabling more accurate risk assessment and eliminating misperceptions, and also enabling business success and reasonable repayment rates. Many factors would likely remain beyond the programme’s influence, however, such as underdeveloped communications and transportation infrastructure, and covariate risks due to rainfall and price fluctuations. This could mean that transaction costs and default rates remain fairly high, once the subsidizing effect of the programme’s concessional financing and technical assistance is removed, such that interest rates offered by the agricultural and other banks post-project increase again, indicating that there has not been a fundamental shift in how the sector operates.

12. In response, the AE highlighted that “With the IGREENFIN programme, IFAD aims to jump-start this market, which is still in the incipient phase in many countries. By building the [local national agricultural banks] LNABs’ capacity to offer affordable green finance for smallholder agriculture and the capacity of smallholder farmers organized in cooperatives, farmers organizations and MSMEs to design and implement bankable green projects, the programme will begin to fill the gap in financing for agriculture in the region and alter the general perception of smallholder agriculture as a risky business to one that views this sector as a vast market with great potential and opportunities. Throughout the programme’s lifecycle, studies and policy papers will be produced on climate risk management to support institutional capacity-building and the design of lending products. This will help encourage the financial sector to invest more in green businesses and reduce the cost of the risks, which is usually the motive for charging high interest rates on loans to smallholder farmers. The partnership with central banks and governments will contribute to maintaining these specific provisions for this specific targeted group after the programme ends.”

13. According to IFAD the determinants of interest rates on loans include: “(i) cost of resource, (ii) the management fee, (iii) the cost of risk, and (iv) the margin. In the targeted countries, the first three determinants are high for the simple reason that there is lack of capital. Target groups submit very few green projects for financing because of the lack of incentives and the general misperception of this group, despite the fact that they generate most of the agricultural production in their countries. This leads banks to focus on medium and big farmers only. To substantially reduce interest rates and reach smallholder farmers, IGREENFIN will support the strengthening of the banks’ know-how to raise capital on external markets and reduce the cost of resource, and to develop bankable pipelines, which will reduce the management cost for a limited number of projects, as well as the high risk perception. By responding to those determinants of the interest with the programme interventions, after the programme ends, accredited banks will build on the experience to develop this market further and continue working on green business projects and green bonds, in accordance with their vision and strategy already shared with the AE. Additionally, the target countries’ central banks will be better equipped to support green finance sector reform and policies and promote specific incentives to support green growth within the financial sector as well as financial inclusion of smallholder farmers.”

14. The AE also pointed to “the rapid development change obtained in current or similar experiences in the region such as ‘Accueil | DER/FJ - Délégation Générale à l’Entrepreneuriat Rapide pour les Femmes et les Jeunes’ in Senegal; SUNREF program, with AfDB initiatives with MSMEs and a similar programme backed by World Bank, and the current movement ‘Home | Finance in Common’, which IFAD and partners are supporting. These initial initiatives have helped get the process started in many sectors, including in smallholder agriculture with local public agricultural banks ... Since then, they have attracted more resources and the portfolios

are being sustained with innovative instruments.” It remains unclear to the independent TAP whether the interest rates offered in these cases have remained low, after the removal of the effective subsidy by the various programme interventions. In general, it is to be hoped that interest rates will be lower after the IGREENFIN programme ends, but it seems optimistic to expect a complete transformation of the agricultural loan finance sector in the five countries, based on a portfolio of 800 loans in each country (4,000 target split across five countries).

15. The paradigm shift involved in the proposed programme is rated as medium.

1.3 Sustainable development potential

Scale: High

16. In addition to the planned adaptation and mitigation benefits, the farming, renewable energy and business activities to be supported through the loan portfolio of the IGREENFIN I programme (Components 1 and 2) are expected to have a number of co-benefits for sustainable development, including environmental, social and economic co-benefits, and benefits in terms of gender sensitivity and social inclusion.

17. **Environmental co-benefits:** Assuming that the loan programme’s Technical Assistance Facility is successful in promoting a combination of traditional knowledge on woodland and pasture regeneration with modern best practices in agroforestry, crop and livestock farming, there is potential for multiple environmental benefits – including increased biodiversity, enhanced soil fertility and stability, maximization of soil moisture and groundwater resources, and improved storage and sequestration of carbon in regreened agroecological landscapes.

18. Insofar as the Regional Support Programme will help to maximize the collective impact of the GCF Great Green Wall portfolio of 29 existing projects, there is potential for significant global environmental benefits in terms of the GGW’s overall objectives of combating desertification. Scientists have observed an expansion of the Sahara Desert by 10 per cent over the past century,¹ affecting regional food and water security, and also influencing global weather patterns and human health, as huge seasonal dust clouds are carried across the Atlantic as far as Central America. The ambitious Great Green Wall concept represents humanity’s collective effort to stem the encroachment of the desert into the Sahel by supporting sustainable livelihoods on restored and regreened land that also reduces atmospheric emissions.

19. Building on the brief references in the funding proposal to the five pillars of the Great Green Wall Accelerator, including Pillar 2 on “Sustainable management of ecosystems and land restoration”, the IGREENFIN I programme’s specific approaches to promoting regenerative, restorative and sustainable approaches for management of cattle and grazing lands, and for agroforestry, have been outlined in detailed response documents from the AE.

20. The independent TAP would like to put on record two risks associated with loans through the IGREENFIN programme for production of commodity tree crops: the risk that loans could be used for monocrop plantations without the environmental benefits of mixed-use agroforestry, and the risk that areas of intact or partial indigenous woodland could be cleared for tree-crop production, contributing to deforestation. Table 16 in the funding proposal, “Preliminary selected sustainable value chains”, highlights the benefits of mixed agroforestry – growing indigenous tree and other species in combination with the focus commodity tree crops of mango, cashew and shea. The independent TAP notes, however, that use of other species is not mentioned by the example loan applicants in the indicative loan pipeline in annex 23, who reference only the cultivation of the commodity crops.

¹ Thomas, N. and Nigam, S. (2018). “Twentieth-Century Climate Change over Africa: Seasonal Hydroclimate Trends and Sahara Desert Expansion”, *Journal of Climate*. Vol. 31.

21. In response to follow-up questions from the independent TAP, the AE provided assurances that the technical assistance facility will include awareness-raising on the multiple benefits of applying agroecological principles in diverse agroforestry systems, and that the investment screening process will ensure that tree crops are grown on already degraded land, and will be monitored to ensure that they do not expand into areas of remaining woodland. This is essential, given the current high rates of deforestation in parts of the target countries for commodity production, for example deforestation of savannas and dry forests in Cote d'Ivoire in recent years for monocrop cashew production fetching high global prices.²

22. **Social co-benefits:** The project has a number of important social co-benefits, including: strengthening cooperative organizations and businesses and creating expanded employment for women and youth within them; providing economic opportunities for unemployed youth in rural villages and small towns, thus reducing the phenomenon of rural exodus; improving the food security of beneficiaries through increasing yields of millet and nutritious tree crops, and enhancing livestock productivity; diversifying household income through creating opportunities for cash crops and value addition to subsistence crops; enhancing communities' access to agricultural extension through the technical assistance facility, particularly for women not previously exposed to such opportunities, and including female-headed households; and improving continuous, affordable, clean and safe energy access for agriculture and agro-processing through new mini-grids.

23. **Economic co-benefits:** The implementation of the new "green credit lines" in the five target countries to finance the business projects of smallholder farmers, mini-grid providers and agro-processing businesses should have a significant effect, both in terms of direct job creation and through stimulating economic activity in rural areas and linking them to markets in urban centres and beyond. Enhanced yields and new income streams will be combined with savings on diesel costs to free up cash for operations and maintenance, and, for some cooperatives with capacity, to take loans to further expand production and processing facilities. The programme aims to reduce poverty through increased productivity of subsistence and cash crops and livestock production, as well as income generation activities such as the production of and value addition to different fruit and non-timber forest products, including tamarind, Arabic gum, baobab leaves, as well as fodder and seedlings sales. The programme will build up links in the selected agricultural value chains from production to markets, enabling value addition and preventing food spoilage and waste. The loans taken through the programme will contribute to creating an estimated 703,600 jobs in total on farms, in renewable grid operations and agri-businesses. The Regional Support Programme will include the creation of a digital innovation platform that could spur economic development across all the countries of the Great Green Wall.

24. **Gender-sensitivity and social inclusion:** The funding proposal and gender analysis acknowledge the constrained socioeconomic position of women in most of the rural regions of the five target countries, as well as the high rate of youth under- and unemployed, which increases the risk of youth radicalization and involvement in jihadist groups. The analysis shows that women play a central role in feeding their families and ensuring other basic needs (water, firewood for fuel, etc.) and that these responsibilities are becoming more onerous with declining crop yields due to climate change. Although women represent 60 to 70 per cent of the on-farm work effort, they often lack access to productive assets, finance and knowledge. The programme will increase opportunities for livelihood improvement and provide concrete benefits to smallholder farmers and pastoralists, both men and women. Injection of loan funds and technical assistance will aim to increase yields, improve soil fertility, make more animal fodder available, and reduce time needed for wood and water collection.

² See, for example, civil society commentary - <https://news.mongabay.com/2021/10/forests-falling-for-cashew-monocultures-a-repeated-mistake-in-cote-divoire-commentary/>

25. The programme will likely face many challenges in reducing entrenched gender inequalities, but the loans will be made without collateral requirements and with flexibility on proving ability to repay, all of which will enhance women's chances of accessing finance to grow their own lines of business. The programme has also set clear targets: 50 per cent of loans are to be granted to women-led MSMEs, cooperatives and farmer organizations, with a special focus on female-headed households, which make up approximately 15 per cent of households in the region. The other 50 per cent will be dedicated to men, of which 50 per cent will be youth-led MSMEs, cooperatives and farmer organizations in which young women participate. Specific actions will be developed to strengthen the technical and managerial capacities of women, aimed at providing them with tools and skills for identifying and developing bankable business plans and improving their level of financial education. As stated in the gender action plan, gender-disaggregated data will be assessed against the appropriate indicator to measure women's enhanced access to loans. At least one-third of the National Steering Committee members will be women, to ensure that they influence the main strategic decisions.
26. The programme's sustainable development potential overall is rated as high.

1.4 Needs of the recipient

Scale: High

27. The needs of the five target countries of the western Sahel, proposed to be involved in IGREENFIN I, and the 11 countries of the entire Great Green Wall, proposed to be involved in the Regional Support Programme, have been considered – both in relation to vulnerability to climate change, and in relation to the level of development of the financial sector.
28. As outlined in the funding proposal, “The Sahel stretches from the Atlantic Ocean to the Horn of Africa across all eleven countries of the Great Green Wall and the northern part of coastal countries such as Benin, Côte d’Ivoire, Ghana, Guinea and Togo. One of the largest semi-arid and arid areas in the world, it is wedged between the Sudanian savannah to the south and the Sahara Desert to the north, which is expanding southward by up to 10 km per year, depending on the country. It is considered one of the poorest and most environmentally degraded regions in the world and therefore one of the most vulnerable to climate change, as the increase in temperatures, rainfall variability and the frequency of extreme weather events will only worsen the already harsh living and growing conditions. According to the IPCC [Intergovernmental Panel on Climate Change] 6th Assessment Report, monsoon precipitation is projected to increase over the central Sahel and decrease over the far western Sahel, and the monsoon season is expected to have a delayed onset and retreat. The IPCC also reported an observed increase in river flooding, drying and agricultural and ecological droughts, and projected increases in meteorological droughts at GWL [groundwater level] 4° (mostly in seasonal timescales), as well as mean wind speed, heavy precipitation and pluvial flooding.”
29. Scientists agree that there has been a gradual warming that is spatially variable, but has averaged out at 0.5 °C per decade across West Africa, and at 1 °C per decade in the Sahel, since the 1970s. Data on precipitation across the region is much more complex. In response to questions from the independent TAP about the specific experience of the five IGREENFIN I target countries in relation to climate change and drought, IFAD provided historical meteorological data on normal, dry and very dry years for at least 30 years for four of the countries – data which showed no trend in terms of increasing number of drought years per decade, but showed a high degree of interdecadal variability. It is therefore inaccurate to state, as the funding proposal does, that “climate change has increased the frequency, intensity and duration of droughts” in the region.
30. The AE highlighted a number of publications analysing meteorological data from the Sahel more broadly, all of which conclude that the region has experienced extreme decadal variability in rainfall during the twentieth century, with severe droughts during the 1970s and 1980s following a wet period in the 1950s and 1960s. Since the late 1990s, rainfall has partly

recovered to earlier levels, but has been accompanied by changes in the intra-seasonal time scale compared to the 1950s and 1960s, for example delayed onset and later retreat of the main rainy season in Mali's Sahelian regions, resulting in uncertainties regarding when to plant crops.

31. The AE shared one study which investigated the recent precipitation changes in the West African Sahel (corresponding more closely to the five target countries) using a high-resolution Climate Hazards Group InfraRed Precipitation with Station product over the period 1981–2014. The study found that “the recent increase in precipitation results principally from an increase in the number of wet days (+10 d[ays] compared to the normal) over the entire West African Sahel band, along with an increase in the precipitation intensity over the central part of the West African Sahel (+3 mm d(-1)). However, this overall increase in precipitation is associated with dry spells that are becoming more frequent but on average shorter over the entire West African Sahel band (on average by 30%), and with precipitation intensity that is decreasing (around 3 mm d(-1) during the study period) in the western part of the West African Sahel.”³

32. The independent TAP acknowledges that there is significant complexity in observed meteorological data in the West African Sahel, as well as significant uncertainty in climate scientists' rainfall projections for the region over the coming decades. Scientists agree, however, that the high inter-annual and inter-decadal variability that results from the effects of the Inter-Tropical Convergence Zone, is likely to continue as the climate evolves, and increased temperatures will enhance evapotranspiration.⁴

33. It seems fair to conclude that increasing temperatures combined with unpredictable rainfall patterns are likely to increase the vulnerability of rural communities and crop production in the Sahelian regions of the target countries, and that the needs of these regions for investment by rural communities in climate-smart agriculture are high. With small-scale producers and MSMEs largely excluded from the commercial financial sector historically, because of high risks and transaction costs, as well as a lack of understanding of the sector's potential (with the right technical assistance and value chain strengthening), there is a clear need to use concessional funding from GCF and co-financiers from development finance institutions to de-risk loan finance in this sector, in order to enable the building of adaptive capacity.

34. Despite the ongoing efforts of the Governments of the target countries and their financial and technical partners to strengthen resilience of the agriculture sector and enhance climate resilience and food security, new and additional approaches are needed, particularly in the provision of affordable finance provision for small-scale agriculture, agro-processing and renewable energy. The fiscal gap and balance of payments deficit for the three LDCs in particular, make the highly concessional GCF loans necessary. The programme will see investment in improving the enabling environment for central banks to access affordable capital for future loans by the state agricultural banks, and will unlock the necessary conditions for scaling up commercial finance to the agricultural sector in these countries, paving the way for greater private sector engagement in climate change adaptation and mitigation.

35. The needs of the recipients in this programme are rated as high.

1.5 Country ownership

Scale: Medium to high

³ Bichet, A. and Diedhiou, A. (2018). West African Sahel Has Become Wetter during the Last 30 Years, but Dry Spells Are Shorter and More Frequent. *Climate Research*. 75. 10.3354/cr01515.

⁴ Christensen, J.H., Hewitson, B., Busuioc, A. et al (2007). Regional Climate Projections. IPCC WG II Chapter 11.

36. Country ownership has been demonstrated for both the IGREENFIN programme Phase 1 in the five target countries (Components 1 and 2), and the Regional Support Programme for the 11 Great Green Wall countries (Component 3).
37. IGREENFIN I was designed at the request of the GCF national designated authorities (NDAs) of the five countries, following the approval of the simplified approval process IFAD-Niger IGREENFIN project (SAP012) “Inclusive Green Financing for Climate Resilient and Low Emission Smallholder Agriculture project for Niger” in late 2019. At an event in Dakar in February 2020, the NDAs, the Chief Executive Officers of the agriculture and investment banks, and government representatives from Burkina Faso, Côte d’Ivoire, Ghana, Mali and Senegal requested IFAD’s support to pursue similar projects with GCF. The meeting led to the formation of a coalition of local state-owned agricultural and investment banks seeking to shift into providing financing for climate-resilient, low-emission smallholder agriculture. IFAD then engaged further with the NDAs of the five target countries in developing the programme, including consultations with civil society organizations. The strong engagement by the five Ministries of Economy and/or Finance as national executing entities (EEs), and by the five local national agricultural banks is an indicator of country ownership that, together with the revolving fund structure, bodes well for the loan facilities being sustained post project.
38. The IGREENFIN I programme is aligned with the five countries’ national development plans and national commitments on climate mitigation and adaptation, as included in their national adaptation programmes of action, national climate change policies and strategies, nationally determined contributions (NDCs), national communications to the UN Framework Convention on Climate Change (UNFCCC), and national strategies for disaster risk reduction, as well as to several of the Sustainable Development Goals for Agenda 2030.
39. The independent TAP notes that the NDCs of four of the five target countries have recently been revised, with new versions submitted to the UNFCCC between December 2020 and October 2021. An analysis of the new NDCs should be completed by the AE, to ensure that the transfer of climate resilient, low carbon technologies and the development of innovative green business models are well aligned with the priorities and adaptation and mitigation targets contained in each country’s NDC.
40. For the Regional Support Programme, country ownership has emerged through the process of developing the programme. In September 2020, at an online meeting organized by the United Nations Convention to Combat Desertification (UNCCD) and involving the responsible Ministers of each of the 11 Great Green Wall countries, these countries issued a joint Ministerial Declaration inviting GCF and other international entities to increase their support to accelerate the pace of activities contributing to the GGW’s objectives. In response, the January 2021 One Planet Summit saw a range of development partners committing to the launch of a Great Green Wall Accelerator, to be led by the Pan Africa Agency for the Great Green Wall (PAAGGW) and temporarily coordinated by UNCCD, and aiming to strengthen coordination, collaboration and monitoring of GGW efforts.
41. In this context, IFAD was proposed to lead a new Great Green Wall Regional Support Programme of the GCF, and will work in partnership with other agencies including the PAAGGW, UNCCD, United Nations Food and Agriculture Organization (FAO), Global Environment Facility (GEF) and the World Bank. No-objection letters were received by the AE for all 11 Great Green Wall countries. Component 3 of this programme, the RSP, is designed to build on the institutional structures already in place to support implementation of the GGW initiative. At the country level, the RSP aims to enhance the capacities of GCF NDAs, as well as of the GEF and Adaptation Fund focal points, through support from geographical information systems and knowledge management specialists, and through south-south knowledge exchanges to help in the planning of future GCF, GEF and Adaptation Fund projects and their implementation.

42. Country ownership of the proposed programme is rated as medium to high.

1.6 Efficiency and effectiveness

Scale: Medium

43. The total project cost is USD 216,900,720, including a GCF grant contribution of USD 36,900,720. The GCF USD 90 million loan contribution is matched with USD 90 million of co-finance – including grant contributions for technical assistance from IFAD and AfDB totalling USD 51 million. Concessional loan finance totalling USD 39 million, from IFAD, Islamic Development Bank and the five national agricultural banks of the Phase I countries, will be blended with the highly concessional GCF loan funds to provide an affordable interest rate for loans to organizations of producers and small businesses, many of whom have been historically excluded from the market for loan finance.

44. The cost of the GCF investment in the five target countries is USD 110,283,232 (Components 1 and 2) which represents USD 44 per beneficiary, taking into account the total of 2,494,000 household members whose adaptive capacity and/or food security will be enhanced through loans made possible by the programme. The project's cost-effectiveness comes also from its cross-cutting nature – with, on the one hand, climate resilient investments in agriculture and selected agricultural value chains contributing to climate change adaptation, and on the other hand, energy efficiency and the switch from diesel to solar contributing to climate change mitigation within those same value chains. Component 3 is designed to add value to the entire GCF portfolio in the Great Green Wall – covering 29 separate projects at a cost of USD 513,055 per project. It is noted that 65 per cent of the project's benefits, corresponding to USD 82.5 million of GCF funds invested, are considered to be contributing to adaptation, with 35 per cent of GCF funds, or USD 44.4 million, to mitigation from improved land and forest management, as well as switching to solar PV for powering rural agricultural and agro-processing activities.

45. In response to a question from the independent TAP about how the high cost of the grant portion of the programme is justified – USD 19.8 million for Component 2 and USD 14.9 million for the Regional Support Programme in Component 3 – the AE clarified that “grant resources are needed to address key barriers in each country and the region, such as: (i) capacity and knowledge gaps hindering the transfer and adoption of effective climate adaptation and mitigation practices in green business projects; (ii) limited access to green finance for green business projects; and (iii) limited policy and regulatory frameworks to increase the economic feasibility of potential green business projects in the GGW area. Component 2 supports technical assistance and is USD 3.96 million per country as grant to support the implementation of a loan portion of about USD 30 million per country [GCF and blended loan co-finance]. Comparing to other similar programmes and projects approved by the GCF, such as SAP Niger, where the grant portion is USD 3 million vs. USD 7 million; the AfDB AFAWA programme, where the grant portion is about USD 6.8 million vs. USD 13.3 million in loans.” The independent TAP notes that this technical assistance loan ratio for IGREENFIN I of 0.13:1 is much more cost-effective than those for the comparable programmes, which have ratios of 0.42:1 and 0.51:1.

46. With regard to the other grant portion, Component 3 – the Regional Support Programme – the AE highlighted “The need to address the barriers related to the absence of a regional support programme and an overarching framework for GCF-funded projects to strengthen institutional capacity on reporting, generating and sharing innovative solutions and knowledge, and increase private sector financial participation in green business projects. Building on the lessons learnt from the GGW initiative and past programmes such as the World Bank SAWAP [Sahel and West Africa Program], the new momentum on the GGW area and commitment by all leaders to support the GGW agenda, it is strategic to provide the Regional Support Programme with enough grants to play a key role in: (i) enhancing knowledge management and exchanges

to accelerate the uptake of good practices, increase learning and inform policy and investments across GCF and other projects; and (ii) mapping innovation and digital transformation technologies and building a digital and innovation ecosystem.” The independent TAP notes that with 11 countries (Burkina Faso, Chad, Djibouti, Eritrea, Ethiopia, Mali, Mauritania, Niger, Nigeria, Senegal and Sudan) to be supported under the grant linked to Component 3 – Regional Support Programme (USD 1.35 million each) – the investment amounts to USD 225,000 per country per year, over the programme’s six-year implementation period. Nonetheless, the investment remains high and needs further definition to ensure value for money, in particular, defining clearly the deliverables involved in each activity.

47. When the mitigation portion of the GCF investment alone is taken into account, the estimated GCF cost per tCO₂eq removed, is USD 7.88 which is very reasonable. It is noted, however, that the mitigation benefits are based on an assumption that 250,000 hectares of land across the five target countries will be managed in a new manner, such that significant greenhouse gas emissions will be avoided, compared to the without-project scenario in which this area of land continues to become ever more degraded, resulting in increasing nett emissions. In response to a question from the independent TAP on how this would be monitored, the AE stated that, “The cumulative positive impact of restorative activities will be measured through the number of ha [hectares] of land or forests under improved and effective management and that contribute to reductions in CO₂ emissions. In relation to monitoring, IFAD will work closely with ICRAF [International Centre for Research in Agroforestry] to use the digital data reporting and monitoring systems (DRMS) as repositories designed and developed to collect, consolidate, process, analyze and visualize data collected during the implementation of the projects. The full methodology is described in annex 22.” The independent TAP notes that the Technical Assistance Facility and all trainers involved will need to be fully trained themselves by ICRAF to understand in-depth which agricultural practices are truly restorative and regenerative, how these should be undertaken, and how precisely the results of loan-based activities are to be monitored over time.

48. The programme’s overall efficiency and effectiveness is rated at medium.

II. Overall remarks from the independent Technical Advisory Panel

49. This programme meets all the investment criteria of GCF. It represents an important intervention in the collective contribution of GCF to the climate risk management objectives of the Great Green Wall, and to the efforts of the five target countries to support vulnerable rural communities in adapting to climate change and increased climate variability, whilst making a modest but significant contribution to GHG emissions reductions.

50. If the programme is approved by the GCF Board, the independent TAP makes the following recommendations to the AE, in order to maximize the programme’s contribution to the investment criteria:

- (a) The AE should conduct an analysis of the revised nationally determined contributions of four of the five target countries for Components 1 and 2, which have been submitted to the UNFCCC subsequent to the development of the funding proposal, to ensure that IGREENFIN I remains optimally aligned with national adaptation and mitigation priorities.
- (b) The AE should work with the 17 AEs in the inception phase of Component 3, the Regional Support Programme, to clarify the expected deliverables of the activities, as well as the differences and relationships between project-level and RSP-level indicators and results, to avoid any confusion and maximize ability of the programme to effectively deliver and monitor portfolio-wide achievements.
- (c) The independent TAP recommends this funding proposal for approval by the Board.

Response from the accredited entity to the independent Technical Advisory Panel's assessment (FP183)

Proposal name:	Inclusive Green Financing Initiative (IGREENFIN I): Greening Agricultural Banks & the Financial Sector to Foster Climate Resilient, Low Emission Smallholder Agriculture in the Great Green Wall (GGW) Countries – Phase I
Accredited entity:	International Fund for Agricultural Development (IFAD)
Country/(ies):	IGREENFIN I [Components 1 and 2]: Burkina Faso, Côte d'Ivoire, Ghana, Mali and Senegal GCF GGW Regional Support Programme [Component 3]: Burkina Faso, Chad, Djibouti, Eritrea, Ethiopia, Mali, Mauritania, Niger, Nigeria, Senegal and Sudan
Project/programme size:	Medium

Impact potential
The AE welcome this assessment of potential impact of the program as High and thank ITAP for the comprehensive assessment
Paradigm shift potential
The AE thank ITAP for the assessment of the program paradigm shift as Medium. In addition to the initiatives mentioned by the AE, there are other specific initiatives such as the Islamic finance, which applied lower interest rate than the market and fully institutionalized with the financial sector after just few years of its launch. These low interest rates have been set for specific finance products and institutionalized to target specific groups and clients. The program intends to work with Central Banks and Financial Institutions to include Green Finance Products in the special financing products with low interest rates. Similar to Islamic finance products and other special products in the region, The AE believes with this program, low interest rates for low emission and climate resilient agriculture will be institutionalized beyond the program duration.
Sustainable development potential
The AE welcome this assessment of potential impact of the program as High and thanks ITAP for the comprehensive assessment
Needs of the recipient
The AE welcome this assessment of potential impact of the program as High and thanks ITAP for the comprehensive assessment.
Country ownership

The AE takes note of ITAP assessment of the Country ownership .

Efficiency and effectiveness

The AE thanks ITAP for its assessment and takes note of the overall efficiency and effectiveness rating as medium.

Overall remarks from the independent Technical Advisory Panel:

The AE would like to thanks ITAP for its rigorous and solid evaluation and recommendations above. The AE takes note of the overall evaluation. However, on 1.5 Country ownership, the AE would like to inform ITAP that the 11 No-objection letters for all 11 Great Green Wall countries received by the AE. These No Objection letters were submitted to the GCF Secretariat. With this information, The AE suggests ITAP to review the rating Scale: Medium to High to High.

Regarding the two recommendations:

Recommendation 1: Conduct an analysis of the revised national determined contributions of four of the five target countries for Components 1 and 2, the AE would like to reassure ITAP and the Secretariat that this analysis will be conducted during the inception of IGREENFIN. Note that IFAD joined the Africa Nationally Determined Contributions (NDCs) hub which conducts regularly such analysis and support countries to fulfil obligations related to the Paris Agreement through programs/ projects

Recommendation 2: to clarify expected deliverables of the activities, as well as the differences and relationships between project-level and RSP-level indicators and results, to avoid any confusion and maximize ability of the programme to effectively deliver and monitor portfolio-wide achievements. The AE takes good note of this recommendation and would like to reassure ITAP that the AE in all its projects always organize 3 to 5 days start up or inception workshop(mandatory) to inform, train and sensitize the future project/ program team and all stakeholders on all aspects of projects and programs including operational, fiduciary, deliverables, indicators and baselines, targets, safeguards, roles and responsibilities, monitoring mechanisms ect.. For this regional program, the inception workshop will cover the recommendations provided by ITAP, and all the AEs active in the GGW countries will be involved, to clarify the expected deliverables of the activities, as well as the differences and relationships between project-level and RSP-level indicators and results.



Greening Agricultural Banks & the Financial Sector to Foster
Climate Resilient, Low Emission Smallholder Agriculture in
the Green Great Wall (GGW) countries - Phase I

GENDER ASSESSMENT AND ACTION PLAN

Final Version November 2021

IGREENFIN I - Multi-Country - Burkina Faso, Cote d'Ivoire, Ghana, Mali and Senegal

Contents

Contents	2
Introduction.....	4
Key gender indicators for IGREENFIN countries	7
Ghana Gender Profile.....	8
Legal and policy frameworks.....	8
Institutional arrangements.....	9
Education.....	10
Health.....	11
Employment	12
Agriculture	13
Burkina Faso Gender Profile	15
Legal and Policy Framework	15
Institutional Arrangements	16
Education.....	17
Employment	17
Health.....	17
Agriculture	18
Cote d'Ivoire Gender Profile	20
Legal and Policy Framework	20
Institutional Arrangements	23
Education.....	24
Health.....	24
Agriculture	25
Employment	25
Mali Gender Profile	28
Legal and Policy Framework	28
Institutional arrangements.....	30
Education.....	30
Employment	31
Health.....	31
Agriculture	33
Senegal Gender Profile	35
Legal and policy framework.....	35
Institutional Arrangements	36
Education.....	37

Health	37
Employment	38
Agriculture	38
Key barriers common to all target countries	39
Institutional Arrangements	41
Approach to Strengthen Gender within the Proposed Program	42
Stakeholders consultations.....	43
Prevention of gender-based violence.....	45
Gender Action Plan	46

Introduction

In accordance with the GCF's environmental and social policy, a Gender Assessment has been conducted for the Inclusive Green Financing Initiative –IGREENFIN, a regional programme dedicated to greening agricultural Banks and the financial sector to foster climate resilient and low emission smallholder in five West African countries of the Great Green Wall Initiative (GGWI). Women are targeted to be 50% of the direct beneficiaries of the programme. This Gender Assessment aims to provide an overview of the gender issues in five countries, namely Burkina Faso, Cote d'Ivoire, Ghana, Mali and Senegal, for informing the design of the above-mentioned regional programme. Its aim is to redirect financial flows towards the sub-projects of Micro, Small and Medium-sized Enterprises (MSMEs), cooperatives and Farmers Organizations for the development of climate resilient, low emission agriculture. Therefore, the proposed adaptation and mitigation solutions for the GCF project are to improve access to financing and further promote the adoption of climate resilient, low emission agriculture practices, including efficient irrigation and solar energy systems, throughout the value chains to reduce GHG emissions.

The IGREENFIN Regional Programme focuses on 3 countries of the Great Green wall (Burkina, Mali, Senegal) and Cote d'Ivoire and Ghana, one of the most environmentally degraded region in the world, also considered as one of the most vulnerable regions to climate change, regularly hit by droughts and floods. West Africa has been identified among the primary observed climate change hot spots, and the most persistent and early emerging prominent hot spots foreseen for the twenty-first century because of the observed and projected widespread increase in mean temperature and extreme hot season occurrence. According to an IPCC report, West Africa and the Sahel are likely to experience increases in the number of hot nights and longer and more frequent heatwaves even if the global temperature increase is constrained to 1.5°C, with further increases expected at 2°C of global warming and beyond. Most of West African countries are likely to experience decreasing rainfall, despite uncertainties on precipitations projections, combined with coastal erosion and degradation that is also a major challenge in this region, expected to worsen in the future.

The region vulnerability to climate change is exacerbated by its high dependence on rain fed agriculture. Agriculture accounts for 35% to the GDP¹ and 78% of food economy jobs are still in agriculture² in West Africa. Although it is an important source of income and employment for the majority of rural populations, smallholder farmers, particularly women, are still facing key challenges to have access to productive resources. This is especially relevant when it comes to access to financial resources for agriculture. Credit to the agricultural sector accounts for, roughly, 2% of total credit offered³, and when it is available, the interest rate is more than 10 % with short repayment loan period. In the West African Economic and Monetary Union (WAEMU) zone, agriculture receives only 6.10% of total credit in the short term, 4.59% and 2.15% for medium and long terms credits respectively⁴. The difficulties associated with access to credit for smallholder producers to invest in agriculture place women at a disadvantaged and vulnerable position. Women's lack of access to credit reduces their access to input, land, other productive resources and extension services.

Gender inequalities persist in the agriculture sector of the five countries targeted by IGREENFIN Regional Programme. This is particularly relevant when it comes to women's access to credit for agriculture, in addition to gender inequalities for accessing to other productive resources. The gendered division of labour in the households place the burden of securing water and fuel supplies and caring for the children and the elderly entirely on the women, leaving them with very little time to engage in income-generating activities or to further their education. On average, women spend six times more time than men on unpaid care work — cooking, cleaning, collecting water and firewood, and caring for children,

¹ FAO and AfDB. 2015. Agricultural Growth in West Africa: Markets and Policy Drivers. FAO, AfDB, ECOWAS.

https://www.afdb.org/fileadmin/uploads/afdb/Documents/Publications/Agricultural_Growth_in_West_Africa_-_Market_and_policy_drivers_-_OSAN.pdf

² Allen, T., P. Heinrigs and I. Heo (2018), "Agriculture, food and jobs in West Africa", West African Papers, N°14, OECD Publishing, Paris. <http://www.oecd.org/swac/topics/food-system-transformations/>

³ Terfa Abraham, 2019

⁴ ROPPA. 2018. Etude sur les mécanismes/ outils nationaux et régionaux de financement du secteur agricole et rural en Afrique de l'Ouest: synthèse régionale. Novembre 2018. http://roppa-afrique.org/TMG/pdf/004-synthese_regionale-v5_bdef.pdf

the ill and the elderly, and this is far higher for Mali where it is 17 times more⁵. Moreover, women in the region are also having less access to skilled jobs compared to men (8 percent in paid employment against 12 percent of men) and they tend to be marginalized in the labour market outside the agricultural sector. They represent only 8 percent of entrepreneurs. Given such challenges, the majority of West African governments adopt national gender strategies and policies, in addition to implementing legislative reforms, for addressing gender inequalities. However, these inequalities are still persistent, particularly in the Agricultural sector and access to credit and other productive resources.

In this context, The main objective of IGREENFIN 1 and the Regional Support Programme (RSP) is to build and scale up the resilience and adaptive capacity of farmers' organizations (FOs), cooperatives and micro, small and medium-sized enterprises (MSMEs) in Burkina Faso, Côte d'Ivoire, Ghana, Mali and Senegal by removing barriers to access to financial and non-financial services to accelerate the creation of a green financing market to promote the uptake of green agriculture practices and technologies. For the purposes of this funding proposal, the term "green" refers to investments in the selected agricultural value chains that promote climate resilient, low carbon practices in line with the GGWI objectives and the target countries' climate change adaptation and mitigation agenda.

The proposed programme is structured under three components:

Component 1: Green Business Financing Facility – will be established to provide concessional loans to foster best adaptation and mitigation practices for green businesses within the selected agricultural value chains in each target country (refer to Section B.1, Table 8). Operated by local national agricultural banks (LNABs) – the Agricultural Bank of Burkina Faso, the National Investment Bank of Côte d'Ivoire, the ARB Apex Bank Limited of Ghana, the Agricultural Bank of Mali and the Agricultural Bank of Senegal – the facility will offer special lines of financing for green business projects prepared by farmer's organizations (FOs), women and youth organizations, cooperatives and MSMEs (including agribusiness dealers and solar operators). The proposed investment criteria and list of specific eligible green business projects are presented in Annex 21 -Operations Manual. This component will contribute simultaneously to pillar 1 (Investment in small and medium-sized farms and strengthening of value chains, local markets and organization of exports) and pillar 4 (Favourable economic and institutional framework for effective governance, sustainability, stability and security) of the Great Green Wall Accelerator. In this funding proposal, the term "**green business**" refers to investments in the selected agricultural value chains that promote climate resilient, low carbon practices in line with the GGWI objectives and the target countries' climate change adaptation and mitigation agenda.

Component 2: Technical Assistance Facility (TAF) – will be created to address capacity, knowledge and policy gaps hindering the uptake of green business projects in the selected countries. It will target LNABs, central banks and clients (FOs, MSMEs and cooperatives), with a special emphasis on women and youth. The TAF will provide support for specific technical assistance to the following Direct Access Entities: i) the Agricultural Bank of Senegal, which is to serve as a GCF accredited pilot NEE for the other targeted agricultural banks; ii) Attijariwafa Bank (an accredited commercial bank), whose role will be to crowd in additional financing as a strategic partner of the selected LNABs, and iii) the Centre de Suivi Ecologique (CSE), a strategic IGREENFIN I partner that is to perform activities 2.1.1 and 2.1.8 with all selected countries and activity 3.2 with IFAD. The TAF will provide necessary technical assistance and policy and regulatory support to the Central Bank of the West Africa Economic and Monetary Union (WAEMU) for Burkina Faso, Côte d'Ivoire, Mali and Senegal, and the Central Bank of Ghana on the creation of an enabling environment for financing the uptake of green business projects (refer to section B.2 for the theory of change). This component will contribute to pillar 2 (Sustainable management of ecosystems and land restoration), pillar 3 (Climate resilient infrastructures and access to renewable energy) and pillar 5 (Capacity-building) of the Great Green Wall Accelerator.

Component 3: GGW Regional Support Programme - This component will increase the collective impacts of the individual GCF projects and programmes (including IGREENFIN I and II) through two outputs: i) **Output 3.1.** Enhanced knowledge management and exchanges to accelerate the uptake of good practices, increase learning and inform policy and investments across GCF and other projects, and ii) **Output 3.2.** Innovation and digital transformation technologies are mapped and a digital and innovation ecosystem built.

⁵ <https://oecd-development-matters.org/2018/03/08/gender-equality-in-west-africa-the-key-role-of-social-norms/>

This Gender Assessment is based on quantitative and qualitative evidence derived from a literature review and secondary databases from national and international sources. For each country, the gender analysis is conducted addressing gender inequalities at the national level, as well as summarizing the existing legal, policy and institutional frameworks. Moreover, a Gender Action Plan has been proposed to set the tone on how gender issues resulting from the assessment can be addressed through the implementation of project activities. This gender action plan will be updated the first year and resubmitted to the GCF after 6 months of each country implementation

Key gender indicators for IGREENFIN countries

IGREENFIN targeted countries are among the one with low rankings in terms of gender indicators, as well as Human Development Index, as presented in *Table 1*. The Social Institutions Gender Index (SIGI) measures discrimination against women in social institutions across 180 countries. The SIGI shows clear regional trends, Western Africa being one of the region where legislation, social norms and practices create the highest gaps between women and men (44%), making the case for applying a social norm lens to gender equality⁶. Among the IGREENFIN countries, Mali and Cote d'Ivoire are the one that are close to this regional average. In Cote d'Ivoire for example, the first area of concern among the fourth dimensions of the SIGI is women's access to productive and financial resources. The Global Gender Gap Index (GGPI) benchmarks the evolution of gender-based gaps among four key dimensions (Economic Participation and Opportunity, Educational Attainment, Health and Survival, and Political Empowerment) and tracks progress towards closing these gaps over time. In all IGREENFIN countries, the gender gap in Political empowerment and in the Economic Participation and Opportunity remain the largest among the four gaps tracked by the index. Senegal is far better than the four other countries where much progress has been made for closing gender gaps in political and economic dimensions. The Gender inequality that measures the gender inequalities in three aspects of human development index confirms Senegal position. Mali and Cote d'Ivoire scored significantly lower, compared to the other countries. As shown in *Table 2*, the maternal mortality ratios is high in all targeted countries, ranging from 308 deaths per 100,000 live births in Ghana to 617 in Cote d'Ivoire. Regarding the percentage of seats in parliament held by women, it ranges between 9.5 % for Mali and 41.8% for Senegal.

Table 1: Gender indicators and human development index in IGREENFIN countries

Countries	SIGI (value/ category)	GGPI (value/ rank out of 156 countries)	GII (value/ rank out of 162 countries)	HDI (value/ rank out of 189 countries)
Burkina Faso	32%/ medium	0.651/ 124 th	0.594/ 147 th	0.434/ 182 th
Cote d'Ivoire	43%/ high	0.637/ 134 th	0.638/ 153 th	0.516/162 th
Ghana	35%/ medium	0.598/119 th	0.538/ 135 th	0.586/ 138 th
Mali	46%/ high	0.591/ 149 th	0.671/ 158 th	0.427/ 184 th
Senegal	37%/ medium	0.684/104 th	0.533/ 130 th	0.514/168 th
	2019	2021	2021	2021

Table 2: Key indicators on gender inequality in IGREENFIN countries

Country	SDG3.1	SDG3.7	SDG5.5	SDG4.4		Labour force participation rate	
	Maternal mortality ratio	Adolescent birth rate	Share of seats in parliament	Population with at least some secondary education		(% ages 15 and older)	
	(deaths per 100,000 live births)	(births per 1,000 women ages 15–19)	(% held by women)	(% ages 25 and older)		Female	Male
Ghana	308	66.6	13.1	55.7	71.6	63.6	71.9
Côte d'Ivoire	617	117.6	13.3	17.9	34.4	48.2	65.5
Senegal	315	72.7	41.8	10.3	26.5	35.0	57.5
Burkina Faso	320	104.3	13.4	6.1	12.3	58.3	74.8
Mali	562	169.1	9.5	7.3	16.4	61.2	80.6

⁶ <https://www.oecd-ilibrary.org/docs/server/bc56d212-en.pdf?expires=1618853627&id=id&accname=ocid195767&checksum=3ACF040CC31C69F4983D861A4529D0AF>

1. According to the Global Gender Gap Report (2021), Ghana scored 0.666, which places it in 117th position out of 156 countries⁷. More specifically, Ghana scored: (i) 0.598 in economic participation and opportunity, with a ranking of 119th position, (ii) 0.951 for educational attainment, placing it at the 120th position, (iii) 0.978 for Health and Survival, ranked at the 40th position, (iv) 0.135 for political empowerment, placing it at the 110th position. Although Ghana has made progress by demonstrating leadership to mainstream gender at various levels, it is still lagging behind in terms of gender empowerment and gender equality, particularly at political and economic spheres. In addition, restrictive social norms and stereotypes are among the biggest barriers to gender equality and women's empowerment in the country. As for the Gender Development Index (GDI), Ghana's value is 0.911 in 2019, placing it to the penultimate group (group 4) which comprises countries with medium to low equality in Human Development Index (HDI) achievements between women and men⁸. Indeed, female HDI value for Ghana is 0.582, lower compared to males with an HDI of 0.639, showing unequal distribution of Human Development between men and women. With regard to Gender Inequality Index (GII), Ghana has a value of 0.538, ranking it at the 134th position, out of 162 countries in the 2021 index.

Legal and policy frameworks

2. The Government of Ghana has made great progress in improving the legal frameworks for promoting gender equality and women's empowerment. Indeed, Ghana's supreme law, the Constitution, which fundamentally commits the country to the elimination of gender discrimination and provides the constitutional basis for gender mainstreaming and equality, guarantees the promotion of equal rights to all citizen, men and women. In addition, the country set out national legal and policy frameworks and ratified international conventions about gender equality. Through Article 17(1) and (2) of the 1992 Constitution ensures gender equality and freedom of both men and women, girls and boys from discrimination based on social and economic status among others. Ghana engaged in institutional, administrative and legal reforms, while formulating and implementing policies towards gender equality and equal access in various realms, for both men and women. At the national level, there are the Domestic Violence Act (2007), the Criminal Code Amendment Act (1994), the criminalization of harmful traditional practices such as Female Genital Mutilation and *Trokosi* (ritual servitude), the 1998 Children Act criminalizing underage and forced marriage. At the international level, Ghana ratified the Convention on the Elimination of All forms Discrimination against Women (CEDAW), and it is signatory of the International Covenant on Civil and Political Rights, the Beijing Platform for Action, the Commonwealth Plan of Action for Gender Equality and African Women's Protocol. There are other international conventions, treaties and plans to which Ghana is committed at the international level on gender equality and women's empowerment. These include: (i) The African Charter on Human and Peoples' Rights ratified on 24 January 1989 and signed on 3 July 2004; (ii) The Protocol to the African Charter on Human and People's Rights on the rights of women in Africa – 0025, signed on 31 October 2003 and ratified on 13 June 2007; (iii) The UN Security Council Resolution 1325 and 1820 on Women, Peace and Security and violence against women; (iv) The 2015 Sustainable Development Goals (SDGs); (v) The Convention against Torture and Other Cruel, Inhuman or Degrading Treatment or Punishment – New York (10 December 1984, ratified 7 September 2000), etc.
3. Ghana is a party to the CEDAW, Convention on the Elimination of All forms of Discrimination against Women, signed since 1980 and ratified in 1986 without reservations. Ghana has demonstrated its commitment to the tenets of the Convention by ensuring that the Constitution, new legislation and policies are consistent with CEDAW. It defines what constitutes discrimination against women and sets up an agenda for national action to end such discriminatory practices. Ghana has submitted jointly its sixth and seventh report to CEDAW in 2014, highlighting various areas where progress have been made in terms of gender equality and women's empowerment. Moreover, Ghana also recognizes the Beijing Declaration and Platform of Action, which is an agenda for women's empowerment with the

⁷ http://www3.weforum.org/docs/WEF_GGGR_2021.pdf

⁸ http://hdr.undp.org/sites/all/themes/hdr_theme/country-notes/GHA.pdf

commitment to remove all obstacles constraining the active participation of women in all spheres of public and private life. It also try to guarantee that women have full and equal share in economic, social, cultural and political decision-making in alignment with the Sustainable Development Goals, an explicit link to sustainable development and a reference to global warming being made by the Declaration. The achievements in the period 2014-2019 under review have been categorised under four major themes: gender equality and women empowerment; mainstreaming gender into socio-economic development; development of laws; and establishment of Gender Advisory Board.

4. At the policy level, the Ministry of Gender, Children and Social Protection (MoGCSP) has developed the National Gender Policy (NGP). Its overarching goal is to mainstream gender equality and women's empowerment concerns into the national development process in order to improve the social, legal, civic, political, economic and cultural conditions of the people of Ghana; particularly women and men, boys and girls in an appreciable manner and as required by National and International Frameworks⁹. In order to improve the conditions of Ghanaians, particularly women, girls and children, the NGP has identified commitments related to women's rights and access to justice, women's empowerment and livelihoods, accountable governance structures, women's leadership and participation, women's economic justice, and gender roles and relations. The NGP has an implementation plan since 2016, but which lacks good ownership.
5. In fact, there are several challenges that constrain the effective implementation of policies on gender equality and women empowerment. They include competing government priorities and political will; weak gender mainstreaming coordinating role of the Ministry; conceptual clarification of gender equality in the public sector, lack of effective monitoring and evaluation systems and practice within the sector machinery and weak accountable governance. Regional stakeholders reported confusion as to when the implementation plan became operational, leading to delay in its effectiveness, and according to government stakeholders, working with other agencies remains a challenge given that policies and activities frequently do not align with the National Gender Policy¹⁰.
6. Another important challenge to the effectiveness of sector policies and plans on gender equality and women empowerment is the limited budgetary support from the Government. For example, a main challenge to the implementation of the National Gender Policy (2015) seems to be the lack of reliable and sustainable funds. Even though, the Ministry of Gender, Children and Social Protection (MoGCSP) has the mandate to spearhead the achievement of gender equality and women's empowerment, the key challenge is limited budgetary support from Government and which lead to a heavy reliance on donor funds to implement programmes¹¹. The same problem of insufficient budget is also reported at local level, such as municipalities and districts. Metropolitan, municipal and district assemblies (MMDA) are unclear about gender issues, and they do not budget or allocate sufficient resources for gender sensitive development planning and budgeting¹². In addition, a requirement of the National Development Planning Commission (NDPC) for Ministries, Department and Agencies (MDAs) to allocate at least 40 per cent of their budget to gender issues, but which is not being respected¹³.

Institutional arrangements

7. Ghana established the National Council on Women and Development (NCWD), established after the first United Nations Conference on Women in 1975. The NCWD became part of the Office of the President after the Beijing Conference in 1995, operating therefore in close collaboration with relevant MDAs. A fully-fledged sector Ministry, namely the Ministry of Women and Children's Affairs (MoWAC) was created in 2001, with regional departments and integrated gender desks in other MDAs. The MoWAC was renamed in 2013 as the Ministry of Gender, Children and Social Protection (MoGCSP) in

⁹ Ministry of Gender, Children and Social Protection (MoGCSP). 2015. National Gender Policy: Mainstreaming Gender Equality and Women's Empowerment into Ghana's Development Efforts, Republic of Ghana, May 2015.

¹⁰ Britt, Charla, Ivankovich, Megan, Essah, Samuel, and Fiscian, Vivian. 2020. USAID/Ghana Gender Analysis Report. Prepared by Banyan Global.

<https://banyanglobal.com/wp-content/uploads/2020/05/USAID-Ghana-Gender-Analysis-Report.pdf>

¹¹ FAO and ECOWAS Commission. 2018. National Gender Profile of Agriculture and Rural Livelihoods – Ghana. Country Gender Assessment Series, Accra.

¹² Britt, Charla, Ivankovich, Megan, Essah, Samuel, and Fiscian, Vivian. 2020.

¹³ FAO and ECOWAS Commission. 2018.

charge of gender equality and women empowerment, as well as social protection and welfare issues. The MoGCSP' mandate is to promote gender equality and safeguard the welfare and full participation of women in social, political and economic development process. There is also the Domestic Violence and Victim Support Unit (DOVSSU) mandate to respond to and enforce all laws with a special focus on sexual and gender-based violence, child abuse and child protection, and protection of all vulnerable persons from abuse¹⁴.

8. Furthermore, the Ministry of Health implemented additional measures, particularly on GBV. Indeed, this Ministry has developed an action plan to establish protocols for better case management for GBV survivors and to ensure collaboration among the health sector, social services, and law enforcement agencies¹⁵. Despite its strong institutional framework towards gender equality and women's empowerment, Ghana is still facing challenges and significant gaps remain. The designated gender desk officers (GDOs) have limited capacity, high rates of attrition and transfer, and cannot be held accountable by the MoGCSP as their appointments are discretionary, arrangement that limits the effectiveness of GDOs and gender mainstreaming at all levels¹⁶.
9. Further, the MoGCSP Department of Gender (responsible for coordinating and monitoring them) is unable to ensure the necessary follow-up because of its institutional weakness. As such, the contribution of sector policies and plans to gender equality is not adequately monitored and evaluated. The absence of any multi-sector coordination mechanism to ensure the implementation of the NGP within all sectors is also a challenge, difficult to overcome given that the gender desks do not have the necessary capacities and budget to mainstream gender different sectors. Furthermore, competing government priorities, lack of effective monitoring and evaluation systems in the MoGCSP and Department of Gender, weak conceptual clarification of gender mainstreaming in the public sector are key challenges to the implementation of Ghana's national legal frameworks and international commitments¹⁷.

Education

10. ***Gender parity in education is improving in Ghana, although gaps still exist at secondary and tertiary levels.*** In order to close the gap between girls and boys about education, Ghana has made significant progress. Compared to most Sub-Saharan African countries, Ghana has relatively high literacy and net primary school enrolment with more girls attending primary school than boys. Although there are considerable efforts made for the enrolment of girls at the primary school, girls still lag behind at secondary and tertiary education level. The percentage of boys' enrolment at the senior secondary and tertiary schools figured around 72.73 and 18.68, respectively, in 2018 as against 71.72 and 13.53 for girls; and the literacy rate of men in Ghana outstrips that of women¹⁸. The gender gaps for enrolment at secondary level are thought to be higher in rural areas, and the northern part of the country.
11. Furthermore, women and girls are less represented in tertiary education and Science, Technology, Engineering and Mathematics (STEM) majors, which are likely to affect their professional life. Only 22.5 per cent of graduates from STEM programmes in tertiary education are female, and the underrepresentation of women in these programmes affect their ability to secure employment in the formal sector, particularly in the fields traditionally dominated by males. A study on the case of females offering STEM majors in Ghana found a positive correlation between constraints female students have and their motivation to offer STEM related programmes¹⁹. Several factors contribute to the imbalances between girls and boys on STEM and at higher levels of education, including early marriage, the fact that girls are over-burdened by household workload, adolescent pregnancy, and the low importance parents seems to have on the benefits of girls' education.

¹⁴ MoGCSP. 2015

¹⁵ Ministry of Health (MOH). 2009. Health Sector Gender Policy. Republic of Ghana

¹⁶ Britt, Charla, Ivankovich, Megan, Essah, Samuel, and Fiscian, Vivian. 2020.

¹⁷ West Africa Development and Business Delivery Office (RDGW), African Development Bank/African Development Fund. Republic of Ghana Country Strategy Paper (CSP) 2019-2023, June 2019

¹⁸ Donkor, P., Ding, Y., Adu-Boateng, G. 2019. The Effect of Parental Economic Expectation on Gender Disparity in Secondary Education in Ghana: A Propensity Score Matching Approach. Sustainability 2019, 11, 6707; doi:10.3390/su11236707

¹⁹ Amponsah, D.K., Mohammed, M.S. 2019. Perception of learning science: the case of females offering STEM majors in Ghana. African Journal of Educational Studies in Mathematics and Sciences Vol. 15, No. 2., 2019

12. According to the Ghana Living Standard Survey round 6 (GLSS 6), the workload at homes for girls is three times more than that of boys, and the proportion of girls who are out of school due to domestic chores in 2014 stood at 13.1, compared to 3.2 for boys, on the average. Furthermore, Ghanaian parents expect their male children to reap more economic benefits from education than girls, and this attitude culminates in higher investment in boys' education to the disadvantage of their female counterparts at senior secondary school²⁰. According to the SDG progress report, while Ghana has been able to provide access to education for the majority of children, quality of education is lagging behind. Poor learning outcomes are evidenced by proficiency in learning and pass rates. Indeed, Ghanaian women spent almost two hours less than men per day on learning, in addition to leisure and social activities²¹.

Health

13. **Significant progress has been made in improving maternal health.** According to the World Bank data, the maternal mortality ratio has decreased from 484 to 308 deaths per 100,000 live births between 2000 and 2017²². More efforts need to be done in order to achieve the SDG3 target to reduce the global maternal mortality ratio to less than 70 per 100,000 live births. Although fertility remains relatively high in Ghana (3.9 children per woman in 2017), it should be noted that it has declined significantly from 6.4 children per woman in 1998 to just under four according to the Ghana Demographic and Health Survey²³. Fertility varies by residence, education and economic status of women. Indeed, women in rural areas have an average of 4.7 children, compared to 3.3 children for those in urban areas, northern region displaying particularly higher fertility rates. According to findings from 2017 DHS, women with no education have twice as many children than women with more than secondary education (5.5 versus 2.7), and women in poorest households have an average of 5.7 children, compared to 2.8 children among women in the wealthiest households. High fertility exposes women of childbearing age to greater risk of morbidity and mortality. It is also worth mentioning the significant progress made in increasing access to professional medical care for pregnant women. The proportion of women assisted at delivery by a skilled health staff increased from 25 percent in 2000 to 79 percent in 2017, yet wide disparities exist across regions, rural-urban area and income profiles.
14. As for HIV, women comprise nearly 60 per cent of those living with the virus, with an increase noted for the newly infected women. Urban women are three times as likely to be HIV positive as urban men, and rural women are two times as likely to be HIV positive as rural men²⁴. The limited knowledge in terms of prevention and transmission, the low contraceptive prevalence (30%), and the limited access to family planning methods could explain that women are disadvantaged compared to men. However, men aged 15-49 years are more knowledgeable (23%) when compared to 14% women of the same age group²⁵. Child marriage remains a challenge in Ghana, despite recent declines resulting from the considerable progress made. The proportion of children who marry before age 15 and 18 has fallen significantly between 2006 and 2017. According to the 2017 MICS²⁶, one in every five women age 20-24 years were first married before age 18 years; child marriage being prominent in rural areas, in Northern, Upper East and Volta regions, and among the poor and less educated. The higher prevalence of child marriage in rural areas lead to higher rates of dropout of girls, resulting in more gender disparities in education in these areas. About 40% of girls of secondary and tertiary school-going age in Ghana are married, and one-fifth of them are mothers, while the corresponding figures for boys are ten times lower²⁷. Findings from the GLSS7 in 2019²⁸ confirm gender disparities about marriage, in terms of mean age and median age at first marriage. Indeed, the mean age at first marriage is 23.1

²⁰ Donkor et al. 2019

²¹ Rubiano-Matulevich, E. and Violaz, M. 2019. Gender Differences in Time Use: Allocating Time between the Market and the Household. Washington, DC, World Bank. (Policy Research Working Paper 8981.)

²² <https://data.worldbank.org/indicator/SH.STA.MMRT?locations=GH>

²³ Ghana Statistical Service (GSS), Ghana Health Service (GHS), and ICF. 2018. Ghana Maternal Health Survey 2017: Key Findings. Rockville, Maryland, USA: GSS, GHS, and ICF. <https://www.dhsprogram.com/pubs/pdf/SR251/SR251.pdf>

²⁴ Ghana Statistical Service (GSS), Ghana Health Service (GHS), and ICF International. 2015. Ghana Demographic and Health Survey 2014. Rockville, Maryland, USA: GSS, GHS, and ICF International. https://untobaccocontrol.org/impldb/wp-content/uploads/reports/ghana_2016_annex1_dhs_2014.pdf

²⁵ UNICEF. 2019. Ghana Multiple Indicator Cluster Survey 2017/18: Snapshots of key findings. Updated version Jan 2019. UNICEF, The World Bank, UNDP, KOICA, USAID. <https://www.unicef.org/ghana/media/576/file/Ghana%20Multiple%20Cluster%20Indicator%20Survey.pdf>

²⁶ UNICEF. 2019. Ghana Multiple Indicator Cluster Survey 2017/18: Snapshots of key findings. Updated version Jan 2019.

²⁷ Ghana Statistical Services. 2014. GSS. Ghana Living Standards Survey (GLSS 6); Ghana Publication Ltd.: Accra, Ghana, 2014 https://statsghana.gov.gh/gssmain/fileUpload/Living%20conditions/GLSS6_Main%20Report.pdf

²⁸ Ghana Statistical Services. 2019. GSS. Ghana Living Standards Survey (GLSS 7); Ghana Publication Ltd.: Accra, Ghana, 2019. https://www.statsghana.gov.gh/gssmain/fileUpload/pressrelease/GLSS7%20MAIN%20REPORT_FINAL.pdf

years, with females marrying about four years earlier than their male counterparts do, 21.3 and 25.7 years respectively. As for the median age at first marriage, it ranges from 19 years to 22 years; that for females ranges from 19 to 22 years while that of males ranges from 24 to 28 years. Furthermore, girls are also victims of gender-based violence (GBV) and Female Genital Mutilation/ Cutting (FGM/C). In 2016, approximately 27.7 per cent of Ghanaian women had experienced at least one form of domestic violence (physical, economic, psychological, social and sexual violence)²⁹. According to the 2014 DHS, more than one in four women (28 percent) agree that a husband is justified in beating his wife for at least one specified reason. Moreover, seven percent of women agree that a husband is justified in hitting or beating his wife if she burns the food, 16 percent if she argues with him, 17 percent if she goes out without telling him, 21 percent if she neglects the children, and 12 percent if she refuses to have sexual intercourse with him. The school-related gender based-violence is also a concern, with 11 percent of girls reporting sexual harassment (compared to 6 percent of men)³⁰.

Employment

15. **Gender gaps are still a reality in the labour market, even though women are economically active.** Although women's economic participation rates are high in Ghana, men represent the majority of the economically active population. According to the GLSS7, among the population 15 years and older, 71 percent were economically active, with 65 percent employed and 6 percent unemployed. In this population, males (72.3 percent) were more likely to be economically active than female (69.7 percent). The gender gap has increased, because the percentage of economically active men represented 80% and that of women 74.9 percent according to the GLSS6. The employment gap by gender is evident in almost all the occupational groups with higher proportions of males engaged in various employment activities than females³¹.
16. Gender disparities in employment can be illustrated by the high presence of women in low-skilled work, the low female participation rates in the formal sector, the higher likelihood of working as unpaid family workers in agriculture, and women underrepresentation in most of the technical, professional and managerial positions. Women's economic participation is mostly in self-employment and the informal sector, such as catering and lodging, food services, textiles, garments, and beauty, while only 3.4 percent of females are professionals, compared to 6.6 percent of men³². Women are more likely to be engaged in vulnerable employment. About 77.8 percent of females were engaged in vulnerable employment (own-account workers – 55.7%; contributing family workers - 22.1%) compared to 54.2 percent of males (own-account workers - 42.3%; contributing family workers - 11.5%)³³. Gender disparities are also a concern regarding incomes. Ghana is among the countries in Africa where gender gaps in income are large, estimated to 68.9 percent³⁴.
17. Women are mostly involved in self-employment in informal businesses in the textile, garment and beauty industries. The European Union reports that household enterprises (HHEs) are the major source of employment in Ghana, engaging 3.6 million households (43% of all households in the country) that run a microenterprise. The majority of HHEs (54%) engage in wholesale and retail trading activities, but also catering and lodging, food services (making and selling snacks or meals), transport, and personal services such as barbering and hairdressing. In Ghana, evidence shows that the majority of women business owners have small or micro businesses (over 70 percent starts with capital less than 100 USD), and they are operating in less profitable areas than men are. Women face higher constraints than men to expand their HHEs. They are more likely to operate and remain operating one- or two-person enterprises sometimes for years, with no prospect (and intention) to grow. The main challenges that these women face include lack of skills in business management and entrepreneurship, access to

²⁹ https://ghana.un.org/sites/default/files/2020-04/COVID-19_UN%20Ghana%20Briefing%20Notes_No.2_2020_04_14.pdf

³⁰ Britt, Charla, Ivankovich, Megan, Essah, Samuel, and Fiscian, Vivian. 2020. USAID/Ghana Gender Analysis Report. Prepared by Banyan Global. <https://banyanglobal.com/wp-content/uploads/2020/05/USAID-Ghana-Gender-Analysis-Report.pdf>

³¹ Ghana Statistical Services. 2019. GLSS7.

³² West Africa Development and Business Delivery Office (RDGW), African Development Bank/African Development Fund. Republic of Ghana Country Strategy Paper (CSP) 2019–2023, June 2019

³³ Ghana Statistical Services. 2019. GSS. Ghana Living Standards Survey (GLSS 7); Ghana Publication Ltd.: Accra, Ghana, 2019.

³⁴ https://www.statsghana.gov.gh/gssmain/fileUpload/pressrelease/GLSS7%20MAIN%20REPORT_FINAL.pdf

³⁴ http://www3.weforum.org/docs/WEF_GGGR_2020.pdf

finance and assets to grow their working capital, poor quality of products, lack of standardization and poor branding of products. The biggest barrier to access to finance for women is due to lack of collateral. In the northern region, men dominate control of household assets including those owned by women. The Institute of Economic Affairs reports that men have absolute control over land (87.7%), motorbike (86.3%), residential buildings (81.4%) and farmland (67.5%).

Agriculture

18. ***Although women are active in the agriculture sector, they still lag behind in terms of income, access to productive resources, and finance.*** Women participate in rural agricultural labour markets, but gender disparity is a major concern. Indeed, nearly two-fifths (38.0 percent) of females were self-employed without employees in the non-agricultural sector compared to 13.7 percent of males, according to the GLSS7. However, the proportion of males who were self-employed without employees in the agricultural sector (28.7 percent) is relatively higher than that for females (13.7 percent). This means that males are more likely to be employed in agricultural activities as against their females counterparts who prefer to engage in non-agricultural activities. In rural areas, we found more males in the skilled agricultural workers (65.9 percent), than females (60.7 percent).
19. Apart from participation to labour markets, gender-based inequalities in agriculture also apply to access to and control over resources, which affect women's agricultural productivity. These gender inequalities exist in land ownership, access to inputs, access to financial resources, and training among others. In spite of a progressive land ownership system, customary land use systems prevail whereby property rights of the assets is normally with the household heads³⁵. Household heads are most often men in Ghana and women have temporary usufruct rights that can be lost when they marry or lose their husband. Significant gender disparities exist in land holdings, with men holding 3.2 times more of the total farms than women and 8.1 times more of the medium-large farms (of 5 acres and more)³⁶. Female headed households have less access to land for farming (89% in Northern Ghana vs a 96% national average). Female-headed households are also more likely to be smallholders (farming five acres or less). Approximately 87% of female-headed households are smallholders compared with 60% of male. About 4% of farms are more than 11 acres compared with 16% of men heading a household³⁷. Limited access of women to land (i.e. collateral) also results in access that is more limited to financial services.
20. As for gender-based disparities regarding earnings and yields, a study on gender and cocoa in Ghana points out the differences. Female-headed households earn less net income from cocoa (USD 960) than male-headed, typical households (USD 1,275) after hired labour and input costs are deducted³⁸. As for yields, there is a difference of around 58 kg/ha, between male-headed households (432kg/ha) and female-headed households (374kg/ha). For the cocoa production in Ghana, evidence shown that differences in productivity between women and men disappear when women have the same access to productive inputs and sell their produce in the same way as men³⁹. The gender differences in agricultural productivity is not specific to cocoa. Indeed, a study conducted in 2008 concluded to a gender differential in farm productivity, which they identified to be the result of lower exogenous quality of women's farm plots than that of their husbands, for instance a lower investment in improving soil quality⁴⁰. Women are disadvantaged when it comes to cash crop for generating significant incomes.
21. Regarding agricultural financing, it is important given the major role women are playing in the agriculture value chain, from farming to food processing and marketing. However, they face obstacles to access funding for agribusinesses due to the limited access to and ownership of land, limited knowledge and capacity in elaborating bankable business plans, limited access to information on how and where to access finance, lack of guarantee for climate change related risks, mind-sets about the agriculture;

³⁵ FAO. 2018

³⁶ FAO. 2018

³⁷ IFAD. 2019

³⁸ Bymolt, R., Laven, A., Tyszler, M. (2018). Demystifying the cocoa sector in Ghana and Côte d'Ivoire. Chapter 14, Gender and cocoa. The Royal Tropical Institute (KIT). <https://www.kit.nl/wp-content/uploads/2019/09/Demystifying-cocoa-sector-chapter14-gender-and-cocoa.pdf>

³⁹ World Bank. 2015. The costs of the gender gap in agricultural productivity in Malawi, Tanzania and Uganda. UNWOMEN, World Bank, UNEP, UNDP.

<http://documents1.worldbank.org/curated/en/847131467987832287/pdf/100234-WP-PUBLIC-Box393225B-The-Cost-of-the-Gender-Gap-in-Agricultural-Productivity-in-Malawi-Tanzania-and-Uganda.pdf>

⁴⁰ Goldstein, M. and Udry, C. (2008). "The profits of power: land rights and agricultural investment in Ghana." Journal of the Political Economy, 116, 6, 981–1022.

considered as for households needs, etc. Access to credit and other financial services is a serious constraint to many actors in the agriculture sector especially the smallholder farmers, who are mostly women, for whom funding from the banks is simply out of reach. Around 90 percent of women-owned businesses started with personal savings, as opposed to credit from formal financial institutions, women are the predominant customers of informal micro-credit institutions, but are thought to borrow very small amounts⁴¹.

22. According to the Ghana Statistical Services, women have limited access to bank accounts, compared to men, especially in rural areas. Only 30.8% of women compared to 69.2% of men have an individual saving account. Furthermore, the main source of financing for women in agriculture comes from the traditional *Susu* Scheme (71.6%) compared to their male counterparts (28.4%). With respect formal credit channels such as Banks, less than 43.9% of women have access to formal credit facilities. Regarding the purpose of loan, the overall percentage of credit from the Financial Institutions for agriculture's sector is less than 4%. With only 13.7% targeting women lending for land, 27.9% for agriculture's equipment is to women and less than 23.5% lending to women for agriculture's inputs. In addition to this, women received the largest number of loans refusal with urban/rural disparities of 61.3% to 65.9% respectively. This is due to the lack of collateral and due to the limited capacity in producing high quality proposals. Female-led agribusinesses women exhibit weak financial record of accomplishment and knowledge gaps that negatively affect their ability to prepare bankable business plans.
23. Women play a key role in agricultural value chains in Ghana, despite the gender inequalities that exist in the agricultural labour markets, along with access to and control over resources. Women are mainly engaged in producing about 70% of food crops, such as rice, maize, cassava, cocoyam, groundnuts, soybean and vegetables⁴². In the production level of the value chains, women dominate small-scale agricultural production of most commodities, except when the product has a comparatively higher value-added or is traditionally a "male-cultivated" product⁴³. Moreover, women act as aggregators most often, traveling to the north of the country to buy food products from various producers and re-sell them in bulk to predominantly male wholesalers in the south, because men unlike women participate in the aggregation process but at lower volumes. Women play also the role of small-scale market vendors of unprocessed agricultural products, purchasing their products from male wholesalers and selling them at limited profit margins. Indeed, women are most of the time unable to buy important quantities of agricultural products, because of insufficient capital and inadequate access to storage facilities. Exceptionally, in the case of "market queens", women may control all transactions pertaining to a particular commodity in a market, tend to have comparatively greater access to capital resources, storage facilities, and other assets, and are able to use their resources to control the marketplace and influence supply and demand⁴⁴. Despite their role in agricultural value chains, women are still facing several constraints to operate and expand within these value chains. These constraints are: (i) access to capital, including land ownership, (ii) access to credit and financial services, in part due to limitation or lack of capital, (iii) skills and knowledge of agro-technology, (iv) extension services, (v) storage facilities and, (vi) primarily for aggregators, transportation infrastructure⁴⁵.

⁴¹ World Bank. 2013

⁴² FAO. 2018

⁴³ WFP. 2017. Value Chain Development, Gender and Women's empowerment in Ghana. VAM Gender and Markets Study #1 (2016-2017). WFP, USAID. <https://docs.wfp.org/api/documents/WFP-0000022433/download/>

⁴⁴ WFP. 2017

⁴⁵ Sarpong, D. B. Department of Agricultural Economics (University of Ghana). Written Interview. April 2016.

Burkina Faso Gender Profile

24. In Burkina Faso, there is significant improvement to be done in order to achieve gender equality. The proportion of women aged 20-24 years old who were married or in union before age 18 is 51.6%. The adolescent birth rate is 104.3 per 1000 population as of 2021, up from 129 per 1000 population in 2014. There are only 13.4% of parliament seats held by women. In 2010, 9% of women aged 15-49 years reported that they had been subject to physical and/or sexual violence by a current or former intimate partner in the previous 12 months. Moreover, women of reproductive age (aged 15-49 years) often face barriers with respect to their sexual and reproductive health and rights: the proportion of women who have their need for family planning satisfied with modern methods in 2017 year stood at 56.4%⁴⁶. This situation is the result of inequalities based on perceptions and social practices that restricts women's rights and reduces their access to economic and social opportunities in relation to men, such as financial inclusion or political commitment.
25. The patriarchal system perpetuates lineage and inheritance favourable to men, and impacts the fertility preferences. More than 44% of Burkinabè would like or would have liked their eldest to be a boy, compared to 17% a girl. Gender bias in resource allocation and under-investment by households in girls' health and nutrition has negative consequences for their health and for society as a whole.

Legal and Policy Framework

26. The government's action revolves around the implementation of the National Gender Policy and the 2017-2019 Operational Action Plan. In addition, the 2016-20 National Plan for Economic and Social Development (PNDES) aims to reduce gender inequalities by increasing the number of female business owners from 21% in 2015 to 50% in 2019, by promoting decent jobs opportunities for women and increasing their access to technical and vocational training, etc. Furthermore, in its Policy Framework, Burkina Faso has also developed the Economic and Financial Sector Policy (POSEF) for the promotion of microfinance and bank financing, the National Employment Policy (PNE) for the creation of jobs for men and women, supporting specifically micro and small enterprises, and the national strategy for promoting women's entrepreneurship. However, in general, sectoral policies insufficiently integrate the gender dimension, which is reflected on the planning of activities, the results of achievements and the evaluation of the results of effect and impact. It should be noted that there is a scarcity of up-to-date disaggregated statistics by gender in the different development sectors.

27. In addition to the Convention on the Elimination of All Forms of Discrimination against Women (CEDAW) and its optional Protocol, Burkina Faso has ratified the Convention on the Rights of the Child (CRC) and the African Charter on Human and Peoples' Rights on the Rights of Women in Africa (Maputo Protocol). Moreover, the country has also ratified the International Covenant on Civil and Political Rights, which contains provisions prohibiting discrimination and protecting women. Burkina Faso has adopted laws and Codes, that should guarantee equal rights to both women and men for accessing services and resources. For instance, there are the Law No 034-2012/ AN on agrarian and land reorganisation, ensuring equal access to land for all; the Law No 034-2009 on rural land tenure, advocating equitable access to rural land for men and women; the Law on violence against women, and the Labour Code which prohibits among others gender-based discrimination in employment and occupation. Regarding land access, the law 034-2009 on rural land tenure advocates that managed land be granted to women. The government has put in place reforms to combat violence against women. Thus, the application of prison sentence for rape, introduced explicitly in the penal code in 2015, began to be applied in 2016. Furthermore, the Constitution recognizes the promotion of gender as a factor for realization of the equality of law between men and women.

⁴⁶ UN WOMEN, 2020. <https://data.unwomen.org/country/burkina-faso>

28. The law allows a girl to be married at 17, against a legal minimum age of 20 for men. No less than 44% of married women were under 18, compared with 3% of men. The stability of these rates between different age groups suggests that this practice is not decreasing. On the contrary, its persistence seems linked to a high level of social acceptance, since 44% of Burkinabè believe that it is acceptable for a girl to be married before the age of 18. More than one in three women (37%) have been victims of domestic violence in their lifetime, compared with one in five men (16%). The law prohibits all forms of violence against women and girls, however domestic violence is not criminalized. Social norms enable the average person to state that a man can beat his wife for one reason or another, while only 4% believe that a woman can beat her husband. Despite its prohibition, the practice of excision is also common. Two-thirds (63%) of women aged 15 to 45 are circumcised.

Table 3: Gender Policies and Institutional frameworks at national level:

Burkina Faso	Ministry for the Advancement of Women	• Constitution 1991, revised successively in 1997, 2000, 2002, 2009, May and June 2012, • National Gender Policy (2009) • Law n ° 034-2012 / AN of 02 July 2012 • Gender Quota Act (2009) • Law No. 034-2009 / AN of 16 June 2009 on rural land tenure • Individual and Family Code (CPF) of 1989 (under revision) • Law No. 043/96 / ADP of November 1996 on the Prevention and Punishment of FGM
--------------	---------------------------------------	--

Institutional Arrangements

29. At the institutional level, the Ministry of Women, National Solidarity, and Family is the institution responsible for promoting gender equality and empowering women, while ensuring the implementation and monitoring of the Government's Policy in that regard. It has the responsibility to coordinate line ministries activities towards gender equality and women's empowerment. The Ministry is carrying out its missions through the support of different directions, among which the General Directorate for Gender Promotion and the General Directorate for the Promotion of Women's Entrepreneurship, and the Permanent Secretariat of the National Council for Gender Promotion (SP/CONAP Genre). There is a National Commission to Follow up on the Commitments of Burkina Faso regarding the Advancement of Women (CNSEF), an advisory body consisting of representatives of ministries and CSOs, which proposes recommendations about the fulfilment of the country's commitments on women's advancement. Furthermore, there are gender-mainstreaming units created within ministries and institutions in order to ensure that gender aspects are well mainstreamed while elaborating policies at the sectoral level. At the local level, the Ministry is represented through decentralised services in charge of the coordination, monitoring and evaluation.
30. Despite all these institutional arrangements, the initiatives undertaken since the creation of the Ministry in terms of gender mainstreaming have not produced the expected results. There is a lack of mechanisms for consultation and coordination between the key players: 1) the national council and the regional and municipal councils for the promotion of gender, 2) the gender units provided for in each government institution, 3) the consultation framework of the PTFs (including the Swiss Cooperation, World Bank, UNDP, UNFPA, etc.) and the TFP Common Gender Fund. The main reasons are the low level of ownership of gender issues by policy makers, the lack of technical and institutional capacity and the lack of allocations to drive the implementation process. Civil society organizations of women in civil society suffer from their weak structure and the absence of a national coordinating umbrella structure.

Education

31. Girls' education in Burkina Faso has been limited due to gender divisions and much-needed improvements in infrastructure. However, several efforts by aid organizations have worked to improve the access that girls have to education in the country. The government in Burkina Faso recognizes the issue well as by aid organizations, and improving girls' education in Burkina Faso has been a goal of these organizations. The Pan-African Conference on the Education of Girls as early as 1993 as well as the more recent Ten-Year Plan on the Development of Basic Education and the National Policy of Integrated Development of Children, which outlined a plan for 70 percent enrollment by 2015, have been specifically designed to address issues of education in the country over the last two decades⁴⁷. In 2010, the boy-to-girl student ratio at primary school level stood at 0.94, up from 0.7 in 2000. However, from 2008 to 2012, in a longer analysis of net participation, the female net enrollment ratio stood at only 50 percent⁴⁸.
32. According to UNICEF the education system is characterized by geographical disparities both in terms of enrollment rate and in infrastructure coverage. There are also disparities related to gender – 65.7 percent of boys attend school against 54.5 percent of girls⁴⁹. The security situation in the North and East is extremely volatile and unrest in the region around Burkina Faso has driven 25,000 refugees to camps within its borders. Dozens of teachers have been attacked and some even killed by militant extremists⁵⁰. In that environment, students are afraid to go to school, and girls are among the most vulnerable. While approximately 1,000 schools have been closed, leaving 120,000 children without access to formal education, the government works hard to reopen the schools.

Employment

33. Unequal access to education limits women's opportunities in the area of skilled employment while they are highly integrated in the informal sector. Women are responsible for all the production, processing and marketing of their agricultural products. In addition to these economic activities, the burden of children and the exclusivity of domestic tasks are also devolved to them. The multiplicity and size of these tasks greatly reduce women's free time, which does not allow them to devote themselves entirely to certain income-generating activities that can raise their standard of living and that of their families. Thus, the number of hours of work per week in rural Burkina Faso is 96 hours while that of men is 56 hours.
34. According to the National Survey of Manpower in the Public and Private Formal Sectors (2018), the participation rate is higher for men than for women: 76.8% for the former and 60.2% for the second. Women generally occupy positions at the bottom of the ladder because of socio-cultural burdens. Inequalities in access to employment persist as women continue to experience more difficulties than men because of cultural and social considerations. Thus, despite the high activity rate, the quality of employment remains precarious. In addition, 9 out of 10 informal workers are women and youth, but the lack of social protection, skills upgrading and productive income often traps these groups in poverty and the exclusion of economic growth and development.

Health

35. The government of Burkina Faso has adopted a National Policy of Health (Politique nationale de santé) and has implemented activities based on the National Development Plan for Health (Plan national de développement sanitaire) for 2011-2020. The plan positions the issue of maternal health as an important issue and clarifies that the cause of maternal death is HIV/AIDS, malaria, anemia, and hemoglobin anomalies. It also presents the practices that have a negative impact on women,

⁴⁷ <https://borgenproject.org/girls-education-in-burkina-faso/>

⁴⁸ UNGEI, 2011.

⁴⁹ https://www.unicef.org/infobycountry/burkinafaso_39159.html

⁵⁰ <https://www.globalpartnership.org/blog/breaking-down-barriers-girls-quality-education-burkina-faso>

including FGM, early marriage, levirate and demands for a widow to marry her husband's brother, abortion, difficulties in accessing health facilities, and the quality of health service⁵¹.

36. Prenatal checkups are free and include HIV/AIDS testing, which helps increase the number of people who receive prenatal checkups. Women face difficulties in accessing health and medical service in part because of geographic location, but also, their inferior social and economic position prevents them from receiving appropriate health and medical services. For example, women need advance permission from their husbands before they take their children to a health center. If they go without permission, they will have to pay for the transportation and medication.
37. At the level of health indicators, it should be noted that the total fertility rate (5.6 children per woman in 2014) and the growth rate (around 3%) are among the highest in the world. Investigations show that almost all women in Burkina Faso receive prenatal care from a trained provider. Maternal mortality was 371 deaths per 100,000 births in 2017.⁵² It depends on a number of factors, including authorization from the spouse, the level of financial resources, distance from the health service, and so on. Infant mortality rates seem to be improving overall.

Agriculture

38. The agro-forestry-pastoral sector employs nearly 90% of the population and accounts for 33.7% of GDP in 2016 (ADB, OECD, UNDP, 2017). Women account for 52% of household farm workers and work several hours on family land before attending to their own fields, whose crops are for home consumption or sale. The General Agricultural Census (GAM) indicates that small ruminant breeding is practiced by 42.2% of women. As for fishing activities, women are present in the processing and marketing of fish. Access to land ownership: This remains a challenge for women. They make up 55% of the agricultural labour force, but not more than 40% of land owners. When they are land owners, their decision-making power remains limited. Only 14% of female land owners have the opportunity to sell their land, compared to 32% of men, because of customary law and community land management practices. Households headed by women use traditional equipment (hoes, dabas ...), in 96% of cases, while 21.5% of households headed by men in comparable situation, use plows. In addition, women's access to agricultural extension services remains low. Access to financial services: Two-thirds of those with an account at a bank or other financial institution are men, while that 15% of the population say that men and women should not have the same decision-making power with regard to financial services

Access to land ownership

39. This remains a challenge for women. They make up 55% of the agricultural labour force, but not more than 40% of land owners. When they are land owners, their decision-making power remains limited. Only 14% of female land owners have the opportunity to sell their land, compared to 32% of men, because of customary law and community land management practices. Households headed by women use traditional equipment (hoes, dabas ...), in 96% of cases, while 21.5% of households headed by men in comparable situation, use plows. In addition, women's access to agricultural extension services remains low.

Access to financial services

40. Two-thirds of those with an account at a bank or other financial institution are men, while that 15% of the population say that men and women should not have the same decision-making power with regard to financial services Employment: Unequal access to education limits women's opportunities in the area of skilled employment while they are highly integrated in the informal sector. Women are responsible for all the production, processing and marketing of their agricultural products. In addition to these economic activities, the burden of children and the exclusivity of domestic tasks

⁵¹ https://www.jica.go.jp/english/our_work/thematic_issues/gender/background/c8h0vm0000anj6-att/burkinafaso_2013.pdf

⁵² UNDESA, 2020.

are also devolved to them. The multiplicity and size of these tasks greatly reduce women's free time, which does not allow them to devote themselves entirely to certain income-generating activities that can raise their standard of living and that of their families. Thus, the number of hours of work per week in rural Burkina Faso is 96 hours while that of men is 56 hours.

41. The government of Côte d'Ivoire adopted the National Policy for Equalities of Chances, Equity and Gender (Politique Nationale de l'Egalité des chances, l'Équité et le Genre) in 2009 to promote gender equality and subsequently implemented various activities. The strategy aims to ensure equal opportunities for women and men in development and decision-making processes. Four priorities are set: governance and human rights, macro-economics and budget, reconstruction and access to basic social services (education and health) as well as capacity building and monitoring and evaluation mechanisms. However, the environment facing women is harsh. Though the rate of girls' enrollment in primary education is increasing, the disparity between urban and rural areas is large, and few girls enroll and complete higher education. The environment surrounding girls' education is not favorable enough. Therefore, further advocacy activities are needed to make parents understand the importance of girls' education. Moreover, many girls drop-out of school due to gender based violence by teachers and boys, and there is a growing need to establish schools for girls.
42. About 60% of women worldwide work in the informal economy; compared to men, these women earn less, save less and are more likely to fall into poverty. In Côte d'Ivoire, women suffer many inequalities and the informal sector accounts for 89% of employment, with women making up 44%. In rural areas, women remain confined to their reproductive roles. However, they also participate, at certain levels, in the production of cash for export, without necessarily being paid. In spite of the heavy burdens and workload, she engages in other activities, such as trading in foodstuffs or handicrafts, in order to earn an income to meet her needs, such as dressing and adorning herself, clothing her children and possibly helping her husband to feed the family.

Legal and Policy Framework

Government Policy on Gender:

43. Since the end of the socio-political crisis, Côte d'Ivoire has reformed many texts and adopted several new public policies to address gender inequalities. The 2016 constitution provides that the state must promote equality between men and women and that it must ensure that no one is discriminated against on the basis of gender. It also provides that the State shall promote parity between men and women in the labour market, and encourage the promotion of women to responsibilities in public institutions and administrations, as well as at the corporate level. National Development Plan (2012-2015): the objective of this strategy is for people to live in harmony in a society where good governance is guaranteed. In terms of social cohesion, the participation of women in local and national institutions is promoted; and in terms of the "standpoint of justice" the protection of vulnerable groups, especially women, is foreseen. 1) The government of Côte d'Ivoire has adopted policies and action plans to promote gender equality. It is important to assure the implementation, monitoring, and evaluation of these actions. 2) The strategy for the fight against gender based violence was established in 2012, and it is now in the implementation phase of the action plan. 3) A project to develop a database of talented women is ongoing under the initiative of the Presidential Office as a way to evaluate and utilize the capacities and talents of women across the country. The National Policy on Equal Opportunities, Equity and Gender was adopted in 2009 by the Ivorian government, and its objective is to ensure equal opportunities for women and men in development and decision-making processes.
44. In 2008, the government adopted the Action Plan to Implement the Resolution 1325 of the Security Council on Women, Peace and Security as well as the National Policy for Equalities for Chances, Equity and Gender (Politique Nationale de l'Egalité des Chances, l'Équité et le Genre) in 2009. This policy aims at assuring just and equitable development and permits both men and women equal chances in the development and decision-making processes. Four priorities are set: governance and human rights, macro-economy and budget, reconstruction and access to basic social services (education and health), and capacity building and monitoring and evaluation mechanisms. The

challenges for the future are to assure the implementation of the 30% quota for elections and the review of the National Gender Policy adopted in 2009. From 2009, the National Strategy for the Fight against the Gender Based Violence (Stratégie nationale de lutte contre les Violences Basées sur le Genre: SNLVBG) has been established, and implementation began in July 2012. SNLVBG prioritized the prevention and protection of sexual violence over the other various kinds of gender based violence. The following five priority axes were selected for action.

45. Regarding the National Strategy to Fight Gender-Based Violence (SNLVBG): it is implemented since July 2012 by the Ministry for Solidarity, Family, Women and Children, which is in charge of coordinating donors on GBV issues. The strategy also includes actions related to the care of victims of gender-based violence, the protection of orphans due to HIV/AIDS and the reintegration of victims of gender-based violence. Significant progress has been made in recent years in creating a legislative framework to encourage women's participation in the economy. Although the legal, regulatory and policy framework in Côte d'Ivoire is largely in line with international best practice regarding the treatment of women's rights, awareness and enforcement of these rights remains low.

Gender related laws and regulations:

46. The Côte d'Ivoire government has adopted various laws and regulations in order to reduce the inequalities between men and women.

Constitutional Law of 08 November 2016

47. The law no. 2016-886 of 8 November 2016 establishing the Constitution of the Republic of Côte d'Ivoire which promotes and protects women's rights through its articles 4, 36 and 37 prohibiting discrimination and promoting equality of women and men in political, public life and in the labour market. The country has also adopted law no. 2014-388 of 20 June 2014 on the promotion and protection of human rights defenders, including women human rights defenders, through its article 9 and Decree No. 2017-121 on its application. Moreover, the government also adopted the Interministerial Circular No. 016/MJ/MEMIS/MPRD of 4 August 2016 and Circular No. 005 of 18 March 2014 of the Minister of Justice and Human Rights related to the reporting of cases of gender-based violence. Although Côte d'Ivoire has ratified the CEDAW, as well as the Protocol to the African Charter on Human and Peoples' Rights on the Rights of Women in Africa (Maputo Protocol), the Government has yet to ratify the Optional Protocol to CEDAW⁵³.

Women's rights to property and land use in the Civil Code, Labour Code and Family Code

- The Civil Code is a set of codified laws such as:
- Law No. 83-800 of 2 August 1983 amending and completing the provisions of Law No. 64-375 on marriage of 7 October 1964:
- Law No. 98-748 of 23 December 1998 amending and completing Law No. 64-376 of 7 October 1964 on divorce and legal separation, amended and completed by Law No. 83-801 of 2 August 1983.
- Law n° 97-400 of 11 July 1997 as amended by Law n° 2015-532 of 20 July 2015 on the Labour Code
- Law n° 99-477 of 02 August 1999 as amended by Ordonnance n° 2012-03 of 11 January 2012 on the Social Security Code
- Law n°98-750 of 23 December 1998 amended by the law of 28 July 2004 on rural land tenure
- Law No. 81-640 of 31 July 1981 establishing the Penal Code

⁵³ <https://www.wikigender.org/wiki/africa-for-womens-rights-cote-divoire/>

- Law No. 98-756 amending and supplementing Law No. 81-640 establishing the Ivorian Penal Code. No difference in treatment between men and women.
- Law No. 98/757 of 23 December 1998 prohibits the practice of excision in Côte d'Ivoire

Legal mechanisms of inheritance/succession

- - Law No. 64-379 of 7 October 1964 on inheritance
- - Law No. 98-750 of 23 December 1998 on rural land

Women's rights to property and land use in the Civil Code, Labour Code and Family Code

- The Civil Code is a set of codified laws such as:
- Law No. 83-800 of 2 August 1983 amending and completing the provisions of Law No. 64-375 on marriage of 7 October 1964:
- Law No. 98-748 of 23 December 1998 amending and completing Law No. 64-376 of 7 October 1964 on divorce and legal separation, amended and completed by Law No. 83-801 of 2 August 1983.
- Law n° 97-400 of 11 July 1997 as amended by Law n° 2015-532 of 20 July 2015 on the Labour Code
- Law n° 99-477 of 02 August 1999 as amended by Ordonnance n° 2012-03 of 11 January 2012 on the Social Security Code
- Law n°98-750 of 23 December 1998 amended by the law of 28 July 2004 on rural land tenure
- Law No. 81-640 of 31 July 1981 establishing the Penal Code
- Law No. 98-756 amending and supplementing Law No. 81-640 establishing the Ivorian Penal Code. No difference in treatment between men and women.
- Law No. 98/757 of 23 December 1998 prohibits the practice of excision in Côte d'Ivoire

Legal mechanisms of inheritance/succession

- - Law No. 64-379 of 7 October 1964 on inheritance
- - Law No. 98-750 of 23 December 1998 on rural land

Governance and security

48. The first strategy in the National Development Plan (Plan national de développement) for 2012-2015 is that people can live in harmony in a secure society where good governance is ensured. This strategy includes several components: consolidation of peace and social cohesion; army, gendarmerie, and police for the nation building; reform of justice; rule of law and public liberty; communication; and development of national statistic system. In regards to social cohesion, women's participation in national and local institutions is being promoted. From the standpoint of justice, the protection of vulnerable groups, including women, is expected. [Security] In Cote d'Ivoire, DDR (Desarmament, Demobilization, and Reintegration) has been implemented following the end of the conflict. In the course of DDR, it is important to take into consideration the women who are ex-combatants or supported armed force. Generally, the term DDR tends to refer to male ex-combatants; however, women also handled logistical support and conducted espionage, even though they were not engaged in actual combat. There are no exact statistics, but it is estimated that about 450 female ex-combatants could be targeted under DDR. Although the conflict has ended, security is a big concern for the country. Small arms are common across the country, and many assaults using these arms still persist. In areas that saw intense combat, sexual violence occurs frequently, and the victims are generally women. When the assailants are ex-combatants, the victims are unable to take action because of their fear. Consequently, the assailants are not punished and repeat the same offenses.

Access to justice

49. In Côte d'Ivoire, since the cost of judiciary procedures is high, those who do not have economic power, including both men and women, do not have access to the courts. However, since women

are vulnerable and have a low literacy rate, they have much greater difficulty in accessing justice than men. The protection of victims is a big concern as well. Many victims do not know about their rights, so few resort to legal procedures. Also, few lawyers exist, and they tend to be based in Abidjan, not in rural areas. In the case of GBV, the judicial proceedings mandate special care for the victims. For example, the trial should take place behind closed doors, and the victims should not be at the same room as the assailant. Nevertheless, trials are not closed, and during the testimony, the assailant is allowed to be in the same room as the victim. In addition, the victims of GBV need a medical certificate as evidence of the violence they suffered. However, it costs 50,000 FCFA, an extremely high amount, to have this certificate issued. It is impossible for a victim of GBV to go to a medical facility to have this expensive medical certificate issued, and moreover, bringing the case before the court is also expensive. Due to the limited access to justice, the assailants of GBV are not brought to justice, which substantiates the idea that acts of violence are not punished. This phenomenon trivializes the violence and creates a vicious cycle of impunity and violence. In this regard, the juridical clinics run by the Association of Female Jurist of Côte d'Ivoire (Association des Femmes juristes de Côte d'Ivoire) with assistance from donors are noteworthy example. These juridical clinics provide necessary information and support to victims of GVB.

Institutional Arrangements

50. The Ministry of Solidarity, Family, Women and Children is the institution responsible for the promotion of gender equality and women's empowerment. The Ministry has a Directorate of Equality and Gender Promotion (DEPG), whose specific missions include to: (i) ensuring respect for equality and equity between women and men; (ii) implementing the national policy on equal opportunities, equity and gender; (iii) fighting against violence against women and girls, in liaison with the Ministry of Justice and Human Rights; (iv) monitoring the actions of the National Programme to Combat Gender-Based Violence, particularly for women and girls in difficulty (girl mothers, widows and women victims of domestic violence); (v) ensuring systematic integration of gender in the elaboration, implementation and evaluation of national development framework documents, programmes and projects, etc. The National Committee to Combat Violence against Women and Children (CNLVFE) is active and produces much appreciated results. The Women's Training and Education Institutions (IFEFE) created since 1958 remain an excellent tool for equality.
51. Furthermore, there are gender units within Technical Ministries, following the ministerial decree under the Ministry of Solidarity, Family, Women and Children. Those units are responsible for ensuring that gender mainstreaming at the sectoral level. They have been very active and have contributed to the production of various national gender promotion documents. An independent evaluation of those units conducted in 2013, confirms the relevance of setting up of gender units in the technical ministries as an approach to promoting gender mainstreaming in the sectors. There are Civil Society Organizations and associations active in the field of gender promotion, women's empowerment, and the fight against gender-based violence.
52. Despite the existence of institutional mechanisms towards gender mainstreaming and gender equality, there are some limitations hindering the institutional framework. They include the lack of a system of accountability to the different gender structures for gender promotion and the limited impact of the gender units in the sectoral ministries. A Gender Observatory is one of the solutions proposed to ensure this accountability, through the monitoring and permanent evaluation of policies, plans, programmes, projects and activities of public or private organisations. Moreover, the coordination is difficult, because of the multiplicity of stakeholders, leading to conflicts of competence, especially in the case of overlapping missions. In addition, the framework for collaboration with civil society is not very restrictive on the obligation to report to the authority in charge of gender authority. In the Ministry in charge of gender, the gender aspect is drowned in the missions of the Ministry of Solidarity, Family, Women and Children, which also include issues of solidarity and social cohesion, the family, children and war victims. Another weakness is also related

to the lack of synergy at the field level between the interventions of different ministries, or those supported by different partners working towards gender equality.

Education

53. In 2017, the enrollment parity index (girls to boys) was 0.98 in primary, 0.87 in lower secondary, and 0.81 in upper secondary. The low enrollment of girls affects labor force participation: about 27 percent of working-age women are inactive in the labor market, compared to 19 percent of men. School completion rates for girl was lower than boys in 2017 due to higher dropouts on account of early marriages and adolescent pregnancies—about 30 percent of girls aged 15 to 19 years were pregnant. According to the latest 2018 household survey, working-age women are much less likely to participate in the labor force compared to their male counterparts (60 percent vs. 78 percent), they also have significantly lower wages in salaried employment.
54. The proportion of female teachers is also low at 13.3%⁵⁴. The government has set a policy to increase the number of junior high schools and high schools. Presently, schools are usually located in urban areas, so many girls must leave home to attend school. This situation hinders girls in rural areas from attending school. In cases where girls leave home to go to school, there are occasions where they get pregnant and drop out of school. In these cases, the parents would never want their other girls to go to secondary school. Given this situation, the government has laid out a policy to promote the establishment of girls' high schools with dormitories in order to prevent pregnancy during school and to assure a favorable environment in which girls can study and live. Also, at least one high school for girls is planned to be built in each region. In regards to junior high schools, the government recognizes that they should be built near residential areas, and around 40 junior high schools are to be constructed across the country.

Health

55. Access to prenatal and maternal care and contraceptive methods is a major public health challenge. In 2015, only 57% of births were attended by skilled health personnel, 89% of women received medical attention during their pregnancy and only 18% of married women or women in union used any contraceptive method (GGGR 2015). The maternal death rate is high in Côte d'Ivoire, and improvements in women's health and equitable access to services are primary objectives. 2) HIV/AIDS is prevalent in young women and men ages 50 and older. It is expected that the government will assume a leading role in the measures against HIV/AIDS. 3) The ratio of FGM had been decreasing since 1998. However, this downward trend reversed during the period of internal conflict, and currently, there are even cases of FGM in Abidjan, an area where FGM was not previously practiced. The maternal mortality ratio has decreased by only 13.4% over the past 25 years, from 754 maternal deaths per 100,000 live births in 1990 to 645 in 2015 (World Bank 2015), it remains high and far from the target set in MDG 5A1. Every day, 16 women die from complications related to pregnancy or childbirth in Côte d'Ivoire, mostly due to preventable or treatable medical problems.
56. The government of Côte d'Ivoire aims at making the country a place where “no woman dies when giving birth and every person is born healthy and lives healthy sexual and reproductive lives.” Therefore, the mission of the government is to assure people the right to health and equal access to all services. Specifically, matters related to women's health have been included in the National Policy of Reproductive Health (Politique nationale de la santé de la reproduction). In order to address the prevalence of HIV/AIDS, the government holds a National Council for the Fight against HIV/AIDS (Conseil National de Lutte contre VIH/SIDA), which was last held in September 2009. [Nutrition] Children, pregnant women, HIV carriers, and orphans are considered to be vulnerable groups. It is important to pay sufficient attention to nutrition before becoming pregnant as it is too late after becoming pregnant. In regards to women, 33% are obese, 12% are underweight, and 59% lack sufficient iron. Most cases of obesity are found in urban areas. Against

⁵⁴ From documents by ministry of education

this backdrop, the Ministry of Agriculture supports female cooperatives and distributes food and seeds in the regions that face malnutrition. In addition, the government plans to support mothers with HIV as breastfeeding can pass the HIV virus to the child.

Agriculture

57. The food crop sub-sector occupies 85% of the agricultural labour force, 90% of whom are women. In addition to food crops, women are also involved in small cattle breeding, processing and marketing of by-products, and they represent 2/3 of the agricultural labour force. Agricultural products make up a large portion of the country's exports and food products. Generally, women grow crops for self-support, such as cassava, maize, bananas and other vegetables. On the other hand, men grow crops to convert into money, such as cacao, palm, and rubber, as they own sufficient land. Women do not own any land or resources, and they do not have money to buy the equipment needed to process the agricultural products. Furthermore, without any collateral and a low literacy rate, they have difficulty in obtaining small loan.

Access to land

58. The law on landownership of 1998 permits equal access to land for both men and women. Law No. 98-750 of 23 December 1998 on rural land ownership (Article 1) stipulates that any Ivorian individual is entitled to own land. This law establishes equal access to land for men and women of Ivorian nationality. In reality, women can only acquire land in urban areas. In rural areas, local customs do not generally allow women to own land. Furthermore, women have difficulty in borrowing land for long term. In addition, the law of succession permits equal rights for both sexes, as customarily, women are regarded as having no inherent rights. The government of Côte d'Ivoire has laid out a policy to promote land ownership by women and conducted advocacy activities. As a result, some women now own land in rural areas. While women engage in self-sufficient food production, they cannot engage in large-scale agriculture such as plantations. Since women do not own any land, when the operators of plantations wish to buy land, women are usually not involved in the acquisition process. Moreover, they cannot object, even if the acquisition is prejudicial to them. Mostly farmers (in rural areas), women gain access to land through the customary law system, which excludes the possibility of women growing perennial crops on the land they are given. Women who grow food crops (in particular) have easier access to land for off-season crops, unlike for perennial crops. Also, regardless of the mode of acquisition, women's access to land is less than 10% of the national area

Employment

59. The informal employment rate in Côte d'Ivoire, according to the Integrated Regional Survey on Employment and the Informal Sector (ERI-ESI 2017), was 91.9 percent 2017, reflecting a small decline from 96.6 percent in 2014. Informal, low-productivity occupations are disproportionately occupied by the poor, female workers, and for those living in rural areas. Agricultural households are highly vulnerable to adverse market and weather conditions and a drop in their income has an immediate impact on the surrounding rural economy, including very limited access to capital. In the private sector, many women work in agriculture, informal sectors, and service sectors. Younger women tend to work as secretaries. Thus, the government plans to increase the number of women in management-level positions and conduct advocacy activities promoting women's employment in industrial rather than service sectors. In the public sector, the proportion of women according to rank is: 18.6% for entry level positions (6,908 female), 27.6% for middle management positions (16,870 female), 37.3% for management positions (7,840 female), and 36.1% for executive level positions (1,875 female). However, only 12.2% of women make up the highest positions in the public sector, which is outside of these rankings⁵⁵. An in-depth analysis of the mechanisms that lead to the unfavourable position of women in the labour market shows that customs and traditions have delayed women's and girls' access to formal education, which in some cases limits their

⁵⁵ République de Côte d'Ivoire, Rapport pays de suivi des objectifs du millénaire pour le développement, version finale, août 2010, p.36.

competitive abilities compared to men. In addition, the triple role of women (social, reproductive and productive)

Access to credit

60. With limited access to land and no resources, women face difficulties in accessing credit. Many micro credit institutions start out offering loan conditions favorable to women, but when the micro-credit operations become successful, they enact much stricter conditions. Thus, there are few micro credit programs that maintain loan conditions favorable for women, and this is one of major concerns in regards to women's access to credit.

Representation in the decision-making process

61. Stereotypes against women, general apolitical attitude among women, low literacy rate, and inadequate education have been pointed out as reasons why women's participation in the decision-making process has not advanced very far⁵⁶. Until 2012, in Côte d'Ivoire, the man was legally the head of the family. In January 2013, the Ivorian parliament abrogated and replaced various articles of the marriage law to establish equality between men and women in the family (Law No. 2013-33). The new law provides that the spouses, who provide for the education of the children and contribute to the household expenses according to their respective abilities, manage the family jointly. It also provides that the domicile of the family is chosen by mutual agreement between the spouses. The reform also gives each of the spouses the right to exercise the profession of their choice. Côte d'Ivoire has set up institutional centres within the state structures so that the issue of women is no longer considered a secondary aspect, but is integrated into public policies and sectoral programmes, as well as to coordinate the efforts of the various ministries in this area and create synergies between them. Under the impulsion of certain UN agencies, several actions are being carried out to promote gender in public institutions, but this is still very timid, even if certain ministries (Women, Youth, and Trade) have action plans.

Traditions and social norms

62. There are various ethnic groups in Côte d'Ivoire, each with a different language, culture and customs, and the social status of women is different in each group. For example, Akan is matrilineal and women hold a high social position, and there are female village chiefs. However, generally, the social and economic position of women in Côte d'Ivoire is low, and their access to social services and economic power is inferior to men. In Côte d'Ivoire, girls are exposed to multiple forms of harassment and violence throughout their education. The sexual abuse of which they are victims is frequent and often treated as something banal. It occurs in or near educational institutions. The perpetrators are usually men or other employees of the school, soldiers, drivers, shopkeepers or other students. According to a 2016 study by the Ministry of Education, 4 out of 10 students, especially in secondary schools, are victims of sexual harassment, and 2 out of 10 of rape, sometimes committed by teachers (UNICEF 2016b)⁵⁷. The government has clearly stated that this custom has no base in religion and has an adverse effect on health, and it has prohibited the practice of FGM by law. The government also conducts advocacy activities such as a special week against FGM. There are many factors in FGM, such as the low literacy rate and insufficient knowledge on health and hygiene. In 2014, Cote d'Ivoire developed a National Strategy Document on the fight against gender-based violence (DSNLVBG) through the Ministry of Solidarity, Family, Women and Children (MSFFE) and supported by technical ministries, UNFPA, UN-ACTION, NGOs and civil society. A Gender-Based Violence Information Management System (GBVIMS) has been set up as a tool to monitor the implementation of the cross-cutting theme related to GBV data

⁵⁶ République de Côte d'Ivoire, Rapport pays de suivi des objectifs du millénaire pour le développement, version finale, août 2010, p.37.

⁵⁷ <http://www.cnpc-mcc.ci/index.php/fr/actualites-media/news/item/236-evaluation-de-la-participation-economique-des-femmes-en-cote-d-ivoire#ftn42>

collection. As of April 1, 2019, Côte d'Ivoire has fifty-nine (59) platforms for the fight against GBV in different localities.

63. According to the Gender Inequality Index –GII, Mali is one of the ten African countries where the disparities between the sexes are the highest. The GII decreased significantly over the 2000-2010 period, but no significant improvement has been noted since 2010. This index offers a composite measure of gender inequality in three areas, empowerment, the labor market and reproductive health. 48. The country of Mali is an agro-sylvo-pastoral land with about 15.8 million inhabitants (51% are women). About 80% of the population lives in rural areas where the national electrification rate in 2016 is 19.39% (39% national, 86% in urban areas). The energy balance in total energy consumption in 2016 established 77% of biomass use mostly by women, the more vulnerable group. The Government has adopted the National Gender Policy that includes an important component to improve women's access to appropriate technologies that are less costly and less wood energy use. This national policy sets the country's vision and strategic directions in terms of priorities and strategic directions for strengthening gender equality and equity. So, all interventions and back donors at the national level must be aligned in order to optimise the results and changes in this area. This gender policy developed in 2011 defines the national vision of gender promotion as follows: "A democratic society that guarantees the development of women and men through the full exercise of their fundamental equal rights, active and participatory citizenship and equitable access to resources, with a view to making Mali an emerging 5 country strong in its growth and proud of its values of justice, peace, solidarity and social cohesion" (Mali, BOAD). The climate issues determine the quality of men and women life, and globally the future of the planet. In this case, Mali is currently facing significant environmental problems such as desertification, floods, and rapid deforestation. Women are more victims of the adverse effects of climate change, but on the other, certain women's activities, such as the artisanal transformations of agricultural products, the sale of wood and coal have negative impacts on The environment. Concrete measures should accompany the ongoing awareness-raising campaigns: development of collection pits for dyeing wastewater, development of soaps workshops, etc. The national priority is to strengthen efforts on the extension of appropriate technologies that are less costly and less energy-consuming for all men and women in rural areas. (Mali, BOAD).

Legal and Policy Framework

64. At the international and regional levels, the country has ratified several conventions. For instance, Mali has ratified the CEDAW, the 2003 Protocol to the African Charter on Human and Peoples' Rights on the Rights of Women in Africa (Maputo Protocol), and the 2008 Protocol on the Statute of the African Court of Justice and Human Rights. However, it has yet to enact key legislation at the national level. Indeed, the African Court on Human and Peoples' Rights found the Malian Government in violation of the Maputo Protocol in May 2018 for allowing child marriage for girls, not always requiring consent for marriage, and discriminating against women in inheritance statutes in the Malian Family Code⁵⁸. Concerning Agriculture, the Agricultural Law of 2006 (*Loi d Orientation Agricole*) takes a strong approach to gender equity, food sovereignty and support for small-scale farming, however some weaknesses exist with regards to enforcement. Moreover, the National Plan of Priority Investment in the Sector of Agriculture (PNIPSA), following the Agricultural Law, focuses on certain value chains where women dominate in all stages of production, like milk and fish. There is also the Law No. 2015-052 instituting measures for promote gender in access to functions nominative and elective (quota of at least 30% representation of the other sex in appointments and on the lists of candidates for nominative and elective functions). The Constitution of Mali guarantees the same rights to citizens of both sexes without discrimination. The principle of the primacy of Universal rights over national laws is affirmed in the Constitution. This is how the article 116 of the Constitution grants duly ratified and published international treaties and agreements binding force greater than that of national laws. Mali has launched its National Gender Policy, providing equality

⁵⁸ https://www.sipri.org/sites/default/files/2019-12/sipriinsight1912_6.pdf

between men and women to be institutionalized in the country's national, sectoral and local planning systems, and to be taken into account in Malian policy-making, institutions and reforms.

Poverty

65. The poverty rate is estimated at 43.8% in 2018, compared to 44.9% in 2017 and 47.2% in 2015 according to the results of the EMOP. During the first decade of the 2000s, a reduction in poverty had also been noted with a rate which went from 55.6% in 2001 to 43.6% in 2010 according to the data from INSTAT. According to the World Bank, the results obtained in terms of poverty reduction over the period 2001-2010 are attributable up to 82% to a better distribution of consumption between households⁵⁹. This reduction could also be explained by the growth of the economy estimated at 5% on average over this period, and by the reduction in inequalities. These tightened in Mali where, between 1995 and 2015, the incomes of the poorest 40% increased by 60 to 80 percentage points more than the average⁶⁰. However, the country experienced an increase in poverty over the 2011-2013 period, from 45.4% to 47.1%, accentuated by the political and security crisis of 2012 and the drought of 2011. According to World Bank data, the percentage of households with a female head is 17.4% in 2018. Taking into account gender considerations, SDG 1 related indicators show that the employed population below international poverty line, disaggregated by sex and age, is higher for women (46.1%) compared to men (40.9%)⁶¹. In 2014, 75% of male headed households and 49% of female headed households in Mali live below \$3.10/day (2011 PPP)⁶².
66. The regional distribution shows that monetary poverty disproportionately affects rural populations. Indeed, if poverty has recently decreased in Bamako and in other urban centers, it has increased in rural areas⁶³. Analysis of INSTAT data on poverty from 2001 to 2017 shows that the poorest populations are in the Sikasso and Mopti regions, where almost 6 out of 10 people are poor, followed by Ségou and Koulikoro. The 2018 poverty incidence map confirms this geographic distribution. UNICEF report on child monetary poverty in Mali shows highest rates in Sikasso (85%), Mopti (49%) and Ségou (49%)⁶⁴. The northern regions and that of Kayes appear to be the least poor, on the one hand, because of the importance of financial transfers from emigrants and, on the other hand, the extent of informal cross-border trade⁶⁵.
67. The subjective poverty rates and the multidimensional poverty index confirm the spatial inequalities; rural areas are always more affected by poverty. More than half of rural households (56.4%) consider themselves poor, compared to 46.6% in urban areas and 53.4% at national level. As with subjective poverty, multidimensional poverty affects rural areas more (46.7%), compared to Bamako (10.6%) or other cities (28.9%). This precarious living conditions and the extent of poverty translate into poor human development. Indeed, Mali ranks among the last countries in the World in terms of Human Development Index - HDI, occupying 184th place out of 189 countries with an HDI of 0.427 in 2018. However, the evolution of the HDI since 2000 shows an increase of 27.9% between 2000 and 2018; even if there is a constant increase between 2000 and 2010 which is followed by a slowdown from 2011. The decline in the HDI could be explained by the weak economic growth and the recession that marked the country from this year.

Government representation

⁵⁹ Banque Mondiale. 2016. Poverty and shared prosperity 2016: Taking on inequality. Washington.

⁶⁰ PNUD. 2019. Rapport sur le Développement Humain 2019. Au-delà des revenus, des moyennes et du temps présent : Les inégalités de développement humain au XXI^{ème} siècle. http://hdr.undp.org/sites/default/files/hdr_2019_fr.pdf.

⁶¹ <https://data.unwomen.org/country/mali>

⁶² <https://dataafrica.io/profile/mali>

⁶³ République du Mali. 2018. Rapport National Volontaire sur la Mise en Œuvre des Objectifs de Développement Durable. Forum Politique de Haut Niveau sur le Développement Durable, New York, Juillet 2018

⁶⁴ de Milliano, M. et S. Handa. 2014. Pauvreté et privation des enfants au Mali : les premières estimations nationales. Document de travail Innocenti no 2014-20, Bureau de recherche de l'UNICEF.

⁶⁵ INSTAT. 2018. EMOP- Rapport d'analyse deuxième passage (juillet-septembre 2018) : Accessibilité aux soins de santé, appréciation de la population sur les actions prioritaires à entreprendre et dépenses de consommation des ménages. Octobre 2018.

68. The empowerment is measured by the proportion of seats held by women and the proportion of the population with at least secondary education started by gender. The female parliamentary representation decreased from 18 women between 1997 and 2002 to 15 women from 2002 to 2007 out of a total of 147 deputies. For the Vth legislature 2014-2018⁶⁶, there were 14 women deputies for 147 seats in the National Assembly. In Mali, as of February 2021, 27.3% of seats in parliament were held by women⁶⁷. Inequalities are also observed at the level of elective functions, where the representation of women is below that of men, and at the level of nominative functions, despite Law No. 2015-052 instituting measures to promote gender in these functions with a minimum representation quota of 30%. This quota is hardly respected for nominative functions. In the government formed in May 2019, there were 09 women ministers out of 38 (23.68%), while no woman occupies the post of Regional Governor in the A of Territorial Administration. In addition, women represent less than 2% of mayors and less than 9% of municipal councilors. The low visibility of women at the level of local governance, the judiciary, and also at the head of civil society organizations is a visible phenomenon in Mali⁶⁸.

Institutional arrangements

69. The Ministry for the Advancement of Women, Children and the Family was established by Decree No. 97-282/P-RM (16 September 1997). Its responsibilities include: (i) formulating and implementing measures to ensure the wellbeing of women, children and the family; (ii) preparing and implementing actions to ensure greater economic, social and cultural integration of women and children by meeting their specific needs; (iii) promoting the rights of women and children; (iv) promoting the family. The government institutional framework for managing gender issues in Mali is coordinated by this Ministry, which has three central departments. The National Women's Advancement Directorate (NWAD), the National Child and Family Promotion Directorate (NCFPD), and the Department of Financial Affairs and Equipment (DFAE). They are supported by the Regional Directorate for Advancement of Women, Children and the Family (RDAWCF) and includes the National Centre for Documentation and Information on Women and Children (CNDIFE) and the Children's Centre. This institutional framework, established by the government, is reinforced by the presence of a strong female civil society committed to supporting government actions (CMWAN, Réseau Yiriba Suma, CWNWRPG/WRC, MWA, AWAR, NFRW, ECTC, etc.). Regional women's organizations, including the PanAfrican Women's Organization (PAWO), the Network of African Women Ministers and Parliamentarians - Malian Section (REFAMP-Mali) and the West African Women's Association (WAWA) also support the activities of MAWCF.

Education

70. While there have been notable gains in overall enrolment levels and steps made to decrease gender disparities, there are still considerable gaps between literacy levels and enrolment rates between males and females, especially in rural areas. There are gender disparities about Education, particularly a lower retention of girls in the 2nd cycle of basic education, compared to the first cycle where they have more or less the same chances as boys in terms of access and completion. Indeed, despite the similar chances of access of the order of 70% at the beginning of basic education, these chances are reduced to 25% for girls at the end of basic education, compared to 41% for boys⁶⁹. As a result, girls are proportionally less likely to end up in secondary, technical and vocational education, and in higher education. The school dropouts, which are frequent in the second cycle of basic education, are also linked to the adolescent crisis in addition to other social factors.

71. The significant withdrawal of girls from the education system is the consequence of, among other things, child marriage, rural exodus, parental poverty or the remoteness of school infrastructures

⁶⁶ Un projet de loi organique a été adopté, prorogeant au 02/05/20 le mandat des députés pour réunir les conditions optimales à la bonne organisation des élections.

⁶⁷ <https://data.unwomen.org/country/mali>

⁶⁸ https://budget.gouv.ml/Loi-des-finances-2020/fichiers/pdfs/anb/Etat_V_A-G.pdf

⁶⁹ <https://www.globalpartnership.org/sites/default/files/document/file/2020-8-Mali-Program%20Document.pdf>

that, if they are present, still do not have amenities that facilitate retention (canteens, toilets, etc.). Girls are withdrawn from school to participate in domestic work, or in case of lack of results. They are married at an early age and forced to drop out of school to look after their households and children. Voluntary dropout and school failure are also factors in school wastage.

72. Reducing gender disparities in education should therefore start at the 2nd cycle of basic education, and in this perspective. Therefore, it is envisaged that actions should be carried out more on the demand side, given that the context is marked by a low level of interest in school on the part of parents and that choices are not always made in favour of the continuation of schooling. Furthermore, it is worth mentioning that despite these challenges, the Government of Mali has declared education a priority and devotes 32.6% of total government expenditure to this sector in 2015. This share is about 30% on average over the average over the last 5 years. The Government and Ministry of Education have also developed policies to address the gender disparity, including the National Policy for Girls' Education, with an ambitious action plan.

Employment

73. There are inequalities between men and women in the labor market. In Mali, four out of ten women (40.1%) aged 15-64 were employed in 2017, just over half (51.2%) of women aged 15-24 and 47.7% of women 15-34 year olds were also employed. According to the results of the integrated regional survey on employment and the informal sector, active women are concentrated in the Agriculture, Livestock, Hunting and support activities branch. However, the percentage of active women employed in this branch differs by age, with 99.0% for 15-24 year olds, 84.3% for 15-34 year olds and 74.4% for 15-64 year olds⁷⁰. In addition, women make up 50% of total employment in the food system, but they dominate in the non-agricultural segments, representing 51% of the catering staff, 62% in food processing and 66% in marketing⁷¹. This preponderance of women in the non-agricultural segments is confirmed by the differences in income compared to men. Households headed by women earn on average 2,233,467 FCFA per year from non-agricultural work, compared to 1,383,488 FCFA for households headed by men⁷². There are regional disparities in the contribution of women to economic activities. The regions of Timbuktu and Gao have the lowest percentages of employed working women, compared with Sikasso, Koulikoro, Mopti and to a lesser extent Ségou. Women are underrepresented in the formal private sector and the public sector, unlike the informal sector, which indicates a low level of qualification and a certain precariousness of their jobs. Gender inequalities are also illustrated by the differences in remuneration between men and women. According to the results of the above-mentioned regional survey, the average monthly income of women (40,387 FCFA) is 2.5 times lower than that of men (103,274 FCFA). In addition, a comparison of the salary at the SMIG shows that more than three-quarters of employed women (76.4%) are paid below the SMIG, compared to just over half for men (51.2%). The same is true for the informal sector where women have a lower average monthly income compared to men, with FCFA 21,925 and FCFA 53,203 respectively. These inequalities between men and women also prevail for unemployment and underemployment of the labor force, women are still the most affected.

Health

74. Efforts have been made in the area of reproductive health, given the decline recorded By its two measurement indicators, the maternal mortality rate and the adolescent birth rate. According to World Bank data, the maternal mortality rate is 368 deaths per 100,000 live births in 2013, compared to 465 and 582 in 2006 and 2001 respectively. There is also a drop in the

⁷⁰ Institut National de la Statistique et AFRISTAT. 2019. Enquête Régionale Intégrée sur l'Emploi et le Secteur Informel, 2017 : Rapport final. Bamako, Mali : INSTAT et AFRISTAT

⁷¹ Allen, T., P. Heinrigs et I. Heo (2018), « Agriculture, alimentation et emploi en Afrique de l'Ouest », Notes ouest-africaines, N°14, Éditions OCDE, Paris.

⁷² République du Mali. 2019. Agriculture et sources de revenu au Mali : Etat des lieux à partir des données de l'EAC-I 2017. Ministère de l'Agriculture, Ministère de l'Elevage et de la Pêche, Ministère de la Population et de l'Aménagement du Territoire. Septembre 2019.

adolescent birth rate, from 188 births per 1,000 women aged 15-19 in 2000, to 181 in 2010 and 169 in 2017.

Gender Based Violence

75. Gender-based violence (GBV) constitutes a major risk for women and girls in Mali. According to EDSM-VI ⁷³, 13% of women aged 15-49 have experienced sexual violence. There is an upward trend in the phenomenon because, 4% of women of the same age group were victims of sexual violence in 2006. Through the GBV information management system, the number of reported cases has increased from 1,462 in 2015 to 2,164 in 2016 and 2,882 in 2017. The regions most affected are Kayes (19%), Koulikoro (15%), Bamako (15%), Ségou (14%) , unlike Tombouctou and Kidal with 4% and 3% respectively. Genital mutilation f éminines / the excision part of sexual forms of discrimination the most widespread in Mali, with more than eight out of ten women concerned by the DHS-VI. The rate has increased over the past five years, from 83% (MICS, 2015) to 89% (EDSM-VI, 2019), and for three-quarters of women who have been cut, the practice was done before the age of 14 years old. The prevalence is high in the regions of Koulikoro and Sikasso (96%), followed by Kayes (95%), Ségou (92%) and the district of Bamako (91%). The practice remains marginal in northern regions such as Kidal and Gao, where the prevalence is at most 1%.

Early Marriage

76. Furthermore, early marriage is one of the most widespread discriminatory practices against girls in Mali. Although the legal age for marriage is set at 16 for girls (18 for boys), the results of the MICS survey (2015) show that 16.1% of women aged 15-49 have married before the age of 15. Among women aged 20-49, almost half (48.9%) married before age 18. The prevalence of early marriage for those under 15 increased from 14% to 16.1% between 2010 and 2015 according to MICS. This practice is observed more in rural than urban areas ; the region of Kayes recording the highest rates (29% for marriages before 15 years and 66% for those before 18 years). Mali is among the West and Central African countries with the highest rates of child marriage, knowing that the rate of this region is higher than that of all the other regions of the world . The early entry of women into conjugal life exposes them to the risks of dropping out of school, and there is a strong correlation between child marriage and the level of education in Mali, represented by the rate completion of lower secondary education. Reducing child marriage would have significant economic benefits for Mali, including welfare gains from reduced population growth, equivalent to \$ 1.6 billion by 2030 ⁷⁴. To this discrimination is added that of the division of domestic tasks between girls and boys, especially in rural areas. Young girls spend 0.6 hours a day collecting water and 0.2 hours collecting wood, compared to 0.1 hours a day for young boys. Women are also disadvantaged in terms of credit and other financial services. Despite the improvement in the number of women over 15 with a bank or mobile account (6.91% in 2011 to 25.71% in 2017), there is still a gap between men and women of 20 percentage points . In the area of agricultural finance, it is important to emphasize that women with access to campaign credits make less than 5% ⁷⁵. Added to this is their poor access to other productive resources such as land to develop certain speculations such as rice. In 2017, for example, female owners of rice plots were 14.71% in the Office Rice Mopti zone and 8% in the Office Riz Ségou zone ⁷⁶.

⁷³ INSTAT, Cellule de Planification et de Statistique Secteur Santé-Développement Social et Promotion de la Famille (CPS/SS-DS-PF) et ICF. 2019. Enquête Démographique et de Santé au Mali 2018. Bamako, Mali et Rockville, Maryland, USA : INSTAT, CPS/SS-DS-PF et ICF

⁷⁴ Banque Mondiale. 2018. Réduire les inégalités de Genre au Mali, Tchad, Niger et Guinée. Bilan économique AFCW3. Groupe de la Banque Mondiale, Automne 2018. 66 pages

⁷⁵ FAO et Commission de la CEDEAO. 2018. Profil National Genre des Secteurs de l'Agriculture et du Développement Rural – Mali. Serie des Evaluations Genre des Pays, Bamako

⁷⁶ https://budget.gouv.ml/Loi-des-finances-2020/fichiers/pdfs/anb/Etat_V_A-G.pdf

Agriculture

77. Mali's economy is based on agriculture. It employs 80 per cent of the active population and contributes about 40 per cent of GDP and 30 per cent of export revenues. Livestock production is the third largest contributor to export earnings, after cotton and gold; it is the largest livestock producer in the WAEMU. Mali has considerable potential in arable land, estimated at 43.7 million ha, and significant water resources. In addition to significant non-perennial surface waters, the Niger and Senegal rivers and their tributaries drain an annual average of 70 billion m³ of water and offer an estimated irrigable potential of more than 2.2 million hectares. The government allocates more than 15 per cent of its national budget to agriculture and significant progress has been made in managing the sector at the legislative, regulatory, and institutional levels.
78. Mali's vast potential for agricultural and rural development has barely been tapped. Just 7 million ha of the 43.7 million ha available for agriculture (representing 4.5 per cent) is now being farmed. Less than 300,000 ha of the 2.2 million ha of irrigable land is being irrigated or semi-irrigated. In the absence of adequate management, a high proportion of the available surface water is not benefiting agriculture but being lost to runoff into the ocean. The agriculture sector faces several development constraints that explains its low performance: (i) highly variable climate conditions; (ii) inadequate water and other natural resources management; (iii) insufficient or poor condition of production, processing and marketing infrastructure; (iv) limited access to financing; (v); limited access to markets; (vi) limited use of technical innovations; (vii) weak capacity of public, private and community-based institutions; (viii) insufficient incentives for private sector engagement; (ix) lack of secure land tenure, in particular access to land for women; (x) underequipped producers; and (xi) the impact of the security crisis on agricultural activity in the northern and central areas of the country.
79. Mali has adopted a Law (n ° 2017-001) on agricultural land, which indicates the modalities of access to agricultural land and prioritizes conflict prevention. It pays particular attention to women and young people, allocating them 15% of the State's land development. It has also enabled women and young people to become more involved in land management and the defense of their interests in decision-making bodies. For example, to operationalize this law, Village Land Commissions (COFO) have been created in the Kayes region, involving women and young people, and contributing to the prevention of land disputes⁷⁷. This is all the more important since land disputes are cited as one of the main causes of conflicts and inter-community violence, especially in central and southern Mali. Another positive discrimination measure relates to the increase in the representation of women in land commissions, rural women in particular. One of the innovations of this Law lies in the decentralization of land management and the creation of local land institutions with which the customary authorities must 'deal'⁷⁸. Also, the Law recognizes the right of rural communities to collectively own certain lands, the management of which would be based on the customs and traditions existing in these communities. Such a provision could constitute a limit to tenure security for women since they are disadvantaged by the discriminatory nature inherent in certain customs and traditions.⁷⁹ However, despite the legal provisions, women continue to be disadvantaged compared to men. According to EAC-I 2017, in most regions, land managed by women is three times smaller than the average size of farms in these regions. In addition, female heads of household own land that is about twice as small as male heads of household of the same age group : 4.36 ha cultivated by male heads of household under 30 years of age, against 1.45 ha for women of this age group⁸⁰.

⁷⁷ https://www.ipar.sn/IMG/pdf/note_de_capitalisation_-_gouvernance_du_foncier_dans_le_bassin_du_fleuve-srb__fao_ipar.pdf

⁷⁸ FAO et Commission de la CEDEAO. 2018.

⁷⁹ <https://www.iisd.org/blog/une-nouvelle-loi-pour-s-curiser-les-terres-au-profit-des-paysans-maliens>

⁸⁰ République du Mali. 2019. Agriculture et sources de revenu au Mali : Etat des lieux à partir des données de l'EAC-I 2017. Ministère de l'Agriculture, Ministère de l'Elevage et de la Pêche, Ministère de la Population et de l'Aménagement du Territoire. Septembre 2019.

Senegal Gender Profile

80. Senegal is a West African country with a population of about 14 million of which 51% is female, with women representing 70% of the main workforce in the agricultural sector. More than half of the households live in rural areas. The heads of households are mostly men, while households headed by a woman are only 15%. Children, women and the elderly are the most vulnerable to climate shocks and to their harmful consequences. Nevertheless, climate shocks affect also men, as they are subjected to the exhaustion of their food reserves and to the decline in agricultural production over time. Facing shocks and their negative effects on their lives and livelihoods, poor households struggle to address their food needs, education and health expenses.⁸¹ In the recent years, Senegal has made progress in terms of gender equality. The Gender Inequality Index (GII) decreased at the national level from 0.540 to 0.528 between 2012 and 2014 to reach 0.521 in 2015. Senegal occupies the first place with the lowest GI, compared to the other countries of the world. 'UEMOA⁸². The SIGI indicator⁸³ from the OECD Development Center, measuring the discrimination against women in social institutions, has a value of 37% for Senegal lower than the regional average. With the application of the law on gender, the share of women parliamentarians increased from 22.7% in 2011 to 42% in 2017; ranking Senegal 7th in the world and 1st in UEMOA and ECOWAS. The Economic, Social and Environmental Council (EESC) has 43 women out of 120 advisers, which represents more than a third. At the local government level, the representation rate of women is 47.2%.

Legal and policy framework

81. The National Strategy for Gender Equality in Senegal was implemented between 2005 and 2015, concentrating on increasing women's status in society, improving their capability, improving their economic position and setting up workshops to start the conversation in order to raise awareness about the issues that are prevalent to Senegalese society⁸⁴. In 1999, the Criminal Code was revised to make tougher penalties for crimes against women. This revision allows for the punishment of previously unrecognized crimes, such as incest, rape, sexual harassment, excision and domestic violence. Legal reform in Senegal has included a law on equality in elected public bodies and a stipulation that Senegalese women can pass their citizenship to their foreign spouse and children. New policies provide free caesarean operations in public hospitals and extend social security grants to help the most vulnerable. An essential plank of the emergency community development programme is to lighten burdens on women working in rural areas. Senegal is committed to giving meaning to the empowerment of women, ensuring that women's participation in society is not measured just in terms of numbers, but in terms of quality so that true equality is achieved⁸⁵. Ratification of the CEDAW in 1983 and the additional protocol to the African Charter on the rights of man and people relative to rights of women in 2004. • Article 7 of the 2001 Constitution recognises equality between the sexes. • Article 15 of the Constitution gives women and men the same rights in matters of access to land. In practice, customary law prevails and hinders women's access to land: access to land is familial/collective and the management is the responsibility of the head of the family, always a man. • The law adopted on 14 May 2010 establishes "absolute parity" in the elective functions, specifying that women and men must be represented in equal shares on candidate lists.
82. Senegal has gained international recognition for its efforts towards political participation of women through the adoption of the Gender Parity Law (2010). As a result of this law, which demands parity on electoral lists, the proportion of seats held by women in national parliament has increased

⁸¹ Philippe De Vreyer and Sylvie Lambert *Intrahousehold inequalities and poverty in Senegal* 2016 <http://pubdocs.worldbank.org/en/407221466181742385/intrahh-inequality-and-poverty-juin2016-2.pdf>

⁸² Rapport national sur le développement humain. 2017

⁸³ Social Institutions and Gender Index

⁸⁴ <https://borgenproject.org/what-is-being-done-and-needs-to-be-done-to-improve-womens-rights-in-senegal/>

⁸⁵ UN Women- <https://www.unwomen.org/en/get-involved/step-it-up/commitments/senegal>

significantly from 19.2% in 2001 to 43% in the 2017 elections. This marks a major step forward in the struggle for gender equality and the promotion of women's rights and makes Senegal a leader in women's political participation in Africa. On 30 July 2017, about 6 million Senegalese citizens voted at the legislative elections. As a result, 70 women (42%) out of 165 parliamentarians were elected compared to 64 women (39%) in 2012. Women represent 21% of ministers in the National Government (8 women out of 39 members). Women represent 49% of the world's population and 52% of Senegal's and it is therefore crucial to ensure that decisions affecting women's lives reflect their priorities and views. For example, women tend to be more sensitive to certain issues of development such as education, nutrition or health than men. Therefore, women's participation in decision-making is not only essential for the democracy and the legitimacy of a political system—it is also necessary to ensure the quality of decisions taken and improve social development. Leading together, men and women can better reflect on and respond to the diverse needs of a society. Therefore, women's political participations bring benefits for the whole society. In fact, countries with more women in politics tend to be more successful in several areas such as the quality of governance, the quality of educational systems, infrastructure investment, and perceptions of corruption. Despite these progresses, regarding governance and women political participation key challenges remain in Senegal:

- * It remains unclear to fully engage men in gender parity and, more generally, in the struggle for gender equality;
- * The Gender Parity Law is quite unpopular among male Parliamentarian and not followed;
- * The parliamentarians do not have the same understanding of the parity law and particularly the parity concept.

Institutional Arrangements

83. There are several institutions in charge of monitoring gender equality in Senegal. In particular the National Observatory of Women's Rights (established by Decree No. 2008-1047 of 15 September 2008), which monitors all violations of women's rights. However, this body cannot receive complaints directly. There is also the National Parity Observatory (established by Decree No. 2011-819 of 7 March 2011), which is tasked with monitoring, evaluating and making proposals for promoting parity between men and women in public policy. Other initiatives for the advancement of women have been taken through the collection and processing of gender-sensitive statistics in accordance with the National Strategy for Gender Equality and Equity (CEDAW, 2013).⁸⁶ The improvements in gender can be explained by the strategies developed by the government and the financing mechanisms put in place to reduce inequalities. In the agricultural sector, incentives have been taken in addition to increasing the rate of representation of women in decision-making bodies to at least 20%. These measures consist in allocating to women a quota of at least 15% of the installations to be carried out from surface water, 20% of the installations to be made from groundwater, at least 20% of subsidized fertilizers and at least 40% of the funding⁸⁷. For a better integration of gender in sectoral interventions, gender cells have been created in key ministries and the budget process incorporates a Gender Budget Document which accompanies the finance bill. A pilot gender-sensitive budgeting exercise has been initiated since 2016, and to which 18 ministerial departments have worked to prepare the 3rd Gender Budget Document as part of the 2019 draft budget⁸⁸. Women may uphold the norms that harm them because the social costs of doing otherwise would be unacceptably high. Even though a social norm may be harmful, it may give women status in their communities, and some women may tolerate loss of control and agency in exchange for economic support (Sen and Ostlin, 2008). The social norms approach to elimination has been key to the success of initiatives such as the Community Empowerment Programme, launched in 1988 in Senegal and implemented by Tostan, a non-governmental organization (Diop

⁸⁶ OECD, 2019 - <https://www.genderindex.org/wp-content/uploads/files/datasheets/2019/SN.pdf>

⁸⁷ République du Sénégal. 2018. Document budgétaire genre. Direction Générale du Budget, Ministère de l'Economie, des Finances et du Plan

⁸⁸ République du Sénégal. 2018

and others, 2004). The programme supports changes in social norms by stimulating personal and collective reflection and critical thinking through community dialogues, education sessions, and “organized diffusion” (UNICEF, 2010).

Education

84. Senegal has made enormous strides in its education system, with primary school enrollment rates increasing nearly 30% since 1994. In that time, Senegal has almost reached gender parity in terms of primary school enrollment, with 77.3 percent girls enrolled as compared to 80% of boys.⁸⁹ Gender gaps in primary education in both enrollment and completion rates have closed and even reversed in Senegal. According to UNESCO, primary education completion rates rose from 33 percent for girls and 43 percent for boys in the year of 2000 to 64 percent and 54 percent, respectively, in 2016. However, gender gaps in secondary and tertiary education persist, with girls’ completion rates in secondary education and enrollment rates in tertiary education lower than boys’ rates. The Demographic and Health Survey (DHS) for that country reported that, in 2012, the average female completion rate for secondary education was only 13 percent, compared to 21 percent for boys.⁹⁰ According to the United Nations Educational, Scientific and Cultural Organisation (UNESCO), although gender parity has been achieved in favour of girls in primary education, where for every 100 boys enrolled, there are about 104 girls, the dropout rate at secondary school among female learners is high. The scale of illiteracy, especially among women in rural Senegal, is also symptomatic of the poor access to education. According to UNESCO, Senegal’s literacy rate for the population aged 15 years and above is 64.81 percent for males and 39.8 percent for females. This phenomenon remains recurrent among women in rural areas where only 25.9 percent of them are literate. A 2014 regional analysis of the phenomenon shows that the regions of Ziguinchor (62.3 percent) and Dakar (61.9 percent) have the best literacy rates. In contrast, the regions of Matam (24.9 percent), Tambacounda (26.6 percent), Diourbel (29.8 percent) and Kolda (33.1 percent) stand out with the lowest rates⁹¹.

85. Key gender-based constraints in the education sector include: Social beliefs about appropriate life choices for girls that encourage early marriage and childbearing and discourage continued education for girls; Social beliefs about appropriate life choices for boys that encourage parents to send them to marabouts who then deploy the boys to beg on the streets; Social beliefs about the appropriateness of child labor (boys and girls) to assist at home; Lack of adequate infrastructure (running water and latrines) that discourage girls from attending school when they are menstruating; Educational policies that discriminate against pregnant girls attending school; Social beliefs as well as poor infrastructure that inhibits many women teachers from taking positions in rural schools.

Health

86. The 2005 Law in Relation to Reproductive Health, recognizes reproductive health as a “fundamental and universal right guaranteed to all individuals without discrimination based on age, sex, wealth, religion, race, ethnicity, matrimonial situation or any other situation.” There are no legal restrictions on young people’s access to contraceptives and other basic services, such as pregnancy and STI testing, apart from a requirement that one be 15 or older to consent to HIV testing. Abortion is legal only to save a woman’s life, and the Penal Code imposes hefty prison and financial penalties. However, many young women resort to illegal abortion services, which often put their health at risk.⁹²

⁸⁹ <https://www.borgenmagazine.com/womens-education-in-senegal/>

⁹⁰ IMF, 2019- <https://www.imf.org/en/Publications/WP/Issues/2019/11/08/A-Quantitative-Analysis-of-Female-Employment-in-Senegal-48716>

⁹¹ <http://www.ipsnews.net/2020/07/providing-education-favour-senegals-girls/>

⁹² International Planned Parenthood Federation (IPPF) and Coram Children’s Legal Centre, Over-Protected and Under-Served: A Study on Legal Barriers to Young People’s Access to Sexual and Reproductive Health Services in Senegal, London: IPPF, 2014.

Gender Based Violence

87. Gender-based violence (GBV) remains a serious problem in Senegal. According to the 2005 DHS, ⁹³65% of adult women believe that men have a right to beat them, ranging from 57% in urban areas to 73% in rural areas. Among wealthier and more educated women, the percentage that believes this is reduced, but still hovers around 50%. There are few programs supporting women who have been abused, and there are some serious institutional and social constraints surrounding medical and police producers. For example, the law apparently requires that to file a claim of rape, a woman must have a medical certificate from a doctor to present to the police so that an inquiry can be initiated. That certificate has a fee of 10,000 CFA. In some cases, an NGO might pick up the cost for a woman, but an advocacy effort is needed to address these costs and this requirement.

Employment

88. In rural areas, the distribution of employment in different economic sectors reveals women's involvement in agriculture, livestock farming and the environment where they represent 70% of the workforce.⁹⁴ Challenges arise regarding women's participation in the labor market, their financial inclusion and salary levels. Results from three ANSD surveys show that women have lower activity and employment rates than men. Women represent only 22.8% of workers in the formal sector, with a larger presence in small businesses (30.1%). Financial inclusion is 13.4% for women compared to 21.9% for men, a difference that remains statistically significant even when integrating income, education and other individual characteristics. These results are partly explained by the fact that women are less involved in household financial and budgetary decisions. Women are generally paid less than men. On average, a male "senior manager" earns 933,333 FCFA / month compared to 697,286 FCFA / month for a woman; the same is true for the category "middle managers and senior technicians" where the monthly remuneration is 684,493 FCFA for men against 569,559 FCFA for women. Such inequalities, both in land, financial and employment resources, increase the vulnerability of households headed by women. The rate of food insecurity is higher in households headed by women (40.4%) than those headed by men (29.4%). According to World Bank (2018) women in Senegal still face gender-based discriminatory legal restrictions that prevent non-pregnant, non-nursing women from performing the same job as men. Furthermore, there are no laws in Senegal mandating equal remuneration for work of equal value nor laws censuring discrimination based on gender in hiring, in job's promotions or in dismissal.⁹⁵ Moreover, discrimination in access to finance based on gender or marital status is not prohibited by law and the constitution does not formally recognize, nor does it prohibit discrimination against women that may result from customary laws.⁹⁶ The working population is predominantly female, yet the unemployment rate for women is double that of men. While performance on the United Nations Development Programme Gender Inequality Index has improved, inequalities remain, especially in rural areas due to issues with access to land meaningful employment and technology. Most of the much-needed job creation will have to come from the private sector⁹⁶.

Agriculture

89. Despite these efforts to promote gender, disparities continue to exist, especially in the area of Agriculture and Rural Development. Access, control and securing factors of production, such as land, remains problematic for women. Indeed, the percentage of women who own their plots is only 13.8% compared to 86.2% for men; 23.2% of women are renting land compared to 76.8% for men; 40.7% of women borrow land compared to 59.3% for men, and 14.3% of women do tenant

⁹³ <http://www.measuredhs.com/pubs/pdf/FR177/03Chapitre03.pdf>

⁹⁴ UNDESA, 2020- <https://worlds-women-2020-data-undesa.hub.arcgis.com/pages/regions>

⁹⁵ World Bank's "Women, Business, and the Law", 2018

⁹⁶ IFAD - Republic of Senegal, COSOP, 2019.

farming compared to 85.7% of men in 2014⁹⁷. Gender inequality remains pronounced regarding participation in the labor market. The results of the 2015 National Employment Survey in Senegal showed that the activity rate is much higher among men (72.1%), compared to women (52.3%). The unemployment rate is higher among women (16.7%) than among men (9.5%). Furthermore, the last census in 2013 shows that 84.7% of agricultural households in Senegal are headed by men against 15.3% by women; the difference being greater in rural areas (88.7% for men against 11.3% for women)⁹⁸. Climate change and climate variability are already changing the agricultural practices and opportunities of rural smallholders in Senegal. ⁹⁹It may soon change the crop mix appropriate for cultivation. New crops will likely shift farm and household labor patterns of allocation, possibly reduce household investments in agriculture, and/or result in an even greater level of migration to urban areas within and outside of Senegal. While men and women may both seek non-farm opportunities, socio-cultural and historical patterns suggest that men will first seek off-farm employment. All of these shifts have gendered consequences that call for greater involvement of women in the study of agricultural sciences and agri-business.

Women Economic Empowerment¹⁰⁰

90. Achieving gender equality and equity in Senegal may seem daunting in a complex socio-cultural environment marked by a strong preponderance of traditional values. However, despite women's lower social status compared to men, joint efforts paved the way for significant progress that led to a greater recognition of women's place and contribution to socio-economic growth. Women living in rural areas are highly active in the processing and marketing of agricultural, livestock and fishery products. Nonetheless, they are confronted with several hurdles of various nature, including a number of constraints that are yet to be overcome include:

- * Access to land and land tenure security;
- * Access to financing mechanisms;
- * Access to factors of production and extension services;
- * Effects of climate change;
- * Access to markets.

In rural areas, the distribution of employment in different economic sectors reveals women's involvement in agriculture, livestock farming and the environment where they represent 70% of the workforce.

Key barriers common to all target countries

91. The country assessments above indicate a series of common factors that must be taken into account in the programme's design and implementation to ensure that it is effective in building women smallholder farmers' resilience to climate change. Many are important barriers to women's access to financial services and credit:

- **Gendered division of labour.** Cultural beliefs and norms govern daily life in the rural areas. Women are expected to assume full responsibility for housework and care for the family, which is unpaid, and certain norms limit their engagement in economic activities outside the home. Only a small percentage of women have paying jobs or their own small businesses. They lack

⁹⁷ FAO et Commission de la CEDEAO. Profil National Genre des Secteurs de l'Agriculture et du Développement Rural – Sénégal. Série des Évaluations Genre des Pays, Dakar.

⁹⁸ ANSD. 2016. Analyse Genre des bases de données existantes. ANSD, Ministère de l'Economie, des Finances et du Plan. ONU-Femmes.

⁹⁹ Gueye 2008.

¹⁰⁰ <https://africa.unwomen.org/en/where-we-are/west-and-central-africa/senegal>

financial resources to invest in machinery, male family labour, technologies and crops that bring higher returns and in sustainable manner.

- **Property rights and control over assets.** Due to cultural beliefs and practices, only a small minority of women own land and few have adequate and stable access to land and agricultural inputs. The lack of assets in their names is an important impediment to developing viable business and obtaining loans, as they have nothing to offer as collateral. This remains a major challenge and prevents financial institutions from providing loans.
- **Lack or no awareness of financing opportunities and new instruments like green financing.** Rural women generally lack knowledge on the financial options available to them with the financial institutions and men control resources. This is mainly due to lack of financial education and prevailing social and cultural norms. Women have no understanding of climate finance, which could strengthen their resilience to climate shocks depending on the intensity and magnitude.
- **Lack of access to knowledge particularly reliable climate information's:** when available , Climate Information Services are critical for the creation of relevant, science-based information to inform decision making, enabling and creating new business opportunities, but women still do not have access to climate information's for decision making processes in agriculture.
- **Lack of education:** Women in rural areas in the Sahel have lower educational, literacy levels than men, and natality rates are high. They also lack access to knowledge and information to develop projects for the banks and MFIs.
- **Biased perception of the financial sector:** banks and MFIs consider smallholder agriculture as a high-risk sector and women are not attractive to banking. In situations where they do have access, interest rates are high and put them into a debt circle.
- Many women are subjected to various forms of gender-based violence: domestic violence, female genital mutilation, child marriage and force-feeding.
- **Access to technologies:** Evidence consistently suggests that male household heads and farmers adopt new agricultural production technologies faster than female heads and farmers across regions¹⁰¹. Female heads of households and plot-managers are less likely to adopt a wide range of agricultural and rural technologies than male heads and plot-managers. The most commonly-cited reasons and results of statistical analyses for this gender gap in technology adoption are greater time and labor constraints; relatively less access to funds and credit; more limited information, education and training; more limited capacity and opportunity for participation in innovation and decisionmaking processes; and more limited access to accompanying inputs and services. These are influenced by weak design or implementation of legislation that protect rights and promote equal playing field and by the persistence of social biases and cultural norms limiting equal access and opportunities for women and men¹⁰².

92. In order to address the gender gap in access to technology IGREENFIN will ensure that women are provided with cell-phones and required extension services. Some other key recommendations to follow will be the following:

- Strengthening capacity of women and men farmers as innovators, evaluators of technologies, and key partners in innovation processes.
- Building measurable targets and strengthening the monitoring and evaluation to ensure that (1) planning and innovation processes addresses women and men's needs, viii preferences and opportunities; (2) women and men can access and use these technologies; and (3) women and men benefit from these technologies.

¹⁰¹ (Doss, 2001; Tiruneh et al., 2001; Bourdillon et al., 2002; Phiri et al., 2004; Kakooza et al., 2005; Jagger and Pender, 2006; Thapa, 2009; the World Bank and IFPRI study on Ghana and India, 2009; Peterman, Behrman and Quisumbing, 2010; FAO, 2011, to name a few)

¹⁰² FAO, 2014. <http://www.fao.org/3/i4355e/i4355e.pdf>

- Holistic and integrated approach of looking at constraints to production and marketing and paying close attention to the complementarities of inputs and services.
- Promoting equal playing field: It will also be important to strengthen women's land, property and water rights.

93. Furthermore, women face multiple barriers when seeking remediation and justice . To ensure women have equal access to the Grievance Redress Mechanism and benefit from it equally, IFAD is conscious about designing a gender-responsive grievance mechanism. This involves, among other things:

- Ensuring gender diversity in grievance mechanism staff and providing them with gender sensitivity and unconscious bias training
- Involving gender specialists from all PMUs within the banks and the baseline projects
- Ensuring visibility of grievance mechanisms to all individuals, including women and marginalised groups, taking into account potential lower literacy levels
- Providing for independent, gender-sensitive investigations of violations and committing to addressing gender-linked power imbalances during dispute resolution processes
- Protecting all people who register complaints from reprisals

Monitoring the grievance mechanism to ensure that access and remediation outcomes are provided on an equal basis.

94. IFAD is recognized for addressing gender inequalities in rural areas. A recent [report](#) by our Independent Office of Evaluation showed that IFAD has been successful in addressing the root causes of gender inequality and women's powerlessness, including in the countries where this initiative will be implemented. The programme will build on this solid experience. It starts with a rigorous analysis of the socio-economic context on which the gender strategy is based. The strategy to address these barriers includes, amongst others, multiple and complementary activities promoting gender equality and women's empowerment, concrete measures to reduce women's workloads, support to women's organizations, working with men is critical as they are often the gatekeepers of customary practices that limit women's access to resources or public spaces, highly participatory approaches and cultural sensitivity.

Institutional Arrangements

95. At the institutional level, it should be noted that the gender mainstreaming approach is taken into account in a number of ways, such as: the involvement of men and women in all program intervention processes, respect for human rights and consideration of the diversity of target groups; the program targeting strategy that allows women and men to enjoy the same opportunities, etc.

96. For greater equity, the gender mainstreaming process will be strengthened through the implementation of specific initiatives to support women's empowerment in order to reduce their economic and social vulnerability. Specific initiatives will be implemented to strengthen the gender dimension in the institutional architecture of the Program. However, it will be important for this program to work at the senior management level to effectively mainstream gender at the institutional and organizational level. Thus, the planned institutional structure will consist of working with the gender specialists already recruited within the project units. At the regional level, a specialist will be recruited and, depending on the needs, he will be supported by consultants recruited on a part-time basis and they will also be able to provide targeted support in the countries. Specific ToRs will be prepared at the appropriate time for the recruitment of these consultants.

Approach to Strengthen Gender within the Proposed Program

97. While the financial inclusion of women-owned SMEs is not the main objective of the program, which is aimed at establishing green credit lines with commercial banks to increase investments in clean energy and sustainable land use, gender is seen as a cross-cutting measure to be included throughout all program components. It is of utmost importance that women are adequately included in all processes of the program at the Local Financial Institutions and final beneficiary level and that financial products are designed and promoted in a way that it encourages uptake by women-owned SMEs. The Programme is anticipated to have primarily positive impacts by improving the access of women to finance for climate projects in the energy and land use sectors, building the capacities of women for SME operation, and improving gender-responsive M&E systems within LFIs to promote the empowerment of women and effectively monitor and evaluate the impact of lending operations on women. Additional gender-related trainings will be conducted to support women owned SME development, and to support LFIs to strengthen their gender policy and design targeted financial products for women – helping to address key barriers that limit women’s access to finance.
98. To this end, the program will conduct specific ad hoc studies in each of the five countries to further assess women's current contributions to productive economies, including value chains. It will also assess the roles and challenges of each category (men and women) with a focus on female-headed households, youth and other marginalized groups in the project intervention areas. The study will be launched during the first year of implementation and the results will be integrated into the program's PTBAs from the second year of implementation and thus, the gender transformative approach will be applied in the project's investment areas: seed banks, transformation of agricultural productions, sustainable livestock, forestry, climate smart agricultural practices, climate resilient infrastructure and technologies, solar energy, etc.
99. In addition, measures will be taken with funding and implementing organizations to prevent any risk of exclusion and to ensure compliance with basic gender equality standards, including equal salaries, hiring practices, gender equality policies, workplace safety and anti-sexual harassment, etc. From the first year of implementation, the program will verify that the banks' policies, practices and procedures ensure balanced access to loans for women and if not, mitigating measures will be taken to address these disparities and have full inclusion of all target groups (including women and marginalized groups). The same attention will also be given to persons living with disabilities.
100. This approach will ensure that loans for women's and youth-owned businesses and groups are effective with equitable access to loans. In the financing strategies, the program foresees specific financial products for women, youth or marginalized groups and all necessary measures will be taken to guide the selection process of loan applications. The targeting strategy combined with this GAP will help reach the target groups including marginalized and indigenous people. However, it should be noted that in some of the five countries of the program intervention, there are indigenous peoples such as Fulani or Touareg .
101. To ensure that the partner institutions in charge of implementing the project activities take into account the GCF gender policy, a review of the partner institution's gender strategy will be conducted at the beginning of the program. Subsequently, a gender guidance note will be prepared and shared with each partner institution to ensure compliance with GCF requirements. If necessary, the EA will prepare an action plan that will be monitored according to the logical framework indicators and targeting percentages. The assessment of each partner's strategy will verify whether practices and procedures allow for equitable access for men and women to loans and if not, measures will be proposed (action plan) to remedy the imbalance in women's access to loans
102. In the implementation of IFAD projects, the gender issue is well covered by all implementing partners. Prior to the start of activities, due diligence is always carried out to confirm the partner's capacity to apply IFAD's standards on gender equality and equity. Whenever necessary, the IFAD team assists the partner in developing a gender strategy. This is the case, for example, of the LBA in Senegal, which will be the implementing partner. It has a gender specialist and a strategy in line

with the GCF policy is being prepared. This will be the same approach for all other implementing partners

103. To effectively address the underlying causes of gender inequality (often entrenched in social norms and practices) the communication strategy should include information and awareness-raising campaigns with, among others, community leaders and women's associations

Addressing Gender gap through additional analysis and assessment in the targeted areas.

104. The program will be implemented in the following targeted areas with direct and indirect beneficiaries per country (See Table 4. Targeted areas and number of beneficiaries).

Table 4: Targeted areas and number of beneficiaries

	Project name	Interventions areas for both IFAD baseline investments and IGREENFIN I	Additionally of IGREENFIN I	Beneficiaries	
				Direct	Indirect
IFAD baseline investments (Country)	Agricultural Value Chains Support Project (PAFA) - Burkina Faso	Boule du Mouhoun, Haut Bassin, Cascades	IGREENFIN I introduces innovative green business models to improve the risk-return profile of baseline projects, makes much needed concessional capital available and crowds in private sector capital, while providing technical assistance for subproject preparation, as well as incentives to beneficiaries to adopt best adaptation and mitigation practices and technologies, capacity-building and reporting & monitoring of GHG emissions.	80,000	330,000
	Agricultural Emergency Support Project (AESP) – Côte d'Ivoire	Bagoue, Poro, Tchologo, Hambol and Gbeke Tonpki, Kabadougou, Folon, Bafing, du Worodougou and Bere		53,600	294,800
	Affordable Agricultural Financing for Resilient Rural Development Project (AFFORD) – Ghana	Northern, Savannah and North-east Regions, Bono, Bono East and Ahafo Regions		155,000	930,000
	Inclusive Finance in Agricultural Value Chain Project (INCLUSIF) - Mali	Koulikoro, Sikasso Kayes, Segou,		40,000	440,000
	Rural Youth Agripreneur Support Project (AGRIJEUNES)- Senegal	Louga, Thiès, Diourbel, Fatick, Kaolack, Kaffrine, Sédhiou and Ziguinchor		50,000	500,000

Summary of the main roles, challenges and gender gap faced by men. Women, women headed households, Youth and other marginalized groups

105. During the first year implementation, the program will carry out for each country and project locations a tailored collection and analysis of the main roles, challenges and gender gap faced by men, women, women headed households, Youth and other marginalized groups to better orient the Gender Action Action.

Stakeholders consultations

106. The program already contains a stakeholder engagement plan that clearly outlines the stakeholder consultation strategy and indicates the target groups. The proposed approach is

inclusive and takes into account women, youth and marginalized groups. In the SEP, it states: “the strategy will be followed at all stages to ensure inclusive and effective engagement with all stakeholders. Women, youth and all vulnerable groups that will be directly and indirectly impacted by the programme will be consulted using reliable means of communication adapted to the culture and characteristics of the stakeholders group. The programme targets smallholder farmers, farmers’ organizations, cooperatives and micro, small and medium sized enterprises (MSMEs), in particular women and young people. It will also have an impact on disadvantaged or vulnerable individuals or groups living within the communities such as:

- Elderly
- People with disabilities
- People living in remote or inaccessible areas
- Internal displaced persons living in and/or out camps
- Refugees living in and out camps
- Female headed households

They will be engaged according to their interest in the Programme by taking into consideration the characteristics of their vulnerability.

Table 5: List of stakeholders (the specific list of stakeholders from baseline projects will also be considered) - See SEP

Stakeholder	Topics of engagement	Responsible party	Form and frequency of engagement
Ministry of Economy and Finance (Republic of Burkina Faso, Mali, Côte d'Ivoire, Senegal, Ghana)	Execution of project and coordinating between the implementing and executing agencies	Ministry of economy and Finance	Steering committee meetings Quarterly, continuous
Agricultural Bank of Burkina Faso Agricultural Bank of Mali Agricultural Bank of Niger National Investment Bank of Côte d'Ivoire ARB Apex Bank Limited of Ghana	Execution of project (setting up green credit lines/products)	PMU	Technical assistance trainings Steering committee meetings Products design meetings Project selection/approval process
Ministry of Agriculture in (Republic of Burkina Faso, Mali, Côte d'Ivoire, Senegal, Ghana)	Implementation of project	PMU	Quarterly through the progress reports and consultative meetings
Finance system actors: Central banks, insurances and regulators in each programme country	Project training and technical capacity	PMU	Technical capacity and business development trainings, round table and events Semi-annual/As necessary
Attijariwafa bank, AfDB , Islamic Development Bank	Cofinancing and strategic partnership to crowd in resources	PMU	Strategic partnership and cofinancing
Farmer organisations	Sub Project implementation	Banks in programme countries	Technical training, loans administration Continuous
Cooperatives	Sub Project implementation	Banks in programme countries	Technical training, loans administration Continuous
Micro Small Medium Enterprises (MSMEs)	Sub Project implementation	Banks in programme countries	Technical training, loans administration Continuous
Women and youth organisations	Sub Project implementation	Banks in programme countries	Technical training, loans administration Continuous

Renewable Energy Technologies Operators	Sub Project implementation	Banks in programme countries	Technical training, loans administration Continuous
Great Green Wall Initiative participants	Consultation, Knowledge sharing	RCU	Round table meetings, webinars, conferences Annual
GCF IFAD umbrella programme (IFAD, EU, WB, UNDP, FAO)	Knowledge sharing, creation of synergies	RCU	Round table Annual

Prevention of gender-based violence

GBV is an umbrella term for any harmful act that is perpetrated against a person's will and is based on socially ascribed (i.e. gender) differences between males and females. It includes acts that inflict physical, sexual or mental harm or suffering, threats of such actions, coercion and other deprivations of liberty. Under this program, prevention of sexual violence will be strengthened through (i) the use of Early Warning Indicators on sexual violence, (ii) the use of the early warning monitoring system in communities, (iii) information and awareness campaigns on the regulatory frameworks related to sexual violence and harassment in each country to inform communities of their rights and sensitize traditional leaders and communities (especially women's groups) to refer victims to the appropriate jurisdictions, (iv) ensure that the selected banks take into account specific measures to prevent and fight GBV.

107. The GBV risk assessment conducted by governments and UN agencies will serve as a baseline against which to strengthen or not strengthen GBV prevention and response measures. However, the Gender Action Plan below already includes actions that will be minimally implemented at the country level.
108. The role of community leaders will be essential in the prevention of GBV, especially women leaders who could more easily gain the trust of potential survivors. The protection of victims, witnesses and at-risk individuals will be reinforced. Opinion leaders, traditional and religious leaders, the media, the authorities and civil society have a significant role to play and can strongly influence the population and communities to change their behavior in relation to GBV. They will also be channels to make the complaint mechanism accessible and to collect the grievances of GBV victims within the framework of the program.

Gender Action Plan

Gender Action Plan ¹⁰³ :				
Objective/Impact: Increased climate resilient agricultural practice adoption and enhanced utilization of Renewable Energy Technologies by women led households, women in married households, people with disabilities, minority groups because of their social or cultural origin, FOs, Cooperatives and MSMEs and improvement of women's participation in governance and decision-making in the energy finance sector in 5 of GGW countries (Burkina Faso, Senegal, Mali, Côte d'Ivoire and Ghana). Outcome: Enhanced access to finance for implementation of climate resilient practices and Renewable Energy technologies by women led households, people with disabilities, minority groups because of their social or cultural origin, FOs, Cooperatives and MSMEs and improvement of women's participation in governance and decision-making in the energy finance sector. Means of Verification: Number of loans provide to women led households, women in married households, people with disabilities, minority groups because of their social or cultural origin, FOs, Cooperatives and MSMEs, and improvement of women's participation in governance and decision-making in the energy finance sector project reports providing analysis of data applicable to appropriate indicators				
Activity	Indicator and target	Timeline	Responsibility	Cost (EUR)
Output 2.1 – Generate knowledge, conduct studies, support knowledge transfer, provide training and monitor climate impacts				
1.1.1. Collection and analysis of the main roles, challenges and gender gap faced by men, women, women headed households, Youth and other marginalized groups in each targeted areas / country the first year of the project	Undertake a collection and analysis of gender gap at sites level 1 analysis on the main roles, challenges and gender gap faced by men, women, women headed households, Youth and other marginalized groups in each targeted areas / country the first year of the project	Y1 Baseline data will be collected during the 1 st year	PMU All countries	123,285
1.1.2. Property rights and control over assets	Number of women supported by the program to access secure land (at the	Y1-Y5	PMU all countries	57,533

¹⁰³ IFAD will provide the GCF (no later than the first quarter of the second year) with an updated version of the GAP after the baseline data collection and analysis planned in the first year. This data will allow for the clarification of some unquantified indicators and also specific targets groups that some countries may add.

Support women (individually or in groups) to obtain plots of land with documents that secure their investments and the products of their business. These documents will include deliberations of the cities (mairies), social agreements signed with the landowners and guaranteed by the administrative or customary authorities, sale or rental contracts, etc. 2.1.3: (a) Training of FOs, MSMEs and cooperatives to implement financed business plans for results (b) Establish a formalised institutional framework to ensure female staff within the selected banks (the Agricultural Bank of Burkina Faso, the National Investment Bank of Côte d'Ivoire, the ARB Apex Bank Limited of Ghana, the Agricultural Bank of Mali and Senegal) and the respective ministries of agriculture in Burkina Faso, Côte d'Ivoire, Ghana, Mali and The Agricultural Bank of Senegal and MFIs are part of management and decision making processes of loan products.	family or community level, etc.) and awareness raising At least 50 % (130 800) of women have access to secure land and information • Developed gender inclusive protocols and policy for lending products. • Percentage (%) (TBD) of women participating in the credit process in financing institutions. • <i>Number of households and individuals (623,500 females) with improved access to low-emission energy sources</i> (per country) • 1 gender inclusivity protocol • 1 Sexual Exploitation and Abuse policy • 1 Awareness raising policy • 1 Grievance redress mechanisms established (incorporating Gender Based Violence safeguards)	Y1- Y2 Baseline data will be collected during the 1st year <i>An update will be provided to the GCF after the baseline survey in year one</i>	PMU ALL COUNTRIES	414,500
--	---	---	--------------------------	----------------

(c) Gender values integrated into the selected banks internal guidelines and tools and gender promoted				
2.1.4 Support (training, support and advice, coaching, assistance with administrative procedures) FOs and MSMEs led by women and youth for increased participation <i>thus promoting Increased resilience and enhanced livelihoods of the most vulnerable people, communities and regions and Increased resilience of health and well-being, and food and water security</i>	• Women and youth-led FOs and MSMEs have the capacity to promote resilience and improve the livelihoods of their members. • Two (02) training session per country are completed per year (Y2 – Y4) • 500 MSMEs, FOs, Cooperatives of which 50% lead by women • <i>50% of males and 50% females benefiting from the adoption of diversified, climate resilient livelihood options (including fisheries, agriculture, tourism, etc.</i> • <i>Number of males (50%) and females (50%) benefiting from introduced health measures to respond to climate-sensitive diseases due to the impact of extreme climate-related disasters</i>	Y1-Y5 Baseline data will be collected during the 1 st year and will allow to determine the number of women and youth who will be targeted for trainings	PMU/ consultants All countries	328,760
2.2.2 Technical assistance to SMEs and financing institutions to enhance governance and policy gap	• Technical assistance to reduce the policy gap and non-compliance of activities with gender equality and equity policies • Number of women and men staff from the banks participating in TA • 01 Study is done on energy technologies to determine the appropriateness for and accessibility	Y1-Y5 Baseline data will be collected during the 1 st yaer <i>An update will be provided to the GCF after</i>	PMU/ consultants All countries	493,140

	for women and women headed households and other marginalized groups • Level of satisfaction with technical assistance : 100% of SMEs and financing institutions are satisfied with TA services provided	<i>the baseline survey in year one</i>		
2.3.2 Round table and events	• Ensure equal access to the round tables organized by the program (the number, format and modalities will be defined by the implementation teams within the budget). • 50% of women and 50% men participating in Round table and events	Y1-Y5	PMU/ consultants All countries	82,190
2.3.3: Organize South-South and triangular cooperation visit tour for experience sharing	• 50% of women and 50% men participating in South-South and triangular cooperation visit tour for experience sharing	Y3-Y5	PMU/ consultants All countries	410,950
2.3.4: Knowledge management and information sharing	• Information is available and accessible to all actors • Number of women and men having access to Knowledge and information (50% for each male/female category) • Number of males and females made aware of climate threats and related appropriate response (50% for each male/female category)	Y1-Y5	PMU/ consultants All countries	493,140
2.3.5: Prevention again gender-based violence: organize information and awareness campaigns on GBV	• Gender-based violence is properly addressed through a functional mechanism	Y1-Y5 ¹⁰⁴	PMU/ consultants All countries	123,285

¹⁰⁴ The development of the GRM and its operationalization will be effective in each country in the first year. The monitoring of the GRM will be done throughout the duration of the project until the end of the activities

	• Number of information and awareness sessions on GBV organized • An appropriate mechanism for resolving complaints of gender-based violence is in place and functioning • 100% of banks and other financial institutions have policies or strategies to prevent and fight GBV • Complainants' satisfaction with the performance of the GRM and the level of handling of registered complaints	Baseline data will be collected during the 1st year <i>An update will be provided to the GCF after the baseline survey in year one</i>		
2.2.7: Develop application protocols and framework for the award	• 50% of women and youth receive incentives and implement climate adaptation and mitigation measures and RET along agricultural value chains • Strategies to mitigate the risks of exclusion of women adopted and implemented • People with disabilities targeted to ensure incentives and inclusion in decision-making processes. • Number of women and youth participating to the AWARD • Number of women (TDD) that cover managerial positions in committees	Y1-Y5 Baseline data will be collected during the 1st year <i>An update will be provided to the GCF after the baseline survey in year one</i>	PMU/ consultants All countries	82,190
2.2.8: Enroll women and youth entrepreneurs to compete for the Sahel and Great Green Award.	• At least 30% of women and youth entrepreneurs participate in the competition for the Sahel Award.	Y1-Y5	PMU/consultants All countries	815,916.56

Communication campaign via visits, website, radio message.	• Number of campaigns and visits (PER COUNTRY) • 2 written campaigns • 12 radio campaigns • 2 visits			
Output 3.1- Knowledge Management strategy under the GGW UP Programme Coordination				
Establishing an operations services facility for GCF GGW UP projects. It will also support DAEs in the development of concept notes and request for PPF support	This facility would function as a multi-agency, multi-partner help desk that would convene thematic and sectoral expertise on gender assessment, requested for GCF Funding Proposal, on a demand basis by the 11 countries		PMU/consultants All countries	287,665
Project Management Unit (PMU) established in each bank				
Hire a full time Gender Specialist within each bank	Contract signed	Y1-Y6	M&E team	123,285
Monitoring and Evaluation (M&E) system established and operational				
Collection of gender disaggregated data for reporting on project performance indicators Hire specialists to conduct surveys and carry out data aggregation methodologies to obtain gender disaggregated data	Number of reports displaying gender disaggregated data 7 reports (per country) Number of female hired in the PMUs	Y1-Y6	PMU/Consultants All countries	410,950
3.1.1. Overall coordination and governance of the GGW UP	At least 2 female hired out of the 5 staff to be hired in the GGW RSP coordination mechanisms	Y1-Y6	PMU/Consultants All countries	123,285